

Chapter 5

Epistemology of Generative Relational Economics

This chapter develops the epistemology required for Generative Relational Economics (GRE) after the ontological distinctions of the preceding chapter have been established. Its role within Part II is to move from a view of reality to a disciplined account of knowledge of that reality. The previous chapter distinguished ontological, relational, and manifestational reality while preserving ontological suspension where stronger commitment is unnecessary. The present chapter asks how inquiry can move from manifestations, traces, representations, observations, records, models, and interventions toward warranted claims about relations and generative processes. Its objective is to specify the epistemic conditions under which GRE can make claims that are sufficiently warranted, appropriately scoped, historically situated, contestable, and revisable, while leaving broader theories of knowledge and stronger claims of transparent access outside the commitments required here.

The central difficulty arises from mediated access. Relational realities can become manifest through bodily, institutional, legal, linguistic, statistical, technological, or other conditions of manifestation, yet manifestation does not uniquely reveal the relational structure from which it arises. Similar manifestations may be compatible with different relational configurations, and similar relations may become manifest differently across contexts, scales, historical periods, or representational systems. The epistemic task therefore has the structure of an inverse problem. Inquiry often encounters what has become observable or representable and attempts to reconstruct the relations, fields, trajectories, and generative processes that may have contributed to it. Such reconstruction can be rigorous, but its warrant depends on explicit assumptions concerning manifestation, evidence, representation, boundary, scale, temporality, and alternative explanation.

Ontological suspension makes this epistemic discipline especially important. GRE can investigate relations without deciding in advance whether every analytically useful relation corresponds to an independently fundamental entity. It can also acknowledge relational realities whose persistence and causal efficacy depend partly on institutions, recognition, convention, classification, or symbolic mediation. These freedoms increase rather than reduce the burden of epistemic specification. A relation that is legally recognized, statistically represented, administratively legible,

behaviorally enacted, or computationally inferred can be real in one relevant sense while remaining misrepresented in another. A representation can be causally effective without being epistemically adequate, and a relation can be weakly represented while still being relationally consequential. The chapter accordingly separates questions of existence, manifestation, representation, causal efficacy, generativity, and normative evaluation wherever their conflation would strengthen a claim beyond its evidence.

The chapter's method is architectural. It begins from the target claim and identifies recurring epistemic problems that research designs may have to address when they study relational or generative phenomena, allowing different evidential forms and methodological traditions to enter according to their claim-relative contribution. These include non-invertibility of manifestation, partial observation, unequal access, boundary and scale selection, relational identity through change, higher-order structure, field reconstruction, generative attribution, historical trajectory, evidential heterogeneity, uncertainty, epistemic justice, machine mediation, and revisability. Formal definitions and propositions are introduced where they clarify the structure of these problems, but they remain analytical devices whose usefulness depends on the empirical and interpretive context in which they are applied.

The first movement, Sections 5.1–5.9, establishes the conditions of epistemic access to relations. Section 5.1 formulates the epistemic problem that follows from ontological suspension and identifies relational reality as an epistemic target reached through manifestational and representational mediation. Section 5.2 develops the inverse problem of manifestation and the consequences of non-invertibility, manifestational equivalence, contextual dependence, and multiple relational explanations. Section 5.3 then makes manifestation relations themselves objects of inquiry, because inference from manifestation to relation depends partly on what is known about the processes through which manifestation occurs. Sections 5.4 and 5.5 distinguish relations from their traces and from the symbolic, legal, administrative, statistical, linguistic, and computational representations through which those traces become usable in inquiry.

That movement also treats representation and observation as active conditions rather than neutral windows. Section 5.6 examines circumstances in which representations become performative: classifications, legal categories, administrative records, statistical descriptions, or algorithmic outputs can enter the relational system and affect subsequent relations. Section 5.7 develops partial observation, observer position, unequal access, distributed observation, concealment, and temporal gaps. Section 5.8 then shows that relational claims are indexed to boundaries and scales whose selection can materially alter what becomes visible and what remains outside the analysis. Section 5.9 closes the first movement by addressing identity and persistence, because longitudinal knowledge requires criteria for deciding when a changing relation should be treated as continuous, transformed, newly formed, or dissolved.

The second movement, Sections 5.10–5.14, extends epistemic inquiry beyond individual relations toward relational configurations and generative processes. Section 5.10 distinguishes relations among relations, collective relations, nested relations, multilayer configurations, and hyperrelational forms while preserving the additional uncertainty introduced by higher-order reconstruction. Section 5.11 devel-

ops the epistemology of relational fields and requires a field claim to establish more than relational density: distributed organization and position must matter to the target under investigation. These discussions provide the structural basis for Section 5.12, which defines generative relations relative to specified effects such as capacity formation, relation formation, structural transformation, constraint generation, altered future possibilities, or changes in the conditions of subsequent transformation.

The distinction between relational existence and generative effect is central to the chapter. Evidence that a relation exists does not by itself establish what the relation generates. Section 5.13 therefore develops the epistemic problem of generative attribution through mechanisms, counterfactual dependence, multiple causation, distributed generation, delayed effects, branching pathways, and emergent outcomes. The chapter therefore uses a plural architecture of causal and generative warrant. The appropriate burden depends on the kind and strength of the attribution being made. Section 5.14 then adds historical depth by distinguishing relational trajectories, generative trajectories, path dependence, genealogy, historical counterfactuals, and re-epistemization. Historical knowledge is treated as revisable reconstruction under changing evidence and categories rather than as retrospective access to a fixed and fully recoverable past.

The third movement, Sections 5.15–5.22, turns from the objects of relational and generative knowledge toward the evidential and institutional conditions of warranted inquiry. Section 5.15 differentiates documentary, administrative, legal, behavioral, testimonial, ethnographic, network, longitudinal, and computationally inferred evidence according to the relational properties they can support and the characteristic mediations through which they are produced. Section 5.16 then specifies the additional evidential burden associated with generative claims, including downstream effects, capacity formation, structural transformation, value propagation, recirculation, and foreclosure. Evidential pluralism in these sections requires claim-sensitive judgment about what each evidential form can reveal, what it may conceal, and how convergent or conflicting evidence should alter the warranted claim while preserving differences among evidential forms.

Section 5.17 organizes uncertainty into a multidimensional confidence profile rather than a single undifferentiated score. Relational, manifestational, generative, boundary, scale, temporal, model, and explanatory uncertainty can vary independently, and a defensible claim should preserve those differences where they matter. Section 5.18 adds epistemic justice by examining how visibility, voice, representational authority, classification, institutional legibility, evidence-production capacity, and generative attribution are unequally distributed. Justice-sensitive inquiry retains evidential standards while asking whether the conditions under which evidence becomes producible, preservable, credible, and actionable systematically exclude relevant relations or contributors.

Machine mediation intensifies several of these concerns. Section 5.19 examines machine observation, algorithmic classification, computationally inferred relations, AI-assisted reconstruction, machine-generated representation, epistemic asymmetry, representational authority, and contestability. Computational scale and precision can enlarge epistemic reach while also introducing opacity, inherited categories, uneven

provenance, and new recursive effects when outputs become operational within the systems being studied. Machine systems are therefore treated as components of an epistemic architecture rather than as independent guarantors of knowledge.

The chapter closes by making revisability an explicit property of GRE inquiry. Section 5.20 develops revision through new evidence, intervention, historical transformation, category failure, reflexive observation, and preservation of serious alternative interpretations. Section 5.21 consolidates the chapter into a set of epistemic commitments: partial knowledge, explicit representation, explicit boundary and scale, explicit uncertainty, mechanism-sensitive attribution, evidential pluralism, historical sensitivity, revisability, and epistemic humility. Section 5.22 then translates these commitments into methodological design burdens. The transition is deliberate. Epistemology specifies what a defensible relational or generative claim requires; methodology must organize inquiry so that evidence capable of supporting, qualifying, contesting, or revising such claims can actually be produced.

Across these movements, the chapter advances a position of bounded epistemic ambition. GRE is permitted to ask large questions concerning distributed relations, historically evolving institutions, heterogeneous forms of evidence, and generative transformations across scales. It is also required to state the conditions under which those questions become answerable. Strong formalization, extensive data, institutional recognition, computational sophistication, or representational precision can improve inquiry while the mediations through which knowledge is produced remain explicit. A narrow claim with explicit scope and strong evidential fit can therefore possess greater epistemic warrant than a more comprehensive claim whose boundary, representation, or inferential pathway remains obscure.

The resulting epistemology is demanding while remaining compatible with incomplete knowledge. It permits provisional relational maps, partial reconstruction, competing explanations, graded generative attribution, and historically revisable knowledge while preserving criteria by which stronger and weaker claims can be distinguished. The chapter therefore supplies GRE with an epistemic position appropriate to an open research programme: knowledge can be situated while preserving standards of warrant, plural while preserving evidential differentiation, formally articulated while retaining ontological restraint, and revisable while retaining epistemic content. With this architecture in place, the book can move from the conditions of warranted knowledge to the practical organization of research praxis and methodology.

5.1 The Epistemic Problem after Ontological Suspension

Chapter 4 established an ontological architecture that is deliberately strong enough to support substantive inquiry and deliberately restrained enough to avoid converting analytical usefulness into final metaphysical settlement. Ontological reality, relational reality, and manifestational reality identify distinct registers of analysis; relational reality can precede perception and symbolization; manifestation is generated through historically and materially conditioned relations; and representations can enter the systems they describe and alter subsequent relations. These commitments

create the starting problem of the present chapter. If reality can exceed what is manifest, if manifestation can be selective and non-invertible, and if representation can be both evidential and performative, then epistemology cannot be reduced to the accumulation of visible facts. The role of this section is to formulate that problem precisely. It develops five connected tasks: knowledge under ontological suspension, specification of relational reality as an epistemic target, the evidential role of manifestational reality, representation as mediated access, and disciplined inference across the three registers.

The argument proceeds from a distinction between *ontological settlement* and *epistemic warrant*. GRE does not require a final account of what an object ultimately is before inquiry can establish that some configuration persists, exhibits a sufficiently stable organization, produces causal effects, or changes under specified interventions. Scientific and philosophical debates concerning ontological commitment, empirical adequacy, model-based representation, and the social conditions of evidential judgment already show that useful inquiry can distinguish the strength of an empirical or representational claim from the strength of a metaphysical one (Quine 1948; Fraassen 1980; Giere 2004; Longino 1990). GRE extends that discipline into a relational setting. The absence of final ontological settlement does not erase the possibility of knowledge; it increases the need to state what is being claimed, through which evidential relations, and with what residual uncertainty.

Claim 5.1 (Epistemic consequence of ontological suspension). *Ontological suspension permits substantive epistemic claims when the epistemic target, evidential access, inferential relation, and scope of the claim are sufficiently specified. Suspension limits the strength of ontological conclusion; it does not by itself diminish the evidential strength of a well-supported relational or manifestational claim.*

Claim 5.1 provides the chapter's initial discipline. A researcher may be highly confident that a monetary arrangement structures access to resources, that a legal category alters feasible action, that an institutional ranking changes subsequent behavior, or that a trust relation has stabilized across repeated interactions while remaining agnostic about the ultimate metaphysical constitution of money, law, institutional identity, or selfhood. The claim can be strong at the level it addresses and suspended at another. Epistemic restraint therefore concerns the calibration of claims rather than a generalized weakening of knowledge.

This calibration also prevents a recurrent movement from uncertainty at one level to skepticism at every level. Uncertainty about ontological fundamentality does not imply uncertainty about observed manifestations. Uncertainty about the unique relational structure underlying a manifestation does not imply that the manifestation itself is doubtful. Conversely, confidence in a manifestation does not establish confidence in a unique relational explanation. The three-register architecture therefore supplies an epistemic differentiation: different claims can carry different degrees and kinds of warrant even when they concern the same event, object, or process.

5.1.1 Knowledge under Ontological Suspension

Ontological suspension is epistemically productive when it prevents an inquiry from demanding stronger settlement than its explanatory task requires. The relevant ques-

tion is not whether inquiry can proceed in the absence of any ontology; Chapter 4 has already rejected such a position. GRE identifies objects, relations, persistence conditions, causal effects, manifestations, scales, and historical transformations, and each of these choices carries ontological commitments. The epistemic question is narrower: which of those commitments must be settled for a particular claim to become warranted, and which may remain open without damaging the inference?

This distinction can be understood through the difference between *target commitment* and *ultimate commitment*. A target commitment identifies what the inquiry is attempting to know: for example, the organization of a labor relation, the persistence of a monetary convention, the distribution of access to computational capacity, or the institutional reproduction of a legal status. An ultimate commitment would answer a stronger question concerning what kind of being such an object finally possesses. The first is unavoidable for research. The second becomes necessary only when competing ontological accounts alter explanation, measurement, intervention, or evaluation.

Definition 5.2 (Epistemic target). An *epistemic target* is the explicitly delimited object, relation, process, configuration, mechanism, trajectory, or property about which an inquiry seeks warranted knowledge under specified spatial, temporal, scalar, and conceptual conditions.

Definition 5.2 makes delimitation part of epistemic responsibility. A statement such as “trust increased” remains under-specified until the inquiry clarifies whose trust, within which relational domain, across what interval, through what manifestations, and relative to which criteria of persistence or change. The same requirement applies to economic claims. “Capital became more powerful” can refer to ownership concentration, control over future investment, institutional claims on income streams, influence over technological development, command over generative infrastructure, or several interacting mechanisms. The epistemic target must therefore be constructed with enough precision that evidence can bear on it.

Ontological suspension then operates selectively. If two metaphysical interpretations generate the same expectations for the target under the inquiry’s present scale, evidence, and intervention possibilities, their disagreement can remain suspended. If they imply different persistence conditions, different causal mechanisms, or different responses to intervention, the disagreement becomes epistemically operative and must be reopened. This preserves the distinction developed in Chapter 4 between non-operative and operative ontological disagreement while giving it a direct epistemological consequence.

The approach differs from a simple instrumentalism. GRE does not treat objects and relations as disposable fictions merely because their ultimate ontology remains unsettled. A relational reality may possess stable causal efficacy within the system under study. The monetary unit used in an accounting system, the institutional category through which a person receives or loses an entitlement, or the recognized identity of an organization can produce consequences that remain explanatorily indispensable. Epistemic warrant can therefore attach to a relationally real object without requiring that object to be fundamental in a stronger metaphysical sense.

A second consequence concerns historical change. The categories through which

an epistemic target is identified can themselves have histories. Hacking's work on historical ontology and Kuhn's account of historically organized scientific practice provide different precedents for taking the historical formation of concepts, objects, and problem structures seriously (Hacking 2002; Kuhn 1962). GRE therefore treats target specification as revisable. A category that once captured the relevant organization can become inadequate when new relations, technologies, scales, or manifestations become causally important. Revisability does not make the target arbitrary; it makes the history of its adequacy part of epistemic inquiry.

Remark 5.3 (Scoped knowledge). A claim can be strongly warranted within a specified epistemic target and evidential domain while stronger claims concerning ultimate ontology, exhaustive mechanism, universal scope, or normative validity remain suspended.

Remark 5.3 will recur throughout this chapter. GRE's epistemology is intended to support strong local knowledge without quietly converting local warrant into universal closure.

5.1.2 Relational Reality and Epistemic Targeting

Relational reality introduces a distinctive epistemic target because the relevant object need not be directly observable, consciously recognized, or symbolically articulated. Chapter 4 defined a relationally real object through emergence within a generative-relational system, sufficient persistence and stability across a relevant spacetime domain, and causal efficacy within that system. Each component creates an evidential burden. An inquiry must identify evidence of emergence, evidence of continuity or patterned persistence, evidence that the object retains enough organization to be distinguished across change, and evidence that it makes a difference to subsequent processes.

The epistemic target is therefore rarely an isolated entity. It includes the relational configuration through which the object emerges and the domain within which its causal efficacy is assessed. Consider money. A token, inscription, or account balance can be observed directly, but the relational reality of money depends on broader arrangements of recognition, law, exchange, accounting, expectation, institutional enforcement, and practical use. Observing a note or a number is evidence of a manifestation. Knowing the monetary relation requires reconstruction of the system in which that manifestation acquires persistence and efficacy.

The same point applies to objects with less obvious material manifestation. Trust can become relationally real before either party states that trust exists. A sense of continuing self can organize action before it is reflectively narrated. A religious object can organize obligations, spaces, temporal rhythms, and collective practices while its ultimate ontological status remains suspended. A contingent configuration can shape what later becomes aesthetically salient before the subject recognizes the contingency or articulates an aesthetic judgment. In each case, the epistemic challenge is to avoid identifying the absence of explicit manifestation with the absence of relational reality.

Definition 5.4 (Relational epistemic target). A *relational epistemic target* is an epistemic target whose identity, persistence, causal role, or transformation depends ma-

terially on the generative-relational configuration within which it emerges and acts.

Definition 5.4 does not imply that every relevant cause is relational in the same sense. Bodies, infrastructures, material resources, ecological conditions, computational hardware, and other objects may enter the configuration under domain-specific ontologies. The definition concerns what must be reconstructed for the target to be adequately known. If the economic reality of a firm's authority depends on legal recognition, contracts, organizational routines, labor relations, financing, and expectations, then knowledge of that authority cannot be reduced to inspection of any single component.

Relational targeting also changes how absence is interpreted. Failure to observe an explicit claim of authority does not show that authority is absent. Failure to observe monetary exchange does not show that monetary dependence is absent. Failure to record a relation in an administrative database does not show that the relation does not organize action. At the same time, the framework does not authorize inference of hidden relations whenever evidence is missing. The relevant burden is constructive: an inquiry should specify what manifestations or traces a proposed relational reality would be expected to generate, under what conditions those manifestations may fail to appear, and which alternative relational structures remain compatible with the available evidence.

This requirement places mechanism and history near the center of epistemic targeting. A relational reality can be identified more securely when different evidential routes converge on its persistence and causal role: repeated behavior, institutional records, testimony, counterfactual variation, intervention, historical traces, and changes that follow the relation's stabilization or dissolution. Convergence does not guarantee unique reconstruction, but it can narrow the compatible relational field and strengthen a scoped claim.

The target may also be distributed. No participant need possess a complete representation of the relation in which they participate. Markets, administrative systems, infrastructures, supply chains, reputational fields, and scientific communities can contain locally informed actors whose combined practices generate structures that exceed each actor's explicit knowledge. Epistemology must therefore distinguish the knowledge possessed by participants from the knowledge an inquiry attempts to reconstruct about the relational configuration as a whole.

5.1.3 Manifestational Reality and Evidential Access

Manifestational reality supplies the principal domain through which inquiry encounters relational reality, but manifestation and evidence are not synonymous. A manifestation becomes evidence only relative to an epistemic target and an account of how the manifestation is generated. A market price can be evidence about exchange conditions, but its interpretation depends on institutional rules, market structure, temporal aggregation, strategic behavior, and what the price fails to record. A legal document can be evidence of a recognized status, but knowledge of effective legal power may also require enforcement practice, administrative discretion, and access to remedies. A behavioral trace can be evidence of interaction without uniquely revealing intention, trust, coercion, or preference.

The distinction can be stated explicitly.

Definition 5.5 (Evidential status of a manifestation). The *evidential status of a manifestation* is the warranted relation by which a manifestation is treated as bearing information about a specified epistemic target under an articulated account of its generation, preservation, selection, and interpretation.

Definition 5.5 places the evidential burden on the relation between manifestation and target. The same manifestation can have different evidential significance for different questions. A wage payment can provide strong evidence that a payment occurred, moderate evidence concerning a contractual employment relation, weaker evidence concerning the worker's experienced autonomy, and almost no direct evidence concerning the complete structure of household care or future bargaining power. Evidential strength therefore cannot be assigned to the datum independently of the claim.

This position is compatible with a broader philosophy-of-science distinction between observed phenomena and stronger claims concerning the structures that explain them. Van Fraassen's constructive empiricism emphasizes the difference between empirical adequacy and stronger realist commitment, while Giere's model-based account treats representation as purposive and selective rather than as simple identity between model and world (Fraassen 1980; Giere 2004). GRE does not adopt either position wholesale. Their relevance lies in a shared caution: evidential accessibility and ontological exhaustiveness should not be conflated.

Manifestational access is also conditioned. Instruments, categories, languages, administrative rules, interfaces, bodily capacities, and historical circumstances determine what can become legible. Longino's contextual empiricism is useful here because evidential relevance depends partly on background assumptions and on the critical contexts through which those assumptions can be challenged (Longino 1990). In GRE, such background conditions are not merely external philosophy-of-science considerations. They can themselves be relationally real parts of the system under investigation. A statistical classification can determine which people become visible to an institution; an interface can determine which actions can be registered; a legal category can determine which harms become actionable; a platform metric can determine which activities are recognized as valuable.

Consequently, an inquiry should examine the conditions under which manifestations are produced, not only the manifested content. A stable manifestation relation can support repeated inference. A changing manifestation relation can make an apparently stable time series epistemically misleading. A new measurement technology can reveal previously inaccessible differences while simultaneously changing behavior. The same symbolic category can preserve continuity in records even when the relational reality it once tracked has transformed. Evidential access therefore has a history.

Claim 5.6 (Manifestation-evidence distinction). *A manifestation acquires evidential force for a relational claim only through a warranted account of the relations connecting the manifestation to the epistemic target. Manifestational presence alone does not determine the strength or uniqueness of the inference.*

Claim 5.6 does not make evidence inaccessible. It clarifies where evidential work

occurs. Repeated manifestations, independent traces, intervention responses, mechanism knowledge, and historical continuity can strengthen the relation between manifestation and target. The later sections of this chapter will differentiate direct, indirect, latent, hidden, documentary, behavioral, testimonial, ethnographic, formal, and computational forms of evidence. Section 112 establishes the common structure within which those differences matter.

5.1.4 Representation and Mediated Access

Representation introduces another layer between manifestation and epistemic claim. A manifestation can be recorded, classified, translated, measured, aggregated, modeled, narrated, visualized, or encoded. Each operation can preserve some relations, suppress others, create new comparabilities, and add assumptions that were not present in the manifestational field itself. Representation is therefore an epistemic resource and an epistemic transformation.

A distinction between manifestation and representation is particularly important in economic inquiry. A transaction is one event; an administrative record of the transaction is another object; a statistical aggregation of many recorded transactions is another; a model estimated from the aggregate is another; and a policy interpretation of the model introduces yet another representational layer. These layers can be tightly connected, but they should not be treated as interchangeable. An error, selection rule, classification change, or aggregation procedure introduced at one stage can propagate through the later stages while leaving the final representation internally precise.

Quine's holistic account of empirical testing and Longino's contextual account of evidential relevance both provide precedents for treating observation and inference as embedded within wider conceptual and social structures rather than as isolated pairings between single statements and neutral facts (Quine 1951; Longino 1990). GRE adds a generative-relational concern: the representational apparatus can become causally active within the domain it represents. A credit score can alter access to credit; a ranking can alter applications and funding; a diagnosis can reorganize self-understanding and institutional treatment; a forecast can alter investment; and a legal classification can change the feasible actions of the classified subject. Representation can therefore mediate knowledge and later become part of the relational reality whose future manifestations are observed.

Definition 5.7 (Representational mediation). *Representational mediation* denotes the set of operations through which manifestations or traces are selected, encoded, classified, transformed, aggregated, modeled, narrated, or otherwise organized into forms used for epistemic inference.

The definition makes representation analyzable without assuming that every representation is distorting. Mediation can increase epistemic capacity. A map can reveal spatial relations that are difficult to perceive locally. A statistical model can expose patterns across large populations. An archive can preserve traces beyond the lifetime of their producers. A legal record can stabilize a claim across institutional contexts. A simulation can make counterfactual structure explorable. The epistemic value of representation depends on what relations are preserved, what transforma-

tions are introduced, and what target the representation is used to investigate.

This also means that representational accuracy has several dimensions. A representation can reproduce observed manifestations accurately while mischaracterizing the relations that generated them. It can preserve a causal mechanism while omitting distributive consequences. It can capture average behavior while suppressing heterogeneity. It can represent a historical configuration adequately for one period and poorly after the underlying relations change. Epistemic evaluation should therefore specify the dimensions along which adequacy is claimed rather than treating representation as globally faithful or unfaithful.

In GRE, representational mediation acquires particular importance because the objects of political economy are frequently constituted partly through the representations used to govern them. Prices, property titles, balance sheets, national accounts, credit ratings, legal statuses, employment categories, performance indicators, and algorithmic scores are simultaneously representations and operative institutional objects. The epistemology of such domains must therefore distinguish evidence about a relational reality from representations that participate in producing that reality.

5.1.5 Inference across Registers and Epistemic Restraint

The final problem concerns inference across the three registers. The architecture developed in Chapter 4 permits several legitimate inferential movements, but none is automatically reversible. A manifestation can support a claim about relational reality when the manifestation relation is sufficiently understood. A relational configuration can support predictions about likely manifestations under specified conditions. A representation can preserve selected features of manifestations and permit comparison across cases. Evidence concerning persistent causal organization can also motivate stronger ontological questions. The epistemic danger appears when a warranted movement in one direction is silently converted into identity across registers.

Three forms of overreach are especially important. The first is *manifestational overreach*: treating what appears under one set of conditions as an exhaustive account of the relational reality. The second is *representational overreach*: treating a successful symbolic, statistical, computational, or formal representation as identical with the target. The third is *ontological overreach*: treating a well-supported relational explanation as sufficient proof of ultimate metaphysical constitution. The non-entailment discipline of Chapter 4 becomes an epistemic discipline here.

Definition 5.8 (Cross-register inference). A *cross-register inference* is an inference in which evidence or claims articulated primarily at one register of reality are used to support a claim concerning another register. Its warrant depends on the explicitly or implicitly posited relations connecting the registers under the relevant historical, material, and manifestational conditions.

Definition 5.8 makes the inferential bridge visible. For example, inferring relational trust from repeated cooperative behavior requires assumptions about the relation between behavior and trust. Inferring institutional authority from legal documentation requires assumptions about recognition and enforcement. Inferring preference from choice requires assumptions concerning feasible alternatives, information, strategic behavior, and constraint. Inferring an underlying generative structure

from a statistical pattern requires assumptions about causal architecture and alternative explanations. Each inference can be warranted, but the warrant is located in more than the observed manifestation alone.

Non-invertibility makes this problem systematic. Different relational realities can generate similar manifestations, and one relational reality can generate heterogeneous manifestations across contexts. The inverse problem therefore produces a set of compatible explanations rather than a guaranteed unique solution. Epistemic progress can consist in reducing that set, discriminating among alternatives, identifying invariants, discovering new manifestations, changing the conditions of observation, or intervening on the system. Complete inversion is only one possible epistemic achievement and often an unrealistic one.

Claim 5.9 (Epistemic restraint across registers). *The strength of a claim inferred across registers should be limited by the demonstrated stability, specificity, and scope of the relations that connect the evidential manifestation or representation to the epistemic target.*

Claim 5.9 gives GRE a practical rule for epistemic calibration. A highly specific and stable manifestation relation can support comparatively strong inference. A historically unstable, institutionally mediated, strategically manipulated, or weakly understood relation requires wider uncertainty. Multiple independent evidential routes can strengthen the claim when their errors and assumptions are sufficiently distinct. Apparent convergence among representations generated from the same underlying classification or dataset provides less independent support than its numerical multiplicity might suggest.

The rule also preserves room for knowledge that is historical and partial. A past relation may be reconstructable only through surviving traces whose production was unequal. A marginalized actor's experience may be poorly represented in official archives but partially recoverable through testimony, material traces, later records, or the consequences of institutions. A new scientific instrument can disclose structures inaccessible to earlier observers. An AI system can reorganize vast amounts of evidence while inheriting the exclusions embedded in its sources. Epistemic warrant therefore depends partly on understanding the history and distribution of access to manifestations.

Section 112 establishes the basic epistemic problem that follows from the ontology of Chapter 4. GRE can pursue knowledge without final ontological settlement because epistemic claims can be targeted, scoped, and warranted at different levels. Relational reality supplies an object of inquiry whose evidence may precede explicit recognition; manifestational reality provides access without guaranteeing exhaustive disclosure; representation expands and transforms that access; and inference across registers requires explicit bridges rather than assumed identity. The next section develops the most immediate technical consequence of this architecture: the inverse problem of manifestation, in which an inquiry must reconstruct relational configurations from manifestations that may be compatible with more than one generative history.

5.2 The Inverse Problem of Manifestation

Section 5.1 established that manifestation supplies epistemic access without guaranteeing exhaustive disclosure of relational reality. The present section develops the most immediate consequence of that position: inquiry frequently proceeds in the inverse direction, beginning from what has become manifest and attempting to reconstruct the relational configurations, histories, and generative conditions compatible with it. The role of this section is to formulate that reconstruction problem as an epistemic problem rather than merely a formal property of manifestation mappings. Its objective is to clarify what can be known when several relational configurations remain compatible with the same manifestation, when one configuration can manifest differently across contexts, and when the mapping itself changes historically. The discussion proceeds through relational reconstruction, non-invertibility, manifestational equivalence, multiple relational explanations, contextual dependence, and historical dependence. The method is inferential: it treats reconstruction as a constrained comparison among compatible relational accounts whose warrant can increase without requiring unique recovery of a single complete antecedent configuration.

The inverse problem should be distinguished from ordinary uncertainty about whether an observation is correct. A manifestation can be accurately recorded and still fail to identify uniquely the relational configuration that generated it. A price can be measured precisely while remaining compatible with heterogeneous market structures, expectations, constraints, and institutional conditions. A legal status can be documented unambiguously while leaving open several paths through which it was produced and several relations through which it is maintained. A behavioral regularity can be reproducible while admitting different accounts of trust, coercion, habit, incentives, or institutional filtering. The epistemic difficulty therefore concerns the relation between manifestation and reconstruction, not simply noise in the manifestation itself. This problem belongs to a wider family of underdetermination problems in which evidence constrains explanation without necessarily selecting a unique account (Duhem 1954; Quine 1951; Stanford 2023).

Definition 5.10 (Inverse epistemic problem of manifestation). The *inverse epistemic problem of manifestation* is the problem of determining which claims about relational reality are warranted from one or more manifestations when the relations connecting relational configurations to those manifestations may be selective, many-to-one, one-to-many, context-dependent, historically variable, or only partially known.

Definition 5.10 makes the epistemic target central. An inquiry need not reconstruct every relation that participated in the history of a manifestation. It may seek only the relations relevant to a specified causal question, institutional mechanism, distributional pattern, or generative consequence. Invertibility is therefore too strong as a universal standard. The more appropriate question is whether the target of inquiry is *identifiable enough* under the available manifestations, assumptions, and conditions to support the intended claim.

5.2.1 From Manifestation to Relational Knowledge

Movement from manifestation to relational knowledge requires an inferential bridge. Section 5.1 called such movement a cross-register inference. The inverse problem makes its structure explicit. Let Y denote a selected manifestational domain, X a selected domain of relational configurations, and C the material, historical, institutional, perceptual, or technical conditions under which manifestation occurs. For an observed manifestation $y \in Y$, the epistemic task is to determine which elements of X remain compatible with y under C and which features of those elements can be warranted from the available evidence.

A useful starting object is the compatibility set developed in Chapter 4. For present epistemic purposes, however, the set should include the evidential and modeling assumptions through which compatibility is judged. Let E denote the available evidential corpus and A the explicit auxiliary assumptions of the inquiry. The epistemically admissible compatibility set is represented in Equation 5.1.

$$\mathcal{K}(y \mid C, E, A) = \{x \in X \mid x \text{ remains compatible with } y \text{ under } C, E, A\}. \quad (5.1)$$

Equation 5.1 is intentionally general. It does not assume that compatibility is probabilistic, metric, deterministic, or reducible to one formal model. In one inquiry, compatibility may be defined by a causal model and a likelihood. In another, it may depend on archival consistency, institutional sequence, legal records, qualitative traces, or process evidence. The equation serves to expose the epistemic structure: the observed manifestation constrains the candidate relational domain, but auxiliary assumptions and available evidence also shape what remains admissible.

This formulation makes two kinds of epistemic progress visible. The first is *eliminative progress*: additional evidence or stronger constraints remove relational configurations from $\mathcal{K}$. The second is *structural progress*: even when several configurations remain, inquiry discovers features shared across them. A researcher may be unable to identify one complete generative history while becoming highly confident that every serious account requires, for example, a particular institutional dependency, a threshold effect, a gatekeeping relation, or a temporal ordering. Relational knowledge can therefore increase through identification of invariants across compatible reconstructions.

Definition 5.11 (Relational identifiability). An epistemic target is *relationally identifiable* relative to manifestations, conditions, evidence, and auxiliary assumptions when the admissible compatibility set is sufficiently constrained that the target property or relation takes a determinate, or appropriately bounded, form across the configurations that remain epistemically viable.

Relational identifiability in Definition 5.11 is weaker than complete inversion. A target can be identifiable even when the full configuration is not. If every viable reconstruction contains the same access restriction, the restriction can be well supported despite disagreement about other relations. If several histories produce the same observed institution but all require a durable asymmetry in control over a resource, that asymmetry may be identifiable even when the exact historical pathway

is not. The epistemic aim is therefore calibrated reconstruction rather than maximal reconstruction by default.

This also places model choice under scrutiny. Models can narrow the compatibility set by imposing structure, but narrowing through assumption should not be confused with narrowing through evidence. A representation can produce a unique estimate because the model excludes alternatives by construction. The uniqueness of the output then reflects both evidence and imposed structure. Model-based scientific representation is therefore epistemically powerful precisely when the relation between target, idealization, and representational purpose remains explicit (Giere 2004).

5.2.2 Non-Invertibility

Non-invertibility becomes epistemically significant when the manifestations available to inquiry fail to support unique recovery of the target relational configuration at the required scale. Chapter 4 treated this as a structural property of manifestation mappings. Chapter 5 adds the inferential consequence: non-invertibility creates a disciplined space of possible reconstructions and therefore requires explicit management of underdetermination.

Several sources of non-invertibility should be distinguished. *Compression* occurs when the manifestation preserves fewer distinctions than the relational configuration contains. *Aggregation* occurs when several units or relations contribute to a common manifest value. *Convergence* occurs when different generative trajectories reach similar manifest states. *Contextual transformation* occurs when the same relational configuration becomes manifest differently under changing conditions. *Selective observation* occurs when only some relations generate traces accessible to the inquiry. *Representational standardization* occurs when heterogeneous manifestations are translated into a common administrative, statistical, legal, or computational form. These mechanisms can coexist.

The epistemic implication is not that every explanation is equally plausible. Evidence can sharply constrain the compatibility set while leaving more than one serious relational account. The classical problem of underdetermination is useful here because it separates multiplicity of compatible explanation from unrestricted arbitrariness (Duhem 1954; Quine 1951; Stanford 2023). GRE adds two qualifications. First, the competing accounts may differ not only in theory but in the relational histories they reconstruct. Second, the mappings that produce evidence can themselves be historically generated and causally operative within the system under investigation.

Claim 5.12 (Non-invertibility and calibrated knowledge). *Manifestational non-invertibility limits the uniqueness and scope of relational reconstruction, while remaining compatible with strong knowledge of properties that are stable across the admissible compatibility set or independently supported through additional evidential relations.*

Claim 5.12 prevents two opposite errors. One error treats a determinate manifestation as if it carried a unique relational antecedent. The other treats non-unique reconstruction as if it eliminated knowledge altogether. The appropriate response is to state which parts of the reconstruction are stable, which remain conditional, and

which depend on assumptions that alternative accounts do not share.

Intervention can sometimes improve identifiability by generating new manifestations under deliberately altered conditions. Comparative cases can serve a similar function when they vary relations that competing explanations treat differently. Historical evidence can discriminate between mechanisms that predict the same present manifestation but require different sequences. Qualitative evidence can reveal relations erased by standardized indicators. Formal analysis can characterize equivalence classes of models or mechanisms. These methods do not share a single epistemic form, yet each can reduce or reorganize the compatibility set. Methodological plurality therefore becomes a response to relational underdetermination rather than a generic preference for variety.

5.2.3 Manifestational Equivalence

The inverse problem becomes especially difficult when distinct relational configurations are manifestationally equivalent relative to the criteria of an inquiry. Chapter 4 defined manifestational equivalence as local indistinguishability under a specified manifestation regime. The epistemic task is to determine what, if anything, that equivalence licenses about the underlying relations.

Definition 5.13 (Epistemic manifestational equivalence). Two relational configurations are *epistemically manifestationally equivalent* for an inquiry when the manifestations available under the inquiry's operative conditions and distinctions do not provide sufficient grounds to discriminate between them with respect to the specified epistemic target.

This definition is deliberately inquiry-relative. Two configurations can be equivalent for a narrow target and distinguishable for another. Two firms may be equivalent with respect to an observed quarterly profit while differing sharply in debt exposure, labor organization, ecological dependence, and technological trajectory. Two households may have the same income while differing in unpaid care burdens, legal vulnerability, asset ownership, and access to social networks. Two AI systems may produce indistinguishable outputs on a benchmark while relying on different training histories, architectures, compute requirements, or governance arrangements. The manifestation criterion determines the domain of equivalence; it does not erase relational difference outside that domain.

Manifestational equivalence should also be distinguished from measurement error. Perfect measurement of the operative manifestation may leave the equivalence intact. Increasing precision can reduce uncertainty around y without reducing multiplicity in $\mathcal{K}(y \mid C, E, A)$. If the measurement captures only a projection of the relational configuration, more precise measurement of that projection does not automatically recover omitted dimensions. Epistemic improvement may require a different manifestation, a new intervention, a changed scale, or access to another trace rather than greater precision of the same indicator.

This distinction is important for political economy because standardized measures often become authoritative precisely through their apparent comparability. Prices, incomes, productivity measures, credit scores, rankings, standardized legal categories, and algorithmic classifications create manifestational domains in

which heterogeneous relations become comparable for practical purposes. Such comparability can be administratively or economically powerful while remaining epistemically incomplete. A common scale can organize action without establishing that the underlying relational configurations are commensurable in every dimension.

Remark 5.14 (Equivalence and epistemic humility). Manifestational equivalence warrants restraint concerning relational difference; it does not warrant a presumption of relational identity. The absence of discriminating manifestation is an epistemic condition, not proof that no relevant difference exists.

Remark 5.14 is especially important where some relations are systematically less likely to leave durable traces. Informal labor, care work, coercion, tacit exclusion, undocumented exchange, affective dependence, and relations involving actors with limited institutional voice can be weakly represented in official records. A manifestational regime can therefore produce equivalence partly because it lacks distinctions through which certain relational differences become legible. The inverse problem is consequently linked to the social distribution of epistemic visibility.

5.2.4 Multiple Relational Explanations

When several relational explanations survive the available evidence, GRE requires criteria for comparing them without pretending that one must always be selected immediately. The objective is neither permanent suspension nor premature closure. It is to make the remaining alternatives explicit and identify what additional evidence, interventions, or conceptual distinctions would materially discriminate among them.

Multiple explanations can differ in several ways. They can identify different relations as causally central, assign different roles to the same relation, specify different historical sequences, draw boundaries around the system differently, or treat a manifestation as generated at different scales. They can also agree on a local mechanism while disagreeing about its wider embedding. Two explanations of wage change, for example, might agree that bargaining institutions matter while differing on whether technological change, labor-market concentration, legal structure, organizational strategy, or macroeconomic conditions explain the institutional shift that preceded the observed outcome.

A relational comparison should therefore examine more than goodness of fit to the manifestation. It should ask whether the explanation preserves relevant temporal order, accommodates independent traces, survives comparison across contexts, specifies the mechanism through which relations generate the manifestation, and remains coherent when the analysis changes scale. Explanations should also expose rather than hide their auxiliary assumptions. Longino's account of evidential reasoning is useful here because it emphasizes that the relevance and interpretation of evidence depend on background assumptions that become more reliable when subjected to critical scrutiny from alternative perspectives (Longino 1990).

Proposition 5.15 (Comparison of relational reconstructions). *When several relational reconstructions remain compatible with the available manifestations, GRE should compare them through the target-relevant structure they preserve, the independent evidence they accommodate, the assumptions they require, their sensitivity*

to scale and context, and the discriminating consequences they generate for further inquiry or intervention.

Proposition 5.15 allows several outcomes. One reconstruction may become substantially better supported. Several may remain viable but share a common relational core. Different reconstructions may be useful at different scales or for different targets. A disagreement may remain unresolved because the surviving evidence does not discriminate among the alternatives. These outcomes should be reported differently. The epistemic vocabulary should preserve the difference between identification, bounded inference, comparative support, and unresolved plurality.

The possibility of a common core is particularly important. Suppose several explanations disagree about the exact mechanism through which a platform acquired market dominance, but every viable account requires asymmetric control over access to a critical infrastructure. The broader claim about infrastructural asymmetry may then be better supported than any single complete historical reconstruction. GRE can therefore advance through partial relational invariants even where the full inverse problem remains open.

5.2.5 Context Dependence

Inverse reconstruction is conditioned by the context in which manifestation occurs. A manifestation relation that is highly discriminating in one context can become weakly informative in another. The same behavior can indicate trust where exit is feasible and indicate compliance where sanction is severe. The same price can emerge under competitive exchange, administered pricing, scarcity, speculation, subsidy, or strategic restriction. The same legal category can carry different practical consequences depending on enforcement, institutional capacity, and adjacent rights. Context therefore enters the inverse problem as part of the generative conditions of manifestation.

Context should not be treated as an unlimited residual category. For epistemic purposes, relevant context consists of conditions whose variation materially changes the relation between the target configuration and the manifestation. These conditions can be institutional, technological, cultural, ecological, legal, spatial, temporal, or relational. Identifying them is part of the reconstruction problem. A good explanation should specify which contextual differences are expected to alter manifestation and which are irrelevant to the target claim.

This requirement also limits unrestricted transfer of inference across populations or domains. A manifestation relation established in one institutional setting cannot be assumed to retain the same specificity elsewhere. A survey item, performance indicator, transaction record, or behavioral proxy can acquire different meanings when incentives, classifications, norms, or available alternatives change. Perspectival and model-based accounts of scientific representation are relevant because representations are constructed for purposes and from positions that select some relations while omitting others (Giere 2004; Longino 1990).

Claim 5.16 (Contextual portability of inverse inference). *An inverse inference is portable across contexts only to the extent that the conditions governing the relevant manifestation relation remain sufficiently stable, or the differences among contexts are*

explicitly modeled and supported by evidence.

The claim prevents an observed regularity from being elevated too quickly into a context-independent relational law. It also permits legitimate generalization when invariants are demonstrated. GRE is not committed to contextual uniqueness. It is committed to making the conditions of portability part of the epistemic claim.

Context can also be actively altered by inquiry. Publication of a metric can change the behavior measured by that metric. A legal classification can reorganize conduct. A forecast can alter expectations. An AI-generated representation can modify subsequent decisions. In such cases, the manifestation relation at a later time partly reflects earlier epistemic activity. The inverse problem must therefore account for reflexive contexts in which observation, representation, and intervention become part of the generative field.

5.2.6 Historical Dependence

Historical dependence gives the inverse problem a further dimension. The same manifest form can arise from different historical sequences, and the same symbol or category can change its relational meaning over time. Reconstruction therefore concerns not only which configuration is compatible with a manifestation, but which trajectory could have generated and sustained that configuration under historically specific conditions.

A present manifestation frequently contains only compressed traces of its history. An institution can preserve rules whose originating conflict is no longer visible. A market convention can remain stable after the conditions that first produced it have changed. A category can continue to be recorded even while its institutional meaning shifts. A relation can become self-reinforcing, so that its present persistence depends less on the mechanism that initiated it than on later feedbacks. Present-state evidence therefore may be insufficient for distinguishing origin, stabilization, and reproduction.

Historical reconstruction expands the evidential field by incorporating sequence, timing, archival traces, testimony, material residues, prior classifications, and changes in the manifestation regime itself. It also introduces losses that cannot always be repaired. Records can be destroyed, never created, selectively preserved, or generated through institutions that gave some actors far greater capacity to leave durable traces than others. The compatibility set for a historical inverse problem can therefore remain permanently wider than the set that would have been available to a contemporary observer with richer access.

Definition 5.17 (Historically indexed inverse problem). *A historically indexed inverse problem reconstructs a relational configuration or trajectory from manifestations whose production, preservation, interpretation, and evidential significance are themselves functions of historical conditions that may change across the period under study.*

Definition 5.17 prevents current interpretive categories from being projected backward without qualification. A record must be understood partly through the manifestation and representational relations under which it was produced. A category

recorded identically at two dates can refer to different institutional realities. A silence in an archive can mean absence, non-recording, suppression, destruction, or lack of institutional recognition. Historical evidence therefore requires reconstruction of the conditions under which traces became possible.

Historical dependence also makes revision an ordinary feature of knowledge. New archives can become available, new techniques can make previously inaccessible traces legible, later events can reveal relations whose earlier significance was unclear, and new concepts can permit distinctions that prior inquiry did not formulate. Such revision does not imply that historical claims are arbitrary. It means that the admissible compatibility set and the criteria for comparing reconstructions can change as evidence and epistemic resources change.

The inverse problem of manifestation consequently provides a bridge between the ontology of Chapter 4 and the more detailed epistemology developed in the remainder of Chapter 5. Relational knowledge begins from manifestations that constrain without necessarily determining their antecedent configurations. Non-invertibility produces underdetermination but still permits strong claims about identifiable properties and shared relational structure. Manifestational equivalence is local to an epistemic regime rather than proof of relational identity. Multiple explanations can be compared through preserved structure, independent evidence, explicit assumptions, and discriminating consequences. Context governs the portability of inverse inference, while history governs both the production of manifestations and the survival of traces. The next section turns from the inverse problem itself to knowledge of the manifestation relations on which such reconstruction depends.

5.3 Knowledge of Manifestation Relations

Sections 5.1–5.2 established that manifestation can provide evidential access to relational reality while leaving the inverse reconstruction problem underdetermined. The present section develops a further requirement that follows from this position: inquiry must seek knowledge not only of manifestations and relational targets, but also of the *relations through which those manifestations are generated*. The role of this section is to make manifestation relations themselves objects of epistemic inquiry. Its objectives are to distinguish knowledge of manifest content from knowledge of the manifestation relation, identify evidence that can support claims about such relations, specify when manifestation rules may be treated as sufficiently stable for inference, analyze changes in those rules, reconstruct them historically, and represent uncertainty about the mapping between relational and manifestational reality. The discussion proceeds methodologically from object specification to evidence, stability, change, history, and uncertainty. Throughout, manifestation relations are treated as historically situated and potentially causally operative components of generative-relational systems rather than as transparent channels standing outside the realities they mediate.

The need for this distinction can be seen whenever the same observed quantity, category, behavior, or representation changes its epistemic meaning because the process that produces it has changed. A stable unemployment rate can arise under al-

tered definitions of labor-force participation. A legal classification can retain the same label while the procedures through which it is assigned change. A platform metric can preserve its numerical form after an algorithmic ranking system has been revised. A recurring expression of trust can be generated by voluntary confidence in one setting and strategic conformity in another. In each case, accurate knowledge of the manifestation does not by itself establish knowledge of the manifestation relation. The epistemic relation between a manifestation and its target therefore depends partly on whether the inquiry understands how, under what conditions, and with what transformations the manifestation is produced.

Definition 5.18 (Manifestation relation as an epistemic target). A *manifestation relation as an epistemic target* is the delimited set of conditions, mechanisms, transformations, selections, mediations, and response regularities through which a specified relational configuration contributes to a specified manifestation within a stated spatial, temporal, scalar, and institutional domain.

Definition 5.18 avoids treating the manifestation relation as a universal function whose form is known in advance. In some domains the relation may be well approximated by a measurement model, response function, or probabilistic mapping. In others it may involve legal procedures, classificatory conventions, organizational routines, interpretive practices, material affordances, or historically accumulated symbolic forms. The epistemic task is to identify the relevant mediations at the level needed by the inquiry. A manifestation relation can therefore be known partially and locally without being reduced to a single formal expression.

This formulation also preserves the graph architecture established in Chapter 4. A manifestation relation is not merely a one-way conduit from relational reality to appearance. The institutions, instruments, categories, and practices through which manifestation occurs can themselves be relationally real and causally consequential. Quantification, standardization, and classification are especially clear cases: the procedures that make social or economic realities legible can organize later conduct, stabilize categories, and alter the very relations subsequently observed. Historical studies of quantification and statistical classification show that apparently neutral representational infrastructures can be institutionally produced and historically variable (Porter 1995; Desrosières 1998). Knowledge of manifestation relations must therefore include, where relevant, knowledge of their participation in subsequent generation.

5.3.1 Manifestation Relations as Objects of Knowledge

Treating manifestation relations as objects of knowledge changes the structure of epistemic inquiry. A researcher ordinarily asks what a manifestation indicates about a target. GRE adds a prior and recurrent question: what warrants the assumption that this manifestation is related to that target in the way the inference requires? The question is not answered merely by naming an indicator, instrument, category, or observation protocol. It requires an account of the relation that makes the manifestation evidentially relevant.

At least four components should be distinguished. The first is the *source domain*: the relational configurations or properties capable of contributing to the manifestation. The second is the *mediating domain*: the material, institutional, perceptual,

computational, symbolic, or procedural conditions that transform or select aspects of the source. The third is the *manifest domain*: the traces, measurements, classifications, behaviors, records, or appearances accessible to inquiry. The fourth is the *scope condition*: the domain within which the inferred relation is expected to hold. These components need not be independent. A legal procedure can belong to the mediating domain while also changing the source relations it classifies; an observer can generate a record while participating in the relation being observed; a statistical standard can produce comparability and thereby alter institutional decisions.

Knowledge of a manifestation relation is therefore not equivalent to possession of a fixed codebook. A codebook can state how categories are currently assigned while leaving the historical emergence, practical implementation, discretionary variation, and feedback effects of classification unspecified. Likewise, an algorithmic documentation page can state an output rule while omitting data-generation processes, upstream filtering, model updates, organizational overrides, and strategic responses by users. The epistemic object is the operative relation in context, not only its official representation.

Claim 5.19 (Manifestation-relation dependence of evidential inference). *The warrant of an inference from a manifestation to a relational target depends partly on the warrant for the manifestation relation presupposed by that inference. Strong confidence in the manifest content cannot by itself compensate for weak knowledge of how that content is generated, selected, transformed, or recorded.*

Claim 5.19 does not require complete knowledge of the manifestation relation before any inference is possible. Many successful inquiries proceed with partial knowledge of measurement mechanisms, institutional processes, or behavioral responses. The claim instead makes the missing knowledge explicit. Where the inference is insensitive to plausible variation in the manifestation relation, partial knowledge may be sufficient. Where small changes in the relation would reverse the conclusion, the relation becomes an especially important epistemic target.

This dependence can itself be scale-relative. At one scale, a standardized price index may function as a sufficiently stable manifestation relation between sampled prices and an aggregate inflation measure. At another, the inquiry may concern how substitution rules, quality adjustments, sampling frames, or consumption heterogeneity shape the index. The first inquiry may legitimately treat the mapping as provisionally fixed; the second makes its internal construction the object of analysis. Epistemic adequacy therefore depends on whether the manifestation relation is specified at the scale at which it materially affects the claim.

5.3.2 Evidence for Manifestation Relations

Evidence for a manifestation relation differs from evidence carried through that relation. Observing that a metric changes is evidence about the metric's value. Establishing that the metric reliably tracks a relational property requires additional evidence about the production of the metric and its connection to the target. This distinction is familiar in scientific modeling and measurement, where representational adequacy depends on purposive relations between models, targets, and selected respects of similarity rather than identity between representation and world (Giere 2004). GRE

generalizes the distinction to institutional, historical, symbolic, and social manifestations.

Several evidential routes can support a manifestation relation. *Repeated covariation* can show that changes in a target-relevant configuration are regularly accompanied by changes in the manifestation. *Intervention evidence* can test whether changing a source condition alters the manifestation in the expected direction. *Mechanism evidence* can identify the mediating process through which the source contributes to the manifest outcome. *Cross-context comparison* can reveal which components of the relation remain stable when surrounding conditions vary. *Calibration* can compare the manifestation against an independently accessible reference. *Process and documentary evidence* can reconstruct institutional procedures, classificatory decisions, instrument design, and implementation. *Negative and discrepant cases* can identify circumstances in which the presumed relation fails. No single route has universal priority; their relevance depends on the epistemic target and the type of manifestation relation involved.

The evidential problem becomes particularly important when the manifestation relation is partly strategic. Survey respondents can adapt answers to perceived expectations. Firms can change reported categories after regulatory thresholds are announced. Platform participants can optimize behavior to ranking metrics. States and organizations can reorganize records in response to performance indicators. In such settings, the manifestation relation contains endogenous responses to observation or evaluation. What appears to be a measurement of an antecedent relation can become part of a feedback process that changes the target itself.

Evidence for manifestation relations should therefore be triangulated where possible. A behavioral trace, an institutional record, a participant account, and an intervention response may each be incomplete, yet their conjunction can constrain the plausible mapping more strongly than repeated observation of one manifestation alone. Longino's emphasis on the role of background assumptions in evidential reasoning is relevant here: evidence becomes more reliable when the assumptions linking observation and conclusion are exposed to criticism from alternative perspectives and bodies of knowledge (Longino 1990). For GRE, manifestation relations are among the central background structures that should be made contestable.

Definition 5.20 (Manifestation-relation warrant). *Manifestation-relation warrant* is the degree to which available evidence supports a specified account of how a relational configuration contributes to a manifestation under stated conditions, including the relevant mediations, scope conditions, and known failure modes.

The definition deliberately avoids reducing warrant to a single numerical confidence score. Some manifestation relations can be estimated probabilistically; others may be supported by process evidence, institutional documentation, historical comparison, or qualitative reconstruction. What matters is that the basis of the warrant is inspectable. A claim about a stable manifestation relation should state what kind of evidence supports stability, what contexts have been examined, and which plausible changes could undermine the inference.

5.3.3 Stability of Manifestation Rules

Inference commonly relies on some degree of stability in manifestation rules. A rule is stable when sufficiently similar relational configurations under sufficiently similar conditions continue to generate manifestations in a way that preserves the distinctions relevant to the epistemic target. Stability does not require exact invariance. Measurement error, contextual modulation, observer variation, and stochastic response may coexist with a mapping that remains stable enough for a particular inference.

The epistemic importance of stability is straightforward. If the relation between target and manifestation changes unpredictably while the inquiry assumes it is fixed, temporal comparison can become misleading. Apparent change in the target can be generated by a changed manifestation rule; apparent continuity can be produced by compensating changes in the source and the mapping. Conversely, a stable manifestation relation can support stronger comparison across observations even when the target itself is dynamic.

Stability should be specified with respect to what is preserved. A classification rule may preserve category membership while changing the threshold by which membership is assigned. A statistical series may preserve its name while changing data sources. A linguistic expression may remain lexically stable while its institutional meaning changes. A market price may remain continuously observable while the microstructure generating it changes. Epistemic claims should therefore avoid treating nominal continuity as evidence of mapping stability.

Definition 5.21 (Epistemic stability of a manifestation rule). A manifestation rule is *epistemically stable* for a specified inquiry when variation in the rule across the relevant domain remains sufficiently bounded that the target-relevant distinctions carried by the manifestation are preserved to the degree required by the claim.

Definition 5.21 makes stability claim-relative. The same mapping can be stable for one inquiry and unstable for another. A survey item may be adequate for tracking broad directional change while too unstable for fine cross-cultural comparison. An administrative category may support short-run monitoring while failing across a major legal reform. A model may preserve local ranking while failing to preserve cardinal magnitude. The relevant standard is whether the manifestation relation preserves the information required by the epistemic target.

Repeated successful use can increase confidence in stability, but historical repetition alone is insufficient. A rule can appear stable because the range of conditions encountered so far has been narrow. Stability claims therefore gain strength when they survive meaningful variation in context, scale, population, institutional setting, or intervention. This is one reason comparative and historical analysis matter even for apparently technical measurement systems.

5.3.4 Change in Manifestation Rules

Manifestation rules can change because instruments change, institutions revise procedures, categories acquire new meanings, observers alter practices, populations adapt strategically, or the underlying relational system reorganizes the mediations

through which it becomes manifest. Such change is epistemically consequential because a discontinuity in the mapping can be mistaken for a discontinuity in the target, while transformation in the target can be hidden by a compensating change in the mapping.

Several forms of change should be separated. *Parametric change* alters the strength or threshold of a familiar relation while leaving its broad structure recognizable. *Procedural change* alters the steps through which the manifestation is produced. *Categorical change* alters the distinctions encoded by the manifestation. *Infrastructural change* alters the technical or institutional system that makes the manifestation possible. *Reflexive change* occurs when actors adapt to the manifestation regime itself, thereby changing the relation between underlying activity and observed trace. These changes need not occur independently.

Historical accounts of scientific and administrative categories provide useful precedents for taking such transformations seriously. Kuhn's account of historically organized scientific practice and Hacking's work on historical ontology both emphasize, in different ways, that concepts, objects of inquiry, and styles of reasoning can change with the practices through which they are constituted and investigated (Kuhn 1962; Hacking 2002). Statistical and administrative histories similarly show that classification and quantification regimes can change the terms through which populations and activities become comparable (Desrosières 1998; Porter 1995). GRE treats these observations as reasons to make the history of manifestation rules part of epistemic analysis rather than as grounds for generalized relativism.

Claim 5.22 (Manifestation-rule change and apparent target change). *Observed change in a manifestation can reflect change in the relational target, change in the manifestation relation, or interaction between both. Epistemic attribution of change therefore requires evidence capable of distinguishing these possibilities to the degree relevant to the inquiry.*

The claim has direct consequences for time-series reasoning. Longitudinal comparability requires more than repeated use of the same label or numerical format. The inquiry should identify whether definitions, sampling frames, measurement technologies, legal procedures, data-generating practices, or strategic responses have changed. Where they have, continuity may need to be reconstructed rather than assumed. In some cases no defensible adjustment can restore comparability, and the epistemically appropriate result is a documented discontinuity.

Rule change can also be a substantive object of GRE. A new manifestation regime may redistribute visibility, recognition, and institutional consequences. When a platform introduces a new reputation score, when a state changes a legal classification, or when an organization adopts a new performance metric, the alteration can change who becomes visible, which relations count, and which actions become consequential. The epistemic history of the mapping therefore intersects with the generative history of the system.

5.3.5 Historical Reconstruction of Manifestation

Historical reconstruction becomes necessary when present manifestations are interpreted through relations that have changed over time. The task is not merely to re-

cover earlier observations. It is to reconstruct, as far as evidence permits, the conditions under which those observations were generated and the distinctions that they could legitimately support at the time. A historical series assembled from surviving records may contain values whose apparent uniformity conceals changes in classification, collection, administrative purpose, or institutional coverage.

Historical reconstruction should therefore treat manifestation relations as trajectories. An archive records not only events but the effects of selection, preservation, categorization, institutional purpose, and survival. A legal register reflects the rules under which statuses were recorded. A price series depends on which exchanges became observable and comparable. A corpus of published research reflects publication practices, disciplinary categories, and technical capacities. Present access to these traces is further mediated by later preservation, digitization, indexing, and interpretation. The epistemic object is thus a historically layered relation between past relational configurations, past manifestation practices, surviving traces, and present reconstruction.

This layered view prevents two opposite errors. The first is *presentist projection*: assuming that a present category had the same meaning, boundary, or measurement relation in the past. The second is *historical skepticism*: concluding that because manifestation relations changed, disciplined comparison is impossible. Reconstruction can often identify periods of relative stability, explicit breaks, overlapping regimes, conversion rules, and residual uncertainty. The goal is to preserve warranted comparability where possible and preserve documented non-comparability where it is not.

Definition 5.23 (Historical reconstruction of a manifestation relation). The *historical reconstruction of a manifestation relation* is the evidence-guided recovery of the conditions, rules, mediations, scope, and transformations through which a past relational configuration became manifest and through which its traces became available to present inquiry.

The definition includes the history of preservation because surviving evidence is itself selective. An event may have generated many manifestations, only some of which were recorded; only some records may have survived; and only some surviving records may be accessible or legible under present categories. Knowledge of the historical manifestation relation therefore includes knowledge of loss, silence, and differential preservation where those factors materially affect inference.

This is also where historical reconstruction connects to later questions of epistemic justice. Some manifestation systems systematically privilege actors with greater institutional visibility, literacy, administrative recognition, or control over record-making. The resulting archive can reproduce those asymmetries. Section 115 will distinguish relation, trace, and evidence more finely; the present point is that the evidential status of a historical trace depends partly on knowledge of the manifestation and preservation relations through which it exists.

5.3.6 Uncertainty about Manifestation Mapping

Manifestation relations are often only partly known. Uncertainty can concern which source relations matter, which mediators intervene, whether the rule is stable, how

actors respond to observation, whether different contexts share the same mapping, or how the mapping changed historically. This uncertainty should be represented explicitly because it propagates into the inverse reconstruction problem developed in Section 5.2.

At least three forms of uncertainty are useful to distinguish. *Structural uncertainty* concerns the form of the manifestation relation: which mediators and transformations belong in the account. *parametric uncertainty* concerns the magnitude, threshold, probability, or sensitivity of a relation whose broad form is accepted. *historical uncertainty* concerns when and how the relation changed, including uncertainty generated by incomplete documentation. A fourth form, *scope uncertainty*, concerns the domain across which a relation can be transported without substantial alteration.

These uncertainties can combine. A researcher may know that an administrative indicator is generated from a defined set of records while remaining uncertain about discretionary implementation across offices. A model may be well calibrated in one period while facing scope uncertainty after institutional reform. Historical sources may document a category change without revealing how quickly practice changed on the ground. GRE should therefore avoid representing mapping uncertainty as a single generic error term when the structure of the uncertainty matters to interpretation.

Claim 5.24 (Propagation of manifestation-mapping uncertainty). *Uncertainty about a manifestation relation propagates into claims inferred through that relation. The propagated uncertainty need not invalidate the claim, but the claim's strength should reflect the range of target-relevant conclusions compatible with serious alternatives concerning the mapping.*

Claim 5.24 creates a direct connection between Sections 5.2 and 5.3. The compatibility set in the inverse problem should expand when several plausible manifestation relations remain open and contract when evidence rules them out. A reconstruction that appears unique under one fixed mapping may become non-unique once mapping uncertainty is admitted. Conversely, some target properties may remain invariant across plausible mappings and can therefore retain strong warrant despite uncertainty about the complete relation.

Epistemic progress can accordingly take several forms. Inquiry can improve knowledge of the target while holding the mapping provisionally fixed; improve knowledge of the mapping while leaving the target uncertain; identify properties robust across several plausible mappings; or discover that an earlier inference depended on a manifestation relation that no longer holds. Such revision is not an embarrassment to relational epistemology. It is one of its expected modes of development.

Knowledge of manifestation relations thus occupies a central position between ontology and empirical method. The three-register architecture makes manifestation possible without reducing reality to what appears; the inverse problem shows why manifestations do not uniquely reconstruct their sources; and the present section shows that the mapping connecting source and manifestation is itself historically situated, evidentially investigable, and uncertain. The next section turns from knowledge of the mapping to the traces through which relations become evidentially accessible. That transition shifts the emphasis from the rules of manifestation to the heteroge-

neous traces that those rules produce, preserve, obscure, or fail to generate.

5.4 Relation, Trace, and Evidence

The preceding sections established that GRE often encounters relational reality indirectly: manifestations are generated through historically variable manifestation relations, and reconstruction from manifestation toward relational structure is generally non-invertible. The present section introduces the evidential vocabulary required to proceed under those conditions. Its role is to distinguish the relation under inquiry from the traces through which that relation becomes accessible and from the evidential status later assigned to those traces. The section develops eight connected distinctions: relation and evidence of relation, direct and indirect traces, observable and inferred relations, latent and hidden relations, and the difference between absence of evidence and evidence of absence. The method is deliberately relational. A trace is treated neither as a transparent fragment of reality nor as an arbitrary sign; its evidential force depends on the generative relation through which it arose, the conditions under which it was preserved or observed, and the inferential target for which it is being used.

This distinction matters because the same material or symbolic object can occupy different epistemic roles. A contract, message log, transfer record, testimony, administrative category, network edge, or repeated behavioral pattern may be produced by a relation, record a relation, mediate a relation, or become part of the relation's later reproduction. Its existence alone therefore does not determine what it evidences. The evidential question concerns how that object was generated, what relation it bears to the target, what alternative relations could have generated a similar trace, and which relevant traces could have failed to appear or survive. Work on scientific objectivity, statistics, historical classification, and contextual empiricism likewise shows that evidence acquires significance through conventions of production, background assumptions, and institutionalized practices of comparison rather than through mere presence alone (Porter 1995; Desrosières 1998; Hacking 2002; Longino 1990).

Definition 5.25 (Trace). A *trace* is a material, behavioral, institutional, symbolic, statistical, computational, testimonial, or other manifestation whose existence or form was generated, conditioned, or made more probable by a relation, process, event, or configuration relevant to an epistemic target.

Definition 5.25 intentionally keeps the category broad. A trace can be intentionally produced, as in a signed agreement or recorded payment, or unintentionally left, as in a pattern of repeated co-presence, a change in resource flows, a sequence of revisions, or a residue in a technical system. It can be contemporaneous with the relation or generated later by consequences of that relation. It can also be partial: a trace may preserve one dimension of a relation while omitting the dimensions most important to another inquiry.

Definition 5.26 (Evidence). *Evidence* is a trace, manifestation, observation, record, testimony, measurement, comparison, or intervention result that is assigned an inferential role with respect to a specified epistemic target under an articulated account of relevance, reliability, and alternative explanation.

The distinction between Definitions 5.25 and 5.26 is foundational for this chapter. A trace can exist without yet functioning as evidence. It becomes evidence only relative to a claim and an inferential practice. Conversely, evidence need not consist of a single preserved object. A pattern across several weak traces can support a relational claim even when none is individually decisive. Evidence is therefore not a substance attached permanently to an item; it is a role that an item or pattern can occupy within an inquiry.

5.4.1 Relation and Evidence of Relation

A relation and evidence of that relation belong to different analytical categories. The relation is part of the target reality under inquiry; the evidence is part of the epistemic access through which a claim about that reality is evaluated. Confusing the two can produce both false confidence and false skepticism. A recorded edge in a database can be mistaken for the social relation itself, while a relation lacking a convenient record can be treated as if it did not exist. GRE therefore preserves a strict distinction between relational existence and evidential manifestation.

The distinction remains necessary even when the relation is highly formalized. A legal record may be constitutive of a legal relation under a given institutional order, yet the record and the relation are still analytically distinguishable. The record may establish status within one domain while failing to capture informal dependence, effective control, lived meaning, or later transformation. Likewise, an account balance can participate in constituting a monetary relation while still remaining only one manifestational component of a wider system of claims, expectations, institutional enforcement, and access. Constitutive traces are therefore possible, but constitution does not eliminate the need to specify which dimension of the relation the trace supports.

Claim 5.27 (Relational–evidential distinction). *Evidence that a relation exists, persists, or changes should be distinguished from the relation itself and from representations that merely encode or summarize it. The stronger the inferential step from trace to relational claim, the more explicit the inquiry should be about the manifestation relation, plausible alternatives, and the dimensions of the target that remain unobserved.*

Claim 5.27 gives GRE a way to handle heterogeneous records without collapsing them into a single evidential hierarchy. A transcript may provide strong evidence for an utterance, weaker evidence for the speaker’s durable intention, and still weaker evidence for the long-term structure of a relation. A transfer record may strongly support that a transaction occurred, while providing only partial evidence about coercion, trust, dependence, or the distribution of alternatives surrounding that transaction. The same trace can therefore carry different evidential weight for different targets.

This target-relativity should not be confused with arbitrary interpretation. The relevant relation between trace and target can itself be investigated. Evidence becomes stronger when the trace-generation process is well understood, competing explanations are constrained, observations are independently corroborated, and the inference remains stable under reasonable alternative codings or models. Evidence becomes weaker when the trace can be generated by many materially different con-

figurations, when preservation is selective, or when the observation system systematically excludes parts of the relation under study.

The relation–evidence distinction also matters for historical inquiry. Present researchers often encounter past relations only through surviving traces. Survival adds an additional manifestation relation: what was produced, what was recorded, what was preserved, what was destroyed, what became legible, and what later institutions chose to index or digitize. Historical evidence therefore bears the history of both the target relation and the evidential infrastructure through which present access becomes possible.

5.4.2 Direct Traces

A direct trace is relatively proximal to the relation or event targeted by the inquiry. Proximity can be temporal, procedural, institutional, material, or causal. Examples include a contemporaneous transaction record for a transfer, an authenticated message for a communication event, a legal filing for a formally registered status, a sensor record for a physical interaction, or an observed sequence of actions within a specified setting. The term *direct* is therefore relational and scoped. It does not mean unmediated access to reality.

Definition 5.28 (Direct trace). A *direct trace* is a trace generated through a relatively short and specified manifestational path from the target relation or relational event to the evidential object, such that the principal transformations between target and trace are sufficiently known for the inquiry’s purpose.

The definition makes directness a property of the evidential path rather than of a privileged class of objects. A signed contract may be direct evidence that specified parties entered a recorded agreement under a particular legal procedure. The same contract is not direct evidence that the parties experienced equal bargaining power. An interaction log may be direct evidence that two accounts exchanged messages while remaining weak evidence about friendship, authority, dependence, or trust. Directness therefore does not automatically confer broad scope.

Direct traces can be especially useful for establishing occurrence, timing, sequence, and formally recorded participation. They are less automatically informative about unrecorded motivations, counterfactual options, tacit norms, or distributed conditions. GRE should therefore resist a documentary bias in which formally captured relations appear more real simply because their traces are easier to stabilize and inspect. Institutional systems often produce abundant direct traces for what they are designed to recognize and sparse traces for relations outside their categories. The history of statistical and administrative classification illustrates how standardized records can increase comparability while simultaneously organizing what becomes visible (Porter 1995; Desrosières 1998).

Direct traces can also be strategically generated. A declaration, report, score, or official record may be created precisely because actors anticipate later evaluation. The trace then remains direct with respect to the recorded act while its evidential relation to broader conduct requires additional scrutiny. A compliance certificate can directly establish that a certificate was issued; its value as evidence of substantive compliance depends on the inspection regime, incentives, and possibility of strategic

adaptation. Directness thus reduces some inferential distance without eliminating the need for a manifestation-relation analysis.

5.4.3 Indirect Traces

Many economically significant relations leave no single proximal record. Their existence is reconstructed from downstream effects, repeated patterns, correlated changes, institutional responses, or combinations of traces produced through several mediating steps. Such traces are indirect relative to the target relation. Indirectness does not make them inferior by definition. In some inquiries, a distributed pattern across multiple indirect traces can provide stronger evidence than a formally direct record that captures only one declared aspect of the relation.

Definition 5.29 (Indirect trace). An *indirect trace* is a trace connected to an epistemic target through one or more intervening relations, transformations, consequences, or preservation processes whose contribution must be reconstructed for the trace to support the target claim.

Examples include changes in prices used to infer supply constraints, migration patterns used to investigate changing opportunity structures, repeated exclusion outcomes used to investigate an institutional barrier, or downstream changes in behavior used to investigate a trust relation. In each case, the trace may be compatible with several candidate explanations. The epistemic burden therefore shifts toward discriminating among plausible generative paths.

Indirect traces are particularly important for relations that are distributed, tacit, stigmatized, historically remote, or institutionally unrecorded. A relation of dependence may be inferred from repeated asymmetries of exit, access, and response even when no actor explicitly records the relation as dependence. A field-like constraint may be reconstructed from patterned local outcomes that cannot be attributed to a single dyadic command. A historical relation may become visible only through later consequences because contemporaneous record-making omitted the affected actors. These cases show why GRE cannot limit evidence to explicit self-description or formal documentation.

At the same time, indirect traces invite over-attribution when analysts move too quickly from pattern to relation. Several relational structures can generate similar downstream manifestations, as Section 5.2 established. The disciplined response is triangulation across traces generated through partially independent pathways. Documentary evidence, behavior, testimony, longitudinal change, institutional records, and intervention results can constrain one another when their errors and selection processes are not identical. Contextual empiricism is relevant here because what counts as evidentially probative depends partly on background assumptions that themselves require critical examination (Longino 1990).

Remark 5.30 (Combination of weak traces). Several individually weak traces can jointly support a stronger relational claim when their dependence structure is understood and when the traces constrain overlapping alternatives through sufficiently different manifestation pathways. Simple numerical accumulation of mutually dependent traces does not provide the same warrant.

Remark 5.30 prevents a common mistake in large evidential systems. Thousands of records generated from the same classification rule may contain less independent information than a small set of traces produced through distinct mechanisms. Evidential quantity and evidential diversity are therefore analytically different.

5.4.4 Observable Relations

Some relations can be observed in a relatively direct operational sense. A transfer can be witnessed, a conversation can be heard, an exchange can be recorded, a contractual status can be queried, or a physical connection can be inspected. GRE nevertheless treats observability as relation-specific and regime-specific rather than as a universal boundary between known and unknown objects. Observation itself occurs through a manifestational configuration involving an observer, instrument, category, temporal window, and criterion of recognition.

Definition 5.31 (Operationally observable relation). A relation is *operationally observable* for an inquiry when the relation or a constitutive relational event becomes available within the inquiry's observation regime with sufficiently specified criteria of recognition and sufficiently limited inferential distance for the target claim.

This definition allows an exchange event to be observable while the durable economic relation within which it occurs remains partly inferred. Observing repeated transfers does not by itself settle whether the relation is reciprocal, exploitative, charitable, coerced, strategic, or embedded in a longer obligation. Similarly, observing two persons collaborate can establish interaction while leaving the distribution of authority, dependence, recognition, and future obligation underdetermined.

Operational observability can also be created institutionally. Administrative systems render some relations queryable by assigning identifiers, categories, and records. Network platforms can make interaction histories inspectable. Sensors and digital infrastructures can transform previously ephemeral events into persistent traces. Such changes expand evidential access while also altering the relation between observation and conduct. Once actors know that a relation is observed, the observation regime can become part of the relational system. The epistemology of performative observation will therefore require later attention in Sections 116–118.

A further distinction is useful between observing a relational event and observing a relationally real object. A single communication event can be observed, while a stable trust relation requires evidence of persistence, organization, and causal efficacy across a relevant interval. The GR criterion for relational reality therefore places a higher evidential burden on durable relational claims than on the occurrence of isolated relational events.

5.4.5 Inferred Relations

Most relations studied by GRE are at least partly inferred. Inference becomes necessary whenever the target exceeds any single observation: persistent authority, institutional dependence, social trust, exclusion, access asymmetry, generative capacity, field position, and many forms of economic power are reconstructed from multiple manifestations and traces. The inferred character of these relations does not

weaken them ontologically. It describes the epistemic route through which they become known.

Definition 5.32 (Inferred relation). An *inferred relation* is a relational target whose existence, organization, persistence, or causal role is supported through an explicit inferential synthesis of traces, manifestations, observations, models, or interventions rather than through a single sufficiently proximal observation.

Inference can take qualitative, historical, statistical, computational, formal, or mixed forms. A historian may reconstruct an institutional dependence from correspondence, legal records, and resource flows. An ethnographer may infer a tacit hierarchy from repeated interactional patterns and participants' accounts. A network model may infer ties from event data. A causal analysis may infer a relational effect from variation under a specified design. GRE does not assign automatic priority to one form. The relevant standard is whether the inferential method is appropriate to the target and whether its assumptions and residual alternatives are made visible.

Inferred relations should therefore carry an account of identification. An inquiry should state which traces would be expected under the proposed relation, which serious alternatives remain compatible with the evidence, and which additional observations could discriminate among them. This requirement follows directly from the inverse problem. The relation becomes better identified when competing configurations are progressively constrained, even if unique reconstruction remains unavailable.

The distinction between inferred relation and speculative relation is consequently evidential rather than rhetorical. An inferred relation has articulated support and known inferential conditions. A speculative relation may be worth considering as a hypothesis but lacks comparable constraint. GRE benefits from preserving both categories because early-stage inquiry often needs hypotheses that can guide trace collection without treating those hypotheses as established relational reality.

5.4.6 Latent Relations

Some relations are causally operative without being manifest in the observation channel currently used by an inquiry. GRE describes such relations as latent relative to that epistemic arrangement. Latency is therefore an epistemic condition of access, not a statement that the relation lacks relational reality. A relation can persist, constrain action, and shape subsequent generation while remaining unavailable to a particular observer, instrument, category, or temporal window.

Definition 5.33 (Latent relation). A *latent relation* is a relation that may satisfy the relevant conditions of relational reality while remaining unmanifest or insufficiently discriminated within a specified observation regime, even though indirect traces or later manifestations may permit its reconstruction.

The qualification "within a specified observation regime" is essential. A relation latent to one observer can be manifest to another. Informal care relations may be absent from market statistics while plainly visible in household practice. A dependency may be invisible in an organizational chart while apparent in patterns of consultation and approval. A computational relation may be inaccessible in a user interface while

recorded in backend logs. Latency therefore often reflects the partiality of epistemic access rather than the intrinsic invisibility of the relation.

Latent relations are especially important when manifestational systems privilege what is readily quantified. Standardized categories can improve coordination and comparison while leaving relations outside those categories weakly represented (Porter 1995; Desrosières 1998). GRE should therefore treat the absence of a variable, category, or explicit edge from a dataset as a property of the representational system until additional evidence supports a stronger conclusion about the underlying relation.

The notion also helps distinguish pre-manifestational relational reality from mere analyst invention. A relation can be latent before it is consciously recognized if it already persists and exerts causal efficacy within the system. Its later discovery changes epistemic access, and perhaps subsequent relations once the discovery becomes operative, without implying that the relation began to exist only when it was named.

5.4.7 Hidden Relations

Hidden relations form a narrower epistemic problem. A relation is hidden when the observation regime fails to reveal it because relevant traces are inaccessible, suppressed, displaced, masked, or systematically routed outside the observer's evidential field. Strategic concealment is one important mechanism, but hiddenness can also arise from institutional segmentation, technical opacity, confidentiality, unequal access, or the topology of a distributed system. Section 118 will examine observer position and strategic concealment in greater detail; the present section defines the evidential distinction.

Definition 5.34 (Hidden relation). A *hidden relation* is a relation for which evidential access is materially obstructed within a specified inquiry because relevant traces are unavailable, obscured, selectively preserved, strategically concealed, or generated outside the observer's accessible manifestational domain.

Hiddenness differs from latency primarily in the structure of access. Latency emphasizes insufficient manifestation within an observation regime; hiddenness emphasizes obstruction or exclusion from the evidential field. The categories can overlap. A relation may be latent because its effects are subtle and hidden because actors also conceal the traces most likely to reveal it. GRE should therefore use the distinction analytically rather than treat it as a rigid ontology of relations.

Hidden relations create asymmetric epistemic burdens. Actors inside a relation may possess knowledge unavailable to external investigators; institutions may control records required for verification; affected subjects may experience consequences without access to the infrastructures that produce them; and observers with technical privileges may see relational structures that ordinary participants cannot inspect. These asymmetries later become questions of epistemic justice and machine-mediated knowledge, but they begin here as properties of evidential access.

A hidden relation should not be inferred solely from the fact that evidence is missing. The hypothesis of concealment itself requires traces: discontinuities, inconsisten-

cies, restricted access, unexplained selection, testimony, leaked records, differential visibility, or other observations that make obstruction more plausible than simple nonexistence or measurement failure. This requirement guards against converting every unexplained absence into evidence of an invisible structure.

5.4.8 Absence of Evidence and Evidence of Absence

Negative evidence requires a particularly explicit manifestation model. The nonappearance of a trace can support a claim that a relation is absent only when the inquiry has reason to expect that the relation, if present, would probably have generated an observable and preserved trace within the relevant observation regime. Without that expectation, missing evidence remains weakly informative.

Definition 5.35 (Negative evidential warrant). *Negative evidential warrant* is the support that nonappearance of an expected trace provides for the absence, weakness, interruption, or transformation of a target relation when the trace-generation, detection, coverage, and preservation conditions are sufficiently specified.

The definition makes four conditions especially important. First, *trace generation*: would the target relation normally produce the relevant manifestation? Second, *detection*: would the observation system register it if produced? Third, *coverage*: was the relevant temporal, spatial, institutional, or population domain actually observed? Fourth, *preservation*: could the trace have been lost, deleted, censored, or transformed before inspection? Evidence of absence becomes stronger as these conditions become better understood and as the expected trace becomes more reliable under the hypothesized relation.

This distinction is crucial in economic and institutional research. The absence of a complaint is weak evidence that no harm occurred when complaint mechanisms are inaccessible or risky. The absence of a transaction record is weak evidence that no transfer occurred when informal exchange is common. The absence of a network edge is weak evidence of no relation when the network is reconstructed from a partial communication channel. By contrast, the persistent absence of an expected entry in a highly complete registry under a stable mandatory reporting regime can provide stronger negative evidence for a formally registered relation.

Claim 5.36 (Condition for evidential absence). *Absence of evidence becomes evidence of absence to the extent that the inquiry can justify a sufficiently reliable expectation that the target relation, if present at the relevant strength and scope, would have generated a detectable and preserved trace within the observation regime.*

Claim 5.36 also clarifies why evidential silence can be historically and politically structured. Classification systems, archives, administrative routines, and research infrastructures determine which relations reliably produce traces. Hacking's account of historical ontology and the histories of statistical classification examined by Desrosières provide examples of how categories and recording practices can change the objects that become visible to inquiry (Hacking 2002; Desrosières 1998). Silence in an archive or dataset therefore requires knowledge of the record-making relation before it can support a substantive inference.

The same principle applies to repeated nonobservation. More observations in-

crease evidential strength only when they meaningfully increase the probability of detection under the target hypothesis. Repeatedly inspecting the same narrow channel does little to rule out a relation expressed elsewhere. Expanding observer positions, temporal coverage, trace types, and independent manifestation pathways can be more informative than multiplying observations within a single constrained regime.

The relation–trace–evidence distinction therefore supplies a basic grammar for the remainder of Chapter 5. Relations belong to the reality under inquiry; traces are heterogeneous manifestations or remainders generated through relational and historical processes; evidence arises when traces are connected to specified claims through warranted inferential relations. Direct and indirect traces differ by the structure of their manifestational paths rather than by an absolute hierarchy of reliability. Observable, inferred, latent, and hidden relations describe different conditions of epistemic access rather than different degrees of ontological reality. Negative evidence becomes informative only when trace generation and observation are sufficiently understood. These distinctions prepare the next section’s treatment of symbolization and representation, where traces are transformed into linguistic, legal, administrative, statistical, and computational forms that can both preserve and reorganize relational knowledge.

5.5 Symbolization and Representation

Section 5.4 distinguished relations from the traces through which they become epistemically accessible and from the evidential roles later assigned to those traces. The present section examines a further transformation: traces, observations, and relational configurations are frequently rendered into symbolic forms that can be stored, communicated, compared, aggregated, interpreted, and acted upon across settings. Its role is to clarify how symbolization and representation participate in relational knowledge without being treated as transparent duplication of the realities they concern. The section proceeds through eight components: manifestation and symbolization, linguistic representation, administrative representation, legal representation, statistical representation, computational representation, symbolized relations, and representation error. The method is comparative. Each representational form is analyzed in terms of what it selects, preserves, compresses, transforms, and makes newly available for inference.

Symbolization matters for GRE because relational inquiry rarely proceeds through unprocessed manifestations alone. A spoken description, a category in a registry, a legal status, a statistical indicator, a graph edge, a model parameter, or a machine-readable embedding can make a relation portable across time, space, institutions, and observers. Such portability expands epistemic reach. It also introduces additional conditions of interpretation. The same relational reality can be symbolized differently under different languages, administrative systems, legal regimes, statistical conventions, and computational architectures. Conversely, similar symbols can refer to relations that differ substantially in their histories, mechanisms, or causal organization. Symbolization therefore becomes part of the epistemic pathway through which relational reality is reconstructed.

Cassirer's philosophy of symbolic forms provides one important lineage for treating symbolic articulation as constitutive of intelligibility rather than as a secondary label placed upon already finished experience, while Bourdieu's analysis of language emphasizes that symbolic forms operate within social relations of authority and differential recognition (Cassirer 1955; Bourdieu 1991). GRE draws on these insights without committing itself to a comprehensive philosophy of symbolic form. The narrower claim developed here is methodological: symbolic systems organize access to relational reality by establishing distinctions, equivalences, references, and conventions through which manifestations become interpretable and transportable.

Definition 5.37 (Symbolization). *Symbolization* is the relational process through which a manifestation, trace, object, relation, process, event, or distinction is rendered in a symbolic form that can be recognized, communicated, stored, compared, transformed, or used within an epistemic or institutional practice.

Definition 5.37 keeps symbolization broader than language. A symbol may be a word, category, code, legal designation, numerical variable, visual mark, network edge, database field, model state, or computational token. What unifies these cases is the establishment of a rule-governed relation through which some differences are preserved and others are suppressed, grouped, or transformed.

Definition 5.38 (Representation). A *representation* is a symbolic, material, formal, linguistic, administrative, legal, statistical, computational, or other constructed configuration used to stand for, describe, model, encode, or make inferentially accessible selected aspects of an epistemic target.

Representation therefore has a target-relative structure. A representation can be highly adequate for one inferential purpose while remaining uninformative for another. Giere's account of scientific representation is useful here because it treats models as purposive representations of selected aspects of reality rather than as literal duplicates of their targets (Giere 2004). GRE generalizes this restraint beyond scientific models. Every representational practice selects a relation between a target, a representational form, and a purpose under which similarity, relevance, fidelity, or usefulness is judged.

5.5.1 Manifestation and Symbolization

Manifestation and symbolization occupy different stages and roles within epistemic access. A manifestation is the way in which some aspect of reality becomes available within a relational configuration of perception, instrumentation, practice, institution, or other mediation. Symbolization renders some aspect of that manifestation into a form that can circulate within a symbolic practice. The two can occur nearly simultaneously, especially in highly institutionalized settings, yet the distinction remains analytically useful.

A price appearing on a trading interface is already heavily symbolized at the point of manifestation. A person's bodily discomfort can become manifest before it is linguistically articulated. An institutional status can be symbolically recorded at the same moment that it becomes administratively manifest. These examples show that manifestation and symbolization can overlap while still answering different questions. Manifestation concerns availability; symbolization concerns the rule-governed

form through which what is available is rendered communicable, recordable, or manipulable.

The distinction also clarifies why pre-symbolic or weakly symbolized relational realities remain possible. A pattern of dependence, trust, vulnerability, attraction, exclusion, or coordinated expectation can become relationally real before participants possess a stable vocabulary for it. Later symbolization can increase epistemic accessibility by supplying names, categories, comparisons, and records. Symbolization can also reorganize the field of attention by making some relations easier to notice and others less legible.

Claim 5.39 (Symbolic mediation of epistemic access). *Symbolization can expand, stabilize, and reorganize epistemic access to relational reality by rendering selected manifestations portable across observers, settings, and times. The resulting symbolic accessibility remains dependent on the selection rules and interpretive conventions through which the representation is produced and used.*

Claim 5.39 explains why symbolic systems are both epistemically enabling and selective. A stable vocabulary can support comparison across cases that would otherwise remain locally intelligible. A coding standard can aggregate records across institutions. A legal category can coordinate recognition across courts and agencies. At the same time, each gain in portability is achieved through a representational decision about which distinctions will count as relevant.

5.5.2 Linguistic Representation

Linguistic representation provides one of the most flexible media for relational knowledge. Language can name entities, specify relations, narrate trajectories, mark modality, attribute causes, preserve testimony, and express uncertainty. It can also embed distinctions that become difficult to see once they are stabilized in ordinary usage. Terms such as *worker*, *owner*, *household*, *market*, *dependency*, *care*, or *trust* can function simultaneously as descriptive conveniences and as historically inherited classifications.

For GRE, linguistic representation should therefore be analyzed at two levels. The first concerns semantic content: what relation or object a linguistic expression is intended to describe. The second concerns the social conditions under which the expression becomes authoritative, ordinary, contestable, or unavailable. Bourdieu's account of linguistic markets is useful here because it emphasizes that linguistic efficacy depends partly on social position, institutional authorization, and the conditions of reception (Bourdieu 1991). A description can thus fail to become publicly consequential even when it captures an important aspect of relational reality, while an institutionally authorized description can become widely reproduced despite serious epistemic limitations.

Linguistic representation is especially important for relations whose traces are dispersed across experience rather than concentrated in formal records. Testimony about dependence, exclusion, coercion, care, trust, or tacit norms often requires narrative articulation before the relevant relational pattern becomes available to others. Such articulation can condense a long trajectory into a linguistic form that permits comparison and inquiry. The evidential value of the account still depends on context,

corroboration, target specification, and the possibility of alternative interpretations.

Language also supports second-order representation. A subject can describe the categories through which an institution describes them, contest those categories, or propose a new vocabulary for a relation previously rendered obscure. Epistemic change can therefore involve changes in the representational language itself. This does not imply that reality changes whenever vocabulary changes. It means that available distinctions influence which relational hypotheses can be formulated, communicated, and tested.

5.5.3 Administrative Representation

Administrative representation converts persons, organizations, events, transactions, locations, resources, and relations into standardized categories suitable for coordination across institutional settings. Registries, forms, identifiers, taxonomies, eligibility rules, case files, accounting fields, and reporting systems create representational infrastructures through which large organizations can act upon realities that no single administrator encounters directly.

The epistemic advantage is substantial. Standardization allows cases to be compared, aggregated, retrieved, and transported across organizational boundaries. The same process can reduce heterogeneous relations to a limited set of recognized attributes. Desrosières' history of statistical reasoning and Porter's study of quantitative objectivity both show how conventions of classification and standardized procedures support impersonal and transportable judgment (Desrosières 1998; Porter 1995). Hacking's historical ontology further emphasizes that categories can acquire histories and participate in reorganizing the fields they describe (Hacking 2002).

Administrative representation therefore generates a specific epistemic asymmetry. Relations that fit the administrative schema can become highly visible, while relations that fall outside the schema can remain weakly represented even when they are causally important. A household may be represented by an address and membership field while care dependencies remain unrecorded. Employment may be represented by contract status while effective control is distributed through subcontracting, platforms, algorithms, and informal expectations. A benefit system may record eligibility categories while omitting relational barriers that shape practical access.

Definition 5.40 (Administrative legibility). *Administrative legibility* is the degree to which an object, relation, event, or condition can be represented within the categories, identifiers, procedures, and record structures through which an administrative system observes and acts.

Administrative legibility is epistemically useful because it supports coordination and comparison. It is also selective because legibility is defined relative to a particular representational infrastructure. GRE therefore treats administrative absence with caution: an unrecorded relation may reflect nonexistence, weak trace generation, exclusion from the schema, incomplete reporting, strategic concealment, or a mismatch between lived relations and recognized categories.

5.5.4 Legal Representation

Legal representation renders relations through categories of status, right, duty, authority, liability, ownership, membership, capacity, obligation, and procedure. Legal systems thereby produce highly structured representations of relational positions. These representations can summarize complex histories into recognized forms: a person is registered as an owner, a corporation as a juridical entity, an agreement as a contract, a relation as employment, or an act as a legally relevant transfer.

Hart's account of primary and secondary rules highlights how legal systems depend on socially organized practices for identifying, changing, and applying valid rules, while Searle's account of institutional facts emphasizes how recognized status functions can support rights, duties, powers, and other institutional consequences (Hart 2012; Searle 1995). GRE can use these accounts to study legal representation while preserving a distinction among the represented relation, the legal category, and the wider relational reality in which the category operates.

Legal representation can be epistemically precise within its own domain. A registry can provide strong evidence of formally recognized ownership. A judgment can establish the legal disposition of a dispute. A statute can specify an authorized category. The same representation may provide only partial access to bargaining power, practical dependence, social recognition, informal obligation, or the full generative consequences of the relation. Legal status and relational structure therefore overlap without being interchangeable.

This distinction becomes particularly important when several legal forms can support similar relational effects or when the same legal form operates differently across material and institutional conditions. GRE should ask which dimensions of a relation a legal category preserves, which it abstracts from, and how much inferential weight the category can bear outside the legal question for which it was designed.

5.5.5 Statistical Representation

Statistical representation renders heterogeneous observations into variables, counts, distributions, indices, correlations, rates, and other numerical forms. This transformation can reveal patterns that remain invisible at the level of individual cases. It can also establish equivalences among cases that differ along dimensions omitted from the representation. Statistical knowledge therefore depends on both measurement and classification.

Desrosières emphasizes that statistical categories rely on conventions that make heterogeneous cases comparable, while Porter shows how standardized quantification can support forms of objectivity that travel across institutional settings (Desrosières 1998; Porter 1995). These insights are particularly important for GRE because relational phenomena are often transformed into variables whose apparent simplicity conceals substantial representational work. A variable labeled *employment*, *income*, *trust*, *household*, *network degree*, or *access* may summarize a complex family of relational judgments.

Statistical representation can operate at several distances from the target. Some variables are direct encodings of standardized records. Others are constructed

indices combining multiple indicators. Still others are latent quantities estimated through models. The evidential burden changes across these cases. A relation inferred from a latent variable requires justification of both the measurement model and the substantive interpretation that connects the statistical construct to the relational target.

Claim 5.41 (Statistical representation and relational compression). *Statistical representation can expand relational knowledge by revealing aggregate patterns and structured variation while simultaneously compressing heterogeneity through classification, measurement, and aggregation. The epistemic adequacy of a statistical representation therefore depends partly on whether the dimensions suppressed by that compression are material to the relational claim under inquiry.*

Claim 5.41 provides a practical criterion for GRE. Compression is not itself an error. Every representation omits detail. The relevant question is whether the omitted structure changes the inference. A national average may be useful for one question and deeply misleading for another when important relational differences are concentrated across regions, classes, institutions, or networks.

5.5.6 Computational Representation

Computational representation extends symbolic transformation into machine-processable structures. Databases encode entities and attributes; graphs encode nodes and relations; simulations encode rules and state transitions; machine-learning systems encode statistical regularities in parameters or latent spaces; agent-based models encode actors, environments, interaction rules, and update processes. These representations enable forms of analysis that exceed unaided human inspection in scale, speed, and combinatorial complexity.

Scientific modeling provides a useful epistemic precedent. Giere treats models as purpose-relative representations of selected aspects of reality, while Winsberg shows how computer simulation mediates among theory, model construction, approximation, computation, and evidence (Giere 2004; Winsberg 2010). GRE extends the same caution to computational representations of economic and social relations. A graph, simulation, classifier, embedding, or prediction model should be interpreted through the choices that generated its objects, variables, relations, loss functions, update rules, and evaluation criteria.

Computational representation can create particularly strong appearances of precision because symbolic categories are translated into exact machine-readable states. Exact encoding, however, does not guarantee exact correspondence with the target. A network edge can be precisely stored while remaining an imperfect representation of friendship, authority, dependence, or information flow. A classifier can assign a precise probability while relying on target categories whose social or institutional meaning is unstable. A simulation can execute deterministic rules whose relation to the modeled system remains uncertain.

Algorithmic systems also blur the boundary between epistemic representation and institutional operation. A computational representation can be used to allocate tasks, rank subjects, determine access, flag risk, or structure incentives. Kellogg, Valentine, and Christin show how algorithms in organizations can participate in directing, eval-

uating, and controlling work (Kellogg et al. 2020). The present section records this possibility while reserving the fuller recursive and performative analysis for Section 117. For epistemology, the immediate point is that an operational representation may later affect the data from which its adequacy is evaluated.

5.5.7 Symbolized Relations

A relation becomes symbolized when some aspect of its structure is rendered into a symbolic form that can be identified and manipulated within an epistemic practice. Symbolization can preserve the identity of relata, relation type, direction, intensity, duration, sequence, legal status, uncertainty, or other properties. Different representations preserve different subsets of these features.

A network graph can preserve adjacency while omitting the qualitative meaning of a tie. A contract can preserve specified obligations while omitting the history of bargaining. A statistical matrix can preserve covariance while omitting local narrative context. A linguistic account can preserve experienced meaning while leaving scale and comparability uncertain. None of these forms is intrinsically superior. Their adequacy depends on the target and the question.

Definition 5.42 (Symbolized relation). A *symbolized relation* is a representation in which one or more aspects of a target relation are encoded into a symbolic structure according to an explicit or implicit representational rule.

Definition 5.42 allows GRE to separate the relational target from the representational object used to study it. The distinction is especially important when a symbolized relation acquires institutional durability. A database edge, legal status, score, category, or label can be preserved long after the relation that originally motivated it has changed. Epistemic inquiry must therefore ask whether the representation continues to track the target, records only a past state, or has become part of a new relational configuration.

Symbolized relations can also be multiply realizable. The same underlying relation may be represented through testimony, legal documents, administrative records, statistical variables, graph structures, and computational states. Agreement across heterogeneous representations can strengthen a claim when the representations rely on sufficiently independent manifestation pathways. Apparent agreement among representations derived from the same underlying record system provides weaker triangulation because their errors may be correlated.

5.5.8 Representation Error

Representation error concerns the mismatch between what a representation preserves or implies and the epistemic target for which it is being used. The concept should remain broader than numerical measurement error. Linguistic categories can mischaracterize a relation; administrative schemas can omit relevant distinctions; legal forms can fail to track practical dependence; statistical indicators can compress consequential heterogeneity; computational models can encode inappropriate categories or unstable proxies.

Definition 5.43 (Representation error). *Representation error* is a target-relative discrepancy introduced when a representation omits, distorts, conflates, misclassifies, mis-scales, or otherwise preserves inadequately a feature of the epistemic target that is material to the claim for which the representation is used.

This definition makes error dependent on both target and use. A representation can omit most details of a system and still be adequate for a narrow question. The same omission can become an error when the inquiry shifts. Representation error is therefore evaluated relative to material relevance rather than descriptive completeness.

Several forms deserve distinction. *Selection error* arises when the representation excludes relations or cases important to the target. *classification error* arises when cases are assigned to categories that fail to preserve relevant differences. *aggregation error* arises when combining observations suppresses structure material to the inference. *semantic error* arises when a symbol is interpreted as referring to a different relation from the one it reliably tracks. *temporal error* arises when a representation is treated as current after the underlying relation has changed. *structural error* arises when the representational form itself presupposes a relation type that differs from the target, for example when a distributed field is forced into pairwise edges or a temporally evolving relation is represented as a static attribute.

Claim 5.44 (Representation adequacy under relational inquiry). *A representation is epistemically adequate for a relational claim to the extent that its selection, encoding, scale, temporal scope, and interpretive rules preserve the target-relevant structure sufficiently for the inference being made, with material sources of representational error identified and bounded where possible.*

Claim 5.44 complements the evidential distinctions of Section 5.4. Evidence can be abundant while representation remains poor. A large administrative dataset can provide many precise observations of categories that fail to capture the relation of interest. Conversely, a sparse qualitative representation can sometimes preserve a relational distinction that a large standardized dataset erases. Epistemic strength therefore depends on the relation among target, trace, representation, and inference rather than on volume or formal precision alone.

Representation error can also be systematic. Institutional categories may repeatedly underrepresent certain relations because the relevant distinctions are absent from the schema. Statistical systems may reproduce the same coding convention across decades. Computational systems may inherit labels and proxies from administrative infrastructures. Once such representations become widely reused, correlated errors can appear as cross-source confirmation. GRE should therefore examine the genealogy of representations as part of epistemic evaluation, especially when several apparently independent datasets or models depend on the same upstream categories.

The section's overall result is a layered account of symbolic mediation. Relational reality can generate manifestations and traces; symbolization renders selected aspects of those manifestations into portable forms; representations organize those symbols for linguistic, administrative, legal, statistical, or computational purposes; and representation error arises when the preserved structure is inadequate for the epistemic target. Symbolic forms can dramatically extend relational knowledge by en-

abling memory, comparison, aggregation, modeling, and communication across distance. Their epistemic force remains tied to the histories, conventions, institutions, and technical procedures through which they are produced. This layered account prepares Section 117, which examines what happens when representations move beyond epistemic mediation and enter the generative-relational system as operative classifications, rules, measurements, and algorithmic interventions.

5.6 Performative Representation

Section 5.5 treated representation principally as an epistemic mediation through which selected aspects of relational reality become portable, comparable, storable, and inferentially usable. The present section examines a further possibility: some representations enter the relational system in which they circulate and contribute causally to subsequent relations, opportunities, expectations, classifications, decisions, and institutional responses. Its role is to distinguish representational description from representational intervention while preserving the separation among epistemic adequacy, causal efficacy, relational stabilization, and normative evaluation. The section proceeds through seven components: representation and subsequent relation, classification effects, legal performativity, administrative performativity, statistical performativity, algorithmic performativity, and recursive representation. The method is relational and process-oriented. Each case is analyzed through the pathway by which a representation acquires uptake, becomes connected to action or institutional procedure, and alters conditions that can later become objects of representation in their own right.

The language of performativity has several intellectual lineages. Austin's analysis of performative utterances established a particularly influential case in which words can participate in the performance of actions under conventional and contextual conditions (Austin 1962). Historical and sociological work later broadened attention toward classifications, measurements, and models whose circulation can alter the practices and objects to which they refer. Hacking's historical ontology emphasizes the historical emergence of categories and objects within practices of classification and reasoning (Hacking 2002). Espeland and Sauder analyze reactivity to public measures, showing how rankings can alter expectations and organizational behavior (Espeland and Sauder 2007). MacKenzie documents cases in which financial models became incorporated into market practices and thereby participated in reshaping the markets they represented (MacKenzie 2006). GRE draws on these lineages while giving the term a narrower relational specification.

Definition 5.45 (Performative representation). A *performative representation* is a representation that, through uptake within a social, institutional, legal, technical, or other relational process, contributes causally to the subsequent configuration, persistence, transformation, or generation of relations relevant to its domain of use.

Definition 5.45 makes performativity conditional. Representations vary in their capacity to affect subsequent relations. A privately written description that remains unread can be epistemically rich while having little practical effect on its target domain. A coarse administrative category connected to eligibility rules can have substantial

causal consequences despite limited descriptive resolution. The relevant question is therefore whether a credible pathway connects the representation to subsequent relational change.

Several conditions commonly strengthen such a pathway. A representation may acquire recognized authority, become embedded in organizational routines, trigger a legal or administrative rule, guide allocation, coordinate expectations among multiple actors, alter incentives, or become repeatedly reproduced in later decisions. Persistence and reach also matter. A classification used once in a local setting can produce a narrow effect, whereas a category embedded across interoperable databases can structure relations across institutions and time. These conditions concern performative efficacy rather than epistemic quality. The distinction becomes central to the analysis that follows.

Claim 5.46 (Separation of representational adequacy and performative efficacy). *The epistemic adequacy of a representation and its performative efficacy are distinct properties. A representation can exert substantial causal influence while preserving the target poorly, and a highly adequate representation can remain causally weak when it lacks uptake or operational connection.*

Claim 5.46 extends the discussion of representation error in Section 5.5. An inaccurate score can still affect access to resources when institutions act upon it. A careful scholarly reconstruction can remain practically inert when no relevant institution, participant, or decision process uses it. Causal efficacy therefore supplies neither a truth criterion nor a normative endorsement. Conversely, epistemic adequacy supplies no guarantee of practical consequence.

5.6.1 Representation and Subsequent Relation

A representation becomes performatively relevant when it enters a pathway through which subsequent relations depend partly on its presence, content, or institutional interpretation. The phrase *subsequent relation* refers here to a relation whose configuration is temporally downstream from the representation and whose formation, modification, persistence, or dissolution is causally influenced by the representation's uptake. The representation can concern the same relation, a neighboring relation, or a higher-order structure that reorganizes several relations at once.

Consider a representation of creditworthiness. The representation may summarize traces of income, repayment, debt, or institutional history. Once lenders use the representation in allocation decisions, the score can influence access to credit, the terms of borrowing, subsequent repayment opportunities, and future records. The representation then occupies two roles. It remains an epistemic object intended to say something about a target, while also becoming a causal input into a changing relational configuration among borrowers, lenders, intermediaries, and institutions. The later configuration can generate traces that are subsequently used to evaluate or update the original representational system.

The same structure can occur without centralized authority. A public reputation category can coordinate expectations among many actors. A widely circulated label can alter whom others approach, avoid, trust, or monitor. A market model can influence trading strategies. A scientific classification can redirect data collection and in-

stitutional attention. The relevant mechanism differs across cases, yet the relational structure is comparable: representation enters practice, practice changes relational conditions, and those changed conditions can later manifest through new traces.

This process also clarifies the relation between performative representation and relational reality. The preceding ontological chapter defined relational reality through emergence, sufficient stability over some spacetime region, and causal efficacy within a generative-relational system. A representation that becomes institutionally stabilized and causally consequential can therefore itself acquire relational reality within the relevant domain. Its relational reality concerns the representation's effective presence in the system. The truth or adequacy of what it represents remains a separate epistemic question.

A useful analysis of performativity should therefore identify at least three objects: the *target relation* that the representation purports to describe; the *operational relation* through which the representation enters action, decision, or coordination; and the *subsequent relation* whose configuration is altered through that operation. These objects can coincide only partially. A legal status may represent one relation while changing several adjacent rights and obligations. A statistical ranking may represent performance while altering recruitment, funding, attention, and internal organizational priorities. A computational score may estimate one target while becoming an input into an allocation process with wider relational consequences.

Proposition 5.47 (Operational mediation of performative effects). *A claim of performative representation is strengthened when the analysis identifies a traceable operational mediation connecting the representation to subsequent relational change, including the actors, rules, infrastructures, expectations, or decision procedures through which the representation acquires causal relevance.*

Proposition 5.47 guards against treating temporal succession as sufficient evidence of performativity. A representation can precede a relational change without contributing materially to it. Epistemic warrant requires evidence concerning the transmission pathway, competing causes, timing, scope, and persistence of the effect. The epistemology of performativity is therefore an attribution problem as well as a representational one. Later sections on evidence, generative relations, and attribution develop this problem more systematically.

5.6.2 Classification Effects

Classification is one of the most common routes through which representation becomes performative. A classification groups cases according to distinctions that make them administratively, legally, statistically, socially, or computationally tractable. Once a category is connected to recognition, expectation, allocation, sanction, entitlement, or organizational procedure, category membership can contribute to the relations available to the classified case.

Hacking's historical ontology is relevant because it directs attention to the historical formation of categories and objects within practices of naming, classification, inquiry, and intervention (Hacking 2002). GRE places particular emphasis on the relational consequences of this process. A classification can make a dispersed pattern publicly recognizable, support coordination around a previously weakly articulated

relation, or provide a vocabulary through which subjects can interpret experience. It can also compress heterogeneity, stabilize a contested boundary, or attach institutional consequences to differences that were previously less consequential.

Classification effects can operate through several pathways. First, institutions can route cases differently after classification. Second, other actors can alter expectations toward a classified person, organization, object, or event. Third, subjects can adapt behavior in anticipation of classification. Fourth, the classification can alter data collection, because institutions begin recording variables that matter to the category. Fifth, repeated use can stabilize the category itself, making it appear increasingly natural or inevitable even when its boundary arose historically from specific institutional purposes.

The economic significance of classification is especially clear when category membership affects conversion possibilities. A person classified as eligible can gain access to an entitlement, service, market, or institutional pathway. An organization classified under a particular legal or administrative type can face different reporting duties or financing possibilities. An object classified as a commodity, protected resource, waste product, cultural object, or regulated good can enter different circuits of valuation and exchange. Kopytoff's analysis of commoditization illustrates how object status can vary across social contexts and histories rather than functioning as an invariant property of the object itself (Kopytoff 1986). GRE treats such changes as transformations in the relational organization through which value and exchange become possible.

Classification effects remain heterogeneous. A category can have strong formal consequences and weak practical uptake. Another can have little formal standing while strongly coordinating expectations. A category can also become performative through anticipation before an official decision is made, for example when actors reorganize behavior around an expected classification threshold. The causal unit is therefore the broader classificatory arrangement, including rules of assignment, visibility, authority, anticipated consequences, channels of contestation, and the temporal persistence of records.

The possibility of classification effects intensifies the epistemic stakes of representation error. Misclassification can alter the target environment and thereby generate traces consistent with the category that produced the intervention. An initially weak representation can thus acquire apparent empirical support after institutions reorganize behavior around it. This possibility does not make all classifications self-confirming. It identifies a specific recursive pathway that relational inquiry should test where classification and intervention are tightly connected.

5.6.3 Legal Performativity

Legal representation supplies a comparatively explicit form of performativity because legal systems connect recognized symbolic acts, categories, and documents to rules governing rights, duties, powers, liabilities, statuses, procedures, and institutional competencies. Austin's account of performative utterances is directly relevant to cases in which conventional acts accomplish legally recognized changes under appropriate conditions (Austin 1962). Hart's analysis of primary and secondary

rules provides a complementary perspective on how legal systems organize conduct and establish rules through which legal validity, alteration, and adjudication are recognized (Hart 2012).

For GRE, the important point concerns the generation or transformation of relations through legally operative representation. A contract can specify obligations that become institutionally recognizable. Incorporation can establish an organizational form with consequences for asset ownership, liability, governance, and third-party relations. Property registration can reorganize claims among holders, users, creditors, successors, and public authorities. Organizational law can structure asset partitioning and relations between firms, owners, managers, and creditors (Hansmann and Kraakman 2000). In each case, a symbolic form participates in producing a relational configuration because institutions recognize the form and connect it to durable rules of action and adjudication.

Legal performativity depends on more than textual existence. A document outside a recognized procedure can fail to alter legal relations. A valid rule can have uneven practical consequences when enforcement, awareness, institutional capacity, or social uptake varies. The performative effect therefore travels through a relational infrastructure composed of authorized procedures, offices, records, enforcement capacities, professional practices, expectations, and conventions of recognition.

This distinction also clarifies the relation between legal reality and wider relational reality. A legal status can be causally effective even when everyday practice only partially conforms to it. Conversely, a stable practical relation can exist without corresponding legal recognition. GRE should therefore distinguish legal representation, legally constituted relations, practically enacted relations, and the wider relational field in which both are situated. Their correspondence is an empirical and historical matter.

Legal representation can further transform epistemic conditions. Once a relation receives a legal form, it often becomes easier to record, archive, contest, compare, and transmit across institutions. Legal performativity can therefore generate new traces at the same time that it restructures rights and obligations. This dual role is important for later inquiry: the evidential record available to the researcher can be partly produced by the legal arrangements whose consequences are being studied.

5.6.4 Administrative Performativity

Administrative systems convert heterogeneous situations into categories, records, statuses, files, priorities, thresholds, and procedural states that can be processed at organizational scale. Section 5.5.3 examined these forms primarily as representations. Their performative dimension emerges when administrative records are connected to allocation, access, sequencing, monitoring, recognition, or institutional response.

An administrative category can determine whether a case enters a queue, receives a benefit, triggers an inspection, qualifies for an exception, or becomes visible to a particular office. A record can also coordinate actions across institutions when interoperable systems reuse the same status or identifier. Administrative representation then becomes an infrastructure for relational coordination. It shapes which actors encounter one another, which procedures become available, how quickly an interac-

tion proceeds, and what documentation must be supplied for further recognition.

Institutional analysis has long emphasized that formal rules and organizational structures shape incentives and patterns of interaction (North 1990). GRE adds a representational question: the operation of a rule frequently depends on how a case has first been rendered administratively legible. The category is therefore part of the pathway through which the institution acts. A mismatch between lived relations and administrative representation can have material consequences even when the formal rule is applied consistently to the recorded category.

Administrative performativity can accumulate over time because records persist. A category assigned at one moment can influence subsequent decisions long after the underlying relation has changed. Later institutions may inherit the recorded status without observing the original context. This creates temporal asymmetry between relational transformation and administrative updating. Historical inquiry should therefore distinguish the time of the represented relation, the time of recording, the time of institutional use, and the time at which consequences become visible.

Large-scale data infrastructures extend this process by enabling records to circulate across organizational boundaries. The World Bank's treatment of integrated data systems illustrates the growing institutional importance of administrative and other data sources for public decision processes (World Bank 2021). From a GRE perspective, integration increases both epistemic reach and performative reach. A category used across many institutional settings can coordinate services and reduce repeated informational costs; the same portability can also propagate a representational error through multiple downstream decisions.

Administrative performativity is therefore inseparable from questions of correction and contestability. A system that acts on records requires mechanisms through which outdated, contextually inappropriate, or erroneous representations can be revised. The epistemic significance of such mechanisms is substantial: they help prevent a representational state from becoming self-reinforcing solely because later institutions repeatedly treat the existing record as evidence of its own correctness.

5.6.5 Statistical Performativity

Statistical representation becomes performative when indicators, rankings, benchmarks, forecasts, classifications, or numerical models are incorporated into behavior, evaluation, coordination, or resource allocation. The history of quantitative objectivity shows how standardized numerical procedures can acquire authority partly because they enable impersonal comparison across cases and institutions (Porter 1995). Desrosières likewise emphasizes the historical construction of statistical categories, conventions, and comparability (Desrosières 1998). These representational infrastructures can later become objects around which institutions organize action.

The performative pathway is particularly visible when measurement becomes evaluation. A numerical indicator used only for descriptive analysis can have limited effect on the target. Once the same indicator enters a funding rule, ranking system, performance review, public comparison, or organizational target, actors gain reasons to attend to the metric. Espeland and Sauder's study of law-school rankings demonstrates this form of reactivity: public measures can alter expectations

and organizational behavior, with commensuration and self-reinforcing expectations contributing to the process (Espeland and Sauder 2007). The measure thereby becomes part of the environment within which the measured organizations act.

Statistical performativity can alter both behavior and category meaning. Actors may direct effort toward measured dimensions, redesign internal procedures to improve recorded performance, invest in data practices that affect future measurement, or reorganize activities around threshold effects. Such responses can improve the target of measurement when the indicator tracks a valued objective sufficiently well. They can also increase divergence between the measured variable and the wider target when optimization concentrates on what the metric captures.

MacKenzie's study of financial economics provides a related case at the level of models rather than simple indicators. Financial models became incorporated into market practices and contributed to changes in the markets to which those models referred (MacKenzie 2006). For GRE, this example is important because it shows that representation can participate in the generation of conditions under which its own assumptions become more or less descriptively successful. Model evaluation in such settings requires historical attention to deployment as well as comparison with an independently evolving target.

The epistemic challenge is especially sharp when a statistical representation changes the data-generating relations that later supply evidence about its adequacy. A ranking can change applicant behavior, institutional investment, and reporting practices. A forecast can alter expectations that influence subsequent transactions. A performance target can change which activities are prioritized and recorded. Later observations then describe a world partly reorganized through earlier measurement. Statistical evaluation should therefore ask whether observed stability reflects a stable target, adaptation to the measure, changes in recording practice, or some combination of these processes.

Claim 5.48 (Reactivity of operationalized measures). *The probability and scope of statistical performativity increase when a measure becomes visible to affected actors and is coupled to consequential decisions, comparative evaluation, resource allocation, reputational effects, or repeated organizational routines.*

Claim 5.48 is deliberately conditional. Quantification alone does not establish reactivity. The causal mechanism depends on how the measure enters a relational field and what capacities actors possess to respond to it. This preserves the distinction between statistical representation and statistical intervention.

5.6.6 Algorithmic Performativity

Algorithmic systems intensify several mechanisms already present in legal, administrative, and statistical performativity. A computational representation can be generated rapidly, updated repeatedly, individualized at fine granularity, and connected directly to operational decisions. Scores, rankings, recommendations, classifications, forecasts, and generated outputs can therefore enter relational processes with short delays between representation and consequence.

Algorithmic performativity begins when computational outputs affect subsequent

action. A ranking can determine visibility; a recommendation can redirect attention; a risk score can trigger review; a scheduling system can reorganize temporal relations among workers, tasks, and managers; a classifier can determine which cases receive further human examination. Kellogg, Valentine, and Christin document how algorithmic systems in workplaces can participate in directing, evaluating, and disciplining work through computational mechanisms of control (Kellogg et al. 2020). The broader epistemic lesson is that algorithmic outputs can simultaneously represent a domain and intervene within it.

Several features make this form distinctive in degree even when its relational logic resembles earlier institutional technologies. First, computational systems can operate at high frequency and large scale. Second, their outputs can be personalized, producing different relational environments for different subjects. Third, automated systems can connect representation to action with little deliberative delay. Fourth, models can be updated using data partly generated after previous versions were deployed. Fifth, complex model architectures can make the operative path difficult for affected actors and sometimes difficult for institutional operators to reconstruct fully.

These features create recursive epistemic problems. A model predicts behavior; institutional action follows the prediction; subjects respond to the resulting environment; their response generates new data; and later models learn from data produced under the earlier intervention. The resulting dataset records a mixture of target behavior and system-induced behavior. Relational inquiry should therefore distinguish pre-deployment patterns, intervention pathways, behavioral responses, institutional adaptation, and later representational updates where evidence permits.

Algorithmic performativity also expands the importance of counterfactual evaluation. Observed outcomes after deployment cannot by themselves establish what would have occurred without the representational intervention. A low observed default rate among people granted credit, for example, does not reveal the outcomes of people who were denied access on the basis of a score. Similar selection problems arise in policing, employment, insurance, moderation, education, and other allocation contexts. The epistemic issue concerns missing relational trajectories created by the decision process itself.

The distinction in Claim 5.46 becomes especially important here. An algorithm can be operationally powerful because institutions rely on it extensively, while its target validity remains uncertain or uneven across contexts. Conversely, a well-validated model can have limited performative impact when humans routinely override its outputs or when the modeled decision has weak consequences. Evaluation therefore requires separate evidence for predictive or representational adequacy, institutional uptake, causal effects of deployment, and normative acceptability.

5.6.7 Recursive Representation

The preceding cases converge on a general epistemic structure: representation can enter the system it represents, contribute to subsequent relational change, and thereby alter the conditions from which later representations are produced. GRE calls this structure *recursive representation*. The term does not imply that the same representation repeats unchanged. Recursion can involve successive legal cate-

gories, updated administrative records, revised indicators, changing model versions, strategic responses, or institutional reinterpretations.

Definition 5.49 (Recursive representation). *Recursive representation* is a process in which a representation contributes causally to subsequent relational conditions, and those altered conditions later become part of the evidential or practical basis for further representation.

Definition 5.49 adds time to the representational architecture developed in Section 5.5. The epistemic relation between representation and target can change after deployment because the target itself may develop partly through responses to previous representations. Evaluation at a later moment therefore cannot always assume that the represented system evolved independently of the representational practices used to know and govern it.

Recursive representation can produce several patterns. One is *stabilizing recursion*, in which a representation coordinates expectations and practices in ways that make the represented category or regularity more durable. Another is *destabilizing recursion*, in which actors respond strategically and reduce the representation's later predictive or classificatory usefulness. A third is *transformative recursion*, in which the representation redirects practices sufficiently that the original target changes form and requires a revised vocabulary or model. These patterns are analytical possibilities rather than exhaustive types.

The epistemic investigation of recursion therefore requires a genealogy of representation. Researchers may need to reconstruct when a representation was introduced, which institutions adopted it, how it was operationalized, which actors encountered its consequences, what opportunities for adaptation or contestation existed, how records were subsequently generated, and whether later measures reused the same categories. This historical reconstruction can prevent later observations from being treated as independent confirmation when they are partly products of earlier representational interventions.

Recursive representation also clarifies why knowledge infrastructures can become components of the realities they study. Statistical offices, registries, legal records, classification standards, ranking systems, and algorithmic platforms can initially emerge to observe, coordinate, or describe. Once their outputs influence action repeatedly, the infrastructure becomes part of the relational environment. Its categories can acquire persistence, causal efficacy, and institutional reach. The resulting relational reality may then include both the represented processes and the representational machinery through which those processes are governed and known.

This conclusion preserves ontological restraint. The causal incorporation of a representation establishes a claim about its relational efficacy within a specified system and interval. It does not establish the representation's ultimate ontological status, descriptive completeness, epistemic truth, or normative legitimacy. These dimensions require their own arguments and evidence. The distinction is particularly important for GRE because economic institutions routinely operate through prices, classifications, contracts, accounts, metrics, models, forecasts, and computational systems whose representational and causal roles are intertwined.

Section 5.6 therefore extends the epistemology of GRE from mediated access to

recursive mediation. Representations can preserve and distort relational knowledge, and some can also become causal participants in later relational generation. Their epistemic assessment consequently requires attention to the target they represent, the operational pathways through which they act, the relations they subsequently transform, and the traces produced after intervention. This result prepares Section 118, which turns to observation and partial access. Once representations can participate in producing the relations later observed, the position, scope, and limits of observation become even more consequential for relational knowledge.

5.7 Observation and Partial Access

Sections 5.4–5.6 established three progressively richer conditions of epistemic mediation: relations leave traces that acquire evidential roles, manifestations can be symbolized and represented through selective schemes, and some representations can enter practice and alter subsequent relational conditions. The present section examines a further constraint that cuts across all three processes: observation is ordinarily partial. Its role is to specify how access to relational reality varies with observer position, institutional capacity, distributed observation, hiddenness, concealment, and time. The objective is neither to reduce knowledge to perspective nor to treat partial access as generalized skepticism. It is to make the conditions of observation explicit enough that claims can be calibrated to what an observation regime could plausibly register. The section proceeds through seven components: partial observation, observer position, unequal epistemic access, distributed observation, hidden relations, strategic concealment, and temporal limits. The method is target-relative and relational. Observation is analyzed through the relation among an epistemic target, an observer or observing system, the channels through which traces become accessible, and the conditions that shape what remains outside those channels.

Scientific perspectivism provides one useful precedent for this orientation. Giere argues that scientific observation and representation are perspectival while remaining open to disciplined evaluation against selected aspects of the world (Giere 2006). Longino’s contextual empiricism adds that evidential reasoning is socially organized and can be strengthened through critical interaction among differently situated inquirers (Longino 1990). GRE draws a narrower conclusion from these lineages: the position and capacities of an observer belong to the evidential conditions of a relational claim, yet perspectival mediation does not dissolve the distinction between better and worse warranted claims.

Definition 5.50 (Observational access). *Observational access* is the target-relative capacity of an observer or observing system to register, retrieve, preserve, or otherwise obtain traces that can bear evidentially on a specified relational claim within a given interval and observational regime.

Definition 5.50 treats access as structured rather than binary. An observer may possess direct access to some traces, mediated access to others, and little access to further traces that remain outside the observation regime. Access can also change through instrumentation, institutional permission, record preservation, legal authority, co-operation from participants, technical interoperability, or historical reconstruction.

The epistemic question is therefore not simply whether a relation was observed. It concerns which aspects of the relation could become observable to which observer, through what channels, and under what conditions.

5.7.1 Partial Observation

Partial observation occurs when the accessible evidential field captures only a subset, projection, interval, manifestation, or trace structure relevant to the relational target. The partiality can arise because the target relation extends beyond the observer's spatial reach, persists longer than the period of observation, contains components that do not manifest through available instruments, or participates in neighboring relations that remain unrecorded. Partiality can also be produced institutionally when a registry, survey, platform, archive, or administrative procedure records selected categories while leaving other dimensions unregistered.

The term should be distinguished from simple error. An observation can be accurate with respect to what it registers while remaining incomplete with respect to the target relation. A transaction ledger may accurately record transfers while revealing little about the expectations, obligations, informal sanctions, or prior relations that gave the transfers their meaning. A labor dataset may accurately record wages and hours while only partially registering care responsibilities, informal work, bargaining relations, workplace authority, or dependencies outside the employing organization. A legal file may faithfully preserve formal acts while leaving much of the surrounding social relation outside the record. The problem is therefore one of evidential reach before it becomes one of incorrect measurement.

Partial observation is also distinct from underdetermination, although the two can interact. Partial access concerns the evidential field available to the observer. Underdetermination concerns the possibility that the available evidence remains compatible with multiple relational reconstructions. Expanding observation can reduce underdetermination when newly accessible traces discriminate among alternatives. It can also reveal additional complexity and thereby enlarge the set of plausible reconstructions. More observation therefore does not mechanically produce uniqueness, yet the structure of partial access remains indispensable for evaluating what conclusions the evidence can sustain.

Claim 5.51 (Target-relative partiality). *An observation is partial only relative to an epistemic target and a specified claim. The same evidential field can provide extensive access for one target while providing limited access for another.*

Claim 5.51 prevents completeness from becoming an undefined ideal. A price series may be sufficient for a narrowly specified claim about recorded price variation and insufficient for a claim about the generative history of market power. A network trace may reveal interaction frequency while providing weaker access to the meanings, obligations, or asymmetries carried by those interactions. Observational adequacy should therefore be evaluated against the target relation and inference actually at stake.

Partial observation also has a recursive dimension after Section 5.6. Observation regimes can alter behavior, and those behavioral responses can change what later becomes observable. Once participants know which indicators are measured, which ac-

tions create records, or which traces trigger institutional responses, they may adapt to the observation regime. The evidential field then reflects both the target process and responses to being observed. Partiality is consequently capable of changing through the very practices designed to reduce it.

5.7.2 Observer Position

Observer position identifies the relational location from which traces become accessible and interpretable. Position can include physical location, organizational role, legal standing, technical permission, social membership, temporal location, disciplinary training, and participation in the relation being studied. These dimensions should not be collapsed into a single standpoint. They describe distinct conditions that can shape access to different portions of the evidential field.

An employee, a manager, a regulator, a customer, and an external researcher can encounter the same organization through different trace structures. The employee may observe day-to-day coordination and informal authority while lacking access to strategic records. The manager may possess aggregate reports while seeing less of informal adaptation. The regulator may obtain standardized disclosures without continuous access to internal practice. A customer may observe outcomes and interfaces while remaining outside the organization's production process. An external researcher may combine records across institutions while lacking the situated knowledge held by participants. These positions can generate complementary evidence and distinct blind spots.

Observer position also affects the interpretation of traces. A technically identical record can acquire different evidential significance when observers possess different contextual knowledge about how the record was generated. An entry in an administrative database may look like a direct indicator to an external analyst while practitioners know that it is routinely produced through local workarounds, discretionary coding, or delayed reporting. Conversely, participants deeply embedded in a practice can normalize regularities that become more visible through comparison across settings. Positional proximity and distance can therefore each generate epistemic capacities and limitations.

Giere's perspectivism is helpful here because it permits situated representation without making all perspectives epistemically interchangeable (Giere 2006). GRE likewise treats observer position as a condition to be reported and evaluated, rather than as an automatic source of privilege or disqualification. A positional claim gains warrant when the observer's location is relevant to the target, the access channel is specified, and the evidential contribution can be compared with other available traces.

Definition 5.52 (Observer-position profile). An *observer-position profile* is a structured description of the locations, permissions, capacities, dependencies, and temporal conditions that shape an observer's access to evidence relevant to a specified relational target.

The profile in Definition 5.52 is an analytical device rather than a requirement for exhaustive self-description. Its purpose is to reveal the dimensions of position that materially affect an inference. For a platform study, technical access and data-retention policy may be decisive. For research on care relations, household

participation and temporal proximity may matter more. For historical reconstruction, archival survival and later access rules can dominate. The relevant positional dimensions should therefore be selected in relation to the claim rather than listed mechanically.

5.7.3 Unequal Epistemic Access

Observation is often distributed unequally. Actors can differ in access to records, instrumentation, language, expertise, computational resources, legal discovery, institutional archives, confidential communications, and opportunities to question participants. They can also differ in whether their own observations are preserved, recognized, and treated as credible by institutions. Unequal epistemic access therefore concerns both possession of traces and participation in the processes through which traces become accepted evidence.

These inequalities can follow existing relational asymmetries. An institution may possess detailed records about individuals who possess little reciprocal visibility into the institution's decision procedures. A platform can observe user behavior at fine granularity while users encounter only selected outputs of the platform's internal models. Employers may aggregate workplace data unavailable to workers; financial intermediaries may combine records unavailable to applicants; states can possess administrative information distributed across agencies while residents experience the resulting classifications without equivalent access to the underlying infrastructures. The epistemic relation is then asymmetric even before the resulting knowledge is used for action.

Unequal access can also arise within research communities. Expensive instruments, proprietary datasets, security classifications, language barriers, disciplinary specialization, and institutional affiliations can determine which investigators can inspect which evidence. Longino's account of socially organized criticism is relevant because the epistemic benefits of plural criticism depend partly on whether differently situated participants can actually enter the critical process and receive uptake (Longino 1990). Where access to evidence and channels of criticism are concentrated, apparent consensus can reflect a narrow observational field as well as strong evidential convergence.

Fricker's analysis of epistemic injustice provides a more specifically normative lineage concerning credibility, testimony, and interpretive participation (Fricker 2007). The present section uses that work cautiously. Unequal access is broader than epistemic injustice: differences in access can arise from specialization, confidentiality, temporal position, or technical constraints without already establishing an injustice. Conversely, some access inequalities can be produced or intensified through power and exclusion. Section 118 records the epistemic structure; the systematic normative treatment of epistemic justice is reserved for Section 129.

Claim 5.53 (Asymmetry of observational visibility). *Relational systems can generate asymmetric observational visibility in which one participant or institution can inspect, aggregate, or preserve traces concerning another while reciprocal access remains substantially weaker.*

Claim 5.53 is important for GRE because observational asymmetry can become eco-

nomically consequential even before a later analysis of power or justice. Differential visibility can affect bargaining, prediction, screening, monitoring, coordination, and the ability to contest a representation. It can also shape the evidential archive available to future inquiry. Researchers should therefore record whose traces become legible to whom and whose interpretive resources enter the evidential process.

5.7.4 Distributed Observation

Partial access does not imply that observation must remain confined to the perspective of a single observer. Relational knowledge can be assembled through distributed observation: multiple persons, instruments, institutions, records, and computational systems can contribute different portions of the evidential field. The epistemic problem then shifts from the limits of one observer to the conditions under which heterogeneous observations can be combined without erasing their provenance or falsely treating them as independent.

Hutchins's study of navigation provides an influential precedent for analyzing cognitive accomplishment across persons, artifacts, procedures, and socially organized practice (Hutchins 1995). GRE does not identify distributed observation with distributed cognition. The narrower borrowing is structural: an epistemic accomplishment can depend on a system of differentiated observational roles whose contributions are coordinated through material and institutional arrangements.

Distributed observation can expand access in several ways. Different observers can occupy complementary positions. Instruments can register phenomena outside ordinary human perception. Administrative systems can preserve traces over long intervals. Historical archives can connect temporally separated observers. Computational systems can search volumes of records beyond unaided human capacity. Affected communities can contribute contextual knowledge missing from standardized datasets. Cross-disciplinary research can combine distinct techniques for eliciting, measuring, interpreting, and reconstructing evidence.

The combination process introduces its own risks. Observations can share a common source and therefore provide less independent corroboration than their number suggests. Several databases may reproduce the same upstream record. Multiple witnesses may rely on a shared narrative. Several models may be trained on overlapping datasets. Institutional reports may inherit a common classification scheme. Apparent triangulation can therefore be generated by replication of the same observational dependency.

Distributed observations can also be difficult to translate. Different observers may use categories that overlap only partially, operate at different temporal resolutions, or register different levels of a relational system. Integration then requires an explicit account of correspondence among the observational schemes. The goal is not necessarily to collapse them into a single dataset. In some cases, preserving several partially compatible representations can retain information that forced harmonization would destroy.

Proposition 5.54 (Provenance in distributed observation). *The evidential value of distributed observation increases when the provenance, dependence structure, observational position, and transformation history of the contributing traces remain recov-*

erable during integration.

Proposition 5.54 connects distributed observation to the book's broader concern with future re-epistemization. Provenance allows later researchers to distinguish independent corroboration from repeated transmission, reconstruct disagreements, revise mappings among categories, and incorporate evidence that earlier syntheses excluded. Distributed observation is therefore strongest when integration preserves the conditions under which each contribution became observable.

5.7.5 Hidden Relations

Section 5.4 introduced hidden relations as relations whose relevant traces remain outside an observation regime because they are inaccessible, suppressed, displaced, masked, or routed beyond the observer's evidential field. The present subsection develops hiddenness from the standpoint of partial access. A relation can be hidden from one observer and accessible to another, hidden during one interval and later reconstructed, or visible through consequences while its internal organization remains inaccessible. Hiddenness is therefore relational to an observer-position profile and an observation regime.

This target-relative account avoids treating hiddenness as an intrinsic metaphysical property. A private agreement can be hidden from the public while fully known to its participants. An informal coordination practice can be hidden from an administrative database while directly experienced by workers. A computational dependency can be hidden from users while inspectable by engineers. A historical relation can be hidden during its operation and become partly accessible after archives open. The status of hiddenness can change without the underlying historical event changing.

Hidden relations often reveal themselves indirectly. A pattern of outcomes can be difficult to explain through visible relations alone. Repeated discrepancies among official categories, reported experience, and observed consequences can motivate inference toward an unobserved mechanism. Such inference requires restraint. The presence of unexplained residuals does not identify a unique hidden relation. It provides a reason to widen the compatibility set, seek discriminating traces, examine missing channels, and compare alternative reconstructions.

The distinction among latent, hidden, and merely unrecorded relations should also remain intact. A latent relation is analytically inferred from manifestations without requiring that actors intentionally obscure it. A hidden relation is inaccessible relative to an observation regime, regardless of whether the inaccessibility is deliberate. An unrecorded relation may have been openly visible at the time while leaving weak surviving traces. These conditions can overlap, yet they generate different evidential problems and call for different forms of reconstruction.

Hiddenness can also be institutionally produced without a deliberate decision to conceal. Organizational fragmentation can place mutually relevant records in separate units. Data protection can intentionally restrict access for legitimate reasons. Technical systems can store information in incompatible formats. Informal practice can remain outside reporting categories. Specialized language can make evidence practically inaccessible to outsiders. Treating all hiddenness as deception would therefore obscure the heterogeneous mechanisms through which relational

knowledge becomes unavailable.

5.7.6 Strategic Concealment

Strategic concealment is a narrower case in which an actor, organization, or coordinated group intentionally changes the visibility, accessibility, timing, or interpretability of traces in order to shape what another observer can know. The strategy can involve withholding information, compartmentalizing access, producing selective disclosure, delaying records, routing activity through less visible channels, manipulating categories, or generating misleading traces. The epistemic target is consequently affected by an adversarial or strategic relation between observer and observed.

Simmel's early sociology of secrecy treated differentiated access to knowledge as a constitutive feature of social relations rather than a marginal anomaly (Simmel 1906). GRE extends this insight into an epistemic specification: secrecy and concealment alter the observation relation itself. A researcher, regulator, contractual counterparty, competitor, or affected community may receive an evidential field partly shaped by the target actor's expectations about observation.

Strategic concealment complicates ordinary assumptions about missing data. Absence can result from nonexistence, failed detection, lack of preservation, restricted access, or successful concealment. The evidential meaning of non-observation therefore depends partly on whether the observation regime is robust against strategic behavior. Where actors benefit from remaining unseen, evidence of absence requires stronger knowledge of detection pathways and stronger reasons to believe that relevant traces could not easily have been displaced or suppressed.

Concealment can also be selective rather than comprehensive. An actor may disclose enough information to satisfy a formal requirement while obscuring the relation most relevant to the observer's target. A firm can provide aggregate figures that preserve confidentiality while also limiting inspection of internal distribution. A platform can publish policy descriptions while withholding model details. A political or commercial organization can release favorable records and leave unfavorable episodes difficult to reconstruct. These cases differ normatively and legally, yet they share an epistemic structure in which the observed evidential field has been strategically curated.

The response to strategic concealment should remain evidential rather than presumptive. Suspicion alone does not establish a hidden relation. Researchers can compare independent sources, inspect inconsistencies, analyze incentives for disclosure and concealment, study record-generation procedures, seek traces generated outside the control of the focal actor, and report remaining uncertainty. An adversarial observation setting raises the burden of reconstruction while preserving the requirement that claims remain supported by evidence.

Claim 5.55 (Strategic observability). *When observed actors can anticipate an observation regime and possess meaningful capacity to alter the traces available to that regime, observability becomes partly strategic and should be modeled as a relational condition of evidence production.*

Claim 5.55 extends Section 5.6's account of performativity. Observation need not

merely record a pre-existing evidential field; the expectation of observation can influence which traces are generated, preserved, displayed, displaced, or concealed. The epistemology of GRE therefore requires attention to the interaction between observational design and strategic response.

5.7.7 Temporal Limits of Observation

Observation is temporally situated. An observer encounters a relation during an interval, through traces that have their own durations, delays, retention periods, and rates of decay. Some relations are slow and persistent relative to observation; others are episodic, rapidly changing, or visible only during transition. Temporal position therefore affects which aspects of a relational process can be observed and which must be reconstructed from surviving traces.

Several temporal limits are analytically distinct. First, an observation window can be shorter than the formation or transformation of the target relation. Short windows may then misclassify transient conditions as stable structures or stable structures as isolated events. Second, manifestations can lag behind relational change. An institutional reform can alter practice before aggregate indicators respond, while a financial or administrative record can continue reflecting earlier conditions after the relation has changed. Third, traces can decay or disappear. Informal communication, tacit coordination, and ephemeral digital interactions can leave uneven historical records. Fourth, later observation can benefit from archives and hindsight while losing access to meanings, expectations, and uncertainties that were available to contemporaneous participants.

Temporal access also interacts with recursive representation. A later observer may study a system after earlier classifications, measurements, and interventions have already reshaped it. Present traces then contain historical layers. Some features record earlier relations; some reflect adaptation to prior observation; some result from later reinterpretation or data migration. Reconstructing relational history therefore requires attention to when a trace was produced, when it was recorded, when it became accessible, and whether its meaning changed across those intervals.

The temporal limit of observation should not be answered by privileging either contemporaneous or retrospective knowledge in general. Contemporary observers may possess practical and contextual access unavailable later while lacking knowledge of long-run consequences. Later observers may compare broader trajectories and recover previously inaccessible archives while interpreting the past through categories that emerged afterward. Each temporal position produces characteristic capacities and blind spots.

Proposition 5.56 (Temporal indexing of observational warrant). *A relational claim whose warrant depends materially on the timing, duration, preservation, or retrospective accessibility of evidence should state those temporal conditions as part of its observational scope.*

Proposition 5.56 is especially important for generative inquiry because causal and constitutive processes often unfold across multiple timescales. Evidence gathered at one moment can be adequate for a local state description while remaining weak for a claim about formation, persistence, transformation, or dissolution. Later sections

on relational identity, generative effects, and historical trajectories will develop these temporal problems further.

Section 5.7 therefore places observation inside the relational epistemology rather than outside it. Observation provides access through positions, institutions, instruments, records, and temporal windows whose capacities are uneven and sometimes strategically contested. Partiality can be reduced through distributed observation, provenance-preserving integration, and the widening of access channels, yet it cannot be assessed apart from a specified target and claim. The resulting epistemic discipline is constructive: researchers can identify what their observation regime reaches, where access becomes asymmetric or hidden, how strategic actors may shape visibility, and which temporal limits constrain inference. Section 119 extends this analysis from observer access to the selection of boundaries and scales through which relational systems become objects of knowledge.

5.8 Boundary and Scale in Relational Knowledge

Section 5.7 located observation within a structured relation among epistemic targets, observer positions, access channels, and temporal conditions. The present section addresses a logically subsequent problem: relational inquiry must also specify where an epistemic object begins and ends, and at what scale its organization is being described. Its role is to make boundary and scale selection explicit parts of epistemic warrant. The objective is to distinguish analytical delimitation from claims about the organization of relational reality, while showing how the two can inform one another. The section proceeds through boundary selection, internal and external relations, relational neighborhoods, scale dependence, cross-scale relations, open and permeable boundaries, and boundary uncertainty. The method is comparative and sensitivity-oriented. A relational claim is evaluated not only against available evidence, but also against the boundary and scale conditions under which the evidence has been assembled and the claim has been formulated.

The issue is especially important for GRE because economic objects are rarely isolated by a single natural perimeter. A firm can be analyzed through legal incorporation, ownership, contractual relations, production networks, workplace organization, financing, technological dependencies, ecological inputs, or household effects. A market can be delimited by product categories, geography, institutions, infrastructures, substitution patterns, or regulatory jurisdiction. A household can appear as a consumption unit, a care relation, a property arrangement, a site of labor reproduction, or a node in wider kinship and transfer networks. Each delimitation can be analytically useful, yet each reveals some relations and places others outside the immediate frame.

Systems and network traditions provide several precedents for taking this problem seriously without requiring a single universal ontology of systems. Simon's analysis of complex organization emphasizes hierarchical and near-decomposable structures whose useful description depends on the level at which subsystems and their interactions are examined (Simon 1962). Wellman's structural network perspective similarly directs attention toward patterned relations, clusters, boundaries, and positions

rather than treating social units as self-contained aggregates (Wellman 1983). Multi-scale approaches make the dependence on scale even more explicit: Bar-Yam shows that organization and coordinated variation can differ across scales, so a description adequate at one scale can omit structure that becomes salient at another (Bar-Yam 2004). GRE draws from these lineages a limited epistemic conclusion. Boundary and scale are conditions of representation and inference whose adequacy must be assessed relative to the relational target, rather than assumed in advance.

Definition 5.57 (Epistemic boundary). An *epistemic boundary* is a stated rule of inclusion and exclusion that delimits the entities, relations, events, traces, intervals, or environments treated as immediately constitutive of a specified object of inquiry.

Definition 5.57 concerns inquiry rather than ultimate ontology. An epistemic boundary can follow a legally instituted border, an organizational membership rule, a spatial perimeter, a temporal interval, a network criterion, or a researcher-defined analytical cutoff. Some such boundaries correspond closely to durable relational structures; others are provisional devices used to make inquiry tractable. Their epistemic status therefore depends on how they are justified and how sensitive the resulting claims are to alternative delimitations.

5.8.1 Boundary Selection

Boundary selection is the act of specifying which relations and conditions belong to the immediate object of inquiry and which are treated as part of its environment, background, or adjacent systems. Selection is unavoidable whenever the relational field exceeds what can be observed or represented in a single analysis. The epistemic problem is consequently less whether a boundary has been imposed than whether the chosen boundary is appropriate to the claim and whether its consequences for inference remain visible.

A boundary can be selected through several kinds of criteria. Institutional boundaries can follow formal membership, jurisdiction, ownership, or contractual responsibility. Spatial boundaries can follow physical proximity, administrative territory, ecological region, or infrastructural reach. Temporal boundaries can delimit episodes, policy periods, organizational lifecycles, or generational intervals. Relational boundaries can include actors connected through specified types or strengths of relation. Functional boundaries can follow participation in a production process, financial circuit, care arrangement, or knowledge system. These criteria can overlap without coinciding.

Boundary selection therefore already contains a theory of relevance. When an analysis defines a firm by corporate ownership, it gives one kind of relation priority. When it defines the same productive activity through supply-chain dependence, labor coordination, or financial exposure, different relations become constitutive of the object. The issue is not that one description is automatically more real. Each boundary supports different epistemic targets. Problems arise when conclusions obtained under one delimitation are generalized as though the omitted relations were irrelevant to the target claim.

This point is familiar in social-ecological inquiry. Ostrom's framework for social-ecological systems distinguishes connected resource systems, resource units, gover-

nance systems, users, interactions, and outcomes, thereby making the analyzed system depend on a configuration of linked components rather than a single institutional or ecological border (Ostrom 2009). GRE uses this precedent to emphasize diagnostic boundary construction: the research boundary should be chosen so that relations capable of materially affecting the target claim remain inspectable where feasible.

Claim 5.58 (Boundary-relative warrant). *The warrant of a relational claim is partly conditional on the boundary within which evidence is assembled and relations are reconstructed when materially relevant relations cross, constitute, or alter that boundary.*

Claim 5.58 does not imply that every analysis must expand indefinitely. A narrowly bounded study can be rigorous when its target is correspondingly narrow and its exclusions are justified. The claim instead requires alignment between the reach of the conclusion and the reach of the inquiry. A workplace study can make strong claims about a specified organizational process while remaining cautious about effects that depend on household, regulatory, ecological, or transnational relations outside its evidential boundary.

Boundary selection can also be recursive. Once a boundary becomes institutionalized through law, accounting, statistics, or administration, it can affect the relations subsequently observed within it. A legal definition of an enterprise, a statistical labor-market region, or an administrative service area can organize reporting, eligibility, responsibility, and resource allocation. The analytical boundary then interacts with the performative mechanisms developed in Section 5.6. Epistemic inquiry should therefore distinguish a boundary selected for analysis from a boundary that has itself become relationally efficacious.

5.8.2 Internal and External Relations

Once an epistemic boundary is selected, relations can be described as internal or external relative to that boundary. An internal relation connects elements included within the immediate object of inquiry. An external relation connects an included element to an element, condition, or process treated as lying outside that object. These terms are boundary-relative classifications. They do not establish that an external relation is causally weak, ontologically secondary, or economically peripheral.

The distinction is important because many consequential processes are boundary-crossing. A firm's internal production arrangements can depend on external credit, regulation, energy infrastructures, intellectual-property regimes, platform access, household labor reproduction, and suppliers. A household's internal allocation can depend on wage relations, public services, property markets, kinship transfers, and legal status. A local market can be reorganized by financing, logistics, information, or regulatory decisions made elsewhere. Treating these relations as external for one analytical purpose can be entirely legitimate while still requiring them to be reintroduced when the target concerns formation, persistence, vulnerability, or transformation of the bounded object.

The internal-external distinction can also change with analytical purpose. An employment contract may be external to a study of household interaction and internal to a study of labor-market organization. A national regulation may be external to a firm-level process model and internal to an institutional analysis of the sector. The same

relation can therefore occupy different classificatory positions without contradiction when the epistemic target changes.

Simon's account of near-decomposability is useful for specifying when such boundaries can support tractable analysis (Simon 1962). A subsystem can sometimes be studied with relative autonomy when interactions within it are sufficiently stronger or faster than interactions with its environment over the relevant interval. GRE adopts this as a methodological possibility rather than a universal structural law. Near-decomposability is something to establish for the case at hand. Where boundary-crossing relations are strong, rapid, or constitutive, an analysis that treats them as background will have correspondingly narrower warrant.

Proposition 5.59 (Conditional analytical closure). *A bounded relational object can be treated as provisionally closed for a specified claim when omitted boundary-crossing relations are unlikely to alter the inference materially over the relevant interval and scale, or when their effects are explicitly represented as external conditions.*

Proposition 5.59 gives closure an epistemic rather than ontological meaning. Closure is a controlled simplification. Its adequacy can be tested by widening the boundary, introducing external conditions, comparing cases with different environments, or examining whether residual patterns systematically track omitted relations. This treatment permits focused inquiry while retaining sensitivity to relational openness.

The distinction also disciplines causal language. If a bounded model explains an outcome through internal relations while a major external condition remains unexamined, the explanation should be stated at the level supported by the boundary. The model may identify an internal mechanism without establishing that the mechanism is sufficient across environments. Conversely, evidence that a process depends strongly on external relations can itself be a substantive result about the relational architecture of the object.

5.8.3 Relational Neighborhoods

Many relational inquiries do not require a complete system boundary at the outset. They begin from a focal entity, relation, event, or problem and reconstruct a *relational neighborhood*: the set of nearby relations judged relevant under a specified criterion of connection. Neighborhood reasoning is particularly useful when institutional borders are porous, when the target is locally situated within a much larger network, or when the relevant environment cannot be identified through spatial proximity alone.

A relational neighborhood can be defined through direct ties, repeated interaction, shared institutions, dependency, information flow, legal obligation, resource transfer, common infrastructure, or other target-relevant criteria. The meaning of “near” is therefore relational. Two geographically distant actors can belong to a close financial or informational neighborhood, while adjacent actors can remain weakly connected for the target relation. GRE consequently avoids assuming a common metric space for heterogeneous relations. Neighborhoods can be topological, institutional, temporal, semantic, contractual, or functional without being reducible to a single numerical distance.

Network analysis provides a useful precedent because it directs attention toward

patterned ties and positions rather than beginning from categorical aggregates (Wellman 1983). Yet a relational neighborhood in GRE can include conditions that are difficult to represent as pairwise links: legal regimes, infrastructures, shared classifications, historical dependencies, environmental constraints, or distributed fields. The neighborhood concept is therefore broader than a graph-theoretic ego network, although network measures can be appropriate in cases where the relevant relation admits that representation.

Neighborhood construction should also preserve the reason each relation is included. A researcher studying access to housing may include household income, landlord practices, credit screening, zoning, transport, employment location, family transfers, and public assistance because each bears differently on the target. Treating all of these as generic “connections” would lose evidential structure. The neighborhood should instead record relation type, direction where relevant, temporal status, evidential basis, and the pathway through which the relation can affect the claim.

Definition 5.60 (Relational neighborhood). A *relational neighborhood* is a target-relative set of relations and conditions included around a focal epistemic object because their connection is judged materially relevant to describing, explaining, or reconstructing that object under a stated criterion.

Definition 5.60 keeps neighborhood construction open to revision. Initial inquiry may reveal relations whose importance was not known in advance. The neighborhood can then expand, contract, or change composition. This is especially important for exploratory research, historical reconstruction, and generative inquiry, where the relations responsible for an observed manifestation may initially be uncertain.

Neighborhoods can also overlap. A worker can simultaneously participate in workplace, household, credit, care, migration, professional, and platform neighborhoods. A firm can occupy production, finance, legal, innovation, and regional-development neighborhoods. Overlap can become epistemically important when a relation appearing weak in one neighborhood becomes consequential through another. This observation prepares the later treatment of higher-order relations and relational fields in this chapter.

5.8.4 Scale Dependence

Scale specifies the level of aggregation, spatial extent, temporal resolution, organizational depth, or relational granularity at which an object is observed and represented. A change of scale can alter which regularities become visible, which variations are averaged, which actors remain distinguishable, and which relations appear stable. Scale is therefore an epistemic condition of relational description rather than a neutral change in presentation.

Different kinds of scale should remain distinguishable. Spatial scale concerns extent and resolution in space. Temporal scale concerns observation interval and temporal granularity. Organizational scale concerns nested units such as teams, firms, sectors, jurisdictions, or transnational arrangements. Relational scale concerns the granularity at which interactions are represented, ranging from individual ties to clusters, networks, institutions, or distributed conditions. Statistical aggregation can combine these dimensions but should not be treated as equivalent to them.

Bar-Yam's multiscale analysis is relevant because it explicitly relates coordinated variation to the scale at which system behavior is described (Bar-Yam 2004). Simon's account of complex architecture similarly emphasizes nested levels and the analytical importance of identifying subsystems (Simon 1962). GRE uses these resources to support a modest principle: a relational pattern established at one scale does not automatically retain the same form, explanatory role, or evidential meaning at another.

For example, aggregate wage stability can coexist with substantial worker-level turnover and redistribution. A national increase in output can coexist with regional contraction. A firm-level productivity gain can coincide with greater unpaid work elsewhere in the relational system. A stable annual indicator can conceal rapid short-term volatility. A network that appears integrated at sector level can contain fragmented local clusters. None of these examples implies that the aggregate description is false. They show that aggregation changes the object represented and therefore changes the scope of the claim.

Claim 5.61 (Scale-indexed relational claims). *When a relational regularity, dependency, or generative effect varies materially with spatial, temporal, organizational, or relational granularity, its epistemic statement should identify the scale at which the evidence supports the claim.*

Claim 5.61 is intended to limit unmarked scale transfer. A pattern observed among individuals should not be attributed to organizations without further evidence, and an aggregate regularity should not be assigned to constituent actors simply because they participate in the aggregate. Similar caution applies temporally: a short-run association may differ from a long-run trajectory because adaptation, institutional change, and delayed effects enter at different intervals.

Scale selection also affects evidential availability. Fine-grained data can reveal heterogeneous responses while creating privacy, sparsity, or interpretability problems. Aggregate data can reveal long-run structure while suppressing local variation. Historical archives can support institutional-scale reconstruction while leaving individual interaction weakly observed. High-frequency computational traces can provide detailed temporal access while covering only activities mediated by the relevant platform or instrument. There is consequently no universally privileged scale of observation. The relevant scale follows from the target, mechanism, and claim.

5.8.5 Cross-Scale Relations

Scale dependence becomes more demanding when processes at different scales interact. Cross-scale relations connect phenomena represented at different levels of aggregation, temporal duration, organizational depth, or relational granularity. Their epistemic importance lies in preventing scale-specific descriptions from being mistaken for self-contained explanations.

Cross-scale relations can operate in several directions. Local interactions can accumulate, coordinate, or diffuse into patterns visible at broader scales. Broader institutions can condition local possibilities through law, infrastructure, standards, prices, categories, or resource distributions. Slow processes can alter the environment within which faster processes unfold, while repeated fast interactions can gradually transform slower institutional arrangements. These descriptions should

be treated as candidate relational mechanisms whose presence requires evidence, rather than as a generic doctrine of upward or downward causation.

Complex-systems research provides useful examples of multilevel organization without resolving the epistemology for GRE. Levin's treatment of ecosystems as complex adaptive systems emphasizes localized interactions, historical dependence, self-organization, and multiple levels of dynamics (S. A. Levin 1998). Ostrom's social-ecological framework likewise studies connected components across ecological and institutional levels (Ostrom 2009). GRE draws on these works to emphasize that a scale can be analytically coherent while remaining dependent on processes represented elsewhere.

Cross-scale inference requires bridge evidence. If a household-level practice is proposed to contribute to a regional pattern, the analysis should identify the aggregation or transmission pathway. If a regulatory regime is proposed to alter individual behavior, evidence should connect the institutional rule to local incentives, constraints, interpretations, or enforcement. If a long-run structural change is proposed to emerge from repeated short-run interactions, the persistence and accumulation mechanisms should be specified. Merely observing compatible patterns at two scales provides weaker warrant than tracing the relations connecting them.

Proposition 5.62 (Cross-scale bridge requirement). *A claim that attributes a relational or generative connection across scales gains warrant from evidence concerning the pathway that links the scale-specific processes, rather than from correspondence between scale-level patterns alone.*

Proposition 5.62 will become especially important in the later sections on generative relations and attribution of generative effects, where the chapter turns from relational knowledge toward knowledge of generation. For the present section, the point is epistemic: cross-scale interpretation requires an account of mediation between levels, including the possibility that intermediate relations modify, filter, amplify, or dissipate effects.

Cross-scale relations can also generate apparent contradictions that disappear once scale is specified. An institution can be stable at one organizational level while undergoing rapid change in local practices. A policy can increase aggregate access while intensifying exclusion in a subgroup. A network can become more connected globally while some neighborhoods become more isolated. GRE should preserve these differences rather than forcing them into a single scalar verdict. Divergence across scales can itself reveal the relational organization of the process.

5.8.6 Open and Permeable Boundaries

Many economic and social objects have boundaries that organize relations without fully containing them. GRE describes such boundaries as open or permeable when actors, resources, information, obligations, representations, risks, or generative effects can cross the analytical or institutional perimeter in ways relevant to the target. Permeability is therefore compatible with boundary significance. A boundary can structure access and responsibility while remaining traversed by important relations.

The distinction is useful for avoiding two opposite simplifications. One treats a for-

mally defined unit as though relations outside it were merely contextual. The other treats openness as though any delimitation were arbitrary. In practice, legal entities, jurisdictions, households, organizations, markets, and infrastructures often possess consequential boundaries precisely because crossing them is regulated, costly, asymmetric, or selective. The epistemic task is to identify what the boundary organizes and what kinds of relation pass through it.

Permeability can be typed. Resource permeability concerns flows of money, material, labor, energy, or goods. Informational permeability concerns communication, data, knowledge, and signaling. Institutional permeability concerns overlapping jurisdictions, shared standards, delegated authority, or external governance. Social permeability concerns mobility, membership, kinship, professional affiliation, or collective participation. Generative permeability concerns the transmission of capacities, constraints, opportunities, and downstream effects across a boundary. These forms can vary independently.

Open boundaries also matter for attribution. An outcome observed within a bounded unit can partly arise through incoming relations, and actions originating within the unit can generate effects elsewhere. A balance sheet, administrative dataset, or organizational performance indicator may record the internal manifestation while leaving the external transfer weakly represented. For GRE, this is particularly important when later chapters examine value conversion, return, extraction, externalization, and the distribution of generative conditions. The epistemology developed here precedes those normative and substantive analyses by asking whether the relevant boundary-crossing relations are visible at all.

Definition 5.63 (Boundary permeability). *Boundary permeability* is the target-relative extent and manner in which relations, flows, constraints, representations, or effects traverse a specified boundary during the interval relevant to an epistemic claim.

Definition 5.63 does not require permeability to be measured by a single scalar quantity. In some inquiries it can be quantified through flows or network ties. In others it is better represented through legal pathways, institutional interfaces, qualitative process tracing, or historical reconstruction. The analytical representation should follow the type of boundary-crossing relation rather than forcing heterogeneous forms into premature commensurability.

Boundary permeability can itself change historically. Trade rules, migration regimes, data infrastructures, financial integration, technological standards, organizational restructuring, war, environmental disruption, or institutional reform can alter what crosses a boundary and under what conditions. The evidential meaning of a boundary should therefore be indexed to time. A perimeter that once supported relative analytical closure can become substantially more permeable, and a previously open interface can become restricted.

5.8.7 Boundary Uncertainty

Boundary uncertainty arises when the evidence and research purpose remain compatible with more than one plausible delimitation of the epistemic object, or when the significance of boundary-crossing relations remains incompletely known. It dif-

fers from ordinary observational uncertainty. An observer can possess extensive evidence while still face uncertainty about where the object should be delimited for the target claim.

Several sources of boundary uncertainty recur in relational inquiry. Institutional categories can overlap or conflict. Informal relations can extend beyond formal membership. Networks can lack a sharp cutoff. Historical processes can change the relevant perimeter during the period studied. Different relation types can imply different neighborhoods. A target can also be nested within several larger systems, each of which introduces potentially relevant conditions. These cases make boundary selection an object of inquiry rather than a preliminary housekeeping decision.

Boundary uncertainty should be represented rather than hidden behind a single convenient specification. One strategy is comparative delimitation: reconstruct the claim under several plausible boundaries and inspect which conclusions remain stable. Another is progressive expansion: begin from a focal neighborhood and widen the boundary until additional relations cease to change the target inference materially within the practical limits of the study. A third is layered representation: retain several overlapping boundaries corresponding to legal, organizational, spatial, financial, ecological, or social structures. A fourth is explicit residualization: identify relations known to cross the boundary while recording the limits of evidence concerning their effects.

Definition 5.64 (Boundary-sensitivity analysis). *Boundary-sensitivity analysis* is the systematic comparison of a relational claim across two or more defensible boundary specifications in order to identify which components of the inference are robust, which are boundary-dependent, and which require additional evidence.

Boundary-sensitivity analysis in Definition 5.64 is a general epistemic practice rather than a single quantitative technique. It can be implemented statistically, computationally, historically, ethnographically, legally, or through mixed methods. Its value lies in exposing the dependence of a conclusion on delimitation choices that might otherwise remain implicit.

Boundary uncertainty also interacts with scale uncertainty. A researcher can be unsure both about where to draw the perimeter and about the granularity at which the relevant relations should be represented. Expanding the boundary can reveal a broader-scale structure; increasing resolution can reveal local heterogeneity; changing temporal extent can reveal formation or decay that was invisible in a shorter interval. These operations can alter one another. The appropriate response is therefore iterative: boundary and scale specifications should be revised together as evidence about the relational architecture develops.

The resulting epistemic stance is neither boundary skepticism nor boundary realism in a simple sense. Some boundaries are strongly instituted and causally consequential; others are provisional analytical constructions; many combine both features. Their status should be established from evidence about how relations are organized, how the boundary is produced and maintained, which processes cross it, and how conclusions change under alternative specifications. Ontological suspension remains intact because this procedure does not require a final claim about ultimate system boundaries.

Section 5.8 therefore extends the account of partial access by showing that observation always occurs within a delimited and scaled epistemic object. Boundary selection determines which relations enter immediate analysis; internal and external status remains relative to that selection; relational neighborhoods provide flexible local reconstructions; scale changes what patterns become visible; cross-scale claims require bridge evidence; open boundaries direct attention toward traversing relations; and boundary uncertainty calls for sensitivity analysis. These principles prepare Section 120, which turns from the delimitation of relational objects to their identity and persistence through time. Once boundaries and scales are made explicit, the chapter can ask more precisely under what conditions a changing relation should still be treated as the same relation, and when formation, transformation, or dissolution warrants a change of identity claim.

5.9 Identity and Persistence of Relations

Section 5.8 established that relational claims depend on explicit boundaries and scales. The present section addresses the temporal counterpart of that problem: once a relation has been delimited for inquiry, under what conditions may observations at different times be treated as observations of the same relation? Its role is to provide an epistemic account of relational identity and persistence that can accommodate formation, stabilization, transformation, and dissolution without reducing identity to surface similarity or requiring exact continuity of every constituent feature. The objective is to make persistence claims auditable by specifying which properties are treated as identity-bearing, which variations remain compatible with continuity, and which changes warrant a revised relational classification. The section proceeds through identity, persistence through change, formation, stabilization, transformation, dissolution, and historical continuity and discontinuity. The method is criterion-based and historically comparative. Identity is treated as a claim whose warrant depends on the target, scale, temporal interval, and evidential continuity relevant to the inquiry.

The problem is pervasive in economic and institutional analysis. An employment relation can persist while wages, supervisors, tasks, schedules, and contractual terms change. A credit relation can continue through refinancing, restructuring, partial repayment, and changes in collateral. A household relation can persist through changes in residence, income, caregiving arrangements, and membership. An organization can remain legally continuous while its personnel, technologies, ownership, and internal authority are substantially reorganized. Conversely, the continuation of a name, identifier, legal shell, or statistical category can conceal a transformation so extensive that treating the later configuration as epistemically identical to the earlier one becomes misleading for the target claim.

Relational sociology offers an important precedent for treating social objects through unfolding relations rather than through fixed substances. Emirbayer emphasizes processual and dynamic relations as an alternative to substance-centered explanations (Emirbayer 1997). White similarly develops identities and social formations through networks, control efforts, and emergent relational organization (White 2008). Temporal-network analysis makes a related point in formal terms: the

timing and duration of ties matter to what can be inferred from a network, because an aggregate network can conceal whether relations actually coexisted or followed one another (Holme and Saramäki 2012). GRE adopts a narrower epistemic conclusion from these traditions. A claim that a relation persists across time requires a defensible continuity criterion and evidence that the criterion remains satisfied over the interval at issue.

Definition 5.65 (Relational identity criterion). A *relational identity criterion* is an explicit set of target-relative conditions under which relational configurations observed at different times are treated as instances or phases of the same relation for a specified inquiry.

Definition 5.65 makes identity operationally accountable without reducing it to a single universal property. Depending on the target, continuity may be carried by participating relata, institutional recognition, recurring obligations, organized patterns of interaction, causal dependencies, governance rules, shared history, functional roles, or processes that reproduce the relation. Several such features can jointly support identity, and their evidential importance can change with scale. The same historical sequence can therefore support continuity for one target and discontinuity for another without contradiction, provided that the respective identity criteria are stated.

5.9.1 Identity of a Relation

The identity of a relation concerns whether two temporally separated relational configurations should be treated as belonging to one continuing relational history. This is distinct from the identity of the entities participating in the relation. The participants can remain the same while the relation changes, and the relation can sometimes persist while some participants change. A firm can retain a contractual relation with a supplier after personnel turnover. An office can retain an authority relation after the officeholder changes. A household can preserve some relational organization through changes in membership. Conversely, two actors can remain physically or legally identical while trust, obligation, authority, dependence, or exchange between them is transformed.

Identity therefore requires specifying what is being tracked. For a legal inquiry, continuity of a recognized contract or corporate person may carry substantial weight. For an organizational inquiry, recurring authority and coordination patterns may matter more. For a historical inquiry, the continuity of a practice, inherited rule, or institutional lineage may be central. For a study of dependence, the relevant identity may lie in a continuing asymmetry of access or obligation even when the formal instrument through which it is realized changes. The identity criterion should follow the epistemic target rather than ordinary-language naming alone.

The ambiguity of identity language has long been recognized in the social sciences. Brubaker and Cooper show that the term “identity” can combine analytically different phenomena such as categorization, self-understanding, commonality, connectedness, and groupness (Brubaker and Cooper 2000). GRE applies the same discipline to relations. Claims of relational identity should be decomposed into the dimensions of continuity actually being asserted. A statement that “the same employment relation

continued” can refer to legal continuity, organizational role continuity, economic dependence, recurring interaction, or a combination of these. The evidential burden changes accordingly.

Claim 5.66 (Criterion-indexed relational identity). *A relational identity claim is warranted only relative to a stated or recoverable identity criterion whose continuity is supported by evidence over the interval and scale relevant to the claim.*

Claim 5.66 prevents persistence from being inferred from labels alone. A continuing category can be evidence of institutional recognition, but it does not establish continuity of every relation carried under that category. Likewise, changes in terminology do not establish that the underlying relational organization has disappeared. Names, identifiers, and classifications belong to the evidential field and can themselves become causally important, yet their continuity and the continuity of the relation remain analytically distinguishable.

Identity criteria can also be plural. A relation can satisfy one criterion while failing another. A corporation can remain legally identical while its ownership and productive organization change; a long-standing exchange relation can remain economically continuous while its contractual form changes; a political or administrative relation can preserve formal authority while practical compliance weakens. The appropriate response is often a differentiated description rather than a forced binary decision. Researchers can report legal continuity together with organizational transformation, or interactional continuity together with a change in the distribution of power.

5.9.2 Persistence through Change

Persistence concerns the maintenance of sufficient relational continuity across change. It therefore differs from immobility. Relations can persist through variation in intensity, participants, media, rules, manifestations, and surrounding conditions when the identity-bearing organization relevant to the inquiry remains sufficiently continuous. Chapter 4 developed this idea ontologically through dynamic persistence. The present section translates it into an epistemic requirement: evidence of persistence should show how continuity is carried through the observed changes.

Several continuity carriers are common. Recurrent interaction can reproduce expectations and obligations. Legal or administrative recognition can maintain roles across changes in personnel. Records and memory can preserve commitments across interruptions. Infrastructure can sustain access and dependency despite changes in surface practice. Organizational routines can reproduce coordination through turnover. Financial balances, debts, credentials, property claims, or reputational histories can connect earlier and later states. None of these carriers is universally decisive, but each can provide evidence for a particular persistence claim.

Persistence can also be intermittent. Relations may tolerate gaps without losing identity when the system preserves expectations or mechanisms of resumption. A seasonal employment relation can recur after months of inactivity. A dormant credit line can remain a continuing relation despite periods without transactions. A diplomatic, kinship, or professional relation can persist through sparse interaction because status, memory, entitlement, or recognized possibility remains operative. Temporal-network representations are useful here because they distinguish the

existence of ties over time from their collapse into one static aggregate (Holme and Saramäki 2012). GRE extends that insight beyond pairwise ties by asking what is preserved during inactive intervals and what evidentially supports the claim of continued relational availability.

Definition 5.67 (Persistence profile). A *persistence profile* is a structured account of the mechanisms, records, practices, recognitions, dependencies, and temporal recurrences through which a specified relation maintains identity-bearing continuity across an interval of change.

The persistence profile in Definition 5.67 helps distinguish several forms of continuity that can otherwise be conflated. A relation can persist because actors actively reproduce it, because an institution maintains it, because material or technical infrastructure carries it, because historical commitments remain enforceable, or because several of these mechanisms reinforce one another. A profile can also reveal fragility. If continuity depends on one record, one intermediary, or one recurring practice, the relation may be more vulnerable to interruption than its current stability suggests.

Persistence is also scale-sensitive. A supply relation can appear continuous at an annual scale while exhibiting repeated breakdowns at a weekly scale. A labor relation can persist institutionally while everyday supervisory relations change rapidly. A monetary institution can remain historically continuous while its technical settlement processes are replaced. The persistence claim should therefore specify the temporal and relational resolution at which continuity is being asserted. This follows directly from the scale discipline developed in Section 5.8.

5.9.3 Relation Formation

Relation formation concerns the transition from the absence of a specified relation, or from a looser precursor configuration, to a relational organization that satisfies the inquiry's entry criterion. The epistemic challenge lies in identifying when enough evidence exists to treat a relation as having formed. Formation rarely occurs through a single universal threshold. It can arise through a contract, repeated interaction, mutual recognition, technical connection, accumulated dependence, institutional designation, shared practice, or gradual convergence of several conditions.

A legal relation can have a relatively sharp formal beginning, such as the execution of an agreement or conferral of a status. An interpersonal relation may emerge gradually through repeated interaction and expectation. A market relation can form through recurring exchange and the development of mutual responsiveness. A dependency relation can arise incrementally as alternatives disappear or switching costs accumulate. A platform relation can begin with registration, yet economically consequential dependence may develop only after data, reputation, customers, or complementary investments accumulate. The beginning of the formal relation and the beginning of the substantively relevant relation can therefore differ.

Formation should consequently be analyzed through entry conditions tied to the target. An investigator interested in legal obligation may date formation to a recognized act. An investigator interested in practical dependence may require evidence that one party's feasible options or capacities have become materially conditioned by the other. An investigator interested in trust may require a pattern of expectation and

reliance extending beyond a single encounter. These criteria can identify different formation times while referring to one historical sequence.

Definition 5.68 (Relational entry condition). A *relational entry condition* is the target-relative evidential threshold at which a developing configuration is treated as having formed the relation specified by the inquiry.

Definition 5.68 supports historical precision because it separates the earliest precursor from the point at which the relation becomes analytically operative. A negotiation can precede a contract; repeated encounters can precede trust; technical adoption can precede infrastructural dependence; shared classification can precede institutionalized membership. Formation can therefore be reconstructed as a process even when the inquiry uses a specific threshold for dating the relation.

Relational formation can also depend on recursive recognition. Participants may alter their conduct once they understand themselves to be in a relation. Institutions can classify a relationship and thereby create obligations or expectations that further stabilize it. This connects formation to the performative representation developed in Section 5.6. Recognition is neither necessary for every relational reality nor merely descriptive when it enters subsequent conduct. The evidential task is to establish which role recognition played in the specific formation process.

5.9.4 Stabilization

Stabilization concerns the processes through which a newly formed or previously variable relation acquires sufficient regularity, durability, and causal reliability to support stronger persistence claims. Stabilization can occur through repetition, institutionalization, mutual adaptation, infrastructure, law, routines, expectations, sanctions, learning, investments, classification, or accumulated history. It should be treated as a matter of degree and mechanism rather than as a universal stage through which every relation must pass.

Giddens's account of structuration provides one precedent for understanding how social practices can recursively reproduce rules and resources through continued enactment (Giddens 1984). GRE uses this insight cautiously. Reproduction can stabilize a relation when repeated practices make later continuation easier, more expected, more enforceable, or more materially embedded. The existence of repetition alone, however, does not establish the mechanism. Stability can also be maintained by legal coercion, technical lock-in, unequal dependence, organizational routine, shared meaning, or deliberate maintenance.

Stabilization should therefore be evidenced through the carriers that make continuation probable or resilient. Repeated successful coordination can increase confidence that a relation will recur. Complementary investments can raise the costs of exit. Records can preserve obligations across personnel turnover. Standardization can reduce ambiguity in interaction. Institutional recognition can make enforcement available. Shared narratives and categories can provide interpretive continuity. Infrastructure can make some interactions routine while making alternatives costly. Each mechanism leaves different traces and carries different implications for later transformation.

Claim 5.69 (Stabilization through reproduced conditions). *Evidence for relational stabilization is strengthened when the conditions that support continuation are themselves shown to be reproduced, maintained, or renewed across time rather than inferred solely from repeated outcomes.*

Claim 5.69 distinguishes stabilization from mere recurrence. Two actors can transact repeatedly because external circumstances repeatedly bring them together without producing a durable relation between them. Conversely, a stable relation can persist through temporary inactivity when institutions, expectations, or obligations maintain the conditions for resumption. The relevant evidence therefore concerns both observed continuity and the mechanisms that sustain it.

Stabilization is normatively indeterminate. Trust, cooperation, care, exploitation, exclusion, hierarchy, monopoly, debt dependence, and discriminatory practice can all become stable. The epistemic task in the present chapter is to establish whether and how stabilization occurs. Evaluation of the desirability or justice of that stabilized relation requires separate normative analysis. Preserving this distinction is especially important because durable relations can become easier to observe precisely because institutions record and reproduce them, while unstable or marginalized relations may remain less legible.

5.9.5 Transformation

Transformation arises when change affects the identity-bearing organization of a relation sufficiently to require either a qualified continuity claim or a revised relational classification. The analytical difficulty lies between two errors. One error treats every change as evidence of a new relation. The other treats a continuing label or participant set as proof that the relation remains substantially unchanged. A defensible transformation claim compares the earlier and later configurations against the identity criterion, persistence profile, and scale of the inquiry.

Transformation can concern the relata, the structure connecting them, the distribution of capacities and obligations, the governing rules, the mechanisms of reproduction, the medium of interaction, or the relation's position within a wider field. Some changes modify only parameters within an established relation. Others alter the conditions through which the relation operates. A loan can be repriced while remaining the same debt relation; a restructuring can change priority, collateral, counterparties, or enforcement conditions enough to alter the relevant relational architecture. Employment can continue through a new schedule, while a transition from direct employment to platform-mediated contracting can alter authority, risk allocation, visibility, and access to protections.

Historical sociology provides useful resources for identifying consequential transformation. Sewell's analysis of historical events emphasizes that events can transform structures by reorganizing schemas, resources, and subsequent action (Sewell 1996). Pierson's account of path dependence emphasizes how timing, sequence, and self-reinforcing processes can make earlier developments consequential for later possibilities (Pierson 2000). GRE draws an epistemic implication from these approaches: transformation should be reconstructed through changes in the organization and reproduction of the relation, with attention to how earlier configurations condition the

later one.

Definition 5.70 (Relational transformation threshold). A *relational transformation threshold* is the target-relative point at which accumulated or event-driven change alters one or more identity-bearing dimensions sufficiently that an unqualified persistence claim would obscure a materially relevant difference.

The threshold in Definition 5.70 need not be numerically sharp. Historical and qualitative inquiry can identify transformation through converging evidence that the relation now operates through different rules, capacities, dependencies, or reproduction mechanisms. Quantitative inquiry can specify thresholds where measurable structure changes. Legal or administrative inquiry can identify formal reconstitution. The appropriate threshold follows the target and evidence rather than a universal metric.

Transformation can preserve lineage. A relation may be historically continuous in the sense that the later configuration descends from and is causally shaped by the earlier one while differing enough to warrant a new subtype, phase, or category. This is particularly important for institutions. A monetary regime, organization, profession, or governance arrangement can transform repeatedly while preserving records, assets, memberships, obligations, or inherited constraints. Lineage and identity should therefore be reported separately when they diverge.

5.9.6 Dissolution

Dissolution concerns the point at which a relation ceases to satisfy the identity and persistence conditions relevant to the inquiry. Dissolution can be abrupt, gradual, formal, practical, partial, or contested. A contract can terminate on a legally specified date. A trust relation can erode through repeated breaches. A market connection can disappear as transactions and expectations cease. An organizational role can remain formally recorded while losing practical authority. A dependency can weaken as alternatives become available. These cases show that termination of one manifestation of a relation and dissolution of the relation itself can occur at different times.

The epistemic burden of dissolution resembles the problem of evidence of absence developed in Section 5.4. Non-observation after time t does not by itself establish that a relation has ended. Activity can become latent, hidden, intermittent, or inaccessible to the observation regime. A stronger dissolution claim requires evidence that the mechanisms, obligations, expectations, channels, or institutional conditions carrying the relation have ceased, become ineffective, or been replaced in ways that make continuation implausible under the stated criterion.

Formal termination can nevertheless be highly probative when the target relation is institutionally constituted by formal status. Dissolution of a corporation, revocation of a license, cancellation of a contract, or abolition of an office can change rights and obligations immediately. Even then, historical effects can remain. Debts, records, reputations, infrastructures, property distributions, learned routines, and legal precedents can survive the relation that generated them. The end of relational persistence therefore differs from the disappearance of all consequences.

Proposition 5.71 (Residual efficacy after dissolution). *The dissolution of a relation*

does not imply the disappearance of effects generated during its persistence; later relational configurations can remain conditioned by residues of a relation whose own identity criteria are no longer satisfied.

Proposition 5.71 matters for historical explanation. Former institutions can continue to shape distributions, infrastructures, expectations, records, categories, and strategic positions. A relation can therefore cease to exist as a continuing relation while remaining causally relevant through its historical products. This distinction prevents retrospective disappearance: a dissolved relation can have been relationally real and economically consequential during the interval in which it persisted.

Dissolution can also be uneven across scales. A formal organization may dissolve while local networks formed within it continue. A legal relationship may end while social obligations persist. A production network may fragment while technical standards remain shared. An exchange relation may end between two firms while industry-level dependencies continue. Researchers should therefore specify which relational object has dissolved and which inherited relations remain operative.

5.9.7 Historical Continuity and Discontinuity

Historical continuity and discontinuity bring the preceding distinctions together. A longitudinal relational account should be able to state where a relation forms, what stabilizes it, which changes remain compatible with persistence, when transformation becomes analytically consequential, and under what conditions dissolution is inferred. These judgments depend on the identity criterion and scale of inquiry, yet they remain constrained by evidence. Historical reconstruction therefore requires more than chronological ordering. It requires a defensible account of what is continuing across the ordered observations.

Continuity can take several forms. There can be continuity of participants, legal recognition, institutional lineage, infrastructure, practice, causal dependence, memory, or generative conditions. Discontinuity can likewise affect some dimensions while leaving others intact. A revolution can transform formal authority while preserving administrative routines. A merger can terminate corporate identities while preserving production systems and employment relations. A technological transition can replace the medium of exchange while preserving creditor–debtor relations. An institutional reform can preserve names and offices while changing the distribution of decision authority. Historical interpretation should therefore identify the dimensions in which continuity and discontinuity occur.

This multidimensional approach also improves periodization. Historical periods are often constructed around events, reforms, crises, regime changes, or technological transitions. Such boundaries can be useful when they correspond to changes in the relations relevant to the inquiry. They become misleading when the period boundary is treated as if every relation changed simultaneously. Different relational processes can have different turning points, rates of change, and persistence mechanisms. A single chronology may therefore contain overlapping continuities and discontinuities.

Definition 5.72 (Relational continuity judgment). *A relational continuity judgment is an evidence-based assessment that specifies which identity-bearing dimensions of a*

relation persist, transform, or dissolve across a stated historical interval and analytical scale.

Definition 5.72 replaces an undifferentiated same-or-different judgment with an auditable historical account. The judgment can conclude that a relation persists substantially, persists with transformation, divides into several relations, merges with another configuration, becomes dormant, or dissolves while leaving residual effects. These possibilities are analytically distinct and can be supported by different evidence.

Historical continuity can also be path-dependent. Earlier stabilization can alter later transition costs, available categories, institutional expectations, and distributions of capability. Pierson's treatment of increasing returns and path dependence highlights how sequences can become self-reinforcing under specified mechanisms (Pierson 2000). For GRE, this means that persistence is sometimes explained partly by the history of the relation itself. Continued existence can result from accumulated investments, learned coordination, legal precedent, reputational capital, sunk costs, institutional complementarities, or the narrowing of alternatives. A persistence claim should nevertheless distinguish such mechanisms from mere duration. Longevity is evidence of survival; explanation requires evidence about what reproduced the relation.

Discontinuity likewise requires historical specificity. Sewell's event-centered account shows how concentrated episodes can transform structures and later trajectories (Sewell 1996). Other transformations accumulate gradually without a single decisive event. The evidence should therefore be open to both event-driven and cumulative change. A discontinuity claim gains strength when earlier persistence mechanisms cease to operate, when new mechanisms become constitutive, or when the relation's causal capacities and obligations change enough that the previous identity criterion loses adequacy.

Section 5.9 therefore establishes relational identity as a temporally and epistemically disciplined claim. Identity depends on explicit criteria; persistence requires evidence of continuity carriers and reproduction mechanisms; formation requires target-relative entry conditions; stabilization concerns the reproduction of conditions supporting continuation; transformation marks materially relevant change in identity-bearing dimensions; dissolution concerns the failure of persistence conditions while allowing historical residues to remain effective; and historical continuity judgments specify which dimensions persist and which change. This framework prepares the following section's move from single relations through time to knowledge of relations among relations. Once the identity and persistence of individual relational objects can be stated with greater precision, higher-order relational configurations can be reconstructed without assuming that their constituent relations are timeless or internally homogeneous.

5.10 Higher-Order Relational Knowledge

Section 5.9 established criteria for judging the identity, persistence, transformation, and dissolution of individual relations through time. The present section extends

the epistemic problem from relations considered one at a time to configurations in which relations themselves become mutually dependent, collectively organized, nested, typed across several layers, or jointly constituted through interactions involving more than two relata. Its role is to specify what GRE may reasonably claim when the relevant object of knowledge is a relational configuration whose organization cannot be recovered by inspecting isolated dyads alone. The objective is to distinguish several forms of higher-order relational knowledge and to state the evidential requirements appropriate to each form. The section proceeds through relations among relations, collective relations, nested relations, higher-order relations, multilayer relations, hyperrelational configurations, and the limits of higher-order reconstruction. The method is reconstructive and representation-sensitive: higher-order claims are treated as inferential constructions constrained by observed traces, relational identities, boundary and scale choices, and the semantics of the representation through which the configuration becomes epistemically accessible.

Higher-order inquiry is economically important because many consequential arrangements are composed of several relations whose effects depend partly on one another. A household's credit relation may depend on employment income; employment may depend on an organizational authority structure; the organization may depend on financing, supply, and regulatory relations; and each of these may be conditioned by infrastructures that connect many actors at once. Studying each relation separately can identify important local mechanisms while leaving the architecture through which they become mutually reinforcing, substitutable, conflicting, or jointly vulnerable insufficiently reconstructed. Relational sociology and network analysis provide established resources for moving beyond atomized attributes toward patterned relations (Emirbayer 1997; Wellman 1983). GRE extends that epistemic orientation by asking what additional warrant is required when the object of inquiry is itself composed of several identified relations or relation types.

The resulting problem is partly representational. A configuration can be encoded as a set of pairwise ties, a membership structure, a multilayer network, a higher-order interaction, or another relational representation. These encodings illuminate different features and suppress others. A richer representation can therefore increase analytical capacity while also increasing assumptions, data requirements, and opportunities for misclassification. Higher-order knowledge should consequently be judged by correspondence between the target claim, the chosen relational representation, and the evidence available for reconstructing the relevant dependencies.

Definition 5.73 (Higher-order relational knowledge). *Higher-order relational knowledge* is warranted knowledge about configurations in which the epistemic target concerns relations among relations, organized collectivities, nested relational structures, multiple relation types, or interactions whose relevant organization is not exhausted by independently described pairwise relations.

Definition 5.73 is intentionally broad. It does not imply that every economic phenomenon requires a higher-order representation. Pairwise relations remain appropriate when the target can be adequately reconstructed at that level. The definition instead identifies cases in which the evidential object includes organization among relations. The burden of inquiry then shifts from establishing that several relations exist to establishing how they are connected, coordinated, conditioned, or jointly con-

stituted.

5.10.1 Relations among Relations

Relations among relations arise when the existence, operation, strength, interpretation, or consequences of one relation depend materially on another relation. This dependence can be institutional, causal, informational, normative, temporal, or infrastructural. A collateral relation can condition a credit relation; an employment relation can condition eligibility for insurance; a legal relation can authorize an administrative relation; a supply relation can be activated only when a financing relation is maintained. The epistemic target is therefore a dependency between relational objects rather than a new attribute of one participant.

Knowledge of such dependencies requires the constituent relations to be identified with sufficient clarity before their interdependence is inferred. Section 5.9 matters directly here. If the identity or temporal continuity of one relation remains uncertain, a claim about its dependence on another relation inherits that uncertainty. The problem also inherits the boundary and scale conditions developed in Section 5.8: two relations may appear independent within a narrow boundary while sharing an external institutional condition at a wider scale.

A relation among relations should not be inferred merely because two relations share a participant. A worker may have an employment relation and a tenancy relation without the two being materially dependent. Conversely, a dependency can exist even when the relations do not share the same immediate relata. A lender's financing relation with a firm may alter the firm's employment relations with workers through covenant conditions, liquidity constraints, or restructuring decisions. Evidence for a relation among relations therefore concerns a mechanism or patterned dependency connecting relational processes, rather than simple co-occurrence.

Claim 5.74 (Non-aggregation of relational dependence). *The observation of several relations within the same boundary does not by itself warrant a claim about relations among those relations. Higher-order dependence requires evidence that specifies how variation, persistence, activation, interpretation, or consequences in one relational object are connected to another.*

Claim 5.74 places an evidential threshold between relational inventory and relational architecture. Such evidence may come from institutional rules, temporal sequencing, intervention, comparative cases, organizational records, contractual conditions, recurring covariation supported by mechanism, or testimony from participants positioned within the relevant processes. Several forms of evidence can converge. The appropriate combination depends on whether the higher-order claim concerns causal dependence, legal dependence, functional coordination, symbolic coupling, or another specified relation between relations.

The distinction is especially important when relations can substitute for or compensate for one another. A household may respond to reduced wage income through borrowing, intra-household transfers, public benefits, asset sales, or informal support. A firm may replace one supplier, financing source, or distribution channel with another. In such cases, knowledge of the relations individually does not yet explain the resilience or fragility of the larger arrangement. The higher-order target con-

cerns the pattern of substitutability, complementarity, or mutual constraint among relations.

5.10.2 Collective Relations

Collective relations concern relations in which at least one relatum is an organized collectivity whose relational capacities cannot be reconstructed adequately by treating it as a mere list of individual members. Firms, households, associations, agencies, cooperatives, classes of officeholders, and other collectivities can act or be treated as relational participants through rules that organize membership, authority, representation, obligation, and continuity. The epistemic problem is to determine when a collective attribution is warranted and what features of the collectivity carry the relation.

The duality between persons and groups provides a useful precedent for this problem. Breiger shows that membership structures connect interpersonal and intergroup organization rather than reducing one level straightforwardly to the other (Breiger 1974). White's relational sociology similarly treats identities and social formations as emerging through patterned relations and control processes rather than as pre-given substances (White 2008). GRE adopts a narrower epistemic lesson: a collective relation should be reconstructed through evidence about the organization that enables the collectivity to enter, sustain, interpret, or modify the relation.

For example, a corporation may owe a debt even though the individuals working within it change. The relation is carried through legal personality, accounting systems, governance rules, authorized representatives, and organizational continuity. A household may participate in a mortgage relation through jointly organized income, ownership, legal responsibility, or caregiving arrangements. A labor union may bargain with an employer through procedures that authorize representatives to act on behalf of members. In each case, the relation involves individuals, yet the collective relational capacity depends on an organization of those individuals that must itself be evidenced.

Definition 5.75 (Collective relational attribution). A *collective relational attribution* is an epistemic judgment that a specified collectivity may be treated as a relatum because identifiable organizational, legal, procedural, representational, or practical mechanisms enable the collectivity to enter and sustain the relation relevant to the inquiry.

Definition 5.75 guards against two opposite simplifications. The first treats collective entities as self-evident actors whose internal organization is irrelevant. The second reduces every collective relation to a sum of individual ties and thereby loses the mechanisms through which the collectivity persists, represents itself, and exercises relational capacities. A warranted account can move between these levels while specifying which evidence supports the collective attribution.

Collective relations also require attention to internal disagreement. A collective can enter a relation without all members sharing the same preferences, knowledge, or benefits. Organizational authorization, voting, hierarchy, delegation, legal representation, custom, or coercion may generate the effective collective action. The existence of the collective relation therefore does not establish internal consensus. GRE

should keep the external relation and the internal relations that constitute collective agency analytically separable, even when they are mutually consequential.

5.10.3 Nested Relations

Nested relations occur when a relation is organized within, enabled by, constrained by, or constitutive of a wider relation or relational arrangement. The nesting can be organizational, institutional, legal, spatial, temporal, or functional. A supervisory relation can be nested within an employment relation; an employment relation can be nested within an organizational arrangement; the organization can be nested within a supply regime, regulatory order, or ownership structure. Nesting therefore concerns relational containment or dependency across analytically distinguishable levels rather than mere physical inclusion.

The epistemic value of nesting lies in showing that local relations often inherit conditions from wider arrangements while retaining locally specific mechanisms. A supervisor's authority over an employee can depend on an employment contract and organizational office, yet the concrete exercise of authority can vary substantially by team, task, workplace, and historical moment. A transaction conducted within a platform can inherit platform rules, ranking systems, payment infrastructures, and dispute procedures while remaining a distinct exchange between participants. Knowledge at one level may therefore condition interpretation at another without determining every local outcome.

Nesting should be specified rather than assumed from organizational diagrams or administrative labels. Formal hierarchy provides one source of evidence, but practical relations can cross or circumvent formal nesting. Informal advice networks can connect units that are administratively separate. Contractors can be economically integrated into production while remaining legally external. Family, employment, ownership, and debt relations can overlap without forming a single neat hierarchy. The relevant question is which wider relation changes the conditions under which the nested relation exists or operates.

A useful reconstruction identifies at least three elements: the nested relational object, the wider relational object, and the mechanism connecting them. The mechanism can involve authorization, resource dependence, rule inheritance, infrastructural access, monitoring, sanction, identity, or another specified pathway. This structure helps prevent scale labels such as "micro," "meso," and "macro" from doing explanatory work by themselves. Nesting is established through the relation between levels, not simply by assigning them names.

Nested relations can also be recursive. A local relation can contribute to reproducing the wider arrangement that conditions it. Employment relations reproduce organizations through labor and coordination; organizations contribute to industries and institutional orders; those wider orders influence the terms under which employment is organized. Such recursion does not erase level distinctions. It makes the epistemic task more demanding because evidence must track both downward conditioning and upward contribution without presuming that one direction exhausts the relation.

5.10.4 Higher-Order Relations

The term *higher-order relation* can be used more narrowly than higher-order relational knowledge. It refers here to a relation whose relata include one or more relations or relational configurations. The objects connected by the higher-order relation are therefore themselves relationally constituted. Examples include a legal rule that subordinates one contractual relation to another, a governance arrangement that coordinates several organizational relations, or a dependency in which the continuation of one relational configuration conditions another.

Definition 5.76 (Higher-order relation). A *higher-order relation* is a target-relative relation in which at least one relatum is itself an identified relation or relational configuration, and for which evidence supports the claimed connection among those relational objects.

Definition 5.76 distinguishes higher-order relations from ordinary pairwise ties between persons or organizations. The distinction concerns the level at which the epistemic target is specified. A bank can have a contractual relation with a firm; that is a first-order relation at the chosen level. A covenant linking continuation of that financing relation to specified debt ratios, ownership conditions, or other financing relations can create a higher-order dependency among relational arrangements. Similarly, a collective-bargaining framework can regulate relations between workers and employers while itself depending on legal recognition and organizational representation.

Higher-order claims can reveal architecture that disappears when all relational objects are flattened into one network of undifferentiated edges. White, Boorman, and Breiger's blockmodel work illustrates how multiple networks can be used to identify roles and positions from patterned relational structure rather than from isolated attributes (White et al. 1976). The broader epistemic lesson is that relational organization can be expressed through recurring configurations among ties and positions. GRE should nevertheless distinguish a formally detected pattern from the stronger claim that the pattern corresponds to a substantively meaningful higher-order relation. Interpretation requires domain knowledge and evidence about the mechanisms represented by the formal structure.

Higher-order relations can also carry asymmetries that are invisible at the first-order level. Two contractual relations may each appear bilateral while one is institutionally subordinate to the other. Several apparently independent exchange relations may be coordinated through a common platform rule. Separate organizational relations may be jointly conditioned by ownership or financing arrangements. The higher-order reconstruction becomes warranted when the evidence shows that the relations' terms, continuation, or consequences are linked through such an organizing mechanism.

5.10.5 Multilayer Relations

Multilayer relational analysis becomes useful when a system contains several distinguishable types of relations, contexts, or connectivity structures whose interaction matters to the target claim. Kivelä and colleagues synthesize a formal literature on

multilayer networks that includes multiplex relations, interconnected networks, and other settings in which nodes and links are organized across several layers (Kivelä et al. 2014). For GRE, the epistemic importance of this literature lies in making relation type and interlayer connection explicit rather than collapsing heterogeneous ties into a single undifferentiated network.

An economic configuration can contain ownership, employment, credit, supply, information, kinship, legal, and administrative relations among overlapping sets of actors. These relations can differ in direction, duration, institutional form, observability, and causal capacity. Aggregating them into one edge set can create false equivalence. Keeping every layer fully separate can produce the opposite problem by obscuring interdependence. Multilayer reconstruction therefore requires both layer-specific semantics and evidence about connections across layers.

Definition 5.77 (Multilayer relational reconstruction). A *multilayer relational reconstruction* is an evidentially supported representation in which distinct relation types or relational contexts are modeled separately and the mappings or dependencies connecting those layers are specified according to the epistemic target.

The definition emphasizes semantics because a layer is an analytical representation, not an ontological domain established merely by being drawn separately. One layer may represent legal ties, another financial exposures, and another information flows. Their separation is justified when the distinctions matter to the target and are supported by the evidence. Interlayer connections then require their own interpretation. Shared actors can connect layers, but cross-layer dependence may also arise through rules, infrastructures, or institutions that connect otherwise different sets of relata.

Multilayer analysis can improve epistemic precision in several ways. It can identify actors whose prominence differs by relation type; distinguish redundancy within one layer from dependence across layers; reveal whether apparent resilience in one network depends on fragility in another; and examine whether a relation that seems peripheral in one domain becomes central because it connects layers. These possibilities are particularly relevant to GRE because value, resources, authority, information, and capabilities may circulate through different relational channels whose conversion interfaces matter economically.

The advantages also create new risks. Layer definitions can reify administrative categories, impose sharp distinctions on relations that are historically entangled, or multiply degrees of freedom beyond what the evidence supports. A multilayer model can appear richly realistic while remaining poorly identified. The number of layers should therefore follow the analytical need and evidential capacity rather than a presumption that more layers necessarily yield better knowledge.

5.10.6 Hyperrelational Configurations

Some interactions involve several relata jointly in ways that lose important information when decomposed into independent dyads. Contemporary higher-order network research formalizes this problem through hypergraphs, simplicial complexes, and related representations of interactions involving groups of three or more units (Battiston et al. 2020). GRE uses the broader term *hyperrelational configuration* for

epistemic targets in which the jointly constituted interaction matters to the economic relation being reconstructed.

A multilateral contract provides a simple illustration. If the obligations of three parties are defined jointly, replacing the arrangement with three bilateral contracts may misrepresent the conditions under which obligations arise. A consortium can allocate authority, contribution, and return through rules that operate at the level of the group. A clearing arrangement can connect several exposures through netting procedures whose consequences cannot be inferred by evaluating each gross bilateral exposure independently. A household decision can likewise involve joint obligations and dependencies that are altered when represented as separate pairwise ties.

Definition 5.78 (Hyperrelational configuration). A *hyperrelational configuration* is a relational object in which three or more relata participate in a jointly constituted interaction and where decomposing that interaction into independent pairwise relations would omit information material to the target claim.

Definition 5.78 distinguishes hyperrelationality from higher-order relations in Definition 5.76. A higher-order relation relates at least one relational object to another object; a hyperrelational configuration concerns a joint interaction among several relata at the same specified analytical level. The two can coexist. A multilateral arrangement can itself become a relatum in a higher-order relation with a legal, financial, or administrative structure.

The epistemic threshold for hyperrelational reconstruction is correspondingly demanding. Evidence should establish that the joint participation is constitutive rather than merely simultaneous. Co-presence at an event or membership in the same organization does not by itself establish a hyperrelation. Relevant evidence can include joint rules, multilateral obligations, collective decision procedures, shared technological protocols, coordinated production processes, or outcomes that depend on the configuration of several participants together.

Pairwise decomposition can remain useful as an approximation when the omitted joint structure is weak relative to the target. The choice therefore remains target-relative. A bilateral exposure analysis can be appropriate for one question even when a multilateral representation is needed for another. GRE should treat the representational gain from hyperrelational modeling as an empirical and analytical question rather than as a general hierarchy in which higher-order formalisms automatically supersede simpler ones.

5.10.7 Limits of Higher-Order Reconstruction

Higher-order reconstruction increases what relational inquiry can represent, yet it also amplifies the epistemic limits developed throughout this chapter. Observation remains partial; relations can be hidden or strategically concealed; boundaries and scales remain consequential; relational identities can change through time; and representation can be performative. A higher-order model inherits these uncertainties from its constituent relations and adds uncertainty about how those relations are connected.

One limit is combinatorial. As the number of relation types, temporal states, nest-

ing levels, interlayer mappings, and multi-party interactions grows, the number of possible configurations expands rapidly. Richer representations can therefore outrun the capacity of available evidence to discriminate among plausible reconstructions. Computational tractability can alleviate some processing constraints, yet it does not create observations that were never available. A large model can enumerate possibilities while leaving the substantive configuration underdetermined.

A second limit concerns representation dependence. Pairwise networks, multilayer networks, hypergraphs, organizational charts, legal schemas, and qualitative relational maps each foreground different structures. Their outputs are therefore conditional on decisions about what counts as a relatum, what counts as a relation, which relation types remain distinct, how time is represented, and where boundaries are placed. Representation comparison becomes part of higher-order epistemic discipline. Robustness across several plausible encodings can strengthen confidence, while divergence can identify precisely where the conclusion depends on representational choice.

A third limit concerns inferential propagation. If evidence for one constituent relation is weak, higher-order claims relying on that relation should reflect the inherited uncertainty. Confidence should not increase simply because uncertain relations are assembled into an elaborate structure. The same principle applies to classification error. A misidentified legal relation, organizational boundary, or participant can alter several inferred dependencies at once. Provenance from Section 5.7 therefore becomes especially important when higher-order reconstructions integrate heterogeneous data sources.

Proposition 5.79 (Inherited uncertainty in higher-order reconstruction). *When a higher-order relational claim depends on uncertain constituent relations or uncertain mappings among them, the warrant for the higher-order claim should remain conditional on those uncertainties and should identify which conclusions are sensitive to alternative admissible reconstructions.*

Proposition 5.79 does not require a single universal numerical uncertainty measure. Depending on the inquiry, uncertainty can be represented through alternative cases, confidence intervals, posterior distributions, sensitivity analyses, competing qualitative reconstructions, provenance grades, or explicit unresolved ambiguities. The common requirement is that complexity should not conceal fragility of inference.

A fourth limit concerns causal interpretation. Structural position, clustering, interlayer centrality, or membership in a higher-order motif can describe relational organization without by themselves establishing why an outcome occurred. Formal structure can generate hypotheses about mechanisms, exposure, coordination, or vulnerability. Causal claims require the additional evidential work appropriate to the mechanism and historical context. This restraint is especially important for GRE because later sections will distinguish relational knowledge from attribution of generative effects.

A fifth limit concerns ontological overreach. The success of a higher-order representation does not establish that economic reality is ultimately a network, hypergraph, multilayer system, or any other formal object. Under ontological suspension, these representations remain tools for disciplined access to relational organization.

Their value depends on explanatory, descriptive, predictive, comparative, or practical performance relative to specified targets and evidence.

Higher-order reconstruction therefore expands the epistemic vocabulary of GRE without establishing a universal representational hierarchy. Relations among relations identify dependencies among relational objects; collective relations concern organized collectivities with evidenced relational capacities; nested relations connect local and wider relational arrangements; higher-order relations take relations or relational configurations as relata; multilayer reconstruction preserves meaningful heterogeneity among relation types while specifying interlayer connections; and hyperrelational configurations capture jointly constituted interactions that materially exceed pairwise decomposition. Each construction requires evidence matched to its target and remains bounded by observation, history, scale, representation, and uncertainty.

Section 5.10 also marks a limit of configuration-based reconstruction. Even a richly specified set of pairwise, nested, multilayer, and hyperrelational relations can leave some distributed conditions insufficiently represented when effects depend on positions within a broader relational environment rather than on identifiable ties alone. The next section develops the epistemology of relational fields to examine that problem. Its task is to ask what kind of evidence can warrant claims about distributed relational conditions while preserving the same commitments to explicit boundaries, representation-sensitive inference, and ontological restraint established here.

5.11 Epistemology of Relational Fields

Section 5.11 extends the epistemology developed in Sections 5.7–5.10 from identifiable relations and higher-order configurations to distributed relational conditions. Its role is to clarify when the language of a *field* adds epistemic content beyond a dense collection of pairwise ties, a shared context, or an aggregate regularity. The section develops this problem in seven steps. It first distinguishes pairwise relations from distributed conditions, then specifies evidential requirements for field claims, reconstructive procedures, the relation between local position and distributed structure, overlapping fields, field boundaries, and limits of field attribution. The discussion treats field representations as target-relative analytical constructions under ontological suspension. Its method is therefore comparative and evidential: alternative representations are assessed by the differences they make to warranted description, explanation, and later generative attribution.

Field language already has several established meanings in the social sciences. Bourdieu and Wacquant formulate fields through relational positions, forms of capital, and structured struggles, while Martin analyzes field theory more generally through explanations in which regularities in action depend on positions relative to others (Bourdieu and Wacquant 1992; Martin 2003). Fligstein and McAdam develop strategic action fields as mesolevel social orders that can themselves be embedded in wider fields (Fligstein and McAdam 2012). GRE can learn from these traditions without identifying its own relational-field vocabulary with any one of them. The present task is narrower: to establish the epistemic conditions under which a distributed

relational environment becomes a warranted object of inquiry.

5.11.1 Pairwise Relations and Distributed Conditions

Pairwise relational analysis represents a system through identifiable relations between specified relata. This representation can be highly informative. Credit relations connect lenders and borrowers; employment relations connect workers and employers; supply relations connect firms; legal relations connect rights holders, duty bearers, offices, and institutions. Sections 5.4–5.10 have already developed the epistemic requirements for identifying such relations, locating them within boundaries and scales, tracing their persistence, and reconstructing higher-order configurations.

Some targets, however, depend materially on distributed conditions that are not adequately localized in any single pairwise relation. A firm's access to financing can depend on its position within a wider credit environment, prevailing classifications of risk, regulatory expectations, sectoral conventions, and the distribution of liquidity across institutions. A worker's bargaining possibilities can depend on labor-market tightness, occupational segmentation, legal recognition, organizational density, and expectations held by multiple actors who may never enter direct interaction with that worker. A price can be locally negotiated while the range of plausible negotiation is conditioned by distributed information, standards, infrastructures, institutional rules, and the positions occupied by other participants.

Network analysis already demonstrates why structural position can matter beyond individual attributes. Wellman's structural perspective emphasizes patterned relations, clusters, boundaries, dependency, and network position (Wellman 1983). Granovetter's analysis of embeddedness likewise shows that economic action is situated within ongoing structures of social relations (Granovetter 1985). A relational field extends the epistemic problem one step further when the target claim concerns a distributed condition whose effects depend on an actor's or relation's position within an organized environment and whose relevant organization cannot be exhausted by enumerating direct ties.

Definition 5.80 (Relational field). A *relational field* is a target-relative reconstruction of a distributed relational condition in which the position, capacities, constraints, expectations, or possible transformations of relevant relata depend materially on an organized environment of relations, positions, rules, resources, classifications, or other relationally efficacious conditions whose joint significance exceeds the description of isolated pairwise ties.

Definition 5.80 makes three commitments. First, a field is target-relative. A configuration can be field-relevant for one question and adequately represented through local relations for another. Second, field attribution concerns distributed organization, not merely large size or high density. A network with many edges can remain fully tractable as a set of relations if the target depends primarily on those ties. Conversely, a relatively sparse configuration can exhibit field-like properties when positional differences, common institutional constraints, or distributed expectations materially condition local possibilities. Third, the definition remains compatible with ontological suspension. The field may be represented through physical, institutional, legal, symbolic, informational, organizational, or other coordinates without establish-

ing that one formal vocabulary captures the ultimate constitution of the economic world.

The distinction between a dense network and a relational field is therefore epistemic rather than cosmetic. Density describes one property of a network representation. A field claim adds the proposition that distributed organization is relevant to the target and that position within that organization changes what can occur locally. The additional burden of evidence follows from this stronger claim.

Distributed conditions can also operate through absence, scarcity, concentration, or patterned non-connection. A peripheral supplier can face limited alternatives because potential buyers are concentrated elsewhere in the field. A household can encounter credit constraints because institutions classify its location or income profile within a broader risk environment. A profession can reproduce opportunity differences through expectations and credential structures that are widely distributed across organizations. In these cases, the relevant condition is partly expressed through relations that are possible, foreclosed, weak, indirect, or institutionally mediated. Field reconstruction therefore concerns the organization of relational possibilities as well as observed ties.

5.11.2 Evidence for Fields

A field claim requires evidence that distributed organization contributes information that local relations or aggregate attributes leave unexplained. The evidential task is especially demanding because many patterns that appear field-like can also arise through compositional differences, common external causes, selection processes, institutional rules applied uniformly, or correlated responses to a shared event. Field language becomes analytically useful when the distributed relational organization can be identified with enough specificity to discriminate among these possibilities.

Several evidential forms can contribute to such warrant. Recurrent position-dependent differences provide one source. If actors occupying systematically different locations within a reconstructed relational environment encounter different constraints, opportunities, expectations, or response patterns, the evidence can support a field interpretation when relevant alternative explanations are addressed. Historical evidence provides another source by showing how positions, rules, resources, and classifications were assembled and reproduced. Organizational records, legal documents, transaction traces, network data, ethnography, interviews, administrative classifications, and computational reconstructions can each reveal parts of the distributed condition. The evidential strength comes from convergence among sources whose biases and access conditions differ, rather than from the quantity of traces alone.

Field evidence can also arise from patterned responses to changes that propagate through an environment. A regulatory revision may alter behavior differently across positions; a liquidity shock can have heterogeneous effects according to financing structures; a platform rule can reorganize opportunities across categories of participants; a new credential can shift the value of positions throughout an occupational domain. Such patterned variation can reveal distributed structure, although the stronger causal interpretation of propagation remains a later problem for the

chapter's subsequent treatment of generative relations and attribution.

Claim 5.81 (Evidential convergence for field attribution). *Warrant for a relational-field claim is strengthened when heterogeneous evidence supports both the existence of a distributed relational organization and the target-relevant significance of positions or dependencies within that organization, while plausible aggregate and common-cause alternatives receive explicit examination.*

Claim 5.81 separates two evidential tasks. The first concerns reconstruction of the distributed organization itself. The second concerns the significance attributed to that organization for the target claim. A formally elegant map can satisfy the first task only partially and leave the second unresolved. Conversely, repeated local reports of constraint can establish an economically important phenomenon while leaving the wider field architecture underdetermined. Field epistemology becomes stronger when the two forms of evidence constrain one another.

The distinction between field effect and aggregate correlation is central. Suppose firms with a certain attribute exhibit similar outcomes. An aggregate account may explain the pattern through the attribute itself. A field account requires additional evidence that the outcome depends on relations, positions, distributed rules, or organized expectations surrounding those firms. Similarly, geographic clustering can reflect shared exposure to infrastructure or policy without establishing a relational field. Spatial proximity becomes field-relevant when the evidence shows that position within a distributed relational environment contributes to the pattern under investigation.

Field claims also benefit from negative and contrastive evidence. If a proposed field boundary predicts similar conditioning within a domain and weaker conditioning beyond it, observations that violate this pattern can reveal a misspecified boundary. If two positions treated as equivalent show systematically different possibilities, the positional reconstruction may require revision. If field language produces no observable or interpretive difference relative to a simpler representation, its epistemic contribution remains limited. A field concept earns analytical work by supporting distinctions that can be investigated.

5.11.3 Field Reconstruction

Field reconstruction is an inverse problem. Researchers encounter manifestations, traces, records, classifications, interactions, outcomes, and situated testimony, then infer a distributed organization that could have contributed to them. The non-invertibility developed in Section 5.2 therefore remains active. Several field reconstructions can be compatible with the same local observations, especially when relations are partially observed, boundaries are uncertain, or institutional categories already shape the traces from which the field is inferred.

A disciplined reconstruction begins with a specified target. The analyst identifies the phenomenon whose distributed conditioning is at issue, the relevant relata, candidate relations and institutional conditions, the temporal interval, and the boundary and scale assumptions inherited from Section 5.8. The reconstruction then asks which dimensions of position appear consequential for that target. These dimensions can include access to resources, legal status, organizational authority, dependence, in-

formation, reputation, credit, classification, spatial location, or other domain-specific conditions. Their selection should be justified by evidence and theory rather than imported from a preferred formal model.

Definition 5.82 (Field reconstruction). A *field reconstruction* is an explicit, revisable representation of a target-relevant distributed relational environment that specifies the relevant relata, relations or conditioning structures, positional dimensions, boundaries, temporal scope, evidential sources, and uncertainties used to support claims about field organization.

Definition 5.82 places provenance at the center of the reconstruction. A field map assembled from administrative data can inherit institutional classifications. A map reconstructed from interviews can inherit unequal access, standpoint differences, strategic presentation, and memory limitations. Transaction data can privilege recorded exchanges over informal relations and unrealized possibilities. Computational network or embedding methods can detect patterned structure while depending strongly on data construction and representation choices. Section 5.5 therefore remains relevant: the reconstructed field is itself a representation whose relation to relational reality requires evaluation.

Field reconstruction can combine formal and interpretive methods. Blockmodeling, for example, identifies positions and roles from patterned relations across networks (White et al. 1976); field-theoretical traditions can emphasize structured positions, resources, struggles, and the logic of a domain (Bourdieu and Wacquant 1992; Martin 2003); strategic-action-field analysis can emphasize mesolevel orders and relations among fields (Fligstein and McAdam 2012). GRE can use these traditions as methodological resources while preserving distinctions among their objects and assumptions. Similar diagrams or terms across traditions do not establish theoretical identity.

Reconstruction should also compare plausible alternatives. One representation may organize a labor market primarily through firms and workers; another may foreground occupations, credentials, unions, legal categories, households, migration status, or platform infrastructures. Each can reveal relations hidden by the others. The appropriate comparison is target-relative: which representation preserves distinctions needed for the claim, which assumptions it introduces, which traces support it, and how sensitive the conclusion is to alternative admissible constructions.

Field reconstruction is therefore iterative. Local observations can suggest a wider distributed structure; the proposed field can then guide new observations; new evidence can alter positions, boundaries, relation types, or the field hypothesis itself. This recursion also creates the performative possibility developed in Section 5.6. Administrative, legal, statistical, or algorithmic field representations can be adopted by organizations and begin to alter the relational environment they describe. Provenance should consequently record when a representation remains primarily analytical and when it becomes part of the field's subsequent organization.

5.11.4 Local Position and Distributed Structure

Field epistemology links local observations to distributed structure through the concept of position. Position is not simply an intrinsic attribute attached to an actor. It

is a relational characterization of where an actor, organization, relation, or configuration stands with respect to relevant structures of dependence, access, authority, classification, distance, substitution, or other field dimensions. Martin's account of field theory highlights precisely this explanatory move: regularities are related to positions vis-a-vis others rather than reduced to isolated properties (Martin 2003).

Definition 5.83 (Field position). A *field position* is a target-relative characterization of a relatum's location within a reconstructed distributed relational environment, defined by the dimensions of relation, dependence, access, resource distribution, classification, or institutional organization that are evidentially relevant to the inquiry.

The same local relation can therefore carry different implications at different field positions. A supplier contract can have one economic significance when alternative buyers are abundant and another when a dominant purchaser organizes access to the market. A debt contract can generate different vulnerabilities for borrowers occupying different income, collateral, legal, or refinancing positions. An employment relation can provide different bargaining possibilities depending on occupational demand, legal protections, union organization, household reserves, migration constraints, and available alternatives. Pairwise descriptions remain accurate at their own level while field position changes the conditions under which those relations operate.

Position also permits epistemic movement in the opposite direction. Local observations can become evidence for distributed structure when similar position-dependent patterns recur across cases. Testimony from participants located in several parts of a field can reveal differences in opportunity, expectation, or constraint. Transaction traces can show substitution patterns or dependencies. Institutional rules can explain why nominally similar actors occupy different practical positions. The inferential movement remains cautious because local evidence can support several field reconstructions. Distributed structure is inferred through patterned comparison, provenance, and alternative representations rather than read directly from a single local case.

Position should also be distinguished from rank. Some fields contain explicit hierarchies, yet positional structure can be multidimensional. An actor can have high legal authority and low informational access, strong reputation and weak liquidity, central connectivity and limited substitutability, or privileged classification in one institutional setting and constrained access in another. Compressing these dimensions into a single ranking can erase the particular mechanisms relevant to the target. Field reconstruction should preserve multidimensional positions whenever the evidence indicates that the dimensions operate differently.

A further complication arises because local action can modify distributed structure. Coordinated entry by firms, legal mobilization by workers, widespread adoption of a standard, changes in lending practice, or new classificatory technologies can alter the positions and dependencies that constituted the field. Position is therefore historically indexed. A field map should identify its temporal scope and remain open to structural change, consistent with the identity and persistence principles developed in Section 5.9.

5.11.5 Overlapping Fields

Economic actors frequently participate in several distributed relational environments at once. A university can occupy fields of higher education, research funding, labor, professional credentialing, real estate, public administration, and international mobility. A household can simultaneously participate in labor, housing, credit, care, education, welfare, kinship, and migration arrangements. A firm can occupy production, finance, regulation, technology, logistics, and professional domains whose boundaries and positional structures only partly coincide. Field overlap is therefore an ordinary epistemic problem rather than an exceptional complication.

Definition 5.84 (Field overlap). *Field overlap* occurs when a relatum, relation, resource, classification, institution, or conditioning process participates materially in two or more distinguishable relational fields relevant to the inquiry, such that the fields' positions, rules, or trajectories can interact in the target phenomenon.

Overlap matters because an observation that appears anomalous within one field can become intelligible when another field is included. A firm may possess a peripheral position in a production network while occupying a privileged financial position. A worker may have scarce occupational skills while facing legal constraints on mobility. A household can appear income-secure while remaining vulnerable through housing, care, or debt relations. These examples do not require collapsing the fields into one larger undifferentiated structure. The epistemic gain comes from identifying where their conditions intersect.

Overlapping fields should also remain distinct from multilayer representation in Section 5.10. A multilayer model distinguishes types of relation or connectivity within a system. Field overlap concerns partially distinct distributed environments whose membership, boundaries, positional dimensions, and organizing processes can differ. A multilayer representation can be used to model overlapping fields, yet the formal representation and the substantive field claim remain separate epistemic objects.

Interactions among fields can take several forms. Conditions can reinforce one another when advantageous positions in finance facilitate entry into production or when professional recognition increases organizational authority. Conditions can conflict when a legally authorized practice encounters professional opposition or when financial incentives undermine another field's institutional norms. Resources, classifications, standards, and actors can travel across fields, although their meaning and efficacy can change during the transfer. Fligstein and McAdam's account of strategic action fields explicitly emphasizes embedding and inter-field relations, providing one established approach to such problems (Fligstein and McAdam 2012). GRE can use this as a comparative resource while retaining its own target-relative epistemic vocabulary.

Field overlap creates an attribution problem as well. An observed regularity may be incorrectly assigned to the most visible field when the operative conditioning is distributed across several. Research design should therefore track cross-field provenance where evidence suggests that positions or rules interact. Robustness can be tested by comparing reconstructions that separate, combine, or differently partition candidate fields. Divergent conclusions identify where field specification materially affects the epistemic result.

5.11.6 Field Boundaries

Field boundaries specify the domain within which distributed relational conditioning is reconstructed. The boundary problem developed generally in Section 5.8 becomes more demanding here because field claims frequently rely on interactions that cross organizational, legal, geographic, or classificatory borders. A field boundary therefore needs both an analytical rationale and evidence about the reach of the conditioning process relevant to the target.

Some boundaries are institutionally explicit. Licensing regimes, legal jurisdictions, organizational membership rules, trading venues, professional associations, platform access rules, or funding programs can create entry conditions and structured differences between participants and non-participants. Such boundaries can themselves possess relational reality when they persist and causally affect access, obligation, recognition, or opportunity. Other field boundaries are more permeable. Information, capital, standards, reputations, labor, and classifications can travel across nominal domains, making the effective relational field wider than the formal institution.

Definition 5.85 (Field-relevance boundary). A *field-relevance boundary* is the provisional limit of a field reconstruction beyond which additional relata or relational conditions cease to make a material evidential difference to the specified target at the chosen scale and temporal interval, subject to revision when new evidence changes that assessment.

Definition 5.85 makes boundary selection an empirical and analytical judgment. The criterion concerns material relevance to the target, not mere formal membership. A supplier located outside a jurisdiction can remain field-relevant when its behavior changes local bargaining possibilities. A distant funding institution can shape research priorities through grant structures. An external regulatory standard can influence domestic production through certification requirements. Conversely, formal members can be peripheral to a particular target when their participation has little bearing on the distributed condition under investigation.

Boundary-sensitivity analysis is therefore especially important for fields. Researchers can compare nested or alternative boundaries and examine which positional claims, regularities, or inferred dependencies remain stable. If a result changes substantially when a plausible set of external actors or relations is included, the field claim should report that dependence. If several boundary specifications yield similar positional conclusions, confidence in the reconstruction can increase even though the boundary remains analytically constructed.

Temporal boundaries require comparable care. Fields can form, stabilize, transform, and dissolve. Institutional continuity can coexist with changes in position, resource distribution, or organizing rules. A reconstruction that pools several historical periods can therefore create an artificial field whose relations never coexisted in the represented form. The persistence criteria developed in Section 5.9 should be applied to the field itself: continuity claims need evidence about which organizing relations or mechanisms carry the distributed structure through time.

5.11.7 Limits of Field Attribution

Field language becomes epistemically weak when it functions as a residual label for whatever remains unexplained after local variables are considered. A field claim should add discriminating structure: it should identify a distributed organization, specify relevant positions or dependencies, delimit a provisional boundary, connect the reconstruction to evidence, and clarify what observations would favor alternative explanations. Without these elements, the term can remain metaphorically suggestive while contributing little to warranted knowledge.

A first limit concerns observational equivalence. Aggregate correlation, shared institutional exposure, network structure, common external causes, selection, and genuine field conditioning can produce similar manifestations. The inverse problem therefore persists at the level of fields. Evidence for a patterned outcome does not by itself identify which distributed mechanism produced it. Researchers should compare candidate reconstructions and identify which traces discriminate among them.

A second limit concerns scale. A pattern identified as field-level at one scale can decompose into several mechanisms at a finer scale or become one component of a wider field at a coarser scale. Scale dependence does not invalidate the field claim. It requires the claim to specify the level at which the distributed condition is reconstructed and the range over which the associated warrant is expected to travel.

A third limit concerns agency and heterogeneity. Field position conditions possibilities without mechanically determining action. Actors occupying similar positions can interpret, resist, exploit, or transform their conditions differently. Historical trajectories, identities, organizational capacities, resources, and cross-field affiliations can mediate responses. A field explanation therefore gains precision by identifying conditional tendencies or structured possibilities rather than converting position into deterministic prediction.

A fourth limit concerns formal representation. Network measures, spatial models, latent embeddings, correspondence analyses, agent-based models, and other computational tools can represent aspects of field structure. Their formal success remains distinct from substantive validation. The representation must preserve domain semantics, and its inferred positions or distances require evidential interpretation. Ontological suspension remains operative even when a field representation predicts effectively.

A fifth limit concerns causal and generative attribution. Establishing a field and locating positions within it does not yet establish that the field generated a particular capacity, relation, structure, constraint, or transformation. Field knowledge can identify the distributed conditions within which generative mechanisms operate, but the evidential burden for generative effect is stronger. The chapter's subsequent sections develop that distinction directly and address attribution among competing and distributed causes.

Proposition 5.86 (Minimum warrant for relational-field attribution). *A relational-field attribution is epistemically warranted to the extent that the inquiry specifies a target-relevant distributed organization, provides evidence for meaningful positional or dependency structure, states the boundary and temporal scope of the reconstruction, examines plausible simpler or alternative explanations, and preserves uncertainty arising*

ing from partial observation and representation choice.

Proposition 5.86 deliberately states a graded standard. Different inquiries can satisfy these requirements with different combinations of qualitative, historical, documentary, quantitative, network, spatial, legal, administrative, or computational evidence. The standard concerns explicitness and discriminating warrant rather than allegiance to a single method.

The epistemic value of field analysis can now be stated more precisely. Pairwise relations identify local connections; higher-order relational knowledge reconstructs configurations among relations; field analysis addresses distributed conditions in which position within an organized relational environment changes the interpretation of local relations, capacities, constraints, or possibilities. These representations can complement one another. Their usefulness depends on the target, evidence, boundary, scale, and historical interval of inquiry.

This distinction also establishes the transition to generativity. A field may be relationally real and epistemically well reconstructed while remaining only part of the background conditions for a later transformation. Conversely, a generative effect can depend on the interaction between a local relation and the field in which it operates. The next section therefore turns from the epistemology of distributed relational organization to the stronger problem of identifying *generative relations*: relations that participate in producing changes in capacities, relations, structures, constraints, possibilities, or the conditions of subsequent transformation.

5.12 Epistemology of Generative Relations

The preceding sections established how GRE can identify relations, reconstruct their persistence, move toward higher-order relational knowledge, and warrant claims about distributed relational fields. The present section introduces a stronger epistemic target: relations that participate in generating changes that become relevant to subsequent economic life. Its role is to specify what a generative claim means before the chapter turns to the more demanding problem of attributing generative effects among competing causes. The objective is therefore taxonomic and epistemic rather than fully causal. The discussion distinguishes generative effects from ordinary state change, then examines capacity generation, relation generation, structural generation, the generation of constraints and enabling conditions, the transformation of future possibilities, and changes in the conditions under which later transformation occurs. Throughout, the method is target-relative and temporally explicit: a claim of generation must identify what changes, for whom or for what configuration, over which interval, at which scale, and with what downstream relevance.

A relation can be relationally real without being generative with respect to the target under study. It can persist, structure interaction, and possess causal efficacy while leaving the relevant capacity, relation, structure, or option space materially unchanged during the interval of inquiry. Conversely, a relation can be generatively consequential even when its most important effects appear only after the initiating interaction has ended. GRE therefore treats generativity as a property of a relation-in-context relative to a specified effect, boundary, scale, and temporal horizon, rather

than as an intrinsic label attached permanently to a relation type.

Definition 5.87 (Generative relation). A *generative relation* is a relation for which warranted inquiry supports the claim that participation in, exposure to, or operation through that relation contributes materially to the emergence, transformation, stabilization, suppression, or dissolution of a target capacity, relation, structure, constraint, possibility, or condition of subsequent transformation within a specified context and temporal horizon.

Definition 5.87 is deliberately relational and target-relative. Employment, credit, education, care, legal recognition, platform participation, research collaboration, or infrastructure access can be generative in one respect and comparatively inert in another. The definition also preserves directional openness. Generation can enlarge capacities and possibilities, but it can also stabilize dependence, produce constraints, suppress alternative relations, or narrow later option spaces. Epistemic identification of generativity therefore remains distinct from normative evaluation.

5.12.1 Relation and Generative Effect

Knowledge that a relation exists establishes an important part of the evidential basis for GRE, but existence alone does not establish what the relation generates. Two organizations can collaborate without the collaboration materially changing either organization's capabilities. A worker can hold a contract that structures payment and obligation without acquiring new skills or access through it. A legal status can be stable and causally relevant while producing little change in the target outcome during a particular interval. The epistemology of generative relations therefore begins by separating evidence for relational existence from evidence for a downstream generative effect.

A generative effect concerns a change that becomes part of the conditions of subsequent action, interaction, or transformation. Immediate outputs can be relevant evidence, yet they need not exhaust the effect. Training can produce an immediately observable certification while the more consequential effect lies in later competence or access. Collaboration can produce a publication while also generating shared routines, trust, instruments, or further collaborations. Infrastructure can provide current service while changing the set of activities that becomes feasible around it. The epistemic target is therefore temporally extended beyond the initiating relation and its first manifestation.

Granovetter's account of embedded economic action provides a useful precedent for treating ongoing relations as economically consequential rather than analytically incidental (Granovetter 1985). Powell, Koput, and Smith-Doerr similarly show that interorganizational collaboration can function as a locus of learning in biotechnology (Powell et al. 1996). These literatures do not by themselves establish the GRE category of a generative relation. They illustrate why the relation can be an explanatory object when later capacities and opportunities depend on relational participation.

Definition 5.88 (Generative effect). A *generative effect* is a target-relevant change whose emergence or transformation is materially connected to a prior or ongoing relational process and whose significance extends to subsequent capacities, relations, structures, constraints, possibilities, or transformation conditions.

The phrase *materially connected* in Definition 5.88 states the epistemic burden without resolving the attribution problem in advance. Association, temporal succession, or narrative plausibility is insufficient on its own. The following section will examine the stronger evidential requirements involving mechanism, counterfactual dependence, multiple causation, distributed generation, delay, branching, and emergence. The present section specifies the objects among which such attribution must operate.

Generative effects can also be layered. A training relation can generate a capacity that later enables entry into a professional relation; the new relation can then alter organizational position; the changed position can affect later access to resources. Each step can be treated as a separate generative claim or as part of a longer generative trajectory. Epistemic clarity requires researchers to state which link is being claimed rather than treating an entire downstream history as the undifferentiated effect of one initiating relation.

5.12.2 State Change and Generative Change

A further distinction concerns change itself. Economic systems exhibit continual state change: prices move, balances fluctuate, persons relocate, inventories vary, organizations hire and release workers, and networks gain or lose connections. A change becomes generatively significant when it modifies the conditions from which later states can arise. The distinction is therefore temporal and structural rather than simply quantitative. A large change can be transient, while a small change can alter a pathway that later becomes consequential.

Consider a short-lived increase in income. If it changes current consumption and then leaves later capacities, relations, obligations, and access materially unchanged, the relevant claim may remain a state-change claim. If the same increase finances training, prevents the loss of housing, sustains a business through a liquidity shock, or permits entry into a relation that would otherwise have been inaccessible, its significance can become generative because subsequent conditions differ. The epistemic question concerns what persists or propagates from the change, not merely its magnitude at the moment of observation.

This distinction is compatible with established economic analyses of path dependence and increasing returns. Arthur's treatment of competing technologies shows how historically contingent events can become consequential when increasing-return mechanisms make later selection depend on earlier adoption (Arthur 1989). GRE uses such work as one lineage for thinking about how an apparently local change can become a condition of later development. It does not infer that every generative process is path dependent or governed by increasing returns.

Proposition 5.89 (Generative significance of change). *Evidence of state change supports a claim of generative change only to the extent that the inquiry also identifies a warranted pathway through which the change alters subsequent capacities, relations, structures, constraints, possibilities, or conditions of transformation within the stated analytical boundary and temporal horizon.*

Proposition 5.89 prevents generativity from becoming a synonym for dynamism. It also separates generative analysis from a requirement that the generated consequence be permanent. A change can be generative over a finite interval if it modifies

subsequent conditions during that interval. Later reversal does not erase the historical effect; it changes the trajectory through which the effect should be reconstructed.

Generative change can occur through persistence, amplification, redirection, threshold crossing, or reconfiguration. These mechanisms have different evidential signatures. Persistence can be visible through durable capability or institutionalized routine. Amplification can appear through cumulative participation or increasing returns. Redirection can be inferred when later action follows a newly available pathway. Threshold crossing can be relevant when a previously unavailable relation becomes feasible. Reconfiguration can occur when the identities of components remain similar while their relational organization changes. The epistemic category should remain broad enough to accommodate these forms while requiring each empirical study to specify which form is supported.

5.12.3 Capacity Generation

Capacity generation concerns changes in what a subject, organization, collective, or relational configuration is able to do. It is one of the most direct ways in which a relation can alter later economic possibility. Capacities can be productive, epistemic, organizational, political, relational, adaptive, or infrastructural. Some can be attributed primarily to identifiable subjects; others remain distributed across teams, institutions, networks, or technical systems.

The capability approach provides an established conceptual resource because it distinguishes resources from substantive opportunities and directs attention to what persons are effectively able to do and to be (Sen 1999). GRE's epistemic problem is narrower and more processual: when does evidence support the claim that a relation participated in producing or transforming such an ability? A resource transfer and a capacity gain should not be treated as equivalent. The same nominal resource can produce different capacities when legal status, infrastructure, care obligations, prior knowledge, recognition, social support, or institutional access differ.

Capacity generation therefore requires at least a temporal comparison and a relational account of conversion. Evidence can include demonstrated performance, access to previously unavailable activities, durable competence, new organizational routines, successful coordination, changes in adaptive response, or the creation of shared infrastructure. Self-report can also be important, particularly for capacities that are not institutionally measured, although the evidential role of testimony must remain explicit. Where capacity is latent rather than exercised, inquiry should distinguish possession of an enabling condition from demonstrated use.

Some capacities are relational in their very operation. Trust, coordination, collective memory, professional recognition, bargaining power, and access through networks cannot be represented adequately as isolated possessions of one actor. Their generation may therefore require evidence at more than one level. A participant can acquire competence while a group simultaneously develops coordination capacity; an organization can learn through relations among units; a collaborative network can produce shared standards that later participants inherit. The higher-order epistemology developed in Section 5.10 becomes relevant whenever the generated capacity resides partly in a configuration rather than in a single relatum.

Capacity generation is also normatively open. A surveillance organization can acquire stronger monitoring capacity; a monopolistic platform can improve its ability to exclude rivals; a debt contract can increase purchasing capacity while also strengthening creditor leverage. GRE therefore records the generated capacity before evaluating its distribution or desirability. Later normative analysis can ask whose capacity increased, whose decreased, which dependencies were created, and how the generated capacity affects justice or power.

5.12.4 Relation Generation

Relations can generate further relations. A collaboration can produce trust that supports later collaboration; professional training can create mentorship and peer networks; infrastructure can connect previously separated actors; legal recognition can make contractual or organizational relations possible; a platform can create matching relations while simultaneously mediating their terms. Relation generation matters because the relational environment from which later economic action begins can itself be an outcome of earlier relations.

The epistemic burden is to distinguish a newly generated relation from a relation that was merely unobserved, latent, or already available. Temporal order provides part of the evidence but does not settle the issue. Researchers should examine whether the later relation became possible, probable, stabilized, or transformed through mechanisms associated with the earlier relation. Records of first contact, institutional eligibility, referral pathways, shared projects, communication histories, or changes in formal rights can be relevant traces. Where the later relation existed previously, the generative claim may concern strengthening, stabilization, or transformation rather than initial formation.

Interorganizational learning provides one empirical context in which relation generation can become visible. Collaboration can create knowledge and reputational effects that affect later alliance opportunities, while participation in a network can change access to additional partners (Powell et al. 1996). The epistemic lesson is not that networks automatically reproduce themselves. It is that later relational structure can be historically dependent on participation in earlier relational configurations.

Definition 5.90 (Relationally generated relation). A *relationally generated relation* is a later relation whose formation, stabilization, weakening, or reconfiguration is supported by evidence linking that change materially to an earlier or ongoing relational process.

This definition also accommodates negative generation. A relation can generate exclusionary boundaries that suppress alternative ties. Exclusive contracts can make some supplier relations unavailable; professional closure can prevent entry relations; reputational mechanisms can discourage association; conflict can dissolve previously viable collaborations. The generative object is the transformation of the relational landscape, which can involve both formation and foreclosure.

Recursive representation from Section 5.6 provides an important special case. A representation can affect eligibility, ranking, recognition, or visibility and thereby generate later relations that produce new traces and representations. Relation generation is therefore one pathway through which epistemic infrastructures become

participants in the systems they describe.

5.12.5 Structural Generation

Structural generation concerns changes in relatively persistent patterns that organize multiple relations beyond a single dyad or episode. Examples include new organizational routines, durable divisions of authority, market architectures, standards, governance arrangements, infrastructures, classificatory regimes, or recurrent patterns of dependence. The concept therefore requires more than evidence that several individual relations changed at once. It requires evidence that the organization of relations acquired a degree of persistence and causal relevance at the scale specified by the inquiry.

Institutions provide a familiar reference point. North's account of institutions emphasizes rules and constraints that structure interaction and shape incentives (North 1990). GRE can treat institutional change as one form of structural generation when prior relations contribute to creating or transforming such organizing conditions. The category is wider than formal institutions, however. Informal routines, standards, technical architectures, and network structures can become sufficiently stable to condition later interaction even without codified rules.

Structural generation raises a scale problem. A relation can generate a local routine within one organization without transforming the wider institutional environment. Conversely, many local interactions can cumulatively contribute to a wider structural change. The claim should therefore specify the scale at which the generated structure is identified and the bridge evidence supporting movement across scales. Section 5.8 established this requirement generally; generative analysis makes it especially important because downstream structural claims can easily outrun the evidence available at the initiating scale.

Field reconstruction also becomes relevant. A newly generated structure can alter positions within a wider relational field without creating the field itself. A regulatory reform can change access to a profession; a platform rule can reorganize visibility among sellers; a standard can alter compatibility among firms. The epistemic claim can concern transformation of distributed position rather than replacement of the entire field. Researchers should therefore avoid treating every consequential structural modification as the birth of a wholly new system.

Structural generation can be identified through persistence, reproduction, and altered conditioning. A new rule or architecture becomes structurally consequential when later relations systematically encounter it as part of their environment. Evidence may include repeated administrative application, stable technical dependencies, recurrent organizational practices, patterned entry or exclusion, durable network reconfiguration, or sustained changes in the distribution of authority. The relevant test is whether the generated structure becomes an active condition of subsequent relations rather than remaining an isolated output.

5.12.6 Generation of Constraints and Conditions

Generativity includes the production of constraints as well as enabling conditions. A credit relation can create liquidity and repayment obligations. A technical standard can generate interoperability and exclude incompatible designs. A credential can create recognized access while making uncredentialed competence less legible. Infrastructure can lower some transaction costs while creating dependence on maintenance, ownership, or proprietary interfaces. These effects show why generativity cannot be equated with expansion, benefit, or freedom.

The epistemic task is to identify how a relation changes the conditions under which later action occurs. Constraints can concern legal permission, material resources, time, information, mobility, compatibility, obligation, visibility, bargaining position, or access to other relations. Enabling conditions can concern the same dimensions in the opposite direction, although a single change can enable one pathway while constraining another. The object of knowledge is therefore often a reorganization of conditional possibility rather than a simple increase or decrease in one variable.

North's institutional analysis is again relevant because institutions can both enable coordination and constrain behavior through structured rules and incentives (North 1990). Sen's emphasis on substantive opportunity likewise helps separate nominal resource possession from the social conditions under which resources become effective freedoms (Sen 1999). GRE uses these traditions to sharpen an epistemic question about generation: which prior relations produced the conditions that later actors encounter as permissions, resources, dependencies, obligations, or obstacles?

Constraints can themselves be generative. A standard can narrow permissible technical choices while enabling large-scale coordination within the remaining space. A budget ceiling can restrict current expenditure while inducing new organizational routines. A legal prohibition can suppress one practice and stimulate alternative organizational forms. Such cases make simple positive/negative coding inadequate. Inquiry should identify what is constrained, what becomes enabled, which actors experience each effect, and over what period.

The generation of conditions also includes maintenance. Some relations do not create a visibly novel capacity or institution; they reproduce the conditions under which an existing system can continue. Care relations sustain persons, maintenance work preserves infrastructure, administrative routines reproduce legal statuses, and ecological practices can maintain the material conditions of production. Epistemically, maintenance should not disappear merely because the observed state remains stable. Stability can itself be the generated consequence of ongoing relational work.

5.12.7 Transformation of Future Possibilities

Generative analysis reaches beyond realized outcomes to changes in the space of feasible or accessible futures. A relation can open, preserve, redirect, weight, or foreclose later possibilities without determining which possibility will be realized. Education can expand occupational and epistemic options; credit can make investment possible while narrowing later discretion through repayment; migration status can alter mobility and labor-market access; shared infrastructure can support activities

that were not specified when it was built. The epistemic object is a changed possibility structure, not a prediction of one future path.

This distinction is important because realized outcomes underdescribe generative significance. A research facility can expand the range of experiments that could be conducted even if only a fraction are eventually attempted. A legal right can create actionable options that remain unused. Redundant infrastructure can preserve alternatives that become valuable only under disruption. Conversely, an exclusive contract can remove feasible pathways even when the affected actor never attempts to use them. Evidence for transformed possibility therefore cannot rely exclusively on observed realized choices.

The capability literature again provides a useful precedent because it distinguishes achieved functionings from substantive opportunities (Sen 1999). For GRE, evidence of future possibility can include formal rights, eligibility, technical feasibility, available partners, resource access, documented opportunity sets, demonstrated substitutability, or credible pathways identified through historical and institutional reconstruction. Researchers should distinguish possibilities that are merely imaginable from those supported by a sufficiently specified relational and material pathway.

Definition 5.91 (Generative possibility transformation). A *generative possibility transformation* is a warranted change in the set, accessibility, weighting, or durability of future pathways available to a specified subject or configuration that is materially connected to an earlier or ongoing relational process.

Possibility transformation is scale-relative and distribution-sensitive. The same institutional change can expand a firm's technological options while narrowing workers' mobility, or expand aggregate investment opportunities while concentrating access among already well-positioned actors. A system-level increase in the number of possible transactions does not establish that each participant's meaningful option space expanded. Epistemic claims should therefore state whose possibilities changed and at which scale.

Historical contingency remains central. Arthur's analysis of increasing returns illustrates one mechanism through which early events can reshape later technological possibilities and make some alternatives progressively harder to recover (Arthur 1989). Other domains can exhibit different mechanisms, including institutional learning, legal precedent, infrastructure dependence, ecological degradation, reputation, or cumulative capability. GRE therefore treats possibility transformation as a broad epistemic category whose mechanism must be established locally rather than assumed from path-dependence language alone.

5.12.8 Transformation of the Conditions of Transformation

The deepest generative claims concern changes in the conditions through which later change itself is produced. An intervention can alter outcomes while leaving the mechanisms of subsequent adjustment broadly intact. In other cases, it changes the institutional procedures, technical interfaces, learning capacities, classification systems, infrastructures, or selection environments through which later transformations occur. The epistemic target then moves from a generated state or option to a change in the generative conditions of future change.

Examples include platformization of a market, where matching and visibility become mediated by ranking and data infrastructures; legal reform that changes the categories through which later claims become recognizable; organizational learning that changes how future problems are solved; adoption of a common technical standard that reorganizes later innovation pathways; and machine-mediated decision systems that alter how subsequent classifications are produced. Espeland and Sauder's study of rankings demonstrates one concrete route through which public measures can reorganize organizational behavior and thereby recreate parts of the social environment they measure (Espeland and Sauder 2007). Such cases show why recursively performative representations can become relevant to generative epistemology.

A claim at this level should be stated carefully. Economic systems rarely possess a single fixed rule of change. The phrase *conditions of transformation* refers to the locally relevant mechanisms, procedures, infrastructures, capacities, and constraints through which change is generated in the studied domain. Different methods can represent these conditions differently. A legal analysis can reconstruct changing rights and procedures; organizational research can identify routines and learning; network analysis can track changing connection rules; computational work can model altered update mechanisms; historical inquiry can reconstruct how institutional sequences changed what later interventions could accomplish.

Definition 5.92 (Second-order generative change). A *second-order generative change* is a transformation in the mechanisms, capacities, rules, infrastructures, or relational conditions through which subsequent generative changes are produced, selected, constrained, or propagated within the stated domain.

The term *second-order* in Definition 5.92 identifies an analytical level rather than a universal hierarchy. A change can be second-order relative to one target and ordinary structural change relative to another. The classification therefore depends on the research question and on evidence that later processes operate differently because the relevant transformation conditions have changed.

This level also sharpens the distinction between generativity and immediate effectiveness. An intervention can solve a present problem while weakening the capacities needed for later adaptation. Another can have modest immediate effects while building institutions, knowledge, trust, redundancy, or infrastructure that change how future problems are addressed. Generative epistemology therefore requires temporal horizons long enough to observe changes in the production of later change, while preserving uncertainty when those horizons exceed available evidence.

The eight subsections together establish a graded vocabulary for generative knowledge. Relations can generate immediate capacities, further relations, persistent structures, constraints and enabling conditions, altered possibility spaces, and changes in the conditions through which later transformation occurs. These objects can overlap, and one generative trajectory can move through several of them. Their analytical separation nevertheless matters because the evidential traces and temporal horizons differ. Evidence that a capacity was generated does not by itself establish structural transformation; evidence of structural change does not establish that the mechanisms of later change were transformed; evidence of expanded possibilities does not establish which possibilities will be realized.

The resulting epistemic discipline can be stated compactly. A generative claim should specify the initiating relation or relational configuration, the target change, the relevant boundary and scale, the temporal horizon, and the level of generative effect being asserted. It should also preserve the distinction between observed change and attributed generation. The present section has defined the possible objects of that attribution. The following section takes the next step by examining what evidence is required to warrant the claim that a particular relation, configuration, or field materially generated the effect in question.

5.13 Attribution of Generative Effects

Section 5.12 specified the objects of generative inquiry by distinguishing relational existence from claims about what a relation, configuration, or field helps to generate. The present section addresses the next epistemic problem: the attribution of those effects among competing, interacting, and temporally distributed conditions. Its role is to establish standards for moving from evidence that a generative change occurred to a warranted claim that some relation or relational configuration materially contributed to that change. The objective is neither to impose a single universal theory of causation nor to reduce generative inquiry to one identification technique. The section instead develops a plural evidential framework organized around association, mechanism, counterfactual dependence, multiple causation, distributed generation, delayed effects, branching pathways, and emergence. The method is comparative and target-relative. Each attribution claim should identify the effect, candidate contributors, relevant alternatives, boundary and scale, temporal horizon, and the evidential basis for assigning causal-generative responsibility.

Generative attribution is demanding because economic change is rarely produced by an isolated relation operating against a fixed background. Relations are embedded in infrastructures, institutions, fields, histories, material conditions, classifications, and other relations. The same observed effect can therefore be compatible with several causal accounts, while the same relation can contribute differently under different surrounding conditions. Mechanism-based explanation provides one established way to connect cause and effect by identifying the processes through which the connection is produced (Hedström and Ylikoski 2010). Counterfactual approaches provide another by asking how the target would vary under changes to a candidate cause or intervention (Woodward 2003; Gangl 2010). GRE draws on these traditions while preserving the additional problems created by relational dependence, historical sequence, heterogeneous scales, and distributed generation.

Definition 5.93 (Generative attribution). A *generative attribution* is a warranted claim that a specified relation, relational configuration, or relational field materially contributed to the emergence, transformation, stabilization, suppression, or dissolution of a target generative effect within a stated boundary, scale, and temporal horizon.

Definition 5.93 leaves room for several forms of contribution. A relation can operate as a trigger, enabling condition, constraint, amplifier, mediator, channel, stabilizer, or component of a distributed configuration. The epistemic task is therefore

richer than assigning a binary label of cause or non-cause. Attribution should recover the form, scope, and degree of contribution that the available evidence can support.

5.13.1 Association and Generation

Association is often the first observable sign that a generative relation may be present. Participation in a training network can covary with later capability; access to finance can covary with organizational survival; legal recognition can covary with access to services; collaboration can covary with later innovation. Such patterns can direct inquiry toward plausible generative processes. Their evidential role is therefore important, especially when the relation under study is difficult to manipulate directly or when historical evidence is incomplete.

Association alone leaves several attribution problems unresolved. Selection into a relation can reflect pre-existing capacities. A shared institutional change can affect both the candidate relation and the later effect. The relation can be a manifestation of a deeper process rather than a contributor to the effect. Reverse influence can also occur when an emerging capability increases the probability of entering the relation subsequently treated as explanatory. These possibilities do not make association uninformative; they make its inferential role conditional on a broader reconstruction of temporal order, alternative explanations, and the processes linking the relation to the effect.

The distinction between association and generation should also remain sensitive to the object described in Section 5.12. A relation can be strongly associated with an outcome while weakly connected to the particular generative change under investigation. For example, membership in a professional network may correlate with income, while the relevant generative question concerns whether the network produces access to knowledge, legitimacy, referrals, or future collaborations. The attribution target should therefore be stated at the level at which the alleged generative contribution occurs.

Proposition 5.94 (Association as directional evidence). *Evidence of association can support the direction of generative inquiry when temporal ordering, selection processes, plausible common conditions, and alternative relational explanations are examined explicitly. The strength of a generative attribution depends on the additional evidence that connects the association to a target-relevant contribution.*

Proposition 5.94 prevents two symmetrical errors. One error treats correlation as sufficient attribution. The other discards associative evidence whenever experimental identification is unavailable. GRE instead treats association as one evidential layer whose value depends on how it interacts with mechanism, counterfactual reasoning, temporal reconstruction, comparative cases, process tracing, quantitative identification, and other sources of warrant.

This layered view is especially important for historically situated relations. A relation may have existed only once, or under institutional conditions that cannot be reproduced. In such cases, attribution can still improve through triangulation across traces, comparison, sequence evidence, mechanism reconstruction, and carefully scoped counterfactual reasoning. The epistemic standard should reflect the attainable evidence while preserving uncertainty about rival accounts.

5.13.2 Mechanism

Mechanism evidence addresses how a candidate relation contributes to the target effect. A mechanism claim identifies a process or organized sequence through which relational participation changes capacities, incentives, expectations, resources, classifications, obligations, information, positions, constraints, or other conditions relevant to later states. Mechanism-based explanation has a long standing in the social sciences because it can connect observed regularities to the processes that generate them (Hedström and Ylikoski 2010). For GRE, mechanism evidence is particularly valuable when the generative effect concerns a transformation of relations or capabilities that cannot be understood adequately from end-state comparison alone.

A relation can support several mechanisms simultaneously. Credit can relax a liquidity constraint, alter bargaining power, create repayment obligations, and establish a history that affects later access. Legal recognition can change formal entitlement, administrative legibility, third-party expectations, and opportunities for collective organization. Collaboration can transmit knowledge, create trust, establish routines, and modify organizational reputation. A credible attribution therefore benefits from a mechanism profile rather than a single narrative link.

Definition 5.95 (Generative mechanism). A *generative mechanism* is a reconstructable process through which a candidate relation or relational configuration contributes to a specified generative effect by altering one or more target-relevant conditions across time.

The requirement of reconstructability matters. A mechanism should be connected to traces, observations, interventions, institutional records, comparative sequences, behavioral evidence, or other forms of empirical support. A verbal sequence that merely redescribes the outcome offers limited additional warrant. Stronger mechanism evidence shows that intermediate changes occurred, that they were linked to the candidate relation, and that they plausibly carried part of the effect toward the target.

Mechanism evidence also has boundaries. Several mechanisms can produce observationally similar outcomes. A mechanism can operate without being sufficient for the effect. The same mechanism can generate different outcomes under different surrounding relations. Some components can remain latent or only indirectly observable. GRE therefore treats mechanism reconstruction as evidence for attribution rather than as automatic closure of the attribution problem.

The scale of mechanism should also match the scale of the claim. A macro-level institutional shift can depend on micro-level interactions, meso-level organizational routines, and field-level constraints. A complete account need not reproduce every lower-level event. It should, however, identify the cross-scale connections necessary for the claimed contribution. Section 5.8 established the need for bridge evidence when transporting relational claims across scales; generative attribution inherits the same requirement.

5.13.3 Counterfactual Dependence

Counterfactual reasoning asks how the target effect would differ under a relevant change to the candidate relation or its operation. Woodward's interventionist account treats such what-if-things-had-been-different questions as central to causal explanation and emphasizes invariant relations under interventions (Woodward 2003). In empirical social research, counterfactual frameworks likewise clarify causal identification by comparing observed outcomes with potential outcomes under alternative treatments or exposures (Gangl 2010). GRE adopts the epistemic discipline of this reasoning while extending the object of comparison from isolated treatments to relational and configurational conditions.

The relevant counterfactual is rarely a simple world in which one relation disappears while everything else remains fixed. Removing a relation can alter connected relations, classifications, expectations, resource flows, or institutional responses. A counterfactual concerning employment, credit, legal status, or network membership may therefore require a specification of what changes together with the focal relation and what is held approximately stable for analytical purposes. The credibility of the counterfactual depends on this relational specification.

Definition 5.96 (Relational counterfactual specification). A *relational counterfactual specification* states the candidate relation or relational feature to be changed, the connected conditions permitted to vary with that change, the background conditions provisionally held stable, the target effect, and the temporal horizon over which the comparison is evaluated.

Definition 5.96 makes counterfactual reasoning compatible with the open and permeable systems analyzed in Sections 5.8–5.11. It also helps distinguish feasible alternatives from purely verbal ones. Historical and institutional constraints can make some counterfactuals comparatively informative and others remote from any plausible transformation pathway.

Counterfactual dependence can provide strong attributional evidence, yet absence of simple dependence does not always eliminate contribution. Redundant causes can each be sufficient under the observed conditions. An enabling relation may become visible only when another trigger occurs. Distributed systems can contain compensatory pathways that preserve the outcome after one component is removed. Generative attribution should therefore avoid treating a single binary counterfactual test as exhaustive.

Remark 5.97 (Counterfactual evidence and contribution). Counterfactual evidence is strongest when the proposed intervention corresponds to a meaningful relational variation and when the resulting comparison preserves the dependencies necessary to interpret the target. Failure of one simple counterfactual test can reflect redundancy, substitution, compensation, or misspecified intervention boundaries rather than the absence of generative contribution.

5.13.4 Multiple Causation

Generative effects commonly arise from several contributors whose roles differ. One condition can enable a process, another can trigger it, a third can amplify it, and a

fourth can stabilize the resulting relation. Attribution in such cases should identify the contribution structure rather than search automatically for one privileged cause. This is especially important when institutional and economic outcomes depend on conjunctions of resources, permissions, capacities, infrastructures, and expectations.

Multiple causation can take several forms. Contributors can be jointly necessary within a local configuration. Several contributors can each be sufficient under the observed conditions. One contributor can substitute for another. A relation can increase the probability of an effect without determining it. A background condition can become causally salient only when another process changes. These forms have different implications for both explanation and intervention.

Definition 5.98 (Generative contribution profile). A *generative contribution profile* records the evidenced role of a candidate relation within a multi-contributor process, including whether the relation functions primarily as an enabling condition, trigger, amplifier, mediator, constraint, substitute, stabilizer, or component of a jointly productive configuration.

The categories in Definition 5.98 are analytical roles rather than mutually exclusive ontological kinds. A single relation can occupy several roles over time. Credit can first enable investment, later constrain cash flow through repayment obligations, and eventually stabilize market access by building a recognized borrowing history. Contribution profiles therefore benefit from temporal indexing.

Multiple-cause analysis also changes the interpretation of negative evidence. If an effect persists after one relation is removed, the remaining pathways can indicate redundancy or substitution. If a relation produces little effect in one setting, a missing complement can explain the difference. Comparative inquiry should therefore examine configurations in which candidate contributors vary jointly and separately where the evidence permits.

For GRE, this configurational perspective is particularly compatible with the generative emphasis of Section 5.12. The aim is to understand how a target becomes possible, stabilized, redirected, or suppressed through interacting conditions. Attribution remains precise when the analysis specifies each candidate's supported role and avoids assigning the entire effect to the most visible contributor.

5.13.5 Distributed Generation

Some generative effects arise through distributed organization rather than through a compact set of readily isolable contributors. Knowledge production, market conventions, organizational routines, collective trust, infrastructural reliability, and field-level opportunities can depend on many interactions whose individual effects are modest while their combined organization is consequential. Section 5.11 established that field attribution requires evidence for distributed organization and positional significance. Generative attribution adds a further requirement: evidence that the distributed configuration contributes to the target transformation.

Complex adaptive systems provide useful precedents for understanding outcomes produced through localized interaction, historical dependence, and feedback across levels (S. A. Levin 1998; Arthur 1999). Adaptive-network research likewise examines

systems in which node states and network topology change through reciprocal feedback (Gross and Blasius 2008). GRE uses these literatures as analytical resources without assuming that economic relational fields obey one universal complexity model.

Distributed generation creates a localization problem. The effect can be attributable to the organization of the configuration even when no single relation reproduces the effect independently. Conversely, invoking “the system” can obscure concrete pathways if the distributed claim is left unspecified. Stronger attribution therefore identifies the organization that matters: connectivity, complementarity, feedback, positional differentiation, repeated interaction, shared infrastructure, or another target-relevant pattern.

Proposition 5.99 (Distributed generative attribution). *A distributed generative attribution gains warrant when evidence supports both the existence of a relevant distributed organization and a pathway through which properties of that organization materially contribute to the target effect. Evidence concerning isolated components remains informative but need not exhaust the attribution when the effect depends on configurational properties.*

Proposition 5.99 preserves the distinction between distributed contribution and explanatory vagueness. It also permits mixed attributions. A focal actor or relation can have a distinctive role within a distributed process while the realized effect still depends on the surrounding field. Such cases should be represented through layered contribution profiles rather than forced into either purely individual or purely systemic explanation.

Distributed attribution also inherits the uncertainty of field reconstruction. Missing relations, changing boundaries, representation choices, and uneven observation can alter the inferred organization. The confidence attached to a distributed generative claim should therefore remain sensitive to the uncertainty established in Sections 5.7–5.11.

5.13.6 Delayed Effects

Generative effects can appear after substantial delay. A relation may first change a capacity, expectation, obligation, or infrastructure whose significance becomes visible only when later conditions activate it. Training undertaken years earlier can affect later occupational mobility; a legal classification can reshape future eligibility; an early technological choice can alter the costs of later switching; a trust relation can become consequential during a later crisis. Delay therefore weakens simple temporal proximity as a criterion of attribution.

Path-dependent analyses demonstrate how earlier events can acquire later significance when self-reinforcing processes make subsequent development sensitive to sequence (Arthur 1989; Pierson 2000). GRE extends this temporal sensitivity beyond lock-in. Delayed effects can also operate through stored capacity, institutional memory, accumulated obligations, latent options, slowly changing expectations, or deferred thresholds.

The main epistemic difficulty is separating delayed contribution from retrospective storytelling. As the time between candidate relation and observed effect

increases, the number of intervening conditions generally grows. Attribution should therefore identify the carriers through which the earlier relation remains relevant. These carriers can include skills, records, infrastructures, debts, organizational routines, legal statuses, durable classifications, network ties, or other persistent residues.

Definition 5.100 (Generative carrier). A *generative carrier* is a persisting or transmitted condition through which the contribution of an earlier relation remains connected to a later target effect across a temporal interval.

Evidence for a carrier strengthens temporal attribution because it links the earlier relation to later conditions without assuming uninterrupted visible activity. The carrier itself can transform. A credential can become a reputation signal; a technical capability can become an organizational routine; a debt can become a constraint on later investment. The epistemic task is to reconstruct continuity of contribution through these transformations.

Temporal horizon also affects the sign and magnitude of attribution. A relation can generate a short-run capability while producing a longer-run constraint, or impose an immediate cost while creating later options. Generative attribution should therefore state the horizon over which the effect is assessed and, where relevant, separate short-, medium-, and long-run contribution profiles.

5.13.7 Branching Effects

A single relation can open several downstream pathways. Collaboration can generate knowledge, status, additional partners, conflict, dependence, and new organizational commitments. Access to infrastructure can enable several economic activities whose later trajectories diverge. A legal reform can alter rights, administrative practice, strategic behavior, and expectations simultaneously. Such branching makes it difficult to represent the generative effect as one linear chain.

Branching attribution begins by separating the opening of a pathway from the realization of a later outcome. A relation can make a trajectory feasible without determining which branch will later be taken. Subsequent decisions, shocks, complementary relations, and field conditions can select among the opened possibilities. Attributing the entire realized history to the initiating relation would therefore overstate its contribution.

Definition 5.101 (Generative branch). A *generative branch* is a target-relevant downstream pathway made newly feasible, more probable, less probable, or structurally altered by a prior relational process, while later conditions continue to influence whether and how that pathway is realized.

Definition 5.101 connects attribution to the possibility transformations introduced in Section 5.12. The generative contribution can lie in changing the topology of feasible development rather than producing one fixed end state. Evidence for branching can therefore include newly available choices, altered transition probabilities, new organizational forms, emergent partnerships, institutional pathways, or the disappearance of previously available routes.

Branching also changes counterfactual reasoning. The relevant comparison may

concern the set of accessible trajectories rather than a single terminal outcome. A relation can be generatively important because it creates several possible futures even when the observed history ultimately follows only one. Conversely, a realized branch can depend heavily on later conditions that deserve separate attribution.

A useful practice is to decompose attribution across branch formation, branch selection, and branch stabilization. The initiating relation can be credited with opening or constraining pathways; later relations can be credited with selecting among them; still later processes can stabilize the chosen trajectory. This decomposition reduces retrospective compression of a long history into one causal statement.

5.13.8 Emergent Effects

Emergent effects are especially challenging because the target property can arise from interaction among many components without being attributable in the same form to any single component. Market conventions, collective expectations, organizational cultures, congestion, systemic fragility, and coordinated patterns of innovation can exhibit this structure. Complexity research offers well-established examples in which localized interactions generate system-level patterns whose dynamics depend on feedback, adaptation, and organization across levels (S. A. Levin 1998; Arthur 1999). These precedents support attention to emergent effects while leaving the empirical burden of each GRE application intact.

Emergence should not function as a substitute for causal reconstruction. A claim that an effect “emerged” can remain merely descriptive unless the inquiry identifies the interactions, constraints, feedbacks, thresholds, or organizational properties through which the higher-order pattern was produced. The epistemic advantage of an emergence claim lies in showing why the target cannot be attributed adequately to one component considered independently and why configurational organization matters to the realized effect.

Definition 5.102 (Emergent generative effect). An *emergent generative effect* is a target-relevant change whose warranted explanation depends on organized interaction among multiple relations or components and whose explanatorily relevant properties are attributable to that interactional organization rather than to any single component considered independently.

Definition 5.102 remains compatible with ontological suspension. It is an epistemic and explanatory category. It does not require a metaphysical doctrine concerning irreducible levels of reality. The claim is narrower: for the specified target, boundary, and scale, the interactional organization carries explanatory content that is lost when components are analyzed independently.

Several kinds of evidence can support such attribution. Micro- or meso-level traces can show reciprocal adaptation or feedback. Comparative configurations can reveal that similar components produce different outcomes under different organizations. Simulation can test whether reconstructed interaction rules generate the qualitative pattern, while remaining a model-based source of evidence whose assumptions require scrutiny. Historical sequence can reveal how a distributed pattern stabilized. Field reconstruction can show positional dependencies that individual-level accounts omit. Convergence among these forms of evidence can strengthen the attribution

without converting any one method into a universal standard.

Proposition 5.103 (Minimum warrant for generative attribution). *A generative attribution is comparatively well warranted when the inquiry specifies the target effect and temporal horizon, identifies the candidate relation or configuration and its proposed contribution, examines plausible alternative contributors, provides evidence for relevant mechanisms or dependencies, states the boundary and scale of the claim, and preserves uncertainty arising from multiple causation, distributed organization, delay, branching, or emergence where these are materially relevant.*

Proposition 5.103 summarizes the section without imposing a rigid checklist on every empirical design. Different research settings provide different evidential opportunities. Experimental or quasi-experimental identification can be strong in some cases; historical process evidence, comparative reconstruction, network or field analysis, simulation, legal and administrative records, or qualitative inquiry can be more informative in others. The epistemic requirement is that the form of evidence fit the structure of the attribution being made.

The resulting conception is deliberately graded. Generative attribution can be weakly supported, strongly supported, locally identified, historically plausible, mechanism-rich but counterfactually uncertain, or counterfactually well identified while mechanistically incomplete. These differences should remain visible in GRE rather than being compressed into a single binary label. The next section extends this temporal discipline from attribution at a bounded interval to the reconstruction of longer trajectories and historical knowledge, where relations and their generative effects accumulate, transform, disappear, and leave residues across time.

5.14 Trajectories and Historical Knowledge

Sections 5.12 and 5.13 established what counts as a generative effect and how a contribution to such an effect may be attributed within a specified temporal horizon. The present section extends that epistemic problem across longer historical sequences. Its role is to explain how GRE can know relations and generative processes when their significance depends on order, duration, inherited conditions, interrupted traces, and later reinterpretation. The objective is to distinguish several historical objects that are often compressed into a single narrative of “development”: relational trajectories, generative trajectories, path-dependent sequences, genealogies of present configurations, historically situated evidence, constrained historical counterfactuals, and later re-epistemization. The method is reconstructive and sequence-sensitive. Historical claims should preserve temporal ordering, changes in meaning and category, evidence provenance, relevant alternatives, and the possibility that later knowledge revises how an earlier process is understood.

Historical knowledge matters to GRE because generative significance can be temporally displaced. A relation can appear inconsequential at one moment and become important after later complementarities arise. A constraint can disappear formally while continuing to shape capacities accumulated under it. A classification can be revised while leaving institutions built around the earlier classification intact. An intervention can alter a possibility structure whose consequences emerge only after

subsequent actors occupy it. The evidential object is therefore frequently a temporally extended configuration rather than an isolated state or event.

This historical orientation does not imply that every economic outcome is path dependent or that chronological narrative is sufficient explanation. Historical reconstruction becomes analytically valuable when order, duration, inherited conditions, or changing interpretive frameworks materially affect the claim. The task is to identify which parts of a sequence carry explanatory or epistemic significance and which are merely antecedent. GRE therefore treats historical depth as a source of possible warrant that must itself be justified.

5.14.1 Relational Trajectories

A relational trajectory records how a specified relation or relational configuration changes across time. The object may be a contractual relation, a collaboration, an authority relation, a household relation, a credit relation, a network position, an institutional linkage, or a wider relational configuration. What makes the object a trajectory is not simply the availability of observations at several dates. The observations must be connected by an explicit continuity judgment of the kind developed in Section 5.9, together with evidence about changes in participants, rules, intensity, meaning, position, or causal efficacy.

Definition 5.104 (Relational trajectory). A *relational trajectory* is a temporally ordered reconstruction of the formation, persistence, transformation, interruption, or dissolution of a specified relation or relational configuration under an explicit criterion of continuity and a stated observational horizon.

Definition 5.104 makes trajectory reconstruction dependent on relational identity rather than on labels alone. An organization can retain the same name while its authority structure changes substantially. Two actors can remain formally connected while the practical content of their relation shifts from cooperation to dependence. A legal category can persist while administrative procedures alter the relations it produces. Conversely, a relation can retain substantial continuity through changes in personnel, representation, or institutional form. Historical reconstruction should therefore identify the dimensions across which continuity is claimed and the dimensions across which transformation occurs.

A trajectory is also boundary- and scale-dependent. A firm-level supplier relation can appear stable while the wider field in which it is embedded changes sharply; a national policy relation can look discontinuous while local routines preserve practical continuity. The boundary and scale principles established in Section 5.8 therefore remain active throughout historical analysis. A trajectory should state the observational unit, the temporal granularity, and the relations treated as relevant context. Changing these choices can alter where formation, transformation, and dissolution appear to occur.

Temporal granularity deserves particular attention. Annual observations can hide short-lived interruptions, bargaining episodes, or rapid reconfigurations. Fine-grained event data can make enduring structure difficult to see. Historical knowledge therefore requires a resolution appropriate to the target claim. Where several resolutions are plausible, researchers should examine whether the inferred trajectory

remains stable across them. A continuity judgment that exists only after aggressive temporal aggregation should be stated more cautiously than one supported across several resolutions.

Relational trajectories can also contain periods of latency. A relation may cease to produce visible manifestations while its obligations, expectations, reputational effects, or stored capacities continue. Section 5.9 introduced residual efficacy after dissolution; historical reconstruction extends this idea by asking whether later states still carry traces of an earlier relation. The absence of continuous visible interaction therefore does not by itself establish historical discontinuity.

Proposition 5.105 (Trajectory reconstruction and continuity). *A relational trajectory is comparatively well supported when the inquiry states its identity criterion, temporal resolution, boundary and scale, documents relevant transformations and interruptions, and distinguishes continuity of the relation from continuity of its manifestations or representations.*

Proposition 5.105 prevents retrospective smoothing. Historical narratives often become cleaner than the processes they describe. Explicit continuity criteria preserve ruptures, ambiguities, and periods for which the evidence supports several reconstructions.

5.14.2 Generative Trajectories

A generative trajectory tracks a different object. It follows a sequence in which earlier relations or generative effects alter the conditions from which later relations, capacities, structures, constraints, or possibilities emerge. The trajectory can therefore include several different kinds of generative effect from Section 5.12. A training relation can generate capability; that capability can enable entry into a professional relation; the professional relation can create access to information and collaborators; the resulting position can later affect organizational authority. The continuity lies in a chain of historically connected contributions rather than in persistence of one relation.

Definition 5.106 (Generative trajectory). *A generative trajectory is a temporally ordered reconstruction in which earlier relations, configurations, or generative effects materially alter the conditions under which later target-relevant relations, capacities, structures, constraints, or possibility sets emerge or transform.*

This definition separates generative trajectory from simple chronology. Event A occurring before event B is insufficient. The reconstruction should identify how conditions created or modified earlier remain connected to later developments. The generative carriers introduced in Section 5.13 are especially important here. Skills, routines, infrastructures, legal statuses, debts, classifications, expectations, reputations, organizational memory, and material assets can transmit earlier effects across intervals in which the initiating relation is no longer active.

Generative trajectories can branch. An earlier relation may open several pathways, only one of which becomes realized. Later relations can redirect, amplify, suppress, or reverse earlier contributions. Historical reconstruction should therefore avoid reading the realized sequence as if it were uniquely implied by its beginning. The

branching concept introduced in Section 5.13 becomes historical when the inquiry reconstructs both the path taken and evidence about alternatives that remained feasible at relevant moments.

A further complication is sign change across time. A relation that initially generates capacity can later create lock-in or dependence. An infrastructure that expands access can later become a bottleneck. A regulatory privilege can support early investment and later inhibit entry. A historical generative claim should therefore state the horizon over which its evaluative or causal description applies. Long-run significance cannot be inferred by extending a short-run effect indefinitely.

Generative trajectories may also contain second-order transformations. Section 5.12 defined these as changes in the conditions through which later transformation occurs. Historically, such changes are especially consequential because they alter subsequent rates, channels, or admissible forms of generation. A new accounting system can change what organizations can observe and govern; a new legal form can alter the kinds of relations that can be stabilized; a new communication infrastructure can change the speed and reach of association. These transformations can make later history difficult to compare directly with earlier periods because the process generating change has itself changed.

The epistemic burden therefore grows with trajectory length. Each link introduces possible alternative explanations, missing traces, changing categories, and boundary shifts. A long generative trajectory should not be treated as one undifferentiated causal claim. Stronger reconstructions identify which links are directly supported, which are mechanism-rich but uncertain, which remain conjectural, and where later developments cease to be attributable to the earlier process with useful precision.

5.14.3 Path Dependence

Path dependence concerns sequences in which earlier developments alter the relative feasibility, cost, probability, or stability of later alternatives. Economic and political scholarship has used the concept to explain cases in which increasing returns, complementarities, learning, coordination effects, institutional feedback, or quasi-irreversibility make sequence and timing consequential. David's analysis of QWERTY helped establish an economic-historical vocabulary for such mechanisms, while subsequent debate also illustrates why a path-dependence claim should not be inferred from persistence alone (David 1985). Pierson extends increasing-return reasoning to political development and emphasizes timing, sequence, and self-reinforcing processes under specified mechanisms (Pierson 2000).

For GRE, path dependence is best treated as a mechanism-sensitive historical claim. Persistence is not enough. An institution can persist because current conditions continue to favor it independently of its history. A practice can recur because actors repeatedly choose it under unchanged incentives. A technology can dominate because it remains technically superior. Path dependence requires evidence that the route taken has itself modified later conditions in a way that makes alternative routes differently accessible than they would have been under another plausible sequence.

Definition 5.107 (Relational path dependence). *A relationally path-dependent process is a historical process in which earlier relational developments materially modify*

the conditions governing later relational possibilities, so that later outcomes depend partly on the sequence through which the system arrived at its current configuration.

This formulation makes path dependence compatible with several mechanisms. Repeated interaction can accumulate trust or distrust. Standards can create compatibility networks. Institutional rules can generate constituencies that defend them. Learning can make some capabilities cheaper to reproduce. Contracts can create obligations that narrow later choices. Reputational histories can alter access to future relations. The key epistemic task is to identify the carrier and feedback process through which history changes the later opportunity structure.

Path dependence should also be distinguished from historical contingency. A contingent event can redirect a trajectory without generating a self-reinforcing process. Conversely, a path-dependent sequence can begin from deliberate choice rather than chance. The concepts answer different questions: contingency concerns openness among possible events, whereas path dependence concerns how realized history changes subsequent possibilities.

The concept of lock-in requires further restraint. Some path-dependent processes narrow alternatives without making them practically irreversible. New technologies, coalitions, crises, legal changes, or shifts in complementary institutions can reopen pathways. GRE should therefore represent degrees of path dependence and the conditions under which reversal becomes feasible. Such a representation is more informative than treating history as deterministic once an early branch has been selected.

Proposition 5.108 (Path dependence and historical mechanism). *A path-dependence claim gains warrant when the reconstruction identifies a sequence-sensitive mechanism through which earlier relational developments alter later transition conditions and when persistence can be distinguished from contemporaneous causes that would support the same configuration independently of the earlier path.*

Proposition 5.108 makes historical dependence empirically contestable. It invites comparison with alternative sequences, evidence about feedback and switching costs, and attention to moments when the process could plausibly have moved differently.

5.14.4 Genealogical Reconstruction

Genealogical reconstruction asks how a present relation, category, institution, practice, or field became historically possible. Its object differs from a trajectory defined around a stable identity criterion. Genealogy may begin with a present configuration and work backward through heterogeneous predecessors, category changes, institutional transformations, and episodes that do not form one continuous object. The purpose is to reconstruct formation rather than to presume a continuous essence that has merely changed its appearance.

This distinction is particularly important for economic categories. Terms such as unemployment, productivity, household, enterprise, market, creditworthiness, or risk can appear to name stable objects while their operational definitions, institutional settings, measurement practices, and practical consequences change historically. Hacking's work on historical ontology provides a relevant precedent for examining how concepts and objects become possible through historically situated styles

of reasoning, language, and classification (Hacking 2002). GRE draws a narrower epistemic lesson: present categories should not be projected backward without examining whether the relations they organize were constituted and rendered visible in comparable ways.

Definition 5.109 (Genealogical reconstruction). A *genealogical reconstruction* is a historically situated account of the heterogeneous relations, practices, classifications, institutions, material conditions, and interpretive transformations through which a present target became possible, stabilized, or intelligible in its current form.

Genealogical reconstruction can reveal several kinds of discontinuity. The same word can designate different objects across periods. Different words can organize substantially related practices. An institution can inherit routines from predecessors while changing their justificatory language. A statistical category can create comparability where earlier records were organized around different administrative purposes. These discontinuities matter because historical evidence is interpreted through categories that may themselves be products of the history under investigation.

Genealogy also guards against teleology. Present arrangements should not be treated as the latent destination of earlier processes. Earlier actors faced possibility structures that included paths later abandoned, defeated, forgotten, or rendered illegible by subsequent classifications. Sewell's analysis of historical events emphasizes that consequential events can transform structures through linked changes in action, interpretation, schemas, and resources (Sewell 1996). Such transformations make later categories poor guides to what earlier actors could see or expect before the event occurred.

A GRE genealogy should therefore preserve superseded alternatives where evidence permits. Draft laws, failed standards, abandoned organizational forms, dissident classifications, unrealized plans, informal arrangements, and contested interpretations can be epistemically valuable because they reveal the field of possibilities within which the realized configuration emerged. Their historical importance does not depend on eventual success.

Genealogical reconstruction remains distinct from normative debunking. Showing that a category or institution has a history does not establish that it is false, illegitimate, or dispensable. Historical formation can be compatible with present usefulness or normative importance. The epistemic contribution is to make constitution, contingency, and inherited structure visible so that current claims are not mistaken for timeless descriptions.

5.14.5 Historical Evidence

Historical knowledge depends on traces that survive unevenly. Documents, administrative records, legal instruments, correspondence, artifacts, testimony, organizational data, statistics, media, physical infrastructure, and later recollections can each disclose parts of earlier relational configurations. Section 5.7 already established that observational access is partial and position-dependent; historical inquiry adds a preservation filter. What can be observed later depends on what earlier actors recorded, what institutions retained, what survived material loss, what became ac-

cessible, and what later researchers know how to interpret.

The survival of evidence is itself relationally structured. Powerful institutions often generate and preserve abundant records because administration requires documentation. Informal, stigmatized, precarious, clandestine, or rapidly changing relations may leave fewer durable traces. Strategic concealment can further distort the archive. Historical absence from records therefore requires the same caution that Section 5.4 applied to absence of traces more generally: non-observation becomes evidence of absence only when the inquiry has reason to expect that the relevant relation would have generated and preserved an accessible trace.

Historical evidence also changes meaning across contexts. A wage register can support claims about payment while leaving household dependence, bargaining, informal exchange, or care relations largely invisible. A contract can show formal obligation while underrepresenting coercion or practical non-enforcement. A statistical series can enable longitudinal comparison while embedding changing definitions. A retrospective interview can recover experience unavailable in administrative records while being shaped by later memory and interpretation. Evidence should therefore be evaluated relative to the target claim rather than through a fixed hierarchy of source types.

Provenance becomes especially important when historical traces are repeatedly copied, standardized, digitized, or summarized. A later dataset can contain values inherited from earlier administrative systems whose definitions changed across time. A secondary narrative can cite several accounts that ultimately derive from one original report. Digitization can improve access while severing contextual metadata. The provenance principle developed for distributed observation in Section 5.7 therefore extends naturally to historical reconstruction.

Proposition 5.110 (Historical evidential adequacy). *Historical evidence is comparatively adequate for a relational claim when its provenance, production context, preservation conditions, category semantics, temporal coverage, and expected relation to the target are sufficiently explicit for alternative reconstructions and evidential gaps to be assessed.*

Proposition 5.110 does not demand complete archives. Many important historical questions cannot satisfy such a standard. It instead requires that evidential limitations remain part of the claim. A reconstruction can be strong despite missing evidence when surviving traces converge, rival interpretations are tested, and the pattern of missingness is itself historically understood. Conversely, a large archive can still produce weak knowledge if its categories or selection mechanisms remain opaque.

The next section will classify evidential forms for relational claims in greater detail. The present concern is narrower: historical evidence acquires epistemic force through an account of how traces were produced, preserved, transformed, and made available across time.

5.14.6 Historical Counterfactuals

Historical counterfactuals ask how a trajectory might have differed under a plausible alteration to an earlier relation, event, institution, or sequence. Their epistemic role is closely related to the counterfactual reasoning developed in Section 5.13, but historical application adds a demanding constraint: the alternative should be specified from the possibility structure available at the relevant historical moment rather than from later knowledge alone.

A counterfactual can be useful for testing whether an earlier development materially constrained later outcomes. If an institutional rule had differed, would the later relation still have been feasible? If a standard had failed to coordinate adoption, would the same network effects have emerged? If a legal recognition had occurred later, would the relevant organizational form have stabilized? These questions can reveal which links in a generative trajectory carry attributional weight.

Historical counterfactuals become weak when they alter too many connected conditions without specifying the consequences of those changes. Replacing one institution can change coalitions, expectations, resource flows, legal interpretation, and later opportunities. The relational counterfactual specification from Section 5.13 remains relevant: the inquiry should state what is changed, which connected conditions are permitted to change with it, what is provisionally held stable, and the horizon over which consequences are considered.

The criterion of historical plausibility is equally important. A later technology, category, or institutional design may provide an imaginable alternative while having been unavailable to actors at the time. Such a comparison can still serve a theoretical purpose, but it should not be confused with a historically grounded counterfactual. Stronger historical counterfactuals draw on alternatives that were materially, institutionally, or conceptually available, including unrealized proposals, neighboring cases, contested reforms, known technologies, or paths that were attempted and abandoned.

Definition 5.111 (Historically situated counterfactual). A *historically situated counterfactual* is a counterfactual reconstruction whose altered condition and connected consequences are specified with reference to the relations, resources, institutions, knowledge, and feasible alternatives evidenced for the historical period under study.

Definition 5.111 does not require researchers to reproduce the subjective beliefs of every historical actor. The constraint is epistemic: alternatives should be disciplined by evidence about the period rather than by unrestricted retrospective imagination. Counterfactual uncertainty should increase as the imagined branch moves farther from historically evidenced possibilities or extends across longer downstream horizons.

Historical counterfactuals can also test path-dependence claims. If changing an early event while preserving plausible surrounding conditions yields little difference in the later reconstructed process, the alleged path dependence becomes weaker. If the altered sequence changes later transition conditions through a documented feedback mechanism, the claim gains support. Such reasoning remains conditional rather than demonstrative, especially where historical systems contain multiple interacting pathways.

5.14.7 Re-Epistemization through Historical Change

Historical change can alter not only the object of knowledge but also the conditions under which earlier evidence becomes intelligible. New records can be released; classifications can be revised; social practices can make previously invisible relations legible; later consequences can reveal the significance of earlier events; new methods can extract patterns from old data; and normative or conceptual developments can create questions that earlier inquiries did not ask. GRE uses the term *re-epistemization* for the disciplined reconstruction of an earlier epistemic object under such changed conditions.

Definition 5.112 (Historical re-epistemization). *Historical re-epistemization* is the revisable reconstruction of an earlier relational or generative claim when later evidence, categories, methods, consequences, or interpretive conditions materially change what can be warranted about the earlier process.

Re-epistemization differs from simple correction. Sometimes later evidence shows that an earlier claim was wrong. In other cases the earlier claim was reasonable under the evidence then available but becomes too narrow after new traces or categories appear. A third possibility is that the object itself changes: a relation later acquires causal efficacy, legal recognition, or social meaning that it did not possess previously. Historical knowledge must therefore distinguish revision of evidence, revision of interpretation, and transformation of the target relation.

Hermeneutic traditions provide an important precedent for this temporal openness. Ricoeur emphasizes interpretation across distance and the interaction of explanation and understanding in the interpretation of text and action (Ricoeur 1991). GRE adapts a limited epistemic lesson: later interpretation can legitimately reopen earlier material without implying that interpretation is unconstrained. Re-epistemization remains answerable to preserved traces, provenance, historical context, rival interpretations, and the consequences of the distinctions introduced.

The emergence of new categories also requires care. Hacking's historical analysis of categories and styles of reasoning shows why later conceptual resources can reveal historically constituted objects while simultaneously changing the terms through which they are understood (Hacking 2002). A later category can illuminate relations that earlier records rendered poorly visible, yet it can also distort if projected backward as though actors and institutions already organized reality in the same way. Re-epistemization therefore requires a two-sided discipline: exploit the analytical resources of the present while preserving the historical semantics and possibility structures of the past.

Later outcomes can similarly alter retrospective significance. An early technical standard can look minor until network effects accumulate around it. A local legal classification can become historically consequential after administrative systems are built on it. A failed experiment can later become informative when a new theory identifies a mechanism that the original researchers could not articulate. Such retrospective significance does not mean that the later outcome was already contained in the earlier event. It means that historical knowledge can change when the downstream relations through which significance is assessed become visible.

Proposition 5.113 (Historical preservation). *The future re-epistemization of a rela-*

tional or generative claim is strengthened when evidence, provenance, category definitions, alternative interpretations, model assumptions, uncertainty, and relevant negative or unresolved results are preserved in forms that allow later inquiry to reconstruct how the earlier claim was produced.

Proposition 5.113 connects historical epistemology to research design. Preservation is not merely archival housekeeping. It maintains future epistemic options. A dataset without category history, a model without assumptions, a summary without provenance, or a decision record without excluded alternatives can make later reinterpretation substantially harder even when the nominal evidence survives.

Historical re-epistemization also places a limit on claims of final closure. A well-supported historical account can be provisionally stable while remaining open to later evidence or reinterpretation. This revisability does not reduce every account to equal plausibility. Evidence can strongly favor one reconstruction, and some alternatives can be rejected. The point is narrower: the warrant of a historical claim remains indexed to the evidential and conceptual conditions under which it was assessed.

The seven subsections together establish history as an epistemic dimension of GRE rather than merely a background chronology. Relational trajectories track continuity and transformation; generative trajectories reconstruct historically connected contributions; path dependence identifies sequence-sensitive changes in later possibility; genealogy reconstructs the formation of present objects without assuming continuous essence; historical evidence is evaluated through production and preservation conditions; historical counterfactuals are constrained by period-specific possibility structures; and re-epistemization keeps historical knowledge open to disciplined reconstruction. The next section shifts from this temporal architecture to a systematic comparison of the evidential forms through which relational claims are supported.

5.15 Evidence for Relational Claims

Sections 5.1–5.14 have progressively specified the objects and epistemic conditions of relational inquiry: manifestation, trace, representation, observation, boundary, scale, persistence, higher-order configuration, fields, generative relations, causal-generative attribution, and historical trajectory. The present section turns from these conceptual conditions to the evidential materials through which relational claims are actually warranted. Its role is to distinguish major evidence forms without assigning any one of them universal priority. Its objective is to show what documentary, administrative, legal, behavioral, testimonial, ethnographic, network, longitudinal, and computationally inferred materials can establish about relations, together with the characteristic inferential limits attached to each. The structure follows these evidence forms in turn. The method is trace-sensitive and target-relative: each evidential source is evaluated by asking how it was produced, what relational property it can reveal, what conditions mediate its accessibility, and what forms of error or underdetermination remain.

The central premise is that evidence is not interchangeable simply because several sources refer to the same actors or event. A contract, a transaction record, an inter-

view, a fieldnote, a network edge, and a model-inferred tie can each concern the same apparent relation while recording different manifestations of it. A legal instrument may establish an authorized relation without showing how that relation is enacted in practice. Administrative data may record repeated interaction while omitting informal obligations. Testimony can disclose meanings or concealed dependencies that leave few institutional traces, while remaining situated and potentially contested. Computational inference can extend observation beyond manually coded relations while introducing model dependence. Evidential diversity therefore increases epistemic reach only when the relation between source, trace, and target claim remains explicit.

Definition 5.114 (Relational evidence). *Relational evidence* is an observation, record, trace, testimony, representation, or warranted inference that bears on a specified claim concerning the existence, form, properties, persistence, transformation, boundary, position, or configuration of one or more relations.

Definition 5.114 makes evidence claim-relative. A payroll record can provide strong evidence of a formal employment relation during a specified interval while offering limited evidence about authority in day-to-day work. A sequence of messages can support a claim of repeated communication while leaving trust, obligation, or influence unresolved. A participant's account can support knowledge of experienced dependence while requiring additional evidence before the same account is used to establish the prevalence of that dependence across a population. The strength of evidence therefore depends partly on the distance between the trace that is available and the relational property being claimed.

This section accordingly resists a simple hierarchy in which quantitative evidence is treated as intrinsically stronger than qualitative evidence, institutional records as intrinsically more objective than testimony, or direct observation as automatically superior to inference. Each evidence form is generated within particular institutions, practices, categories, instruments, and access arrangements. Administrative and statistical data are especially powerful because they can provide large-scale and repeated observations, but their categories and collection systems also shape what becomes visible (Desrosières 1998; World Bank 2021). Qualitative and ethnographic evidence can recover relational meaning and practice that standardized records omit, while their scope and observer position must remain explicit. The epistemic problem is therefore one of evidential fit, combination, and calibrated inference.

5.15.1 Documentary Evidence

Documentary evidence includes letters, contracts, minutes, reports, correspondence, memoranda, organizational records, policy documents, archival files, diaries, technical specifications, media records, and other durable inscriptions produced within or around a relational process. Such documents can establish that actors communicated, recognized obligations, created categories, recorded decisions, asserted authority, coordinated activity, or represented a relation in a particular way. Their durability also makes them especially important for historical reconstruction, where direct observation is unavailable and relational trajectories must be recovered from surviving traces.

A document should nevertheless be treated as a produced object rather than a transparent window onto the relation. Documents have authors, audiences, genres, institutional purposes, preservation histories, and conditions of access. A memorandum may record the position that an organization was prepared to state formally rather than the full structure of disagreement that preceded it. Meeting minutes can establish that a decision was recorded while compressing uncertainty, dissent, or informal negotiation. A contract can preserve explicit obligations while leaving tacit expectations and practical dependence outside the written text. Ricoeur's hermeneutics of text is useful here because textual inscription creates a degree of distance between an act of writing, its immediate context, and later interpretation (Ricoeur 1991). For GRE, that distance is an epistemic condition to reconstruct rather than a defect that automatically disqualifies documentary evidence.

Documentary evidence is especially strong when the target claim concerns representation, formal acknowledgment, declared intention, authorized procedure, recorded sequence, or institutional memory. It becomes more inferential when used to establish relations that the document was not designed to record. The evidential task should therefore specify whether the document is being used as evidence of the represented relation, evidence of the practice that produced the document, evidence of the institutional categories available at the time, or evidence of a later archival selection process. These are different claims.

Preservation also creates selection. Surviving documents are rarely a neutral sample of past relational life. Powerful organizations often preserve records more systematically than marginal or informal actors. Routine success can be recorded differently from conflict, failure, or activity outside institutional reporting requirements. Section 5.14 treated provenance and preservation as conditions of historical re-epistemization. Documentary inquiry therefore benefits from recording archive provenance, gaps, destruction, access restrictions, and changes in filing or classification practice where these materially affect the relational reconstruction.

Proposition 5.115 (Document–relation distinction). *A documentary source can warrant a relational claim only to the extent that the inquiry states how the document was produced, what aspect of the relation it records or represents, and what inferential steps connect the inscription to the target relational property.*

Proposition 5.115 prevents documentary existence from becoming automatic relational proof. A document can itself be relationally consequential, especially when it authorizes, classifies, obligates, or coordinates, while still remaining incomplete evidence about the wider relation in which it operates.

5.15.2 Administrative Evidence

Administrative evidence is produced through systems that register, classify, monitor, allocate, authorize, bill, tax, employ, enroll, license, inspect, or otherwise administer persons, organizations, resources, and transactions. Contemporary administrative systems can generate relational data at a scale and temporal resolution that would be difficult to obtain through direct observation. The expansion of administrative and private-sector data has materially increased the observability of many economic behaviors (Einav and J. Levin 2014), while public data systems increasingly combine ad-

ministrative, survey, citizen-generated, and machine-generated sources (World Bank 2021).

For relational inquiry, administrative records can reveal repeated transactions, institutional membership, transfers, claims, service use, organizational affiliation, co-location, authorization, sanction, or other patterned connections. Their major strength is often coverage and repetition. When identifiers are stable and collection procedures are known, administrative data can support large-scale reconstruction of relational neighborhoods, trajectories, and institutional positions.

Their major limitation is equally important: administrative systems observe through categories created for administrative purposes. A database records the relations that its schema permits and that its procedures require participants or officials to enter. Statistical and administrative classifications have histories of construction, coding, aggregation, and comparability rather than arising as neutral descriptions of social reality (Desrosières 1998). A household category, employment status, beneficiary identifier, ownership code, or provider relationship can therefore carry institutional assumptions into the evidence base.

Administrative evidence should consequently be evaluated along at least four dimensions: coverage, category validity, record-generation incentives, and linkage quality. Coverage asks who and what enters the system. Category validity asks whether administrative variables correspond sufficiently to the relational property under study. Record-generation incentives ask whether actors gain from accurate reporting, strategic reporting, or avoidance. Linkage quality asks whether records belonging to the same actor, organization, or relation can be connected reliably across sources and time. Failures in any of these dimensions can generate false absence, false continuity, or spurious connection.

Administrative silence is particularly ambiguous. A missing record can indicate that a relation did not exist, that it existed outside administrative jurisdiction, that it was exempt, informal, misclassified, unreported, technically lost, or deliberately concealed. The negative-evidence conditions developed in Section 5.4 therefore apply directly. Administrative nonobservation becomes stronger evidence of relational absence only when coverage, reporting, detection, and preservation conditions make the missing trace genuinely expected under the hypothesis that the relation existed.

5.15.3 Legal Evidence

Legal evidence is especially important when relational claims concern rights, duties, statuses, authority, jurisdiction, ownership, organizational personality, enforceability, or recognized procedures. Statutes, regulations, judicial decisions, contracts, charters, registries, licenses, and legally operative records can establish relations that have institutional force because relevant actors recognize and act through legal rules. Hart's analysis of legal systems through primary and secondary rules provides one established account of how recognized rules structure obligations and authoritative legal change (Hart 2012); Searle's account of institutional facts likewise emphasizes status functions, constitutive rules, and collectively recognized deontic powers (Searle 1995).

Legal evidence has a distinctive epistemic advantage: in some domains the legal

record does not merely describe an independently existing relation but helps constitute the formal relation itself. Incorporation documents and organizational law, for example, can create legally recognized entities and asset-partitioning relations that alter creditor and ownership claims (Hansmann and Kraakman 2000). A legal instrument can therefore be simultaneously evidence of a relation and part of the mechanism through which that relation is stabilized.

This constitutive role should not be extended beyond its evidential scope. Formal entitlement does not establish practical access. A statutory right does not demonstrate that administrative agencies implement it consistently. A contractual duty does not reveal the bargaining conditions under which the contract was accepted. Formal ownership does not by itself identify operational control. Legal evidence is therefore strongest for claims about legally recognized relations and weaker when transported without additional evidence into claims about enacted, experienced, or effective relations.

Legal interpretation introduces a further layer. Rules can contain open terms, conflicts, exceptions, jurisdictional boundaries, and historically changing precedents. The evidential task may therefore require identifying which legal source was authoritative for the time, place, actor, and issue under study. Where several interpretations were institutionally plausible, the reconstruction should preserve that contestability rather than reading a later settlement backward into an earlier period.

A useful distinction is thus between *formal relational validity* and *relational realization*. The first concerns whether a relation was legally recognized, authorized, prohibited, or constituted. The second concerns how that relation operated through practice, enforcement, compliance, evasion, negotiation, and institutional capacity. GRE often needs both because economic consequences can depend on the interaction between formal status and practical realization.

5.15.4 Behavioral Evidence

Behavioral evidence concerns actions and interaction patterns through which relations become manifest. Exchanges, transfers, communication, coordination, compliance, reciprocity, sanctions, referrals, mobility, repeated cooperation, avoidance, and other observable behaviors can provide evidence that actors are connected in ways that formal records do not fully capture. Behavioral approaches to collective action, for example, have examined how reciprocity, reputation, and trust are manifested through repeated interaction rather than inferred solely from abstract preference assumptions (Ostrom 1998).

The evidential strength of behavior lies in enacted relation. Repeated transfers can demonstrate a material connection even when participants never label it formally. Coordinated timing can reveal practical interdependence. Patterns of sanction can reveal authority or norm enforcement. Repeated referral can reveal brokerage or reliance. Behavioral traces can therefore help bridge the gap between formally represented relations and relations enacted in practice.

Behavior remains multiply interpretable. The same observed act can be compatible with several relational meanings. A transfer can represent payment, gift, repayment, obligation, coercion, care, strategic investment, or accounting adjust-

ment. Silence can indicate agreement, exclusion, indifference, deference, or lack of opportunity to respond. Frequent communication can reflect collaboration, monitoring, conflict, or procedural necessity. Behavioral evidence therefore warrants a relational claim most strongly when the mapping from action to relational property is supported by context, sequence, institutional knowledge, comparison, testimony, or other evidence.

Selection also matters. Actors choose whether to enter, continue, or leave relations, and their behavior within a relation may reflect capacities or constraints that preceded the observed interaction. Behavioral evidence can establish that an interaction occurred without by itself establishing why the actors were positioned to participate. This limitation becomes especially important when behavioral data are later used for generative attribution, which the next section addresses separately.

GRE should therefore distinguish *behavioral manifestation* from *relational interpretation*. The former identifies an observed action pattern. The latter identifies the relation for which that pattern is taken as evidence. The inferential bridge between them should remain open to revision when new contextual or comparative evidence becomes available.

5.15.5 Testimonial Evidence

Testimonial evidence includes interviews, oral histories, declarations, participant accounts, expert statements, complaints, witness reports, and other communicated accounts of relations or relational experience. It is indispensable when the relevant relation includes meanings, expectations, fear, trust, obligation, dependence, informal coordination, concealed practice, or other dimensions that may not produce adequate documentary or administrative traces. Participants can possess access to relational processes that external observers cannot recover from behavior alone.

Testimony should be evaluated neither as an unmediated report of reality nor as merely subjective material. A speaker occupies a position within the relation and therefore has both epistemic access and epistemic limits associated with that position. Participants can know tacit routines, informal expectations, coercive pressures, and historical context unavailable to an administrative database. They can also misremember, strategically represent, reinterpret retrospectively, or observe only one part of a distributed relation. Section 5.7 developed the idea of observer-position profiles; testimonial evidence is one setting in which such positionality is especially explicit.

Credibility itself is socially structured. Fricker's analysis of testimonial injustice shows how credibility can be unevenly allocated through identity and power relations, while hermeneutical disadvantage can affect the concepts available for making an experience intelligible (Fricker 2007). For GRE, this means that disagreement among testimonies cannot be resolved simply by assigning equal weight to all accounts or by privileging institutionally authoritative speakers. The inquiry should examine access, incentives, vulnerability, dependence, interpretive resources, corroboration, and the relational conditions under which testimony is produced and received.

Testimony can also be evidence about representation rather than only about the

represented relation. How actors name a relation, justify it, refuse a category, or contest an institutional description can reveal symbolic and classificatory structures that affect the relation itself. Divergent accounts may therefore constitute substantive evidence. A worker and a manager can describe the same arrangement differently because they occupy different positions, possess different information, or contest the normative meaning of the relation. The disagreement need not be reduced prematurely to a single corrected narrative.

Definition 5.116 (Testimonial position). A *testimonial position* is the combination of relational location, access, dependence, incentives, interpretive resources, and exposure to consequences that conditions what a speaker can observe, experience, articulate, or strategically disclose about a target relation.

Definition 5.116 turns positionality into part of the evidence record. Testimonial triangulation then involves comparison across positions, not merely accumulation of speakers.

5.15.6 Ethnographic Evidence

Ethnographic evidence is produced through sustained situated engagement with practices, interactions, artifacts, environments, and participants. Its distinctive contribution to relational inquiry lies in recovering relations in operation: how formal rules are enacted, how actors coordinate without explicit articulation, how meanings are negotiated, how material arrangements organize interaction, and how routines distribute knowledge across persons and artifacts. Hutchins's study of navigation provides a well-known example of analysis in which cognitive work is reconstructed across people, procedures, instruments, and organizational arrangements rather than located within one individual (Hutchins 1995). GRE can draw on this evidential sensibility without treating its own relational categories as identical to distributed cognition.

Ethnographic evidence is particularly valuable when relational properties are tacit, situational, performative, or difficult to separate from context. An organizational chart can record formal authority, while observation may reveal who is actually consulted before decisions. A platform rule can specify an official workflow, while field engagement may reveal workarounds, informal assistance, or avoidance practices. A legal entitlement can exist formally, while situated observation reveals the interactional conditions that make claiming it easy, costly, humiliating, risky, or practically impossible.

The observer is part of the evidential situation. Access can depend on gatekeepers, rapport, timing, language, institutional permission, and the willingness of participants to permit observation. Presence can also alter behavior. Ethnographic evidence therefore requires explicit attention to observer position, field access, temporal coverage, note production, and transitions between observation and interpretation. Reflexivity here is an epistemic practice: it records conditions under which the evidence became available and the ways those conditions may have shaped what was observed.

Local richness does not automatically establish wider prevalence. Ethnographic evidence can strongly support a mechanism, practice, meaning, or relational form

within the observed setting while leaving its distribution across other settings uncertain. Generalization therefore requires an explicit bridge, such as comparative cases, purposive variation, theoretical sampling, administrative prevalence evidence, survey evidence, or other forms of cross-context support. The same caution applies in reverse: population-level regularities can establish distribution while missing the situated processes through which relations acquire practical meaning.

Ethnographic inquiry thus contributes a form of *context-preserving relational evidence*. Its epistemic strength lies less in escaping mediation than in documenting enough of the relational setting that actions, categories, artifacts, and accounts can be interpreted together.

5.15.7 Network and Relational Data

Network and relational data represent actors together with explicitly recorded connections among them. These connections can be directed or undirected, weighted or binary, multiplex, signed, temporal, observed, reported, or inferred. Network analysis developed precisely to study patterned relations rather than reducing social structure to aggregated individual attributes (Wellman 1983), while contemporary network science provides extensive tools for describing connectivity, clustering, centrality, degree structure, community organization, and dynamical processes on networks (Newman 2003).

For GRE, the main advantage of network data is explicit relational representation. A dyadic data structure makes it possible to ask whether actors occupy brokerage positions, whether relations cluster, whether some ties are reciprocal, whether paths connect otherwise separated groups, or whether relational patterns differ across layers. Such data are especially useful for the higher-order and field problems introduced in Sections 5.10 and 5.11.

A recorded edge is nevertheless only as meaningful as its operational definition. Co-authorship, communication, trade, co-location, friendship nomination, legal ownership, financial transfer, shared membership, and algorithmically inferred association are different relations. Combining them without preserving type can erase distinctions that matter for explanation. Conversely, treating every recorded edge as equivalent can make formal network precision conceal substantive ambiguity.

Boundary selection is another major source of uncertainty. Network measures can change when omitted actors, external ties, or missing relation types are restored. Sampling is therefore not merely an issue of sample size. Missing nodes and edges can alter the apparent position of observed actors and the architecture of the network itself. The boundary principles of Section 5.8 should be recorded directly in network-data construction: who could enter the network, what relations were detectable, over what period, through which data-generating procedure, and with what missingness.

Definition 5.117 (Relational data provenance). *Relational data provenance* is the recoverable account of how nodes, relation types, edge values, time stamps, boundaries, missingness rules, and transformations were produced from underlying observations, records, reports, or inference procedures.

Provenance is especially important when relational data are repeatedly trans-

formed before analysis. Deduplication, entity resolution, thresholding, aggregation, projection from two-mode data, temporal binning, and privacy-preserving transformations can all change the represented relation. A reproducible relational analysis should therefore preserve these operations as part of the evidential record rather than treating the final adjacency structure as raw observation.

5.15.8 Longitudinal Evidence

Longitudinal evidence observes relational objects across more than one time point or records events with sufficient temporal information to reconstruct change. It is essential for claims about formation, persistence, transformation, dissolution, sequence, and trajectory. A cross-sectional network can show that two actors are connected at a given moment; longitudinal evidence can show whether the relation is newly formed, long-standing, intermittent, intensifying, weakening, or reconstituted after interruption.

Temporal network research demonstrates why timing and duration can alter the analysis of relational processes: the order of contacts can determine which paths are actually available and how processes propagate through a network (Holme and Saramäki 2012). GRE generalizes this temporal discipline beyond formal network models. The evidential question is whether observations preserve enough temporal resolution to support the identity and sequence claims being made.

Longitudinal evidence creates problems of comparability. Categories, measurement instruments, identifiers, organizational boundaries, and reporting rules can change across waves or periods. Apparent relational change can therefore arise from a changed observation regime. Conversely, harmonization can suppress historically meaningful category change by forcing later and earlier records into one coding scheme. Section 5.3 treated manifestation relations as historically variable; longitudinal inquiry must preserve this possibility rather than assuming that repeated variables have constant meaning.

Attrition and survivorship also matter. Relations that dissolve can disappear from later datasets, actors can leave panels, organizations can cease to report, and archives can become less complete farther into the past. Longitudinal evidence is strongest when entry, exit, censoring, missingness, and changing coverage are themselves documented. When these processes are related to the relational phenomenon under study, treating them as random missing data can bias the reconstructed trajectory.

A further distinction is needed between repeated observation and historical continuity. Multiple records bearing the same label do not establish that the same relation persisted. Continuity requires an identity criterion and evidence of the carriers through which the relation remained connected across time, as developed in Sections 5.9 and 5.14. Longitudinal evidence provides the temporal material for that judgment; it does not replace the judgment itself.

5.15.9 Computationally Inferred Evidence

Some economically important relations cannot be observed directly at the desired scale, speed, or granularity. They are reconstructed computationally from text, im-

ages, transactions, communication logs, sensor data, mobility traces, platform events, registries, or other high-dimensional observations. Computational methods can infer entity matches, latent affiliations, predicted ties, similarity relations, communities, roles, influence patterns, or probable interaction structures. Such inference can greatly expand epistemic access, especially where manual coding is infeasible.

Computational inference should be represented as an evidential layer rather than silently converted into observation. A predicted edge and an observed transaction are different epistemic objects even when both appear in the same graph. The distinction parallels Giere's pragmatic account of models as purposive representations of selected aspects of a target rather than identical copies of reality (Giere 2004). Computational simulation and modelling likewise mediate theory, data, approximation, and evidential judgment rather than eliminating those judgments (Winsberg 2010).

At least four components of computationally inferred evidence should remain recoverable. First, the input evidence should be documented, including its coverage and provenance. Second, the target relation should be operationally specified: the model must be trained or designed to infer something more precise than an unspecified notion of connection. Third, model assumptions and transformations should be visible, including thresholds, embeddings, priors, training labels, entity-resolution rules, or other procedures that materially affect the inferred relation. Fourth, validation should compare inferred relations with evidence that is sufficiently independent of the inference process where such evidence is available.

Error structure also matters more than aggregate predictive performance when the downstream object is relational. A small overall error rate can still distort centrality, community boundaries, brokerage positions, or the apparent isolation of a minority group if errors are concentrated systematically. Validation should therefore examine whether false positive and false negative relations vary by position, relation type, subgroup, time, or observation regime.

Computationally inferred relations can become recursive when institutions act on them. A fraud score, recommendation, risk classification, inferred social connection, or predicted affiliation can alter access, monitoring, visibility, or future interaction. Section 5.6 treated this as a performative-representation problem. The evidence file should therefore distinguish the period in which an inference was merely analytical from periods in which it entered the relational system and began to affect the very relations later used for validation.

Proposition 5.118 (Evidential fit for relational claims). *A relational claim is comparatively well warranted when the evidence form is appropriate to the relational property being claimed, the trace-production process and access conditions are documented, characteristic sources of omission or distortion are examined, and materially different evidence forms converge or their disagreements are preserved and explained.*

Proposition 5.118 summarizes the section without requiring every inquiry to collect every evidence type. Some relations are documented densely but difficult to observe behaviorally; others are enacted repeatedly while remaining legally informal; still others become visible mainly through testimony, ethnography, network traces, or computational reconstruction. The appropriate evidential profile depends on the target relation and the strongest inferential gaps that remain.

The resulting stance is one of evidential pluralism under explicit provenance. Documentary, administrative, legal, behavioral, testimonial, ethnographic, network, longitudinal, and computational materials are partially complementary because they expose different manifestations and dimensions of relational reality. Their combination can strengthen reconstruction when differences among them remain visible rather than being collapsed into a single undifferentiated dataset. The next section applies the same discipline to a stronger epistemic target: evidence that a relation, configuration, or field generated a specified effect rather than merely existed or possessed a particular form.

5.16 Evidence for Generative Claims

Section 5.15 classified major evidential forms for establishing that relations exist, possess particular properties, persist, change, or compose into wider configurations. The present section addresses a stronger epistemic target. Its role is to specify what additional evidence is required when GRE claims that a relation, configuration, field, or trajectory contributed to generating a specified effect. Its objective is to differentiate evidence for immediate effects, downstream effects, capacity formation, relation formation, structural transformation, value propagation, recirculation, and generative foreclosure. The structure follows these generative targets in increasing temporal and structural depth. The method is attribution-sensitive: for each target, the analysis asks what change must be evidenced, what temporal and relational linkage must be reconstructed, what alternative explanations remain plausible, and what uncertainty should accompany the resulting claim.

The distinction between relational and generative evidence follows directly from Sections 5.12 and 5.13. Evidence that a relation existed establishes a candidate generative condition. Evidence that an outcome followed establishes temporal succession. A generative claim adds a further requirement: the inquiry must warrant the proposition that the candidate relation or relational configuration contributed to producing, enabling, suppressing, redirecting, or transforming the target effect. Mechanism evidence, counterfactual reasoning, temporal reconstruction, comparison, and heterogeneous traces can contribute to this warrant in different combinations (Woodward 2003; Hedström and Ylikoski 2010; Gangl 2010). The relevant combination depends on the generative object rather than on a universal methodological hierarchy.

Definition 5.119 (Generative evidence). *Generative evidence* is evidence that bears on a specified claim that a relation, relational configuration, field, or historically connected process contributed to producing, enabling, constraining, redirecting, sustaining, or suppressing a target change under stated boundary, scale, and temporal conditions.

Definition 5.119 makes the target effect explicit. A record of collaboration can support a relational claim; evidence that the collaboration produced a new capability requires additional observations concerning capability before and after participation, the process through which learning or access occurred, and plausible competing sources of the change. A transfer can establish resource movement; a claim of recirculation requires evidence that some generated value returned to and altered

the conditions of further generation. A new institutional rule can establish formal change; a claim of structural transformation requires evidence that the rule reorganized recurring relations or possibilities at the relevant scale.

Generative evidence is therefore layered. At minimum, an inquiry normally requires evidence for the candidate relation or configuration, evidence for the target change, and evidence linking the two. Stronger claims can require additional layers: evidence of intermediate carriers, persistence across time, transformation at multiple scales, altered opportunity structures, or feedback into later generative conditions. These layers can be supplied by different source types. Administrative data may establish temporal patterns, testimony may reveal a mechanism, ethnography may identify changed practice, legal materials may establish altered authority, longitudinal network data may show relation formation, and computational models may test whether a specified mechanism is sufficient under explicit assumptions. Evidential pluralism becomes especially important as generative claims extend beyond immediate and directly observable effects.

5.16.1 Evidence of Immediate Effects

Immediate effects are changes that occur within a temporal interval sufficiently close to the candidate generative process for the inquiry to observe the initial response before extensive downstream mediation accumulates. Examples include a change in access following an eligibility decision, altered behavior following a new rule, a resource gain following a transfer, a coordination response following information exchange, or an initial skill improvement following instruction. The relevant meaning of *immediate* is empirical and process-relative. Minutes can be immediate in a market microstructure study, while months can be immediate in institutional or educational change.

Evidence of an immediate effect should normally establish four elements. First, the candidate generative condition must be identified with sufficient temporal resolution. Second, the target outcome must be measured or reconstructed before and after the relevant exposure or event, where the design permits such comparison. Third, the temporal order should place the candidate condition before the observed change. Fourth, plausible contemporaneous causes should be examined to the extent required by the strength of the attribution being made. Experimental, quasi-experimental, counterfactual, and mechanism-based strategies can each contribute to these elements under different research conditions (Woodward 2003; Gangl 2010; Hedström and Ylikoski 2010).

Temporal proximity is useful because it narrows the interval in which competing processes can intervene, yet proximity alone supplies limited causal information. A policy announcement can coincide with a behavioral response because both reflect a third process. A payment can precede increased activity when seasonal demand also rises at the same time. A newly recorded relation can appear shortly before an outcome because the relation itself became observable only after an administrative change. Evidence of immediate effects therefore benefits from combining temporal ordering with a specified mechanism, a credible comparison, intervention variation, or other evidence capable of discriminating among competing explanations.

Proposition 5.120 (Temporal proximity in generative evidence). *Temporal proximity strengthens evidence concerning sequence and the timing of a response. Attribution of generation additionally requires evidence connecting the candidate relation or process to the target change and examining materially plausible contemporaneous alternatives.*

Proposition 5.120 also clarifies why immediate-effect evidence can be strong while remaining narrow. An intervention may have an identifiable first-order effect while its later consequences remain uncertain. Conversely, weak or absent immediate change does not settle whether delayed or mediated effects will emerge. Immediate evidence should therefore specify its temporal horizon and avoid silently extending a short-run result into a claim about a longer generative trajectory.

5.16.2 Evidence of Downstream Effects

Downstream effects occur after one or more mediating stages separate the candidate generative condition from the target outcome. An educational relation can alter competence, which later changes occupational access; an infrastructure investment can alter connectivity, which changes firm location and then local employment; a classification can affect eligibility, which changes resource access and later household behavior. The epistemic burden increases with each mediating stage because additional processes can enter the trajectory.

Evidence for downstream generation therefore requires more than observing a distant association between an earlier relation and a later outcome. The inquiry should reconstruct at least part of the intervening pathway. This can involve records of intermediate states, repeated observations, process-tracing evidence, institutional sequences, participant accounts, network changes, or other traces showing how the effect was carried through time. Mechanism-based explanation is especially useful here because it asks which processes connect earlier and later states rather than treating temporal distance as an unexplained gap (Hedström and Ylikoski 2010).

Definition 5.121 (Downstream evidential chain). *A downstream evidential chain is a temporally ordered set of evidentially supported intermediate states, relations, carriers, or mechanisms connecting a candidate generative condition to a later target effect.*

The chain need not document every micro-event. Its adequacy depends on whether the omitted intervals contain plausible alternative pathways capable of changing the attribution. A long-term effect can be well supported when several independent traces establish the crucial transitions, while a densely recorded sequence can remain weak if its observations all derive from the same unvalidated administrative category or model. The quality of the chain therefore depends on both continuity and evidential independence.

Delayed effects also require attention to stored or latent carriers. Section 5.13 introduced the generative carrier for cases in which an effect is preserved through capabilities, records, infrastructure, expectations, obligations, or other intermediate conditions before becoming manifest. Evidence of downstream generation should identify such carriers where they are central to the explanation. A later outcome that appears after a long interval can then be linked to an earlier generative process through persistent relational or material conditions rather than through temporal

proximity alone.

A further difficulty concerns branching. The same initial condition can generate several pathways, and a downstream target may depend on only one branch. Evidence should therefore preserve branch-specific sequences where possible. Aggregating all later outcomes into a single average can conceal that one subgroup gained access through a generated relation while another encountered a new constraint. The evidential object is the historically realized generative pathway under the stated boundary and scale, together with credible alternatives relevant to the attribution.

5.16.3 Evidence of Capacity Formation

Capacity formation concerns changes in what an actor or collectivity can effectively do, sustain, access, coordinate, understand, create, or become within the relevant relational environment. The capability approach provides an established reason to distinguish resources from substantive opportunities and achieved functionings (Sen 1999). GRE extends this distinction toward an evidential question: what supports the claim that participation in a specified relation contributed to forming or transforming a capacity?

Evidence of capacity formation should identify a capacity with sufficient operational clarity to distinguish it from the inputs believed to support it. Training hours, funding, tools, credentials, or network membership can be enabling resources, while the target capacity may concern independent performance, coordination ability, access to institutions, problem-solving competence, bargaining ability, or the practical feasibility of undertaking a valued activity. Observing the input therefore provides evidence of exposure. Evidence of formation additionally requires traces of changed ability or changed effective opportunity.

Several evidential strategies can contribute. Repeated performance measures can show changed competence; independent task completion can show that a capability persists beyond the immediate instructional relation; new access to partners or institutions can reveal relational capacity; participant testimony can identify changes in confidence or practical feasibility; administrative or market records can show newly realized activities; and longitudinal observation can establish whether the capability persists, decays, or transfers across contexts. The strongest profile often combines demonstrated ability with evidence about the process through which the capacity was acquired.

Definition 5.122 (Capacity-formation evidence). *Capacity-formation evidence* supports a claim that a specified relation or generative process contributed to a change in effective ability or opportunity by documenting the relevant pre-existing condition, the changed capacity, and a plausible pathway connecting relational participation to that change.

This definition leaves room for capacities that are collective or infrastructural. A research consortium can acquire a capacity for coordinated experimentation that no member possesses independently. A community can develop governance capacity through repeated participation and record keeping. A platform can create technical capacity for exchange while simultaneously concentrating control over access. Capacity evidence should therefore specify the bearer of the capacity and the relational

conditions required for its exercise.

Persistence also matters. A short-lived performance improvement and a durable capacity are different claims. The temporal window should match the intended conclusion. Where capacity depends continuously on a supporting relation, the inquiry should state that dependence rather than representing the capacity as fully internalized by the actor. This is especially important for economic capacities that rely on infrastructure, legal recognition, care, credit, data access, language, reputation, or membership in an institutional field.

5.16.4 Evidence of Relation Formation

Relation formation becomes a generative target when an earlier relation, configuration, or field contributes to creating a later relation. Examples include collaboration that creates later partnerships, membership that produces new professional ties, platform participation that creates buyer-seller relations, institutional recognition that enables contracting, or conflict that reorganizes alliances. Section 5.12 defined relationally generated relations; the present question concerns the evidence required to support that attribution.

The evidential burden has two parts. The inquiry must first establish that the later relation formed or changed in the relevant interval. It must then connect that formation to the prior relation. Longitudinal network data can establish temporal ordering, but the observed sequence can arise through selection: actors who were already likely to form the later relation may also have entered the earlier one. Mechanism evidence can help by showing how participation created information, trust, reputation, compatibility, access, obligation, or shared infrastructure that made the later relation possible or more likely.

Interorganizational research provides an illustrative precedent. Powell, Koput, and Smith-Doerr show how collaboration can operate as a locus of learning and how participation in interorganizational networks can shape later organizational opportunities (Powell et al. 1996). Such findings illustrate the kinds of traces that can support generative relation-formation claims: dated collaboration records, subsequent ties, partner histories, learning processes, reputational changes, and comparison with actors lacking similar prior participation. The specific GRE claim still requires target-relative reconstruction rather than assuming that network participation is generative in every setting.

Evidence is especially strong when the later relation would have been inaccessible, substantially less feasible, or materially different without the earlier generative condition. Formal authorization can provide one such bridge: a credential, legal status, or institutional membership may create eligibility for later relations. Informal processes can also be decisive: repeated interaction can create trust, shared language, recognition, or knowledge of available partners. These mechanisms leave different traces and therefore require different evidence forms.

Relation formation can also be negative. An earlier process can generate exclusionary ties, dependency, or relations of monitoring and control. The evidential standard should remain symmetrical with respect to desirability. Establishing that a relation was generated does not establish that it was beneficial, just, voluntary, or stable. Nor-

mative evaluation enters through separate criteria.

5.16.5 Evidence of Structural Transformation

Structural transformation concerns changes in relatively persistent arrangements that organize multiple subsequent relations. Relevant objects can include institutional rules, recurrent role structures, market architectures, organizational routines, technical standards, infrastructures, category systems, ownership arrangements, or patterned distributions of authority and access. Institutions are an established economic example because rules and constraints can organize interaction over time (North 1990). For GRE, the evidential question is whether a generative process altered such an organizing structure at the scale claimed.

Evidence of structural transformation should therefore extend beyond isolated local outcomes. A new rule can be documented legally, yet structural transformation requires evidence that the rule altered recurring relations or the conditions under which those relations form. A technological standard can be formally adopted, while evidence of structural transformation concerns its effects on compatibility, entry, coordination, or dependence across the relevant system. A reorganization can change an organizational chart while leaving actual authority patterns largely intact. Structural evidence consequently benefits from combining formal change with repeated behavioral, relational, or institutional traces.

Definition 5.123 (Structural-transformation evidence). *Structural-transformation evidence* supports a claim that a generative process altered a persistent organizing condition of multiple relations by documenting the prior structure, the transformed structure, the mechanisms or sequence linking them, and recurring consequences at the boundary and scale specified by the inquiry.

Persistence is relevant because structural claims concern organizing conditions that continue to shape later interaction. The required duration varies by object. A market rule can transform trading immediately, while an educational classification may take years to alter institutional trajectories. Historical institutional research on increasing returns and path dependence shows how earlier arrangements can become progressively consequential when feedback alters the costs or feasibility of alternatives (Pierson 2000). Such work provides one evidential lineage for examining whether a change became structurally reproduced over time.

Structural transformation also requires boundary sensitivity. A change can be structural within one organization and local within an industry; structural within a national legal system and peripheral within a transnational production network. The claim should therefore state the population of relations organized by the transformed condition. Cross-scale claims require bridge evidence showing how the local transformation propagated into wider arrangements, following the scale discipline established in Section 5.8.

A useful evidential profile combines evidence of rule or architecture change, repeated relational consequences, persistence, and altered pathways for actors entering the structure later. This last component matters because a transformed structure often becomes most visible through how it conditions actors who did not participate in producing the original change.

5.16.6 Evidence of Value Propagation

Value propagation concerns the movement, transformation, recognition, or reproduction of a specified form of value across relations or sites. The term requires care because economic, cultural, social, symbolic, ecological, informational, and other forms of value need not share one metric. Throsby's distinction between economic and cultural value provides an established example of multidimensional valuation (Throsby 2001), while Appadurai's analysis of commodities emphasizes that value is organized through socially and politically structured trajectories of exchange (Appadurai 1986). GRE therefore treats value propagation as an evidential problem only after the relevant value form and its indicators have been specified.

Evidence of propagation requires more than repeated appearance of the same nominal quantity. A financial transfer can be traced through accounting records, but a claim that knowledge value propagated through a collaboration may require evidence that information was learned, reused, recombined, or made available to additional actors. Cultural value can persist while monetary price changes; reputational value can propagate through recognition networks without a direct transfer of assets. The evidential object is therefore the continuity or transformation of value-relevant effects across relational steps.

Definition 5.124 (Value-propagation evidence). *Value-propagation evidence* supports a claim that a specified form of value or value-relevant capacity moved, transformed, or became effective across a sequence of relations by identifying the value form, the relational carriers or conversion interfaces, and the traces through which continuity or transformation can be reconstructed.

Conversion creates a major epistemic challenge. If financial support funds education and later competence contributes to scientific collaboration, the sequence contains several heterogeneous objects. Treating the entire sequence as propagation of one conserved quantity would erase the transformations that make the process economically and socially significant. Evidence should instead record where the value form changes, which actors or institutions perform the conversion, what is lost or created in conversion, and which indicators remain comparable across stages.

Propagation also differs from attribution of final benefit. Value can pass through an actor without remaining there; it can be amplified, diluted, redirected, captured, or transformed. A platform can propagate information while capturing monetary value; a community can generate cultural value that later acquires commercial value elsewhere; a public infrastructure can create access whose benefits disperse across many actors. The evidential design should preserve these directional and distributive differences rather than compressing them into one aggregate outcome.

This subsection remains epistemic rather than a completed theory of value. Its purpose is to state what evidence would be needed when later GRE chapters make substantive claims about heterogeneous value propagation. Those claims will require value-specific operationalization and cannot be inferred solely from the existence of a relational chain.

5.16.7 Evidence of Recirculation

Recirculation concerns a more specific temporal structure than propagation. Something generated through a relational process returns, directly or through mediated pathways, to replenish, maintain, repair, expand, or transform the conditions from which further generation can occur. The generative-justice literature provides an important precedent through its emphasis on unalienated circular value flows and the return of value toward the people and communities involved in its generation (Eglash 2016; Eglash et al. 2024). GRE uses this lineage while keeping its broader categories of heterogeneous value, capacity, relation, and generative condition analytically distinct.

Evidence of recirculation should identify at least four components: the generative configuration or condition from which the relevant value or capacity emerged; the generated object or benefit being traced; the pathway through which some portion returns; and evidence that the return affects the later generative condition. Repeated transfer alone establishes circulation. Recirculation, in the GRE sense used here, requires a return relation whose effects can be connected to renewed or transformed generative capacity.

Definition 5.125 (Recirculation evidence). *Recirculation evidence* supports a claim that a generated resource, value form, capacity, or other consequential object returned through a specified relational pathway and contributed to maintaining, replenishing, repairing, expanding, or transforming the conditions of subsequent generation.

This definition allows recirculation to be partial and mediated. Revenue can return through taxation to public infrastructure; knowledge generated in research can return through teaching to a community of practice; cooperative surplus can finance shared equipment; reputational gains can attract new partners whose participation expands later capacity. In each case, the returning object may change form before it affects the generative condition. Evidence should preserve the conversion pathway rather than requiring literal return of an identical object.

The temporal horizon can be long. A generated surplus invested in training may affect productive capacity years later. Recirculation evidence therefore often combines accounting or transfer records with longitudinal evidence of changed capacity, infrastructure, membership, or relation formation. Where several cycles occur, the inquiry should distinguish a single return episode from a self-reinforcing recurrent process. A durable cycle claim requires evidence that the return repeatedly alters subsequent generation and that the cycle persists across more than one iteration.

Recirculation also has distributional structure. Value can return to one component of a generative system while leaving other contributors uncompensated or depleted. Evidence of recirculation therefore does not establish equitable return. The inquiry should identify which actors, infrastructures, ecosystems, or relations receive the return and which generative conditions remain unreplenished. This distinction will later matter for substantive analysis of extraction, capture, regeneration, and justice.

5.16.8 Evidence of Generative Foreclosure

Generative foreclosure occurs when a prior relation or generative process contributes to narrowing, disabling, making prohibitively costly, or otherwise suppressing future capacities, relations, trajectories, or transformation possibilities. The target is therefore an altered possibility structure rather than merely an unfavorable realized outcome. Evidence of foreclosure is demanding because unchosen possibilities leave weaker traces than realized events.

The first requirement is a historically credible account of the alternatives that were feasible before the relevant generative process. These alternatives should be grounded in contemporary resources, technologies, institutional rules, available partners, legal permissions, knowledge, or demonstrated practices. Section 5.14 established the same constraint for historical counterfactuals: later possibilities should not be projected backward into periods in which their enabling conditions were absent.

The second requirement is evidence of a mechanism through which the candidate process narrowed the option set. Increasing returns and lock-in provide one established family of mechanisms. Arthur shows how early contingent adoption under increasing returns can alter the later viability of competing technologies (Arthur 1989), while Pierson extends sequence-sensitive increasing-return reasoning to institutional development (Pierson 2000). GRE treats these as specific mechanisms rather than a complete theory of foreclosure. Other mechanisms can include legal exclusion, resource depletion, reputational penalties, loss of interoperability, concentrated ownership, dependency, destruction of infrastructure, accumulated switching costs, or suppression of alternative relation formation.

Definition 5.126 (Generative-foreclosure evidence). *Generative-foreclosure evidence* supports a claim that a specified relation or generative process contributed to reducing a historically feasible set of future capacities, relations, trajectories, or transformation pathways by documenting the prior possibilities, the mechanism of narrowing, and the subsequent reduction in effective feasibility at the stated boundary, scale, and temporal horizon.

The third requirement is evidence that the narrowing affected effective feasibility. A formally available alternative can become practically inaccessible because costs, skills, infrastructure, recognition, financing, or partner availability have changed. Sen's distinction between resources and substantive opportunities is again useful here because nominal permission can coexist with severely reduced practical capability (Sen 1999). Evidence of foreclosure can therefore include rising switching costs, disappeared suppliers, loss of skills, withdrawn legal recognition, infrastructure incompatibility, shrinking partner sets, increased dependency, or repeated failure of alternatives that were previously feasible.

Reversibility should also be assessed. Some foreclosure is temporary and can be reversed through modest intervention; other forms become increasingly costly to undo as complementary investments, expectations, and institutions accumulate. The strength of the foreclosure claim should reflect this degree of persistence. A reversible local constraint and a historically entrenched reduction in system-level possibility are different generative objects.

Foreclosure remains analytically distinct from normative condemnation. Safety rules can intentionally eliminate hazardous options, environmental restrictions can close destructive extraction pathways, and compatibility standards can reduce certain forms of variation while enabling wider coordination. Evidence can establish which possibilities were narrowed, through what mechanism, for whom, and with what reversibility. Normative assessment requires additional criteria concerning the values, rights, harms, benefits, and distributional consequences involved.

Proposition 5.127 (Minimum evidential burden for generative claims). *A generative claim is comparatively well warranted when the inquiry identifies the candidate generative relation or configuration, specifies and evidences the target change, reconstructs a plausible mechanism or dependency connecting candidate and target, examines materially relevant alternatives, and states the boundary, scale, temporal horizon, and uncertainty under which the attribution is made. Claims involving downstream mediation, structural transformation, recirculation, or foreclosure require additional evidence for the intermediate carriers, recurrent effects, return pathways, or altered possibility structures that define those generative objects.*

Proposition 5.127 summarizes the evidential difference between relation and generation. The burden does not demand one privileged research design. Immediate effects may sometimes be identified through intervention or quasi-experimental variation; downstream processes may require longitudinal reconstruction and mechanism evidence; capacity formation can require demonstrated ability and situated interpretation; relation formation can require temporal network and institutional evidence; structural transformation can require persistent cross-relational effects; value propagation and recirculation require traceable carriers and conversion pathways; foreclosure requires evidence about historically feasible alternatives and their later loss of effective accessibility.

The section therefore establishes a graded evidential architecture for GRE. Stronger claims require evidence that reaches farther into temporal sequence, mediation, structure, and possibility. Evidential pluralism remains compatible with demanding attribution because each evidence form is assigned a specific role in the generative reconstruction. The next section turns from the kinds of evidence that can support relational and generative claims to the uncertainty that remains after such evidence has been assembled, including uncertainty about relations, manifestations, generative effects, boundaries, scales, time, models, and competing explanations.

5.17 Uncertainty and Degrees of Epistemic Confidence

Sections 5.15 and 5.16 specified evidential forms for relational and generative claims. The present section addresses the epistemic remainder that persists after relevant evidence has been assembled. Its role is to organize uncertainty within GRE so that uncertainty is reported at the level where it arises rather than being compressed into a single undifferentiated judgment. Its objective is to distinguish relational, manifestational, generative, boundary, scale, temporal, and model uncertainty, and to specify how competing relational explanations affect degrees of epistemic confidence. The structure follows these sources of uncertainty from the target relation through the

observational and representational process to the explanatory model. The method is claim-relative and profile-based: confidence is assessed with respect to a specified proposition, boundary, scale, temporal horizon, evidence base, and inferential route.

Uncertainty enters relational inquiry through several routes. The relation itself can be incompletely identified; manifestations can underrepresent or distort it; generative contribution can remain only partially attributed; analytical boundaries can be contestable; scale changes can alter the pattern observed; temporal coverage can be insufficient; and alternative models can organize the same traces differently. Partial identification provides one useful precedent for disciplined restraint: Manski argues that researchers should distinguish what data identify from what follows only after stronger assumptions are imposed (Manski 2008). The broader philosophical problem of underdetermination similarly concerns cases in which available evidence remains compatible with more than one theoretical or explanatory account (Stanford 2023). GRE incorporates these concerns without reducing all uncertainty to statistical sampling error or to a single philosophical problem.

Definition 5.128 (Epistemic uncertainty in GRE). *Epistemic uncertainty in GRE* is the residual openness concerning a specified relational or generative claim that remains after available evidence, assumptions, boundary and scale choices, temporal coverage, representational decisions, and competing explanations have been stated.

Definition 5.128 makes uncertainty conditional on an articulated inquiry. The same body of evidence can warrant high confidence in one proposition and limited confidence in another. Repeated administrative records can strongly support the persistence of a recorded eligibility status while providing weaker support for the lived relation between an institution and a household. Longitudinal network data can establish repeated interaction while leaving the meaning, authority, or generative significance of those ties uncertain. A credible quasi-experimental design can support a local causal effect while leaving the long-run trajectory, cross-scale consequences, or mechanism comparatively open.

For this reason, GRE treats epistemic confidence as a profile rather than presuming that every claim can be summarized by one scalar quantity. Quantified probabilities, confidence intervals, posterior distributions, sensitivity ranges, or partial-identification regions can be appropriate when the inferential design supports them. Other uncertainties are better expressed through bounded alternatives, competing reconstructions, graded qualitative judgments, or explicit statements of unresolved assumptions. Numerical expression gains epistemic force only when the quantity being represented and the assumptions generating it are themselves adequately specified.

Definition 5.129 (Epistemic confidence profile). An *epistemic confidence profile* is a structured statement of the degree and location of warrant for a specified claim across the dimensions materially relevant to that claim, including where appropriate relational identification, manifestation, generative attribution, boundary, scale, temporal coverage, model dependence, and competing explanations.

The profile in Definition 5.129 prevents strength in one evidential dimension from silently compensating for weakness in another. A large dataset cannot by its size alone settle whether the recorded variable adequately manifests the target relation.

Rich qualitative material cannot by local depth alone establish prevalence across a wider population. Agreement among several models can strengthen confidence when the models rely on meaningfully different assumptions; agreement among near-identical specifications can provide less independent support. Confidence therefore concerns the architecture of warrant rather than the quantity of evidence in the abstract.

5.17.1 Relational Uncertainty

Relational uncertainty concerns what relation exists, between which relata, with what properties, and with what degree of persistence or efficacy. It begins at the level of the target itself. Two actors may exchange messages, share an administrative category, enter repeated transactions, or appear adjacent in network data while the substantive relation remains uncertain. The inquiry may know that some interaction occurred yet remain uncertain whether the relation is cooperative, hierarchical, contractual, fiduciary, coercive, symbolic, informational, or composed of several of these dimensions.

Relational uncertainty can arise from incomplete access, ambiguous traces, changing participant identities, heterogeneous interpretations, hidden intermediaries, or insufficient evidence about the persistence criteria discussed in Section 5.9. It can also arise when the relation is real but only partially stabilized. An emerging collaboration, contested legal status, unstable alliance, or newly forming market relation can possess causal efficacy before its properties become settled enough for a simple classification.

The epistemic task is therefore to disaggregate the relational claim. Confidence that interaction occurred can exceed confidence concerning its content. Confidence concerning legal obligation can exceed confidence concerning actual compliance. Confidence concerning repeated exchange can exceed confidence concerning trust. Confidence concerning organizational membership can exceed confidence concerning effective participation. Reporting these distinctions avoids treating the relation as epistemically all-or-nothing.

Remark 5.130 (Relational confidence and classificatory resolution). Increasing classificatory precision can lower warranted confidence when the available evidence supports a broad relational category more strongly than a fine-grained subtype. GRE therefore permits a well-supported coarse relational claim to coexist with uncertainty about its more specific characterization.

Remark 5.130 is important for historical and comparative work. A surviving archive can support the existence of a dependency relation while leaving its precise informal obligations unclear. Administrative data can establish repeated service contact while providing limited information about interpersonal recognition or coercion. Computationally inferred communities can reveal clusters of interaction while leaving the substantive relation among their members open. Epistemic discipline sometimes requires reducing classificatory resolution rather than increasing it.

Relational uncertainty should also remain distinct from normative ambiguity. An inquiry can be highly confident that a relation is asymmetric while remaining uncertain about whether the asymmetry is justified under the relevant normative criteria.

Conversely, normative concerns can be substantial even when the exact relational structure remains uncertain. Chapter 5 concerns the epistemic architecture of the claim; later normative analysis can build upon that architecture without substituting moral evaluation for evidential warrant.

5.17.2 Manifestational Uncertainty

Manifestational uncertainty concerns the relation between relational reality and the manifestations through which it becomes accessible to inquiry. Section 5.2 established that manifestation is generally non-invertible, while Sections 5.3–5.5 distinguished manifestation relations, traces, symbolization, and representation. Even when the observed trace is measured accurately, uncertainty can remain about what relational state produced it.

A decline in recorded transactions can reflect weakening exchange, migration to an unrecorded channel, strategic concealment, administrative reclassification, or a change in the recording system. A stable legal category can coexist with changing practice. A survey response can reflect experience, strategic presentation, interpretive framing, or the available answer categories. A model-generated score can aggregate traces that correspond only indirectly to the relational target. The uncertainty therefore lies partly in the manifestation mapping rather than in measurement noise alone.

Manifestational uncertainty can be reduced when the inquiry possesses evidence about the production of the trace. Knowledge of administrative procedures clarifies how categories enter a database. Ethnographic observation can reveal how survey terms are interpreted. Technical documentation can show how platform events are logged. Legal analysis can distinguish formal status from implementation. Repeated measurement across different manifestation channels can reveal whether a relational inference is robust to the particular trace used.

Definition 5.131 (Manifestational uncertainty). *Manifestational uncertainty* is uncertainty concerning how observed manifestations, traces, or representations correspond to the relational state or process targeted by an inquiry, including uncertainty about omission, distortion, selective visibility, translation, and changes in the manifestation process itself.

Definition 5.131 implies that data quality and relational correspondence are related but distinct questions. A dataset can be internally consistent and accurately recorded while providing a limited manifestation of the target relation. Conversely, a noisy set of heterogeneous traces can sometimes support a relational conclusion when their independent production pathways converge on the same underlying structure. The relevant question is how the evidence bears on the target through the specific manifestation relation.

Manifestational uncertainty also changes historically. When reporting rules, platform architectures, legal definitions, or cultural conventions change, the same observed category can acquire a different relation to the underlying process. Longitudinal research should therefore assess continuity in the manifestation relation itself. Apparent change in the target relation can otherwise be produced by a changing observational interface.

5.17.3 Generative Uncertainty

Generative uncertainty concerns whether, how, and to what degree a relation or configuration contributed to producing a target change. Sections 5.12, 5.13, and 5.16 established that generative claims are stronger than claims of relational existence. Even when a relation and an outcome are both well evidenced, uncertainty can remain concerning the contribution that connects them.

Several sources are common. The mechanism can be only partially observed. Multiple sufficient or enabling causes can coexist. A candidate relation can matter only in conjunction with complementary conditions. Counterfactual dependence can vary across units or contexts. Effects can be delayed, distributed, or mediated by several stages. A process can redirect the form of change without determining its occurrence. The same relation can generate different consequences under different surrounding configurations.

Generative confidence should therefore specify the object of attribution. Confidence that a program increased access can exceed confidence that the resulting access produced durable capability. Confidence that a classification altered eligibility can exceed confidence concerning its longer-term effect on household trajectories. Confidence that a network position supplied information can exceed confidence that information was decisive for a later innovation. Local causal identification does not automatically settle wider generative significance.

Interventionist and counterfactual approaches provide tools for distinguishing causal contribution under explicit assumptions (Woodward 2003; Gangl 2010), while mechanism-based inquiry can reconstruct the processes connecting earlier and later states (Hedström and Ylikoski 2010). GRE uses these resources selectively. Confidence can increase when distinct forms of evidence support the same generative attribution, especially when they constrain different plausible alternatives.

Proposition 5.132 (Layered confidence in generative claims). *Confidence in a generative claim depends jointly on warrant for the candidate relation or configuration, warrant for the target change, and warrant for the linkage between them. High confidence in either the relation or the outcome cannot by itself compensate for weak evidence concerning the generative linkage.*

Proposition 5.132 also applies to stronger objects such as recirculation and foreclosure. A recirculation claim inherits uncertainty about the initial generation, the conversion or transfer pathway, the return path, and the effect of the returned value on later generative conditions. A foreclosure claim inherits uncertainty about prior feasible alternatives, the mechanism of narrowing, and the persistence of reduced effective accessibility. Generative uncertainty often accumulates across a chain, although independent evidence at later stages can sometimes narrow parts of that uncertainty.

5.17.4 Boundary Uncertainty

Boundary uncertainty concerns which actors, relations, institutions, environments, and processes should be included in the analytical object. Section 5.8 showed that analytical boundaries can be epistemic constructions while some boundaries are simul-

taneously relationally real and causally efficacious. Uncertainty arises when several plausible boundary specifications produce materially different interpretations of the same process.

Consider a firm-level productivity analysis. A narrow boundary can attribute performance to internal organization. A wider boundary can reveal dependence on suppliers, public infrastructure, family care, ecological inputs, regulatory systems, or knowledge produced elsewhere. Neither boundary becomes adequate simply by being wider. Expanding the boundary can introduce processes irrelevant to the target and reduce analytical tractability. The relevant question is which boundary captures the relations materially necessary for the claim being evaluated.

Boundary uncertainty can be explored through sensitivity analysis. The inquiry can reconstruct the claim under nested or alternative boundary specifications and identify which conclusions remain stable. If an attribution survives reasonable boundary changes, confidence in its portability increases. If the conclusion changes sharply when one neighboring relation is included, the analysis should report that dependency rather than hide it inside a preferred specification.

Definition 5.133 (Boundary-sensitive confidence). *Boundary-sensitive confidence* is confidence in a relational or generative claim stated together with the range of plausible analytical boundaries across which the claim remains adequately warranted and the boundary changes under which its interpretation materially changes.

Boundary uncertainty can also be asymmetric. Excluded relations may matter more for some actors than for others. A household boundary that omits informal care can distort the position of caregivers more strongly than that of other members. A national boundary can obscure transnational production or financial relations for sectors whose operation depends heavily on them. Boundary sensitivity should therefore be evaluated with attention to heterogeneous positional effects rather than only to aggregate model fit.

The objective is conditional closure rather than exhaustive inclusion. An inquiry can legitimately delimit an object when the boundary is explicit, relevant external relations are examined, and the claim remains scoped to the resulting domain. Epistemic confidence weakens when a bounded result is silently generalized beyond the relations that made it possible.

5.17.5 Scale Uncertainty

Scale uncertainty concerns the level or resolution at which a relational pattern should be described and the extent to which a conclusion survives movement across scales. Section 5.8 established that relational claims can be scale-indexed. A regularity visible at an aggregate level can coexist with heterogeneous or contrary patterns at local levels, while a locally important relation can disappear when observations are aggregated.

Scale uncertainty therefore differs from boundary uncertainty. Boundary questions concern which elements are included; scale questions concern the level of organization or resolution at which their relations are represented. The two interact, but they can vary independently. The same bounded population can be studied at indi-

vidual, household, organizational, regional, or system level, and each representation can reveal different dependencies.

Confidence across scales requires bridge evidence. If an individual-level relation is used to explain an organizational pattern, the inquiry should specify how individual interactions aggregate, coordinate, or become institutionalized. If an aggregate relation is used to infer individual experience, the inquiry should establish the distributional or positional mechanism that carries the aggregate condition downward. A cross-scale inference gains strength when these bridge processes are directly evidenced rather than assumed.

Proposition 5.134 (Scale-indexed epistemic confidence). *Epistemic confidence attaches first to a claim at the scale for which the evidence and inferential design provide warrant. Extension to another scale requires additional evidence concerning the relations that connect the two scales and the transformations introduced by aggregation or disaggregation.*

Proposition 5.134 protects against two common errors. The first is ecological extension, in which an aggregate pattern is attributed to individual relations without adequate bridge evidence. The second is atomistic extension, in which locally observed mechanisms are presumed to determine system-level outcomes despite coordination effects, institutional mediation, feedback, or compositional change. GRE's multiscale ambition therefore increases rather than reduces the need for scale-specific uncertainty statements.

Scale uncertainty can also be substantive. Some relations become causally efficacious only after sufficient aggregation or institutional stabilization; others are weakened by aggregation because their significance depends on local position or meaning. The appropriate epistemic response is to state which scale the evidence supports and which cross-scale claims remain provisional.

5.17.6 Temporal Uncertainty

Temporal uncertainty concerns when a relation begins, how long it persists, when an effect occurs, which historical interval is relevant, and whether an observed pattern can be transported across time. Sections 5.9 and 5.14 established that relational identity and historical trajectory require explicit continuity criteria. An inquiry can possess strong evidence about a relation at several time points while remaining uncertain about what occurred between them or whether the same generative mechanism persisted throughout the interval.

Temporal uncertainty is especially important when effects operate on different timescales. A policy can have a rapid administrative effect, a slower behavioral effect, and a much later institutional effect. A relational intervention can initially expand access while gradually creating dependency. An innovation can produce immediate efficiency gains while altering capability formation over a longer horizon. Confidence concerning one temporal window should therefore remain indexed to that window.

Historical evidence introduces additional asymmetry. Later periods can generate records that clarify earlier processes, yet later categories can also distort reconstruction when projected backward. Section 5.14 treated historical re-epistemization as

disciplined revision under later evidence. Temporal confidence should preserve both possibilities: later knowledge can strengthen an earlier reconstruction, and changing categories can make direct comparison more difficult.

Definition 5.135 (Temporal confidence horizon). A *temporal confidence horizon* is the interval over which the available evidence and continuity assumptions provide adequate warrant for treating a relational or generative claim as applicable, together with the conditions that limit extension beyond that interval.

The temporal confidence horizon in Definition 5.135 can be short even when the underlying phenomenon is long-lived. Data availability, institutional change, altered measurement rules, or unobserved transitions can restrict the period for which a claim is warranted. Conversely, sparse records can sometimes support a longer continuity claim when independent historical evidence establishes stable mechanisms and persistent relational carriers.

Sequence also matters. Path-dependent processes can make the same intervention produce different outcomes depending on when it enters the trajectory. Pierson's analysis of increasing returns highlights how sequence and timing can shape institutional development (Pierson 2000). Temporal uncertainty therefore concerns more than duration; it also concerns historical ordering and the timing of relational entry into an evolving configuration.

5.17.7 Model Uncertainty

Model uncertainty concerns the dependence of an inference on the representational, statistical, computational, formal, or conceptual model through which evidence is organized. Every model selects variables, relations, functional forms, priors, categories, mechanisms, spatial or temporal resolutions, and criteria of relevance. These selections can materially affect the relational structure that appears in the analysis.

Model uncertainty includes specification uncertainty, parameter uncertainty, structural uncertainty, and uncertainty concerning the relation between the model and the target system. These forms should be distinguished. A model can have tightly estimated parameters while relying on a contested structural specification. Several models can fit observed data similarly while implying different relational mechanisms. A computational simulation can demonstrate that a mechanism is sufficient to produce a pattern under its assumptions while leaving open whether that mechanism generated the empirical case.

Partial-identification reasoning is especially useful when several assumptions remain plausible. Rather than selecting one strong assumption set merely to obtain a point estimate, an inquiry can report the range of conclusions consistent with weaker assumptions (Manski 2008). GRE generalizes the discipline behind this move: where the model does more inferential work than the evidence alone can support, the added assumptions should remain visible in the confidence statement.

Definition 5.136 (Model-dependent warrant). *Model-dependent warrant* is epistemic support for a claim whose strength or content changes materially with the model class, specification, representational assumptions, or inferential constraints used to connect evidence to the target.

Model dependence does not make a claim unusable. It identifies where confidence resides. An inference can be highly robust within a well-justified model class and less secure across alternative classes. Sensitivity analysis, specification comparison, out-of-sample validation, qualitative cross-checking, mechanism evidence, and comparison with independently produced data can each reduce particular forms of model uncertainty. Their value depends on whether they probe genuinely different assumptions.

GRE should also distinguish model uncertainty from ontological uncertainty. Several models can remain empirically compatible because the evidence underdetermines the relational mechanism. This does not require a conclusion that the underlying relational reality itself is indeterminate. The epistemic claim is narrower: the current evidence and models do not yet warrant a unique reconstruction. Ontological suspension therefore remains compatible with strong comparative judgments among models when the evidence supports them.

5.17.8 Competing Relational Explanations

Competing relational explanations arise when two or more reconstructions remain materially compatible with the available manifestations and evidence while differing in the relations, mechanisms, boundaries, scales, or trajectories they posit. This problem appears throughout Chapter 5. The inverse problem of manifestation in Section 5.2 permits several relational realities to generate similar manifestations. Higher-order reconstruction in Section 5.10 can organize the same first-order ties into different structures. Field attribution in Section 5.11 can compete with aggregate or common-cause explanations. Generative attribution in Section 5.13 can remain open among several causal pathways.

The existence of competing explanations should therefore be represented as structured uncertainty rather than treated automatically as epistemic failure. Rival explanations can differ in their empirical commitments. New evidence can discriminate among them by targeting manifestations on which they diverge. A field explanation and a pairwise-network explanation may make different predictions about positional effects. A direct-causation account and a mediated account can imply different temporal sequences. A legal-institutional explanation and an informal-relational explanation can imply different responses to formal rule changes.

Definition 5.137 (Explanatory competition set). An *explanatory competition set* is a stated collection of materially plausible relational or generative explanations that remain compatible with the current evidence at the relevant boundary, scale, and temporal horizon, together with the evidence that favors, weakens, or could discriminate among them.

The competition set in Definition 5.137 is narrower than an unrestricted list of imaginable alternatives. Plausible competitors should possess some evidential or theoretical basis and should address the same target claim. This discipline prevents manufactured uncertainty in which arbitrary possibilities are invoked merely because they cannot be logically excluded. Underdetermination matters when alternatives remain live under the actual evidence and reasonable background assumptions (Stanford 2023).

Competing explanations can also be partially complementary. An institutional rule can alter incentives while a social network shapes information diffusion and a cultural classification affects recognition. Inquiry should test whether the explanations truly compete for the same generative role or describe different parts of a composite process. Premature selection of a single explanation can be as misleading as premature synthesis. The appropriate reconstruction can retain several contributions while differentiating their domains and evidential status.

Proposition 5.138 (Profiled epistemic confidence). *A relational or generative claim is epistemically stronger when confidence is stated at the level of the specific proposition and accompanied by the material uncertainties concerning relational identification, manifestation, generative attribution, boundary, scale, temporal horizon, model dependence, and competing explanations. Strength in one dimension should not silently erase unresolved uncertainty in another.*

Proposition 5.138 provides the organizing principle for degrees of epistemic confidence in GRE. Some inquiries can legitimately summarize particular uncertainties numerically. Others require intervals, sets of admissible explanations, sensitivity statements, or disciplined qualitative judgments. The common requirement is traceability: a reader should be able to identify which evidence supports the claim, which assumptions extend beyond that evidence, which dimensions remain open, and what additional observations could materially change the conclusion.

This profile-based approach also supports revisability. Confidence can increase when new evidence narrows the relation, manifestation mapping, boundary, scale, temporal sequence, or model class. It can decrease when new traces reveal omitted actors, changing classification rules, previously hidden pathways, or stronger competing explanations. Revision therefore becomes an expected epistemic operation rather than an exceptional admission of failure.

The next section turns from uncertainty within relational inquiry to the distribution of epistemic visibility, authority, and evidential capacity among differently positioned actors. Once uncertainty is recognized as structured and position-sensitive, questions arise concerning whose relations become visible, whose testimony receives uptake, whose classifications organize institutional reality, and whose capacities to produce evidence remain constrained. These questions provide the transition to epistemic justice in relational inquiry.

5.18 Epistemic Justice in Relational Inquiry

Section 5.17 treated uncertainty as structured across relational identification, manifestation, generative attribution, boundary, scale, time, and model choice. The present section adds a normative-epistemic dimension that becomes visible once those uncertainties are situated within unequal relations of observation, classification, testimony, documentation, and evidential production. Its role is to identify conditions under which the organization of inquiry can advantage some actors' capacities to make relations visible, intelligible, and evidentially consequential while constraining others. Its objective is to connect visibility, voice, classification, exclusion, institutional legibility, evidential capacity, representation, and generative

attribution without converting epistemic justice into a substitute for evidential warrant. The structure follows these sites of asymmetry in turn. The method is relational and process-sensitive: it asks how epistemic positions are produced, how they enter the evidential chain, and how inquiry can preserve contestability, provenance, and differentiated authority.

Fricker's account of testimonial and hermeneutical injustice provides an important philosophical precedent for analyzing credibility deficits and unequal participation in shared interpretive resources (Fricker 2007). GRE extends the problem into a wider architecture of relational inquiry. A participant can be disadvantaged before testimony is evaluated because the relevant relation leaves few institutionally recognized traces, because an administrative classification has already fixed the available categories, because access to data or instruments is concentrated, or because the actor lacks the resources required to convert situated experience into a form that travels within research and policy institutions. Longino's account of socially organized criticism is also relevant because epistemic robustness depends partly on whether criticism can enter a community and receive serious uptake (Longino 1990). These precedents motivate a concern with participation and contestability while leaving the specific GRE architecture open to domain-sensitive development.

Definition 5.139 (Epistemic justice in relational inquiry). *Epistemic justice in relational inquiry* is the just organization of opportunities and capacities to become epistemically visible, contribute relevant testimony and interpretation, contest classifications and representations, produce or access evidence, and receive appropriately calibrated epistemic uptake concerning relations and generative processes in which actors are differently positioned.

Definition 5.139 separates justice from automatic deference. Situated actors can possess privileged access to particular manifestations, meanings, histories, or consequences while remaining uncertain about mechanisms outside their observational position. Institutional researchers can possess broad comparative data while lacking access to informal obligations or concealed dependencies. Technical systems can reveal regularities at scales inaccessible to unaided observation while inheriting the categories, missingness, and objectives embedded in their data-generating process. Epistemic justice therefore concerns the organization of access, authority, contestation, and evidential capacity together with disciplined calibration of what each source can support.

The section also remains narrower than a complete theory of social, political, or economic justice. Later GRE chapters can evaluate distribution, domination, exploitation, recognition, participation, and generative return more fully. Here the normative focus concerns the conditions under which relational knowledge is produced and attributed. A procedurally inclusive inquiry can still reach an empirically weak conclusion, and a well-identified empirical regularity can still emerge from an epistemically unjust process. GRE therefore keeps epistemic justice and evidential adequacy analytically distinct while examining the ways each can affect the other.

5.18.1 Visibility and Invisibility

Visibility is an epistemic condition through which a relation, actor, effect, or trace becomes available for observation, recording, comparison, or interpretation. Section 5.7 established that observational access is distributed unevenly across positions. Epistemic justice adds the question of how that distribution is organized and whose relations repeatedly enter the shared evidential field.

Institutional systems often privilege relations that generate standardized traces. Formal employment, registered ownership, taxable transactions, licensed activities, documented service use, and digitally mediated interaction can become comparatively visible because the relevant institutions already produce records. Informal care, tacit coordination, uncompensated maintenance, undocumented exchange, reputational dependence, ecological contribution, and many forms of exclusion can remain less visible because their traces are dispersed, privately held, weakly standardized, or absent from the categories through which administrative systems collect information. The resulting asymmetry can shape which relations appear empirically substantial and which appear marginal.

Visibility nevertheless requires a qualified normative treatment. Greater visibility can improve recognition, accountability, and evidential access, while intensified observation can also produce surveillance, exposure, strategic vulnerability, or new forms of classification. Strategic concealment in Section 5.7 already showed that opacity can be relationally produced. In some settings, limited visibility protects actors whose safety, autonomy, or bargaining position would be weakened by compulsory disclosure. Epistemic justice therefore concerns appropriate visibility relative to purpose, risk, consent, and relational position rather than maximal observability.

Claim 5.140 (Justice-sensitive visibility). *A justice-sensitive account of epistemic visibility evaluates both access to relevant traces and the consequences of making those traces visible, including which actors gain interpretive or operational power from disclosure and which actors bear the resulting risks.*

Claim 5.140 matters for research design because the same data practice can increase knowledge while redistributing relational power. A survey that reveals hidden labor can support recognition and policy analysis; publication of highly granular records can expose vulnerable participants. A transparency requirement can make institutional decision rules contestable; disclosure of individual-level attributes can create new opportunities for profiling. The justice of visibility therefore belongs to the design of observation itself rather than appearing only after data have been collected.

In practical terms, GRE inquiry should record what is visible, through which infrastructure, to whom, and with what consequences. It should also identify relevant domains of invisibility and distinguish among absence of a relation, absence of a trace, lack of institutional recognition, and deliberate protection from observation. Such distinctions reduce the risk that institutional visibility becomes an implicit measure of relational importance.

5.18.2 Voice and Representational Authority

Voice concerns the capacity to contribute descriptions, interpretations, reasons, experiences, and contestations to the epistemic process. Representational authority concerns the degree to which those contributions can shape the categories and accounts through which a relation is understood. These are related but distinct. An actor may be invited to speak while the interpretive frame remains fixed elsewhere; another actor may possess formal authority to define a category despite limited situated access to the relation being classified.

Fricker's analysis of testimonial injustice is directly relevant where credibility is systematically discounted because of identity-mediated prejudice or other power-laden credibility practices (Fricker 2007). Within GRE, credibility assessment also interacts with relational position. A worker can have strong situated access to day-to-day authority relations while management possesses records of formal reporting lines. A patient can describe experienced dependence within a care relation while an institution possesses standardized clinical or administrative records. A platform user can observe recurrent consequences of ranking or moderation while the platform possesses aggregate behavioral data and internal system logs. These sources carry different epistemic strengths, and justice requires that the asymmetry among them be made explicit rather than resolved through institutional status alone.

Situated authority should remain bounded by the target of the claim. First-person or participant testimony can be especially strong regarding experienced treatment, local meaning, tacit practice, and consequences that remain weakly recorded. It does not automatically settle population prevalence, hidden mechanisms, or effects outside the participant's observational horizon. Institutional evidence can support broader comparison while remaining weak on lived interpretation or informal practice. Epistemic justice therefore supports differentiated authority: sources receive weight in relation to the forms of access they possess and the inferential distance between that access and the claim under consideration.

Definition 5.141 (Representational authority). *Representational authority* is the institutionally or socially effective capacity of an actor, group, procedure, or system to shape how a relation or generative process is described, categorized, interpreted, and circulated within an epistemic domain.

Definition 5.141 directs attention beyond who speaks toward whose description becomes operational. A participant account can be heard and archived while a legal, administrative, statistical, or computational representation remains the version that allocates benefits, triggers investigation, determines eligibility, or structures later research. Section 5.6 showed that representations can become performative once they enter subsequent relational processes. Representational authority therefore has epistemic and generative consequences.

A justice-sensitive inquiry should preserve attribution and provenance so that different voices do not disappear into an anonymous synthesis. Where interpretations conflict, the disagreement can itself be evidentially informative. Recording who observed what, from which position, under which incentives and constraints, permits later readers to evaluate the relation between situated access and representational authority rather than treating a merged account as perspective-free.

5.18.3 Classification Power

Classification power concerns the capacity to establish categories that organize observation, comparison, eligibility, intervention, or subsequent interpretation. Sections 5.5 and 5.6 established that representation is selective and can become performative. Epistemic justice asks who participates in constructing those classifications, whose distinctions are preserved, and whose relations become difficult to express once a classificatory system stabilizes.

Statistical and administrative categories are historically produced rather than merely discovered. Desrosières shows how statistical reasoning depends on the construction of equivalences, coding practices, and institutions of comparability (Desrosières 1998). Hacking likewise emphasizes the historical emergence of categories and objects within styles of reasoning and classificatory practices (Hacking 2002). GRE draws a limited implication from these traditions: a classification can be epistemically useful and operationally necessary while still carrying assumptions about which differences matter, which entities can be compared, and which relations fit the available schema.

Classification power becomes justice-relevant when actors affected by a category have little capacity to contest its meaning, correct its application, or introduce distinctions that the schema suppresses. A household can be represented by an income band that is useful for aggregate analysis yet insensitive to unstable obligations or care burdens. A worker can be coded within an occupational category that obscures informal supervisory responsibility. A community can be represented through geographic or demographic units that fail to match the relational networks through which resources circulate. These examples concern mismatch and representational compression; they do not by themselves establish injustice. The normative question arises when the mismatch is consequential, systematically difficult to contest, and distributed through power-laden institutional arrangements.

Claim 5.142 (Classification power and epistemic consequence). *When classifications govern observation, aggregation, or institutional response, power over their construction and revision can affect which relational distinctions acquire evidential visibility and which alternatives remain difficult to formulate within the dominant representational system.*

The claim is compatible with the practical necessity of classification. Research and administration require some reduction of complexity. Justice-sensitive classification therefore emphasizes revision mechanisms, explicit scope, documentation of category construction, preserved residual information where feasible, and channels through which affected or differently situated actors can challenge systematic mismatch. Espeland and Stevens' account of commensuration is relevant here because common metrics can reorganize attention and comparison while carrying political consequences (Espeland and Stevens 1998). The epistemic task is to preserve awareness of what a classification makes comparable and what relational differences it compresses.

5.18.4 Epistemic Exclusion

Epistemic exclusion occurs when actors or forms of knowledge are systematically prevented from entering, influencing, or contesting the processes through which relational claims are formed. Exclusion can operate through institutional membership, language, credentialing, technical access, proprietary data, security restrictions, meeting formats, publication channels, methodological conventions, or credibility practices. Its epistemic significance lies in the loss of observations, interpretations, objections, and alternatives that could materially alter the claim under examination.

This problem can occur without overt prohibition. A consultation may be formally open while requiring technical vocabulary, travel, digital access, unpaid time, or legal familiarity that distributes effective participation unevenly. A dataset can be publicly described while meaningful access requires computational resources or proprietary linkages. A research process can invite testimony while allocating interpretive decisions entirely to professional experts. In such cases, formal inclusion and effective epistemic participation diverge.

Longino's account of critical interaction helps clarify why this matters epistemically as well as normatively. The capacity of a community to expose background assumptions depends on the presence and uptake of criticism from differently positioned participants (Longino 1990). A homogeneous or access-restricted inquiry can therefore appear internally coherent partly because relevant objections never become operational within its critical process.

Definition 5.143 (Epistemic exclusion in relational inquiry). *Epistemic exclusion in relational inquiry* is a patterned restriction on the effective capacity of relevant actors or knowledge-bearing positions to contribute evidence, interpretation, criticism, classification revision, or representational contestation concerning a relational or generative claim.

Definition 5.143 is deliberately process-focused. It does not imply that all viewpoints require equal evidential weight. A contribution can enter the inquiry and subsequently receive limited support because its evidence is weak, its scope is overstated, or stronger alternatives exist. Justice concerns the opportunity for relevant epistemic contribution and fair calibration of uptake; evidential evaluation still applies.

A useful diagnostic is to ask where in the epistemic chain exclusion operates: access to the phenomenon, recording of traces, preservation of records, admission to the dataset, category construction, interpretation, model specification, publication, institutional decision, or later revision. Different locations call for different remedies. Expanding testimony will have limited effect if the classification system cannot encode the relation described; opening a dataset will have limited effect if the relevant traces were never recorded.

5.18.5 Relations without Institutional Legibility

Institutional legibility concerns the extent to which a relation can be represented within the categories, records, procedures, and evidential formats recognized by an institution. Relational reality can exceed that legibility. Informal support networks, undocumented work, tacit obligations, community mediation, reciprocal care, eco-

logical dependence, reputational relations, and emergent coordination can exert substantial causal efficacy while remaining weakly represented in official records.

The epistemic problem is especially important because institutions often possess durable preservation systems. Once a relation enters administrative, legal, or statistical form, it becomes easier to retrieve, compare, cite, and incorporate into later decision making. Relations lacking such representation can leave fragmented traces that are costly to reconstruct. Over time, the archive can therefore make institutionally legible relations appear historically more substantial than relations whose manifestations were less systematically preserved.

Section 5.4 distinguished absence of evidence from evidence of absence, and Section 5.15 treated documentary and administrative materials as produced traces. Epistemic justice adds a distributional question: which actors and relations possess infrastructures that continually produce admissible traces? A relation can be socially important while generating little standardized evidence, particularly when the participants have weak access to documentation systems or when the relation falls outside the institution's operational categories.

Definition 5.144 (Institutional legibility of a relation). The *institutional legibility of a relation* is the degree to which a relation can be recognized, encoded, preserved, retrieved, and acted upon through the categories and evidential procedures of a specified institution.

Institutional legibility is target- and institution-relative. The same care relation may be legible to a household, partially legible to a local organization, and weakly legible to a tax or employment system. A financial dependency can be visible to a lender while remaining obscure to a welfare agency. The same relation can also become differently legible over time as categories, reporting requirements, and data infrastructures change.

Justice-sensitive inquiry should therefore retain space for relations whose evidential form does not match the institution's preferred formats. This can involve qualitative reconstruction, community-held records, participant testimony, archival triangulation, or new measurement designs. Such inclusion still requires provenance and critical evaluation. Its purpose is to reduce the conflation of institutional readability with relational existence.

5.18.6 Unequal Capacity to Produce Evidence

Actors differ substantially in their capacity to produce evidence that travels across institutional settings. Governments, firms, universities, platforms, and large organizations can maintain databases, legal teams, standardized reporting systems, professional communication staff, archives, and computational infrastructure. Individuals, small organizations, informal groups, precarious workers, or dispersed communities may possess rich situated knowledge while lacking comparable means to preserve, aggregate, validate, or circulate it.

This asymmetry can influence the apparent evidential balance. Repeated standardized records often acquire institutional authority because they are searchable, comparable, and auditable. A diffuse set of experiences can remain epistemically weaker

in practice even when it concerns a consequential relation, because the experiences are costly to coordinate and translate into admissible form. The World Bank's treatment of modern data systems illustrates the growing institutional importance of administrative, survey, citizen-generated, and machine-generated data infrastructures (World Bank 2021). GRE emphasizes that access to such infrastructures is itself relationally distributed.

Evidence-production capacity includes more than data collection. It includes the ability to define variables, commission studies, retain experts, set reporting standards, choose temporal windows, maintain archives, control disclosure, and respond quickly when a dispute arises. It also includes the capacity to preserve uncertainty and provenance. An actor with extensive documentation can present a coherent historical record; another actor may rely on memory or fragmented traces partly because preservation resources were unavailable.

Claim 5.145 (Evidence-production asymmetry). *Differences in resources, infrastructure, institutional standing, and access to recognized evidential formats can generate persistent asymmetries in whose relational claims are easiest to document, aggregate, circulate, and defend.*

The epistemic response should avoid compensatory romanticization. Scarce documentation does not make a claim true, just as abundant documentation does not make a representation complete. Justice-sensitive inquiry instead makes production conditions part of evidential assessment. It asks whether one side's documentary abundance reflects greater truth-tracking capacity, greater surveillance and archival capacity, or some combination of both. It also asks whether relevant counter-evidence was difficult to produce because access, preservation, or institutional recognition was unequally distributed.

Research design can partially address these asymmetries through data-access support, participatory documentation, translation, accessible reporting channels, preservation assistance, explicit recording of missingness mechanisms, and resources for independent analysis. The appropriate design depends on risk and context; some settings require strong privacy protections precisely because increased evidence production can expose vulnerable actors.

5.18.7 Justice in Relational Representation

Relational representation selects entities, relation types, directions, weights, boundaries, temporal windows, categories, and levels of aggregation. Section 5.5 treated those choices as epistemic mediation, and Section 5.6 showed that representations can subsequently shape relations. Justice in relational representation concerns how these choices distribute visibility, interpretive authority, contestability, and consequences among actors represented within or affected by the model.

A justice-sensitive representation should begin by making its standpoint and purpose explicit. A network built for fraud detection, a map built for community planning, a legal representation of obligation, and an ethnographic account of dependency can represent overlapping relations while selecting different features. Their differences can be legitimate when the analytical purpose is explicit. Problems arise when a purpose-specific representation travels into a new domain as though it were

a complete account of relational reality.

Representational justice therefore benefits from several practices. First, provenance should remain visible: readers should be able to reconstruct who generated the traces, who selected the categories, and what transformations were applied. Second, consequential category choices should be documented together with plausible alternatives. Third, affected or differently positioned actors should have channels to identify systematic mismatch and propose corrections. Fourth, uncertainty and missingness should remain attached to the representation rather than disappearing during aggregation. Fifth, when multiple representations capture materially different aspects of the same relation, preservation of plurality can be epistemically preferable to premature synthesis.

Proposition 5.146 (Justice-sensitive relational representation). *A relational representation is comparatively more justice-sensitive when its purpose, provenance, categories, exclusions, uncertainty, and consequential uses are explicit; when materially affected or differently positioned actors possess meaningful opportunities for contestation; and when alternative representations with relevant evidential support can remain recoverable for comparison and revision.*

Proposition 5.146 does not establish the representation's empirical adequacy by itself. Contestable and well-documented models can still be wrong. The proposition concerns the epistemic conditions under which error, mismatch, and power-laden compression can become more visible and revisable. This aligns with the chapter's broader commitment to explicit mediation and revision rather than an ideal of perspective-free representation.

Young's critique of reducing justice to distribution is relevant to this process-centered orientation because participation, decision structures, domination, and cultural representation can themselves be justice-relevant (Young 1990). GRE does not import Young's framework wholesale. The narrower implication is that the justice of knowledge production cannot be assessed solely by distributing informational goods after representational structures have already been fixed.

5.18.8 Justice in Generative Attribution

Generative attribution assigns contribution to the relations, actors, infrastructures, institutions, environments, and prior processes that help produce capacities, structures, values, constraints, or later possibilities. Sections 5.12, 5.13, and 5.16 developed the epistemic burdens attached to such claims. Epistemic justice adds a further concern: contribution can be systematically misrecognized when some generative inputs are highly visible and others remain diffuse, backgrounded, informally organized, or weakly documented.

A familiar pattern is concentration of attribution at the most visible point of outcome realization. A product can be attributed primarily to the organization that markets it while prior public infrastructure, care work, maintenance, ecological conditions, upstream knowledge, standards, training, and collaborative networks remain less visible. An institutional success can be attributed to formal leadership while informal coordination and repair remain weakly documented. A research output can foreground named authors while technical, administrative, community, or data-

generating contributions are represented only partially. These examples differ substantively, yet they share an epistemic structure in which outcome visibility exceeds visibility of distributed generative conditions.

Justice-sensitive generative attribution therefore requires the contribution profile introduced in Section 5.13 to remain open to distributed and enabling contributions. It should distinguish direct production, enabling infrastructure, maintenance, reproduction of capacity, coordination, constraint removal, preservation, and downstream recirculation where these are evidenced. It should also preserve uncertainty where contribution cannot be disaggregated reliably. A broad attributional map can be more honest than a precise but unsupported allocation of shares.

Definition 5.147 (Justice-sensitive generative attribution). *Justice-sensitive generative attribution* is the practice of attributing generative contribution in a manner that actively examines visibility asymmetries, evidential-capacity differences, classification effects, and background conditions that can systematically obscure or overstate particular contributors, while retaining the causal and evidential standards required for the underlying generative claim.

Definition 5.147 is intentionally epistemic before it becomes distributive. Establishing a contribution does not by itself determine compensation, ownership, recognition, liability, or return. Those normative questions require additional principles developed later in the book. The present requirement is that downstream evaluation begins from a contribution map whose omissions and asymmetries have been critically examined.

The same principle applies to blame and harmful generation. Highly visible actors can receive disproportionate causal attribution while background institutions, inherited structures, enabling infrastructures, or distributed incentives receive too little attention. Conversely, structural explanation can become so diffuse that identifiable agency disappears. Justice-sensitive attribution therefore preserves several levels simultaneously: actor contribution, relational configuration, enabling conditions, historical trajectory, and field-level structure where the evidence supports them. This layered account resists both individualized simplification and structure-only abstraction.

Proposition 5.148 (Epistemic-justice constraint on generative attribution). *Where generative contributions are unevenly visible or unequally documentable, warranted attribution should examine whether evidential asymmetries systematically track power, institutional legibility, or access to preservation and representation; conclusions should preserve relevant distributed contributions and uncertainty before later normative claims about recognition, responsibility, return, or remedy are introduced.*

The epistemic-justice perspective developed in this section therefore modifies neither the requirement for evidence nor the chapter's commitment to uncertainty. It changes the questions asked about how evidence becomes available and authoritative. GRE inquiry should ask whose relations are visible, who can define categories, whose testimony receives uptake, which relations fit institutional formats, who can produce durable evidence, how representations remain contestable, and whose generative contributions are liable to disappear from the record. These questions strengthen relational inquiry by exposing systematic conditions under which evi-

dential fields themselves can be asymmetrically organized.

The next section turns to machine-mediated relational knowledge. Machine observation, algorithmic classification, computational inference, and AI-assisted reconstruction intensify many of the epistemic issues identified here because they can expand observational reach while concentrating classificatory and representational capacity in systems that operate at scales difficult for observed actors to inspect or contest. The transition therefore moves from epistemic justice in relational inquiry to the specific architecture of machine-mediated observation, inference, and representational power.

5.19 Machine-Mediated Relational Knowledge

Section 5.18 examined epistemic justice in relational inquiry by asking how visibility, voice, classification, evidential capacity, representation, and attribution are distributed among differently positioned actors. The present section narrows that problem to machine-mediated knowledge. Its role is to specify how computational systems enter the epistemic chain between relational reality, traces, representation, inference, and subsequent action. Its objective is to distinguish machine observation, algorithmic classification, computational inference, machine-generated representation, and AI-assisted reconstruction while identifying the epistemic asymmetries and contestability requirements that arise when these operations are concentrated in technical systems. The section proceeds from observation to classification and inference, then to representation and reconstruction, before examining asymmetry, representational power, and contestability. The method remains epistemically plural: computational output is treated as one mediated source of evidence whose warrant depends on provenance, target specification, model assumptions, boundary and scale, and the relation between machine-produced representations and other forms of evidence.

Machine mediation expands the range, speed, and granularity at which relational traces can be collected and organized. Administrative databases can connect records across institutions; platform logs can record interaction sequences at scales inaccessible to unaided observation; computer vision and remote sensing can generate repeated observations of physical or ecological environments; language models can summarize, classify, and recombine large documentary corpora; and network or statistical models can infer candidate relations that are only partially represented in the raw record. These capacities can materially improve inquiry. They also create additional layers at which categories, missingness, optimization objectives, architecture, training data, access privileges, and institutional purposes can shape what becomes visible as relational knowledge.

The distinction between machine mediation and machine authority is therefore fundamental. A computational system can assist observation or reconstruction without receiving final authority over the relational claim. Conversely, a machine-produced classification can become operationally authoritative even when its epistemic basis remains uncertain, especially when institutions embed it in eligibility, ranking, allocation, monitoring, or enforcement. Section 5.6 developed this

possibility through performative representation. Machine-mediated epistemology must therefore evaluate both whether a system tracks a target relation adequately and how its outputs enter subsequent relations.

Definition 5.149 (Machine-mediated relational knowledge). *Machine-mediated relational knowledge* is relational knowledge whose evidential access, classification, inference, representation, reconstruction, or circulation depends materially on computational systems, such that the system’s data-generating conditions, categories, transformations, model assumptions, and operational uses become part of the epistemic conditions of the claim.

Definition 5.149 is deliberately broader than machine learning. A rule-based administrative system, a database query, a sensor network, a statistical model, a graph algorithm, and a generative model can all mediate relational knowledge in different ways. The relevant question is not whether a system is called “AI,” but where computational operations enter the evidential chain and how those operations affect what can be warranted.

5.19.1 Machine Observation

Machine observation concerns computationally mediated acquisition of traces that bear on a relational target. The term includes automated sensors, transaction logs, administrative data pipelines, digital interaction records, remote sensing, platform telemetry, and other systems in which observation is produced through technical infrastructure rather than through unaided human perception alone. The World Bank’s account of contemporary data systems, for example, explicitly includes administrative, survey, citizen-generated, and machine-generated data within wider data infrastructures (World Bank 2021). For GRE, the epistemic importance of machine observation lies in the way these infrastructures extend observational reach while also fixing particular units, events, intervals, and recording rules.

Section 5.7 treated observation as position-dependent and partial. Machine observation does not remove that condition. It relocates observer position into an engineered observation regime. A platform can observe clicks, messages, transactions, or mobility traces because its architecture has defined which events are recorded and at what resolution. A sensor can repeatedly measure a physical variable while remaining insensitive to relations that do not manifest through its detection channel. An administrative system can observe registered interactions while missing informal obligations or unreported exchange. Machine observation can therefore be extensive and still relationally selective.

Definition 5.150 (Machine observation regime). A *machine observation regime* is the organized combination of instruments, event definitions, sampling rules, recording intervals, data schemas, access conditions, and preprocessing operations through which computational infrastructure converts selected manifestations into machine-readable traces.

The definition makes several sources of uncertainty explicit. First, event ontology matters: the system determines what counts as an interaction, transaction, encounter, exposure, or response. Second, temporal resolution matters: repeated observations at one interval may obscure dynamics occurring at another. Third, miss-

ingness matters: a system can record only actors, locations, devices, or activities that enter its observational infrastructure. Fourth, preprocessing matters: filtering, deduplication, imputation, aggregation, and identity resolution can alter the trace before analysis begins. Fifth, access matters: the institution controlling the infrastructure may observe a relation at a level of granularity unavailable to the people whose behavior generates the traces.

Machine observation should therefore be documented as a trace-production process. A relational claim based on logs or sensor data should identify what the infrastructure was capable of recording, what it systematically excluded, how identities or events were linked, what transformations occurred before analysis, and how the observational window relates to the target relation. Large volume does not remove the need for this account. A very large set of selectively generated traces can support precise estimates of the wrong relational object.

5.19.2 Algorithmic Classification

Machine observation becomes analytically consequential when traces are assigned to categories. Algorithmic classification maps observed or derived features into labels, scores, clusters, rankings, risk classes, similarity groups, or other operational distinctions. The epistemic issue is continuous with the chapter's earlier treatment of statistical and administrative representation. Desrosières shows historically how statistical categories and comparability depend on conventions of classification and coding (Desrosières 1998), while Hacking emphasizes the historical interaction between classificatory practices and the objects or persons classified (Hacking 2002). Algorithmic systems intensify this problem because classification can occur repeatedly, automatically, and at scale.

A classification is not yet a relation. A label such as “high risk,” “similar,” “coordinated,” “fraudulent,” “influential,” or “vulnerable” may be used to infer a relational position, but the connection between category and relation requires separate warrant. Two actors placed in the same cluster may share measured features without interacting. A recommendation system may classify users as behaviorally similar while remaining agnostic about whether any socially meaningful relation exists between them. A fraud-detection model may flag transactions as related through a learned pattern while the substantive mechanism remains uncertain.

Algorithmic classification also introduces threshold dependence. A continuous score can become a discrete category through an operational cutoff, and that cutoff can affect which relations appear present or consequential. When categories subsequently control access, review, surveillance, or opportunity, classification becomes performative in the sense developed in Section 5.6. The epistemic and performative questions should remain separate: whether a classification is well supported is one issue; what institutional consequences follow from assigning it is another.

Proposition 5.151 (Classification dependence of machine-mediated relational claims). *Where a relational claim depends materially on algorithmic categories, the warrant for that claim depends partly on the evidential adequacy of the category definition, feature construction, threshold or grouping rule, and portability of the classification to the population, period, and relational target under study.*

Proposition 5.151 does not imply that every classification requires a single uniquely correct category system. Several classifications can be useful for different research purposes. The requirement is that the category's purpose and scope remain explicit and that a purpose-specific classification is not silently treated as a complete description of relational reality.

5.19.3 Computationally Inferred Relations

Computational inference moves beyond recorded traces or predefined categories by estimating relations that are incompletely observed. Link-prediction models, latent-variable models, embedding methods, clustering procedures, probabilistic graphical models, sequence models, and related techniques can identify candidate connections, dependencies, similarities, or relational structures. Section 5.15 already distinguished computationally inferred evidence from directly observed relations. The present subsection clarifies the epistemic status of the inferred relation itself.

Definition 5.152 (Computational relational inference). *Computational relational inference* is the model-mediated estimation of a relation, relational property, or relational configuration from observed traces and specified assumptions when the target relation is not completely recorded in the evidential input.

The inferential distance between data and target can vary considerably. Inferring that two registered accounts exchanged messages may require relatively little modeling when the underlying logs are reliable. Inferring that two actors coordinate strategically, share an obligation, occupy analogous institutional positions, or participate in the same latent community requires additional assumptions about how the observed traces map onto the substantive relation. Section 5.2 described this as an inverse problem of manifestation: multiple relational realities can generate similar observed patterns.

Manski's work on identification is relevant because it separates what data identify from conclusions that become available only after stronger assumptions are added (Manski 2008). GRE applies the same discipline to computational relation inference. Model performance on a prediction task does not by itself establish that the inferred latent object corresponds to the substantive relation named by the researcher. A representation can predict interaction successfully while organizing that interaction through dimensions that differ from the researcher's interpretive vocabulary.

For this reason, computationally inferred relations should carry explicit provenance about training or estimation data, target construction, validation conditions, model class, relevant assumptions, error structure, and sensitivity to alternative specifications. External or qualitative evidence can be especially important when the substantive meaning of an inferred relation exceeds what was directly encoded in the training target. The goal is not to subordinate computational inference to human interpretation, but to prevent predictive success from becoming an unexamined bridge between trace and relation.

5.19.4 Machine-Generated Representations

Computational systems increasingly generate representations rather than only classify existing inputs. They produce summaries, captions, labels, rankings, graph layouts, synthetic descriptions, extracted entities, reconstructed timelines, and natural-language accounts of relational configurations. Generative models can further reorganize large corpora into compact representations that appear coherent and immediately usable. These capabilities increase the speed with which relational material can be explored, but they also increase the distance between the original trace and the representation presented to the investigator.

Section 5.5 treated representation as selective mediation. Machine generation adds a transformation process whose output can contain compression, interpolation, omission, normalization, or unsupported completion. Bender and colleagues' discussion of large language models is relevant because fluent linguistic output does not by itself establish grounded understanding or factual adequacy (Bender et al. 2021). Within GRE, the corresponding point is narrower: linguistic or graphical coherence does not establish that the represented relations are evidentially warranted.

Definition 5.153 (Machine-generated representation). A *machine-generated relational representation* is a computationally produced description, visualization, categorization, summary, or structured encoding of relations whose form is generated through model operations rather than being a direct transcription of the underlying evidential record.

Such representations can be epistemically useful when they reduce search cost, reveal patterns, generate candidate comparisons, or make complex relational structures inspectable. Their use nevertheless requires preservation of the path back to source material. A generated summary of correspondence should permit retrieval of the documents that support each consequential relational claim. A generated network description should remain linked to the records and inference rules through which edges were created. A machine-produced historical narrative should distinguish sourced events from model-generated connective language.

The need for source recoverability grows with generative compression. When hundreds or thousands of traces are compressed into a concise narrative, the representation can become easier to circulate than the evidence that produced it. Repetition then gives the generated representation an appearance of stability even when its evidential basis is mixed. GRE therefore treats machine-generated representations as revisable analytical artifacts whose claims should remain decomposable into traceable evidential units where consequences are substantial.

5.19.5 AI-Assisted Relational Reconstruction

AI-assisted relational reconstruction uses computational systems to help assemble partial traces into a candidate account of relations, trajectories, fields, or generative processes. Relevant tasks include entity resolution across archives, extraction of relation mentions from documents, coding of qualitative materials, identification of temporal sequences, comparison of alternative classifications, detection of anomalous patterns, and generation of candidate hypotheses for human examination. These ap-

plications can expand the feasible scope of relational inquiry while also creating new forms of dependency on model behavior.

The epistemic unit should be the reconstruction workflow rather than the model output in isolation. A robust workflow specifies which tasks are delegated to the system, which outputs are independently checked, how disagreements are handled, how evidence is linked to reconstructed relations, and where human interpretation enters. Longino's account of socially organized criticism is useful here because epistemic robustness depends partly on opportunities for critical interaction and uptake of objections (Longino 1990). AI assistance can widen the space of candidate interpretations, yet it can also narrow that space when one model's categories or summaries become the default entry point to the evidence.

Proposition 5.154 (AI-assisted reconstruction). *AI assistance strengthens relational reconstruction when it expands evidential search, comparison, or pattern detection while preserving source provenance, alternative interpretations, uncertainty, and independent routes of criticism; its epistemic contribution weakens when generated representations become difficult to trace back to the evidence or when model outputs silently replace the inferential steps they were intended to assist.*

The proposition supports a division of epistemic labor rather than a general hierarchy between human and machine judgment. Machines may be superior at repeated comparison across large corpora; situated participants may possess forms of access that the corpus does not record; domain experts may identify institutional distinctions that a general model merges; computational models may reveal regularities that established theory overlooked. GRE therefore evaluates contributions task by task and preserves disagreement as evidence about where reconstruction remains unsettled.

AI assistance also changes the economics of hypothesis generation. When candidate relations and explanations can be generated cheaply, the bottleneck shifts toward validation, source checking, discrimination among alternatives, and preservation of negative results. Abundant candidate explanations should therefore increase, rather than relax, the importance of explicit evidential burden. Machine-assisted inquiry becomes epistemically stronger when it makes comparison easier without making acceptance automatic.

5.19.6 Epistemic Asymmetry between Observer and Observed

Machine-mediated inquiry can create unusually large differences between what an observing institution knows about a relation and what observed actors can know about the observing process. A platform can aggregate behavioral histories across millions of interactions while an individual sees only a local interface. An employer can use computational systems to direct, evaluate, and discipline work at scale, a phenomenon documented in the literature on algorithmic management (Kellogg et al. 2020). A lender, insurer, government agency, or marketplace can combine records from multiple sources while the person classified may not know which variables, historical records, proxies, or model transformations contributed to the result.

Burrell's analysis of machine-learning opacity distinguishes secrecy, technical-literacy barriers, and opacity arising from model characteristics and scale (Burrell

2016). This distinction is useful because epistemic asymmetry can arise for different reasons and therefore require different remedies. Contractual secrecy may require disclosure or independent audit. Technical complexity may require explanation, expertise, or tooling. Scale-related opacity may require sampling, monitoring, or structured documentation rather than an expectation that every individual inspect every model operation.

Definition 5.155 (Machine-mediated epistemic asymmetry). *Machine-mediated epistemic asymmetry* is a relational difference in access to the observations, categories, model operations, aggregate patterns, or revision mechanisms through which computational systems produce consequential knowledge about actors or relations.

This asymmetry has several dimensions. *Observational asymmetry* concerns access to traces. *classificatory asymmetry* concerns knowledge of the categories and thresholds applied. *inferential asymmetry* concerns access to the models and assumptions connecting traces to conclusions. *temporal asymmetry* arises when institutions preserve long histories while observed actors cannot reconstruct the same record. *revision asymmetry* concerns who can correct data, categories, or outputs once an error or mismatch is identified.

Asymmetry is not automatically unjust or epistemically defective. Epidemiological surveillance, fraud detection, infrastructure monitoring, or scientific instrumentation can require specialized access. The relevant question is how asymmetry interacts with consequence, error, vulnerability, and opportunities for challenge. Where outputs materially affect actors' opportunities or standing, the burden for provenance and contestability becomes correspondingly stronger.

5.19.7 AI and the Power to Define Relational Reality

The phrase “define relational reality” requires restraint. Computational systems do not acquire ontological authority merely because they classify at scale. The stronger and more defensible claim is that institutions can grant machine-produced representations *operational authority*: the categories, scores, clusters, or inferred relations become inputs to decisions that reorganize subsequent interaction. Section 5.6 showed how representation can become performative, while Section 5.18 showed that classification and representational authority can be distributed unequally.

Historical work on classification and statistical reasoning helps clarify the structure. Desrosières demonstrates that categories and measures are institutionally produced rather than transparent windows onto a preformatted social world (Desrosières 1998). Hacking likewise examines how classificatory practices can participate in the historical constitution of objects of knowledge (Hacking 2002). Machine systems can accelerate and routinize this process because classification is embedded directly in operational infrastructures.

A recommendation system can alter which social or commercial relations become likely to form. A ranking system can change organizational attention. An automated risk score can redirect scrutiny. Algorithmic management can restructure the temporal and informational relation between workers and supervisors (Kellogg et al. 2020). In such cases, the machine-produced representation participates in the generation of the relational reality later observed. The epistemic danger is recursive confirmation:

a classification changes behavior or opportunity, and the resulting trace is then interpreted as further evidence that the original classification captured an underlying stable relation.

Proposition 5.156 (Operational authority of machine representation). *When a machine-produced representation is embedded in consequential allocation, ranking, monitoring, recommendation, or enforcement, its epistemic role cannot be assessed solely by predictive performance; inquiry should also examine how the representation changes the relations that later generate evidence about its apparent validity.*

This proposition does not imply that performative effects invalidate a model. Some systems are intentionally designed to alter outcomes, as in early-warning or safety systems. The requirement is temporal and relational clarity. Researchers should distinguish evidence collected before deployment, effects produced through deployment, adaptation by affected actors, and later data generated within the altered environment. Otherwise, the system can appear to discover a relation that it has partly stabilized or produced.

5.19.8 Contestability of Machine-Mediated Knowledge

Contestability concerns the practical capacity to question, inspect, test, correct, or replace a machine-mediated relational claim. It extends beyond explanation in the narrow sense. A system can provide an intelligible local explanation while leaving the underlying data, category, target definition, institutional purpose, or appeal process fixed. Conversely, complete publication of source code may provide little meaningful contestability to an affected actor if the relevant training data, operational thresholds, or institutional consequences remain inaccessible.

Kroll and colleagues argue that consequential automated systems require accountability mechanisms capable of verifying compliance with relevant standards rather than relying on ordinary assumptions developed for human decision makers (Kroll et al. 2017). Dataset documentation offers a complementary infrastructure: Gebru and colleagues propose recording dataset motivation, composition, collection processes, and recommended uses so that downstream users can evaluate provenance and limitations (Gebru et al. 2021). GRE draws a broader epistemic lesson from these approaches. Contestability should be attached to the whole relational-knowledge pipeline rather than to the final model output alone.

Definition 5.157 (Contestability of machine-mediated relational knowledge). The *contestability of machine-mediated relational knowledge* is the degree to which relevant actors can identify the target claim, recover its evidential and computational provenance, understand the consequential categories and assumptions at an appropriate level, introduce counter-evidence or alternative interpretations, obtain review of material error or mismatch, and preserve the result of that challenge for subsequent revision.

Contestability is therefore layered. *Data contestability* concerns whether records can be inspected and corrected where appropriate. *category contestability* concerns whether the relation between operational labels and substantive concepts can be challenged. *model contestability* concerns assumptions, validation, uncertainty, and sensitivity to alternatives. *decision contestability* concerns how outputs enter conse-

quential action. *historical contestability* concerns whether earlier versions, disputes, and corrections remain preserved so that later re-epistemization is possible.

Proposition 5.158 (Contestability architecture). *For consequential machine-mediated relational claims, a minimally adequate contestability architecture should preserve enough provenance to reconstruct how the claim was produced, specify the target and relevant categories, expose material uncertainty and known limitations, provide a route for qualified counter-evidence or review, and record substantive corrections or unresolved disputes for future revision.*

Proposition 5.158 is intentionally compatible with differentiated access. Privacy, security, trade secrecy, or protection of vulnerable actors can limit complete disclosure. Contestability can therefore involve trusted independent review, secure audit, controlled data access, or representative challenge rather than universal publication of every technical artifact. The epistemic standard concerns the existence of credible routes through which consequential claims can be tested and revised.

Machine-mediated relational knowledge thus intensifies themes developed throughout Chapter 5 without replacing them. Observation remains partial; classification remains historically and institutionally situated; inferred relations remain model-dependent; representations remain mediated; generative attribution remains burdened by mechanism and alternatives; and epistemic justice remains concerned with unequal capacities for visibility, voice, evidence, and correction. Computational systems can expand the reach of inquiry while simultaneously concentrating the infrastructure through which relational reality becomes visible and actionable. GRE therefore treats machine mediation as an epistemic architecture requiring explicit provenance, calibrated confidence, and contestability rather than as an independent source of authority.

The next section turns from machine mediation to revisability. Once relational knowledge is understood as partial, mediated, historically situated, and increasingly produced through human-machine infrastructures, the ability to update relational maps, categories, interpretations, and attributions becomes a constitutive requirement of epistemic adequacy rather than a secondary procedure applied only after error has been discovered.

5.20 Revisability of Relational Knowledge

This section establishes revisability as a constitutive property of relational knowledge in GRE. Its role is to specify how warranted claims can remain usable while the observed relations, manifestation processes, categories, evidential bases, and conditions of inquiry continue to change. The objective is therefore methodological in an epistemic sense: to distinguish several sources of revision and to identify the documentary and interpretive conditions under which revision can be disciplined rather than improvised. The section proceeds through provisional relational maps, updating through new evidence, revision after intervention, historical revision, category revision, reflexive observation, and preservation of alternative interpretations. The discussion treats revision as traceable change in a claim, representation, or attribution under stated reasons and preserved provenance. This orientation is compatible

with traditions that expose scientific claims to criticism and correction (Popper 2002; Longino 1990), while GRE gives the requirement a specifically relational and historical form.

Definition 5.159 (Revisability of relational knowledge). The *revisability of relational knowledge* is the organized capacity to modify, qualify, extend, contract, replace, or retire a relational or generative claim when changes in evidence, intervention, historical conditions, categories, observer relations, or competing interpretations materially alter its warrant, while preserving enough provenance to reconstruct why the earlier and later formulations differed.

Definition 5.159 separates revisability from epistemic instability. A claim can be strongly warranted at a specified time, scale, boundary, and evidential state while remaining revisable under materially changed conditions. Revisability therefore concerns the architecture of knowledge maintenance rather than a presumption that every conclusion is equally fragile. It also differs from routine textual editing. The relevant change is epistemically consequential: the object, representation, evidential basis, inferential relation, or confidence profile has changed sufficiently that the knowledge claim should change with it.

5.20.1 Provisional Relational Maps

Relational inquiry commonly produces maps of actors, relations, fields, boundaries, flows, dependencies, generative pathways, or historical trajectories. Such maps can be indispensable analytical objects while remaining provisional. Provisionality follows from the conditions developed throughout this chapter: observation is partial, boundaries and scales are selected, relation identity is criterion-dependent, higher-order structures are reconstructed, field claims require distributed evidence, and generative attributions carry additional inferential burdens. A relational map should therefore be understood as a warranted representation under stated conditions rather than as a completed inventory of the relevant relational world.

Definition 5.160 (Provisional relational map). A *provisional relational map* is an explicitly bounded and time-indexed representation of identified relations, positions, fields, trajectories, or generative linkages whose evidential basis, representational choices, uncertainty, and conditions of revision are sufficiently documented for later reassessment.

Provisionality has at least four dimensions. *Observational provisionality* concerns incomplete or unequal access to traces. *representational provisionality* concerns the possibility that another formal, linguistic, legal, statistical, or computational representation preserves different aspects of the same target. *historical provisionality* concerns transformation of the relations themselves. *inferential provisionality* concerns the possibility that new evidence changes the relative warrant of competing explanations. These dimensions can vary independently. A network representation, for example, may remain an adequate summary of recorded interactions while later evidence changes the interpretation of what those interactions mean.

A provisional map should therefore carry metadata about its epistemic conditions. At minimum, the map should identify its target, observational window, boundary, scale, relation criteria, principal evidence types, major representation choices, and

known uncertainties. When generative links are included, the map should also distinguish observed association from stronger mechanism-sensitive attribution. This practice allows a map to be used operationally without allowing its graphic or computational completeness to imply stronger warrant than its evidential basis supports.

Proposition 5.161 (Provisional warrant). *A relational representation may support action, comparison, or further inquiry while remaining provisional, provided that the conditions under which its claims are warranted and the conditions under which they should be reconsidered remain explicit.*

Proposition 5.161 matters because a demand for unrevisable certainty would make most relational inquiry unusable, whereas a refusal to stabilize any working representation would prevent cumulative investigation. GRE therefore permits temporary stabilization under documented conditions. The epistemic task is to make the stabilization inspectable and corrigible.

5.20.2 Updating through New Evidence

New evidence can alter relational knowledge in several ways. It may increase confidence in an existing relation, weaken an earlier identification, reveal a previously hidden participant, extend the temporal duration of a relation, narrow its scope, expose an alternative mechanism, or motivate replacement of the earlier interpretation. Updating is consequently broader than adding observations to an unchanged model. The update can affect the data, the representation, the category system, the causal interpretation, or the confidence profile introduced in Section 5.17.

The relevance of evidence is itself mediated by background assumptions, categories, and standards of criticism, a point emphasized in Longino's account of scientific objectivity and critical interaction (Longino 1990). GRE therefore treats updating as an argued relation between an earlier claim state and a later evidential state. The investigator should identify what changed and why the new material bears on the target claim. A newly available administrative record may change confidence in a formal relation without resolving its enacted intensity. New ethnographic evidence may revise interpretation of an interaction while leaving its recorded frequency unchanged. A newly reconstructed historical source may reopen a genealogy without directly altering contemporary observational evidence.

Definition 5.162 (Traceable epistemic update). *A traceable epistemic update is a documented revision that identifies the prior claim, the new or reinterpreted evidence, the inferential reason for revision, the dimensions of the claim affected, and the resulting change in confidence, scope, representation, or attribution.*

The definition supports a versioned understanding of relational knowledge. Earlier formulations need not disappear when they are superseded. Their retention can show which evidence was available at the time, which assumptions governed interpretation, and where later revision entered. This matters especially when a later claim appears obvious only because intervening evidence has changed the epistemic environment.

Updating also requires proportionate response. Minor evidence should rarely trigger wholesale theoretical replacement when a narrower qualification is sufficient.

Conversely, a large volume of confirmatory observations should not protect a claim when new evidence undermines the relation criterion, identifies a systematic observation failure, or reveals that the apparent pattern was produced by the measurement regime itself. The magnitude of revision should therefore correspond to the part of the epistemic architecture that the new evidence actually affects.

Proposition 5.163 (Dimension-matched revision). *When new evidence alters only a specified dimension of a relational claim, revision should ordinarily be concentrated on that dimension unless the dependence structure of the argument shows that wider conclusions also lose warrant.*

This proposition preserves both responsiveness and continuity. It also makes disagreement more diagnosable: researchers can dispute the evidence, the relation between evidence and target, the affected dimension, or the wider dependencies without treating every update as a judgment on the entire research programme.

5.20.3 Revision after Intervention

Intervention creates a distinctive revision problem because the inquiry may change the relational system whose earlier representation it sought to describe. Policy reform, organizational restructuring, legal recognition, platform redesign, resource redistribution, experimental treatment, public disclosure, or governance intervention can alter participants, incentives, opportunities, classifications, and manifestation processes. Evidence gathered after intervention may therefore concern a transformed relational configuration rather than a cleaner observation of the earlier one.

Interventionist accounts of causal explanation use controlled changes and counterfactual contrasts to clarify causal dependence (Woodward 2003). GRE adopts the value of intervention-sensitive reasoning while adding a historical-relational caution: an intervention can become part of the subsequent relational reality. The epistemic task is then twofold. Inquiry may learn about the earlier system from how it responded, while also needing to reconstruct the new system produced through that response.

Definition 5.164 (Intervention-sensitive revision). *Intervention-sensitive revision is the reassessment of relational knowledge that distinguishes evidence about a pre-intervention configuration from relations, manifestations, or generative pathways produced, suppressed, or reorganized through the intervention itself.*

This distinction is especially important when interventions are adaptive. Actors can learn from monitoring, reorganize around a new rule, create substitutes for a constrained pathway, or redirect activity into less visible channels. A short-run post-intervention change can therefore differ from the later stabilized configuration. The relevant temporal horizon should remain explicit, and the revision record should identify which parts of the earlier map persisted, which transformed, and which newly emerged.

Intervention can also expose relations that were previously difficult to observe. Removing a constraint may reveal latent coordination capacity; altering an institutional interface may expose dependencies that had been absorbed by routine practice; changing access conditions may make hidden relations observable. Such evi-

dence can legitimately revise the earlier account, but the reasoning should separate discovery from creation. The intervention may have revealed a pre-existing relation, created a new relation, or done both in different respects.

Proposition 5.165 (Post-intervention relational reassessment). *When an intervention materially changes the conditions under which relations are formed, manifested, or observed, post-intervention evidence should be interpreted through an explicit reassessment of relation identity, manifestation rules, boundary conditions, and temporal horizon before it is used to update claims about the pre-intervention system.*

Proposition 5.165 limits a common inferential shortcut: treating later observations as if they belonged automatically to the same relational object. It also prepares the methodological discussion of intervention in later Parts, where research and application praxis require designs capable of learning from transformation without erasing the distinction between observation and intervention-produced change.

5.20.4 Historical Revision

Historical revision arises when later evidence, later consequences, changed categories, or a transformed interpretive context alters what can be warranted about an earlier relational configuration. Section 5.14 described historical re-epistemization as a disciplined reconstruction of earlier knowledge under later conditions. The present subsection places that process within the broader architecture of revisability.

Historical revision can proceed from newly recovered sources, improved provenance, reinterpretation of previously known records, changed knowledge of surrounding institutions, or recognition that an earlier category concealed distinctions relevant to the historical actors or relations. Hacking's historical ontology emphasizes that concepts and objects of knowledge can acquire historically specific conditions of possibility (Hacking 2002); Ricoeur's hermeneutics likewise treats interpretation as mediated through distance, text, and changing horizons of understanding (Ricoeur 1991). For GRE, these traditions support a practical requirement: later reconstruction should preserve the difference between revision of present knowledge about the past and transformation of the past itself.

Definition 5.166 (Historical epistemic revision). *Historical epistemic revision* is a documented change in the warranted interpretation, classification, boundary, trajectory, or generative attribution of an earlier relational configuration produced by later evidence or later analytical resources, while preserving the temporal location and evidential conditions of the earlier claim.

The preservation clause is essential. An earlier claim may have been reasonable given the records then available even if later evidence substantially revises it. Conversely, a historically influential interpretation can remain important as an object of study after its descriptive warrant has weakened, because institutions or actors may have acted on it. Revision should therefore distinguish at least three objects: the past relational configuration, the earlier representation of that configuration, and the historical effects produced by that representation.

Later consequences also alter salience. An event that appeared minor at the time

can become analytically significant because it entered a long generative trajectory. This later significance does not establish that the event already contained its later outcome or that contemporaries could have known the trajectory. Historical revision should therefore preserve the chapter's earlier distinction between retrospective significance and retrospective determinism.

Versioned historical knowledge is especially important where records are incomplete or politically uneven. Superseded interpretations, disputed classifications, alternative chronologies, and unresolved evidential gaps should remain visible when they materially affect later reconstruction. Such preservation allows future inquiry to distinguish a settled finding from an interpretation that became dominant because competing records disappeared.

5.20.5 Revision of Categories

Relations become knowable through categories: actor types, institutional statuses, transaction classes, legal forms, network edge definitions, risk groups, occupational codes, value categories, and many other classificatory devices. Categories enable comparison and cumulative research, yet they can become misaligned with the relational system. Category revision is therefore a distinct epistemic operation rather than a special case of adding more data.

Historical studies of statistics show that comparability depends on conventions of classification and coding (Desrosières 1998). Hacking likewise examines the historical interaction between classificatory practices and the objects classified (Hacking 2002). Sections 5.5, 5.6, and 5.19 have already shown that categories can mediate and sometimes alter relational reality. A category that once produced useful analytical compression may later obscure differentiation, become strategically gamed, exclude emergent subjects, or acquire performative consequences that change what the category measures.

Definition 5.167 (Relational category revision). *Relational category revision* is the explicit modification of a classificatory scheme used to identify, compare, or represent relational objects when evidence indicates that the existing categories materially distort, omit, overmerge, or misalign with the target relations or their historical transformation.

Category revision can involve splitting an overly broad category, merging distinctions that have ceased to matter for the target, adding a previously unrepresented relational form, changing membership criteria, altering temporal or spatial resolution, or replacing a category whose operational meaning has drifted. These changes affect comparability. A revised category can improve current fit while making historical series less directly comparable. The revision record should therefore retain category definitions and mapping rules between versions where such mappings are defensible.

Category failure should be diagnosed with target-specific evidence. Low predictive performance may indicate category inadequacy in one setting, while ethnographic mismatch, systematic testimonial disagreement, institutional change, high within-category heterogeneity, or emergent strategic behavior may reveal other forms of failure. None supplies a universal threshold. The relevant question is whether the

category continues to preserve distinctions required by the relational claim for which it is used.

Proposition 5.168 (Category-revision discipline). *When a relational category is revised, the revision should document the target problem, the evidence of mismatch, the semantic change, the consequences for comparability, and any effects that the old and new categories may themselves have on subsequent relational behavior.*

This proposition prevents category revision from becoming an invisible preprocessing step. It also allows later inquiry to determine whether an observed change reflects transformation in the relational world, transformation in the classification scheme, or an interaction between the two.

5.20.6 Reflexive Observation

Observation can become reflexive when the process of observing, measuring, publishing, ranking, classifying, or explaining enters the relations under observation and changes their subsequent development. Section 5.6 treated performative representation, and Section 5.19 analyzed machine-mediated recursive classification. Reflexive observation places these phenomena inside the revision process: once an inquiry becomes consequential to its object, later evidence cannot be interpreted as if the observational relation remained external and unchanged.

Research on rankings documents how public measures can alter expectations and organizational behavior through reactivity (Espeland and Sauder 2007). MacKenzie similarly shows how financial models can become incorporated into market practice and participate in reshaping the processes they represent (MacKenzie 2006). These cases differ in mechanism, yet both illustrate a general epistemic possibility relevant to GRE: the observer, representation, or measurement infrastructure can become a participant in the relational configuration.

Definition 5.169 (Reflexive observation). *Reflexive observation* occurs when an observational, classificatory, representational, or explanatory practice becomes known, anticipated, or operationally consequential within the target system and thereby contributes to changes in the relations or manifestations subsequently observed.

Reflexive observation is more specific than ordinary measurement error. A noisy instrument can distort a reading without changing the underlying relation. Reflexivity requires a pathway through which the inquiry or its representation enters the relational process. That pathway can operate through anticipation, strategic adaptation, stigma, prestige, resource allocation, compliance, resistance, imitation, learning, or redesign of institutional routines.

Revision under reflexivity requires temporal segmentation. The investigator should distinguish observations produced before the inquiry became consequential, the period in which actors learned or responded to it, and the later configuration that incorporates those responses. Where possible, the analysis should also compare differently exposed actors or settings. Such strategies cannot fully restore an external observer position, but they can make the observer relation itself part of the evidential account.

Proposition 5.170 (Reflexive monitoring requirement). *When there is credible evi-*

dence that an inquiry, measure, classification, or published representation has become consequential to the target system, subsequent relational claims should include the observer–system relation among the candidate mechanisms shaping the observed configuration.

The proposition also applies to GRE itself. A relational map circulated among institutions, a generative-foreclosure diagnosis used in policy, or an AI-assisted classification adopted operationally can alter the object that subsequent GRE research encounters. Reflexivity therefore links epistemology to research ethics and governance without collapsing those domains into one another.

5.20.7 Preservation of Alternative Interpretations

Revisability depends on more than the ability to overwrite an earlier conclusion. Future revision requires access to the evidential and interpretive alternatives that made earlier judgment non-trivial. If rival explanations, negative findings, disputed categories, uncertainty ranges, or minority interpretations disappear from the research record, later inquiry may inherit an artificially narrowed possibility space. Preservation of alternatives is therefore part of epistemic infrastructure.

Longino's account of objectivity emphasizes the importance of effective criticism and plural perspectives in the production of scientific knowledge (Longino 1990). GRE translates that insight into a preservation requirement. Alternative interpretations need not receive equal weight, and implausible possibilities need not be retained indefinitely as live explanations. The relevant alternatives are those that were materially plausible given the evidence, represented consequential disagreements about category or mechanism, or preserve information needed to understand why a dominant interpretation prevailed.

Definition 5.171 (Alternative-interpretation record). An *alternative-interpretation record* is a preserved account of a materially plausible competing relational or generative interpretation, including its evidential basis, assumptions, boundary and scale, principal objections, current epistemic status, and conditions under which it could regain or lose warrant.

This record can include rejected hypotheses, unresolved anomalies, competing relation criteria, alternative historical periodizations, different boundary specifications, or interpretations supported by evidence that later became inaccessible. Preservation is particularly valuable when disagreement tracks unequal access to records or representational authority. An interpretation can be weakly institutionalized without being evidentially empty, and future evidence can change its status.

Alternative preservation should also record why an interpretation was set aside. A superseded model that failed because its mechanism contradicted evidence is different from one that lost favor because data were unavailable, categories changed, or research priorities shifted. This distinction allows later investigators to avoid rediscovering abandoned paths without understanding the reasons for abandonment.

Proposition 5.172 (Preservation condition for revisability). *Where competing interpretations are materially relevant to a relational or generative claim, a revisable knowl-*

edge system should preserve enough information about serious alternatives and their evidential status to permit later reassessment without reconstructing the earlier controversy solely from the interpretation that became dominant.

The requirement remains proportional. Routine exploratory alternatives need not all become permanent archival objects. Preservation becomes more important when the claim is consequential, uncertainty remains substantial, categories are historically unstable, evidence is difficult to reproduce, or alternative interpretations are linked to differently positioned observers whose access may later disappear.

Revisability therefore completes the chapter's movement from mediated access to disciplined epistemic maintenance. GRE treats relational knowledge as capable of strong warrant while remaining open to revision through new evidence, intervention, historical transformation, category change, reflexive observation, and preserved alternatives. The resulting epistemic stance avoids both premature closure and unrestricted indeterminacy. Claims are stabilized enough to support inquiry and action, yet their conditions of warrant, histories of change, and routes of correction remain visible. The next section consolidates this architecture by stating the principal epistemic commitments that GRE carries forward into methodology.

5.21 Epistemic Commitments of GRE

This section consolidates the epistemic architecture developed across this chapter into a set of commitments that can guide later research design without collapsing epistemology into method. Its role is synthetic: the preceding sections examined manifestation, trace, representation, observation, boundary, scale, relational identity, higher-order structure, fields, generative attribution, historical reconstruction, evidence, uncertainty, epistemic justice, machine mediation, and revisability as distinct problems. The present section states what GRE commits itself to after those problems are taken together. The objective is to make those commitments sufficiently explicit that later empirical, historical, interpretive, formal, and computational work can be assessed against them. The discussion proceeds through partial knowledge, explicit representation, explicit boundary and scale, explicit uncertainty, mechanism-sensitive attribution, evidential pluralism, historical sensitivity, revisability, and epistemic humility. The method is reconstructive rather than additive: each commitment summarizes an epistemic discipline already developed in the chapter and identifies the constraint it places on subsequent inquiry.

Definition 5.173 (Epistemic commitment of GRE). *An epistemic commitment of GRE is a standing discipline governing how relational and generative claims are formulated, warranted, delimited, represented, challenged, and revised under conditions of mediated access to relational reality.*

Definition 5.173 distinguishes an epistemic commitment from a preferred technique. A commitment such as explicit uncertainty can be satisfied through different research methods; conversely, the use of a sophisticated method does not by itself establish that the commitment has been satisfied. This distinction prepares the transition developed in the next section: epistemology specifies what a warranted claim requires, whereas methodology specifies how inquiry can be organized to meet those

requirements.

5.21.1 Partial Knowledge

GRE treats partiality as a normal condition of relational knowledge rather than as an exceptional defect. The argument follows from the architecture already established in this chapter. Manifestation does not transparently disclose relational reality; inverse inference is generally non-unique; traces depend on production and preservation; observation is position-dependent and incomplete; boundaries and scales delimit what enters the analytical field; higher-order structures and relational fields are reconstructed from partial evidence; and generative attribution adds further inferential burdens. A claim can therefore be warranted without being exhaustive.

Partial knowledge has several forms. *Observational partiality* concerns incomplete access to manifestations and traces. *representational partiality* concerns the selective preservation of some target properties over others. *relational partiality* concerns incomplete identification of relevant relations, participants, positions, or higher-order configurations. *historical partiality* concerns uneven preservation of past evidence and changing categories of interpretation. *generative partiality* concerns incomplete knowledge of mechanisms, mediating conditions, distributed contributors, delayed effects, or unrealized possibilities. These forms can coexist while differing in degree.

Definition 5.174 (Bounded epistemic warrant). *Bounded epistemic warrant* is support for a claim whose target, evidential basis, boundary, scale, temporal horizon, representation, and principal uncertainties are stated with enough precision that the claim can be used without implying knowledge beyond those conditions.

Bounded warrant permits GRE to make substantive claims while preserving the difference between local adequacy and epistemic completeness. A reconstruction of a labor relation, for example, can be well supported within a specified institution and period even when informal dependencies outside the observed boundary remain incompletely known. A field model can be useful for a defined target while leaving some relations latent. A generative attribution can identify a strongly supported contribution without exhausting every contributing condition.

Partiality therefore changes the grammar of strong claims. Strength comes from fit between evidence and target, explicit scope, discrimination among alternatives, and preserved uncertainty. It does not require a declaration that all relevant relations have been observed or that every competing explanation has been eliminated. This commitment protects GRE from converting the breadth of its relational vocabulary into an assumption of total relational visibility.

Proposition 5.175 (Partial knowledge and substantive inquiry). *Partial knowledge can support substantive relational and generative claims when the limits of access are characterized and the claimed warrant remains proportionate to the evidence available within those limits.*

Proposition 5.175 also places a burden on claims of absence. As Section 5.4 established, non-observation becomes evidence of absence only under sufficiently specified conditions of observability. Partial knowledge therefore does not merely counsel modest wording; it changes how positive, negative, and comparative claims are

warranted.

5.21.2 Explicit Representation

The second commitment concerns the visibility of representation. Section 5.5 distinguished relation, trace, symbolization, and representation, while Section 5.6 showed that representations can enter subsequent relational processes. GRE therefore treats representations as analytically consequential choices rather than transparent containers for already given reality.

An explicit representation identifies what is represented, which properties are preserved, which are compressed or omitted, what categories and symbols are used, and what inferential work the representation is expected to perform. The requirement applies to prose categories as much as to formal models. Terms such as “dependency,” “trust,” “capital,” “value,” “field,” or “constraint” already organize observation and comparison; a network diagram, regression specification, legal category, archival code, ethnographic typology, or machine-generated embedding adds further representational commitments.

Definition 5.176 (Representation declaration). A *representation declaration* is an explicit account of the target, representational form, principal selection or compression choices, relation between represented features and the underlying claim, and known conditions under which the representation can become misleading.

The purpose of a representation declaration is not to demand exhaustive philosophical commentary for every variable or diagram. Its function is proportional disclosure. A highly consequential classification requires more explanation than a conventional formatting choice. A representation that materially affects relational identification, comparison, causal attribution, or operational decisions requires enough documentation for readers to understand what the representation contributes to the inference.

Explicitness is especially important when formal precision exceeds epistemic precision. A numerical score can be reported to several decimal places while the mapping from the score to the relational target remains contestable. An algorithm can produce stable clusters while the substantive relation represented by those clusters remains uncertain. A legal category can be formally definite while incompletely describing enacted relations. GRE therefore distinguishes precision of representation from adequacy of representation.

The commitment also includes contestability. When a representation materially shapes what can be seen, classified, compared, or acted upon, affected interpretations should remain examinable against alternative representational choices where feasible. This follows from Sections 5.18 and 5.19: representational authority can be unequally distributed, and machine-generated representations can acquire operational power beyond their evidential warrant. Explicit representation creates the documentary basis on which such authority can be evaluated.

5.21.3 Explicit Boundary and Scale

Relational claims depend on what is included in the analytical field and at what level or resolution relations are reconstructed. Section 5.8 showed that boundaries are neither purely arbitrary analytical conveniences nor automatically given by the object. Some boundaries are institutionally, legally, materially, or organizationally consequential; others are research selections; many combine both properties. Scale similarly affects which relations, variations, and generative processes become visible.

GRE therefore requires boundary and scale to become part of the claim rather than remaining hidden in the research setup. A statement about a firm, household, platform, market, neighborhood, national economy, supply network, or historical period tacitly establishes inclusion and exclusion rules. The interpretation of a relation can change when external dependencies are included, when a local process is examined across a wider field, or when a long-run trajectory is reconstructed from observations that appeared stable at a shorter horizon.

Definition 5.177 (Scope specification). *A scope specification identifies the boundary, scale, temporal window, relevant level of aggregation or resolution, and principal cross-boundary or cross-scale dependencies under which a relational or generative claim is asserted.*

Scope specification serves at least three epistemic functions. First, it prevents local findings from acquiring an unearned universal range. Second, it makes cross-study disagreement more diagnosable: two studies may reach different conclusions because they study different relational neighborhoods or temporal scales rather than because one is methodologically defective. Third, it reveals when the research boundary excludes a mechanism necessary to explain the observed outcome.

Explicit scope does not imply that a single correct scale exists. Some relations are most intelligible at one level; others are produced through interactions across levels. Cross-scale claims therefore require evidence for the bridge by which local and distributed processes are connected. The commitment is to expose the scale logic of the argument and to avoid treating a convenient unit of observation as an ontologically privileged unit of explanation.

Proposition 5.178 (Scope-indexed warrant). *The warrant of a relational claim is indexed to the boundary, scale, and temporal horizon under which the relevant evidence and inferential relations hold; portability beyond that scope requires additional justification.*

5.21.4 Explicit Uncertainty

The fourth commitment follows from Section 5.17. GRE treats uncertainty as structured information about the state of an inquiry. Uncertainty can concern relational existence, manifestation, generative attribution, boundary choice, scale, temporal persistence, model specification, or the relative support of competing explanations. Compressing these dimensions into a single confidence statement can obscure where the argument is strong and where it remains vulnerable.

An uncertainty statement should therefore identify the dimension to which uncertainty belongs. A relation may be well evidenced while its mechanism remains

uncertain. A mechanism may be plausible while the magnitude or temporal reach of its effect remains weakly identified. A historical trajectory may be strongly reconstructed over one interval while continuity across an archival gap remains uncertain. A computational model may predict reliably while its relational interpretation remains contested. These are different epistemic states and should be reported differently.

Definition 5.179 (Uncertainty disclosure). An *uncertainty disclosure* identifies the principal dimensions of a claim whose warrant remains incomplete, contested, model-dependent, boundary-sensitive, scale-sensitive, temporally limited, or dependent on unresolved alternatives, together with the evidence that could materially change that assessment where such evidence can be specified.

Explicit uncertainty supports comparison and revision. It allows later evidence to address the uncertain part of a claim without requiring wholesale reassessment of everything already supported. It also reduces the risk that technical sophistication masks unresolved inferential assumptions. A simulation, causal model, or machine-learning system can quantify one source of uncertainty while leaving classification, representation, boundary, or historical uncertainty largely outside the formal calculation.

GRE consequently favors profiled confidence over rhetorical certainty. This orientation permits strong localized conclusions where the evidence supports them and visible restraint where the evidential architecture remains incomplete. Uncertainty is therefore neither a disclaimer appended to an otherwise finished argument nor a reason to suspend inquiry indefinitely. It is part of the content of the claim.

5.21.5 Mechanism-Sensitive Attribution

Generative claims create a stronger epistemic burden than claims that a relation exists or covaries with an outcome. Sections 5.12 and 5.13 distinguished generative effects from ordinary state differences and developed a plural architecture for attribution involving mechanisms, counterfactual dependence, multiple causation, distributed generation, delayed effects, branching pathways, and emergence. GRE therefore commits to mechanism-sensitive attribution without making one mechanism template a universal criterion.

Mechanism sensitivity asks how the attributed contribution is carried, enabled, mediated, reproduced, or transformed through the relevant relations. In some cases the mechanism can be specified through organizational rules, legal obligations, material flows, incentives, communication pathways, or institutional procedures. In other cases the contribution is distributed across a configuration, making a single linear mechanism an inadequate description. The epistemic requirement is to identify the structure of dependence that makes the generative claim stronger than observed association.

Proposition 5.180 (Mechanism-sensitive generative attribution). A *generative attribution gains warrant when evidence identifies a credible pathway or dependency structure linking the candidate relation or configuration to the target effect, while relevant alternative explanations, boundary conditions, temporal ordering, and distributed contributions are examined in forms appropriate to the case.*

This proposition deliberately allows different combinations of evidence. Interventionist evidence can be powerful where meaningful interventions are possible. Historical sequence can identify enabling conditions or branching processes that cannot be experimentally recreated. Ethnographic or documentary evidence can reveal mechanisms hidden from aggregate data. Longitudinal or computational analysis can identify temporal dependencies or distributed patterns requiring further interpretation. The commitment therefore links mechanism sensitivity to evidential pluralism rather than to methodological monism.

Mechanism-sensitive attribution also requires restraint in normative interpretation. A relation can generate an effect without that effect being desirable, just, efficient, or valuable. Conversely, a desirable outcome does not establish the mechanism that produced it. GRE keeps causal-generative attribution distinct from normative evaluation so that later analyses of value, power, justice, capture, return, and governance can build on rather than substitute for the epistemic argument.

5.21.6 Evidential Pluralism

Sections 5.15 and 5.16 established that relational and generative claims can draw on documentary, administrative, legal, behavioral, testimonial, ethnographic, network, longitudinal, historical, formal, and computational evidence. These evidence forms differ in what they record, how they are produced, whose position they encode, what they systematically omit, and how directly they bear on the target. GRE therefore commits to evidential pluralism in the sense of target-relative complementarity rather than indiscriminate accumulation.

Definition 5.181 (Evidential pluralism in GRE). *Evidential pluralism in GRE* is the disciplined use and comparison of heterogeneous evidence forms according to the relational or generative property each can support, their trace-production conditions, characteristic omissions or distortions, and the way convergence or disagreement among them affects the target claim.

Pluralism does not imply that every research project should use every method or that every evidence type receives equal weight. A legal document can strongly evidence formal obligation while weakly evidencing enacted intensity. Behavioral data can reveal patterned interaction while underdetermining meaning. Testimony can provide access to experience or interpretation unavailable in administrative records while remaining situated and potentially contested. Network data can reveal recorded ties while depending on how edges were defined and observed. Computational inference can extend analytical reach while adding model and category dependence.

The epistemic task is therefore to match evidence to claim. Heterogeneous evidence becomes especially valuable when distinct trace-production processes converge on the same relational proposition or when disagreement reveals that apparently identical measures refer to different dimensions of the relation. Evidential pluralism also protects against systematic blindness produced by a single archive, administrative regime, platform, expert vocabulary, or machine classification system.

This commitment has a justice dimension without collapsing evidential standards into representational inclusion. Section 5.18 showed that actors differ in their ca-

capacity to create durable records, institutional categories, expert interpretations, and machine-readable traces. Evidential pluralism can widen the space of epistemic participation, while warrant still depends on what each source can support and how conflicts among sources are adjudicated.

5.21.7 Historical Sensitivity

GRE treats relations, categories, manifestation processes, evidential infrastructures, and standards of interpretation as historically situated. Section 5.14 developed relational trajectories, generative trajectories, path dependence, genealogical reconstruction, historically constrained counterfactuals, and historical re-epistemization. Historical sensitivity therefore concerns more than adding chronological background to a contemporary analysis.

A historically sensitive relational claim asks whether the relation itself persisted, whether its participants or constitutive rules changed, whether the same manifestation carried the same significance at different times, whether evidence was preserved under comparable conditions, and whether present categories distort earlier forms. A historically sensitive generative claim additionally asks whether the mechanism, opportunity structure, and feasible alternatives changed across the trajectory.

Definition 5.182 (Historical index of a relational claim). The *historical index of a relational claim* specifies the period, trajectory, category regime, evidential conditions, and relevant path-dependent or transformative processes under which the claim is warranted.

Historical sensitivity also limits retrospective determinism. Later consequences can change the significance attributed to earlier events without proving that those consequences were already inevitable or fully contained in the earlier configuration. Historical counterfactuals should therefore be constrained by possibilities evidenced for the period rather than by alternatives that became available only later.

This commitment has implications for concepts used throughout GRE. Terms such as capital, market, household, labor, ownership, value, organization, expertise, and technology may preserve partial continuity across periods while their relational constitution changes. The framework should therefore remain capable of tracing continuity and transformation together. Historical sensitivity prevents a stable label from being mistaken for an unchanged relation.

5.21.8 Revisability

The eighth commitment synthesizes Section 5.20. GRE treats relational knowledge as revisable because new evidence can emerge, interventions can transform the object, historical conditions can change, categories can fail, observation can become reflexive, and previously weak alternatives can acquire stronger support. Revisability is consequently part of responsible stabilization rather than its opposite.

A revisable claim preserves enough provenance for later investigators to reconstruct how its warrant was established. This includes the evidence available at the time, category definitions, boundary and scale choices, major representational decisions, uncertainty profile, competing interpretations, and reasons for selecting the

working conclusion. Versioning is therefore epistemic infrastructure. When a claim changes, the change should be traceable to the dimension that changed.

Definition 5.183 (Revision architecture). A *revision architecture* is the set of documentary, procedural, and interpretive practices through which a relational or generative claim can be reassessed, updated, qualified, replaced, or retired while preserving the provenance of earlier formulations and the reasons for change.

Revisability includes contestability. A claim that cannot be meaningfully challenged because its data, categories, model assumptions, representational choices, or inferential path are inaccessible has a weaker revision architecture even when its current performance appears strong. This point is particularly important for machine-mediated inquiry, where analytical capability can expand faster than human capacity to reconstruct every internal operation. Contestability can therefore be supported through provenance, documentation, comparative testing, alternative representations, appeal procedures, or other mechanisms appropriate to the domain, without requiring that every technical process be reducible to a simple human-readable explanation.

Revisability also supports cumulative inquiry. Earlier claims can remain useful even when later work narrows their scope or revises their categories. Preservation of serious alternative interpretations allows future researchers to distinguish evidence that genuinely resolved a disagreement from conditions that merely made one interpretation institutionally dominant. GRE therefore treats the history of its own knowledge claims as part of the material needed for future research.

5.21.9 Epistemic Humility

The final commitment concerns the relation between theoretical ambition and epistemic limitation. GRE is intentionally ambitious in analytical scope: it seeks to study heterogeneous relations, distributed fields, generative processes, historical trajectories, multiple forms of value, power over relational and conversion structures, and transformations of future possibilities. Such breadth can create a temptation to treat the relational vocabulary as if it offered a privileged view from which the system can be comprehensively known. The epistemology developed in this chapter rejects that inference.

Definition 5.184 (Epistemic humility in GRE). *Epistemic humility in GRE* is the disciplined refusal to infer epistemic mastery from theoretical scope, formal sophistication, computational capacity, evidential volume, institutional authority, or temporary explanatory success, together with the positive practice of stating what is known, how it is known, where its warrant ends, and what could revise it.

Epistemic humility differs from generalized skepticism. It permits strong claims where evidence, scope, mechanism, and alternatives support them. It also permits action under incomplete knowledge. Its function is to prevent a successful analytical framework from erasing the mediations through which its own claims are produced. Ontological suspension remains relevant here: GRE can investigate relational reality rigorously without converting analytical usefulness into a final metaphysical settlement.

Humility also has an institutional dimension. Analysts, experts, organizations, and computational systems can occupy positions that make their representations unusually consequential. The resulting authority does not automatically track epistemic adequacy. Sections 5.18 and 5.19 therefore make contestability, provenance, unequal evidence-production capacity, and representational authority part of the epistemic architecture. Humility requires institutional arrangements capable of receiving correction, especially from positions whose evidence is weakly represented by dominant recording systems.

The commitments developed in this section can now be stated together.

Proposition 5.185 (Minimum epistemic commitments of GRE). *A GRE inquiry that advances relational or generative knowledge should ordinarily make visible, in forms proportionate to the claim and research context:*

1. *the partial character and evidential reach of the knowledge claim;*
2. *the principal representational choices through which the target is rendered;*
3. *the boundary, scale, and temporal scope under which the claim is asserted;*
4. *the principal dimensions of uncertainty and live competing explanations;*
5. *the mechanism or dependency structure supporting stronger generative attribution where such attribution is made;*
6. *the relation between heterogeneous evidence forms and the target they support;*
7. *the historical conditions under which relations, categories, evidence, and mechanisms are interpreted;*
8. *the provenance and contestability required for later revision; and*
9. *the limits beyond which the inquiry does not claim epistemic mastery.*

Proposition 5.185 is intentionally methodological only in the broad sense of disciplining inquiry. It does not select a research technique. The same commitment can be satisfied differently in archival research, ethnography, survey analysis, network analysis, causal inference, formal modeling, computational simulation, or mixed research designs. The proposition therefore completes the epistemic synthesis while leaving the design problem open.

GRE can remain ambitious under these commitments because ambition concerns the range and depth of questions that inquiry is prepared to pursue, whereas epistemic mastery concerns an unwarranted claim to complete, final, or position-independent knowledge. The framework can investigate distributed generativity, historical transformation, heterogeneous value, and complex relational fields while continuing to expose its boundaries, mediations, uncertainties, and revision conditions. This combination of analytical reach and disciplined limitation provides the immediate basis for the transition from epistemology to methodology developed in the next section.

5.22 From Epistemology to Methodology

This section closes the epistemic argument of this chapter by translating the chapter's commitments into requirements for subsequent research design. Its role is tran-

sitional: the preceding sections established what must be made explicit before GRE can claim warranted knowledge of relations, fields, trajectories, and generative effects; the next Part must determine how inquiry can be organized so that those epistemic requirements are practically addressed. The objective here is therefore neither to prescribe a universal research procedure nor to introduce a new methodological canon. It is to identify the bridge between epistemic warrant and research practice. The discussion proceeds from the difference between knowing relations and studying relations, to epistemic requirements for design, claim-sensitive method selection, the additional burden associated with generative claims, and the transition toward relational methodology. The method is architectural: each subsection converts an epistemic commitment already developed in this chapter into a design question that later methodological chapters can answer with domain-appropriate techniques.

Definition 5.186 (Epistemic design translation). An *epistemic design translation* is the process of converting a condition of warranted knowledge into an explicit requirement for research design, evidence production, analysis, documentation, or revision.

Definition 5.186 establishes the limited function of this section. Epistemology identifies what would make a claim defensible; methodology organizes inquiry so that evidence and analysis capable of supporting, qualifying, or revising that claim can be produced. The relation is therefore constraining without being mechanically deductive. A commitment to explicit boundary and scale does not determine one technique, yet it does require the design to identify the relevant inclusion rules, levels of analysis, and conditions of cross-scale inference. A commitment to revisability does not select one empirical method, yet it does require provenance, versioning, preserved alternatives, and procedures through which later evidence can alter earlier claims.

5.22.1 Knowing Relations and Studying Relations

Knowledge of a relation and the practical study of that relation belong to different analytical levels. The first concerns warrant: what supports the claim that a relation exists, persists, changes, or contributes generatively to a target outcome. The second concerns research organization: how observations are obtained, how evidence is produced or recovered, how participants and archives are accessed, how representations are constructed, how comparison is performed, and how alternative explanations are evaluated. The chapter has concentrated on the first level while repeatedly exposing its dependence on the second.

The distinction matters because epistemic adequacy cannot be inferred directly from the label of a method. An interview may provide strong evidence concerning lived dependency, weak evidence concerning population prevalence, and indirect evidence concerning institutional structure. A network dataset may represent a recorded interaction pattern with considerable precision while leaving the substantive meaning of an edge uncertain. An archival series can support reconstruction of institutional categories and historical sequences while preserving systematic silences. A computational model can clarify consistency, sensitivity, or possible dynamics without independently establishing that its represented relations correspond to the empirical world. Method names therefore provide limited information until the target claim and evidential pathway are specified.

Conversely, the same epistemic target can often be approached through different research practices. A relation of dependence may be reconstructed through contractual records, administrative data, longitudinal behavior, testimony, ethnography, historical comparison, or combinations of these materials. Each route reveals some aspects of the target and obscures others. The methodological problem is consequently one of fit between a claim, the traces through which that claim may become accessible, and the procedures capable of recovering and interpreting those traces.

Definition 5.187 (Relational study design). *A relational study design is an organized set of procedures for identifying, observing, reconstructing, comparing, representing, and revising claims about specified relations under explicit conditions of access, boundary, scale, temporality, and uncertainty.*

The definition treats research design as more than a sequence of techniques. It includes the conditions through which a relation becomes researchable. Access arrangements, category construction, sampling rules, temporal windows, field boundaries, recording systems, legal classifications, machine infrastructures, and researcher position can all affect what becomes visible. Sections 5.7, 5.8, 5.18, and 5.19 established that such conditions can shape both evidence and the object represented by that evidence. Studying relations therefore requires methodological attention to the production conditions of knowledge itself.

The distinction between knowing and studying also protects GRE from a common shortcut: moving directly from an epistemic ideal to a preferred technique. A commitment to evidential pluralism, for example, does not imply that every project should use mixed methods. It means that no evidential form receives unconditional authority and that the selection of evidence should be justified relative to the claim. A narrowly designed study can be epistemically strong when its claim is correspondingly narrow and its evidential fit is clear. A methodologically elaborate project can remain epistemically weak when the relation between evidence and target is poorly specified.

Proposition 5.188 (Claim–study distinction). *The epistemic strength of a relational claim depends on the fit between the target, the evidence generated or recovered by the study, and the inferential pathway connecting them; it cannot be determined from the nominal sophistication or plurality of methods alone.*

Proposition 5.188 provides the first bridge into methodology. The next methodological chapters must therefore ask how research practices produce evidence of different kinds, which claims those practices can support, and how their characteristic blind spots can be made visible.

5.22.2 Epistemic Requirements for Research Design

The epistemic commitments consolidated in Section 5.21 become design requirements when inquiry begins to specify an object, field site, archive, dataset, model, comparison, or intervention. These requirements do not form a universal checklist whose every item must receive equal attention. Their relevance and burden vary with the strength and consequence of the intended claim. They nevertheless provide a common architecture for asking whether a design can generate evidence proportionate to its epistemic ambitions.

Partial knowledge becomes a requirement to characterize access conditions and likely blind spots. A design should identify what can be observed directly, what must be inferred, what remains latent or hidden, and which absences are interpretable. Explicit representation becomes a requirement to document how relations are encoded in concepts, categories, variables, graphs, narratives, legal forms, or computational objects. Boundary and scale commitments become requirements to state the domain of inclusion and the level at which claims are expected to hold. Uncertainty becomes a requirement to preserve dimensions of doubt rather than compress them prematurely into a single confidence statement.

Mechanism-sensitive attribution becomes a requirement to collect or reconstruct evidence capable of discriminating among plausible generative pathways when stronger causal or generative claims are intended. Evidential pluralism becomes a requirement to justify why a particular evidence form is suitable for the target and, where consequential uncertainty remains, whether complementary evidence could materially improve discrimination. Historical sensitivity becomes a requirement to reconstruct the categories, opportunity structures, institutions, and trace-production systems relevant to the period rather than projecting contemporary classifications backward without qualification. Revisability becomes a requirement for provenance, versioning, preserved assumptions, and explicit triggers for revision. Epistemic humility becomes a requirement to state the limits of portability, generalization, and authority.

These translations can be summarized through the notion of an epistemic design requirement.

Definition 5.189 (Epistemic design requirement). *An epistemic design requirement is a research-design condition introduced because failure to satisfy it would materially weaken the warrant, scope control, interpretability, contestability, or revisability of the intended claim.*

The definition makes proportionality central. A study of a narrowly bounded documentary relation may require relatively modest procedures for cross-scale inference, while a claim concerning distributed generative transformation across institutions requires considerably stronger boundary, temporal, and mechanism design. The burden should therefore rise with inferential depth, scope, and consequence.

Research design also involves prioritization. Some epistemic problems can be reduced by obtaining more observations or recovering missing records. Others arise from category ambiguity, strategic concealment, historical discontinuity, observer effects, or underdetermination among relational explanations. Additional data can sometimes leave these problems intact. The appropriate response may require interpretive reconstruction, comparative design, historical analysis, participant engagement, category revision, sensitivity analysis, or preservation of competing explanations. Methodology therefore begins by diagnosing the type of epistemic limitation before selecting the procedure intended to address it.

Proposition 5.190 (Design proportionality). *The methodological burden of a GRE study should be proportionate to the inferential distance, scope, temporal reach, and practical consequence of the claims it intends to support.*

Proposition 5.190 prevents two symmetrical distortions. One is methodological

minimalism applied to claims whose generative, historical, or cross-scale scope exceeds the evidence produced by the design. The other is unnecessary methodological expansion in studies whose claims are deliberately local and modest. GRE methodology should be demanding where the claim requires it and economical where additional complexity would not materially improve warrant.

The design stage should therefore make explicit at least four elements: the intended epistemic target; the principal evidence pathway; the major conditions under which that pathway can fail; and the procedures through which those failures will be detected, bounded, or preserved as uncertainty. Later chapters can operationalize these elements differently across qualitative, historical, quantitative, formal, computational, and mixed research practices.

5.22.3 Relational Identification and Method Selection

Method selection follows from the structure of the relational claim rather than from allegiance to a disciplinary technique. The first methodological question is therefore not which method GRE prefers, but what kind of relation the study seeks to identify and which properties of that relation matter for the research problem. A relation may be legal, material, behavioral, symbolic, institutional, affective, informational, ecological, spatial, computational, or composite. Its evidential traces can differ accordingly.

Relational identification usually contains several separable tasks. The study may need to identify the relevant relata; determine whether the relation is direct, mediated, nested, collective, multilayer, or field-like; specify the temporal criterion under which the relation is treated as persistent; reconstruct its boundary and scale; characterize its directionality or symmetry; and determine whether observed manifestations are sufficient to distinguish it from plausible alternatives. The appropriate methods depend on which of these tasks carry the main uncertainty.

A study concerned with formal legal relations may require close analysis of legal texts, institutional procedure, and enacted practice because formal validity and relational realization can diverge. A study concerned with informal care or dependency may require testimony, observation, and contextual interpretation because administrative traces can remain sparse. A study concerned with network structure may require relational data and explicit edge semantics, while a study concerned with historical persistence may require longitudinal or archival reconstruction. A study concerned with a relational field may require evidence that local positions derive meaning or constraint from distributed organization rather than merely from pairwise density.

Definition 5.191 (Claim–method fit). *Claim–method fit* is the degree to which a research procedure can generate, recover, or interpret evidence that bears directly on the properties required to identify and delimit the intended relational claim.

Claim–method fit is therefore property-specific. A method can fit one component of a claim while remaining weak for another. Longitudinal administrative data can support persistence and sequence, yet offer limited access to unrecorded meaning. Ethnographic observation can clarify relational enactment and category mismatch, yet require additional design for population-level prevalence. Formal models can ex-

pose structural consequences of assumptions while requiring independent evidence for empirical correspondence. Method selection should make such asymmetries explicit rather than allowing the prestige or familiarity of a method to substitute for evidential fit.

This orientation also clarifies the role of methodological pluralism. Pluralism does not require the accumulation of methods for its own sake. It permits the research design to combine methods when distinct properties of the claim require different forms of access. Complementarity is strongest when each method addresses a known weakness in another evidential pathway or when heterogeneous evidence provides independent support for a relational reconstruction. Redundancy without additional discrimination can increase workload without materially increasing warrant.

A further implication concerns sequencing. Method selection can be iterative. Early qualitative or historical inquiry can reveal categories that later support structured measurement; computational exploration can identify patterns requiring interpretive reconstruction; formal analysis can expose assumptions whose empirical plausibility must then be examined; surprising field observations can trigger revision of a sampling frame or relational boundary. GRE therefore permits research designs in which identification and method selection co-evolve as evidence clarifies the object under study.

Proposition 5.192 (Relational method selection). *Method selection in relational inquiry should be justified by the properties required to identify, delimit, compare, or reconstruct the target relation, together with the characteristic access conditions and failure modes of the proposed evidential pathway.*

Proposition 5.192 prepares the more detailed relational methodologies developed later. Those methodologies can organize concrete procedures around relational identification without treating one technique as the defining method of GRE.

5.22.4 Generative Claims and Methodological Burden

Generative claims place a heavier methodological burden on inquiry because they extend beyond relational identification toward claims about contribution to emergence, transformation, persistence, suppression, capacity formation, structural change, or alteration of future possibilities. Sections 5.12, 5.13, and 5.16 established that generativity must be specified relative to a target and supported through evidence connecting a relation or configuration to the claimed effect.

The methodological burden has several layers. First, the underlying relation must be identified with adequate warrant. Second, the target change must itself be evidenced. Third, the design must support an inferential pathway linking the relation to that change. Fourth, materially relevant alternatives, enabling conditions, mediators, and distributed contributors must be considered at a level proportionate to the strength of the attribution. Fifth, delayed and branching effects require temporal design capable of tracing carriers across stages rather than relying exclusively on immediate association.

These layers mean that a generative study may require different forms of evidence at different points in the argument. The existence of a relation may be documented

through contracts or network data; its mechanism may require process evidence; its downstream effect may require longitudinal observation; its historical conditions may require archival reconstruction; its distributed contribution may require comparative or configurational analysis; its transformation of future possibilities may require reconstruction of feasible alternatives rather than observation of realized outcomes alone. No single evidential form receives automatic priority across these tasks.

Definition 5.193 (Generative methodological burden). The *generative methodological burden* is the additional design and evidential work required to move from a warranted claim that a relation or configuration exists to a warranted claim that it contributed to a specified generative effect under stated boundary, scale, temporal, and uncertainty conditions.

The burden is graded. A modest claim that a relation plausibly enabled a capacity can require less evidential discrimination than a claim that the same relation transformed an institutional structure or altered the conditions through which later transformation occurred. The vocabulary of attribution should track this difference. Terms such as association, contribution, mediation, enabling, constraint, co-generation, and stronger generative attribution can express different levels of warranted commitment.

Generative inquiry also increases the importance of negative and unrealized evidence. Foreclosure concerns possibilities that became less accessible; path dependence concerns alternatives whose feasibility changed through sequence; capacity formation concerns what actors or systems became able to do; second-order generative change concerns altered conditions of later transformation. Research design therefore cannot rely only on directly realized outcomes. It may need historical comparison, counterfactual discipline, institutional analysis, process tracing, or formal representation of feasible states, with the evidential status of each construction kept explicit.

The methodological problem is especially demanding where generation is distributed. An outcome can depend on background infrastructure, maintenance, tacit coordination, legal conditions, ecological supports, technical systems, and contributions whose traces are unequally preserved. Section 5.18 showed that unequal evidence-production capacity can distort generative attribution. A methodology adequate to distributed generation should therefore examine whether evidential visibility is itself patterned by power, institutional legibility, or recording infrastructure.

Proposition 5.194 (Escalating burden of generative inference). *As a claim moves from relational identification toward stronger claims about downstream, distributed, historical, structural, or second-order generative effects, the methodological burden ordinarily increases because additional mediating conditions, alternative pathways, temporal carriers, and boundary assumptions become relevant to warrant.*

Proposition 5.194 supplies the principal reason for maintaining a distinction between methodologies of relational inquiry and methodologies of generative relational inquiry. The two remain connected: generative inquiry presupposes relational identification. Their methodological burdens differ because the latter adds a further

inferential layer concerning transformation through time.

5.22.5 Transition to Relational Methodology

The epistemic work of this chapter establishes the conditions under which GRE can move responsibly from manifestation and trace toward claims about relations and generative processes. The next Part must convert those conditions into research praxis and concrete methodological architectures. The transition therefore begins with a change in question. The chapter has asked what warrants a relational or generative claim. Subsequent chapters ask how inquiry should be organized when researchers must produce, recover, interpret, compare, formalize, and revise the evidence needed for such warrant.

The first consequence is that methodology must remain relationally reflexive. Research is conducted through relations: between researchers and participants, analysts and institutions, observers and instruments, archives and classification systems, computational systems and datasets, funders and research agendas, disciplines and objects of study. These relations can shape access, categories, recording, interpretation, and the practical effects of published representations. The epistemic concerns developed in Sections 5.6, 5.7, 5.18, 5.19, and 5.20 therefore become problems of research praxis as well as abstract epistemology.

The second consequence is methodological pluralism with explicit division of labor. Historical and interpretive inquiry can recover categories, meanings, trajectories, and institutional transformations that sparse contemporary datasets cannot preserve. Qualitative inquiry can examine enacted relations, observer position, category mismatch, and weakly institutionalized experience. Quantitative inquiry can estimate distributions, comparisons, temporal patterns, and conditional associations where measurement is appropriate. Formal and computational inquiry can clarify structural implications, generate counterfactual scenarios, test sensitivity, and represent complex relational configurations. Their value depends on the target and their relation to evidence; none receives automatic jurisdiction over the others.

The third consequence is that methodology must preserve the distinction between relational and generative questions. Relational methodology concerns how relations are identified, reconstructed, represented, compared, and situated within boundaries and fields. Generative relational methodology adds the study of how such relations contribute to the production or transformation of capacities, relations, structures, constraints, values, trajectories, and future possibilities. Integrated designs can combine the two, yet their inferential steps should remain visible so that evidence for relational existence is not mistaken for evidence of generative effect.

The fourth consequence is that research design itself becomes revisable. If categories change, access conditions shift, interventions transform the field, new evidence appears, or machine-mediated observation alters what can be seen, then the design may require modification. A methodology aligned with Section 5.20 therefore includes procedures for documenting such revision rather than treating the initial research plan as epistemically privileged simply because it came first.

These consequences can be consolidated into one transition principle.

Proposition 5.195 (Epistemology-to-methodology transition). *GRE methodology should organize research practices so that the evidential, representational, historical, relational, generative, justice-sensitive, and revisability requirements established in this chapter become explicit design problems rather than implicit assumptions of analysis.*

Proposition 5.195 marks the endpoint of the chapter's argument. Ontological suspension in the preceding chapter prevented analytical representations from being mistaken for final ontology. The present chapter has added a corresponding epistemic discipline: manifestations, traces, representations, observations, models, and historical records provide mediated and partial access whose warrant depends on boundary, scale, evidence, attribution, uncertainty, justice, provenance, and revisability. The methodological chapters that follow can now proceed without requiring epistemic certainty and without treating technical procedure as a substitute for epistemic justification.

The resulting research programme begins from a demanding but workable position. GRE may study complex relational systems under incomplete knowledge; it may combine heterogeneous evidence; it may use historical, interpretive, quantitative, formal, and computational resources; and it may advance generative explanations where the evidential burden is met. At the same time, each design remains answerable to the limits of its access and representation. Methodology therefore becomes the practical organization of disciplined inquiry under the epistemic conditions established here.

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